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Xiaochang Wang

Lecture Notes in Real Analysis

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Lecture Notes in Real Analysis

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Preface

This book is the collection of my lecture notes for the two-semester graduate level course, Real Analysis, at Texas Tech University. The students are supposed to have an intermediate analysis course, something like W. Rudin's *Principles of Mathematical Analysis* or R. Goldberg's *Methods of Real Analysis*, before taking this course.

I have used both H.L. Royden's *Real Analysis*, 2nd edition, and G.B. Folland's *Real Analysis, Modern Techniques and Their Applications*, 2nd edition, as textbooks for my lectures. I like Royden's approach; the book is quite easy to read. However the measure and integration theories are developed twice in Royden's book using different approaches, first time in Part 1 for Lebesgue measure and integration, and second time in Part 3 for general measures and integrations. When I teach from Royden's book, I combine Part 1 and Part 3, and use Part 3's approach also for Lebesgue measure and integration. I like Folland's framework. A unified approach for all measures and integrations is used. But most students feel that Folland's book is little harder to read. My lecture notes basically follow Folland's framework and borrow many approaches from Royden's book.

Many students have a question in their mind: Why do we need a new fancy way to define integrals? In order to help students to see that the Lebesgue measure and integration, and therefore the general measure theory, come naturally from the process of fixing the flaws of Riemann integrals, many no-traditional interpretations of myself are used.

► Chapters 1–3 are covered in the first semester and ► Chapters 4–6 are covered in the second semester. There is plenty of time to cover the subjects in details and to have exercise sessions. Many of the included exercise problems are modified from Folland's book, with additional steps added to guide students through. I also include many problems from the Ph.D. Preliminary Examinations here at Texas Tech made by my colleagues and myself.

Lubbock, TX, USA
June 2018

Xiaochang Wang

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Prologue

Notations:

$\mathbb{N}$ =the set of natural numbers
$\mathbb{Z}$ =the set of integers
$\mathbb{Q}$ =the set of rational numbers
$\mathbb{R}$ =the set of real numbers
$\mathbb{C}$ =the set of complex numbers
iff: if and only if
$\forall$: for all
$\exists$: there exists
$\in$: belongs to (it is a relation between an element and a set)
$\subset$: is contained in (it is a relation between two sets)
$\bigcup$: union
$\bigcap$: intersection
$E \setminus F = \{x : x \in E \text{ and } x \notin F\}$
$E \triangle F = (E \setminus F) \cup (F \setminus E)$ (symmetric difference of E and F)
Given a universal set X, $E^c = X \setminus E$ (the complement of E)

de Morgan's laws: For any E and $\{E_\alpha : \alpha \in A\}$ in X

$$E \setminus \left(\bigcup_{\alpha \in A} E_\alpha \right) = \bigcap_{\alpha \in A} (E \setminus E_\alpha),$$

$$E \setminus \left(\bigcap_{\alpha \in A} E_\alpha \right) = \bigcup_{\alpha \in A} (E \setminus E_\alpha).$$

In particular,

$$\left(\bigcup_{\alpha \in A} E_\alpha \right)^c = \bigcap_{\alpha \in A} E_\alpha^c,$$

$$\left(\bigcap_{\alpha \in A} E_\alpha \right)^c = \bigcup_{\alpha \in A} E_\alpha^c.$$

Cartesian product:

$$X \times Y = \{(x, y) : x \in X,\ y \in Y\}.$$

Cardinality:
For a finite set E:

$$\text{card}(E) = \text{the number of elements in } E;$$

For infinite sets X and Y, $\text{card}(X) = \text{card}(Y)$ if and only if there is a one-to-one onto map between the sets. If there is a one-to-one map from X into Y, then $\text{card}(X) \leq \text{card}(Y)$, and if there is an onto map from X to Y, then $\text{card}(X) \geq \text{card}(Y)$. $\text{card}(X) < \text{card}(Y)$ if and only if $\text{card}(X) \leq \text{card}(Y)$ and $\text{card}(X) \neq \text{card}(Y)$.

For a given set X, the power set $\mathscr{P}(X)$ of X is the collection of all subsets of X, i.e.

$$\mathscr{P}(X) = \{E : E \subset X\}.$$

Proposition 0.0.1

$$card(X) < card(\mathscr{P}(X)).$$

Proof
Certainly

$$\text{card}(X) \leq \text{card}\ (\mathscr{P}(X)).$$

If $\text{card}(X) = \text{card}\ (\mathscr{P}(X))$, then there is a one-to-one onto map $g : X \to \mathscr{P}(X)$. Define

$$Y = \{x \in X : x \notin g(x)\}.$$

Then $Y \in \mathscr{P}(X)$. So there is an x_0 such that $Y = g(x_0)$.

If $x_0 \in Y$, then by the definition of Y we have $x_0 \notin g(x_0) = Y$. Conversely if $x_0 \notin Y$, then by the definition of Y we have $x_0 \in g(x_0) = Y$. Both cases give us contradiction. Therefore such g does not exist and $\text{card}(X) < \text{card}(\mathscr{P}(X))$. □

Definition 0.0.2

$$c_0 = \text{card}\ (\mathbb{N}).$$

$$c = \text{card}\ (\mathbb{R}).$$

Definition 0.0.3

A set X is called countable if

$$\text{card}\,(X) \leq c_0.$$

In other words, a set is countable if one can "count" all the elements of the set one by one.

Proposition 0.0.4

a) If both X and Y are countable then $X \times Y$ is countable (i.e., a finite product of countable sets is countable.)

b) If X_i, $i = 1, 2, \ldots,$ are countable, then $\bigcup_{i=1}^{\infty} X_i$ is countable (i.e., a countable union of countable sets is countable.)

Proof

a) Let $X = \{x_1, x_2, \ldots,\}$ and $Y = \{y_1, y_2, \ldots,\}$. Then all the elements of $X \times Y$ can be counted in the following way:

$$\begin{array}{ccccc}
(x_1, y_1) & \rightarrow & (x_1, y_2) & & (x_1, y_3) \cdots \\
 & \swarrow & & \nearrow & \\
(x_2, y_1) & & (x_2, y_2) & \cdots & \\
\downarrow & \nearrow & & & \\
(x_2, y_1) & \cdots & & & \\
\vdots & & & &
\end{array}$$

b) Let $X_i = \{x_{i1}, x_{i2}, \ldots,\}$. Then all the elements of $\bigcup_{i=1}^{\infty} X_i$ can be counted in the following way and skip any element which is counted before

$$\begin{array}{ccccc}
x_{11} & \rightarrow & x_{12} & & x_{13} \cdots \\
 & \swarrow & & \nearrow & \\
x_{21} & & x_{22} & \cdots & \\
\downarrow & \nearrow & & & \\
x_{31} & \cdots & & & \\
\vdots & & & &
\end{array}$$

□

Corollary 0.0.5 *$\mathbb{Z}$ and $\mathbb{Q}$ are countable.*

Proof

$\mathbb{Z} = -\mathbb{N} \cup \{0\} \cup \mathbb{N}$ is a finite union of countable sets, so is countable.

For any number $x \in \mathbb{Q}$ there are unique integers $p, q, q > 0$, and $(p, q) = 1$, such that $x = \frac{p}{q}$. The map $\phi : \mathbb{Q} \to \mathbb{Z} \times \mathbb{Z}$ defined by $\phi(x) = (p, q)$ is one-to-one. So $\mathbb{Q}$ is countable. □

Corollary 0.0.6 *$\mathbb{R}$ is uncountable.*

Proof
Define $\phi : \mathscr{P}(\mathbb{N}) \to \mathbb{R}$ by

$$\phi(E) = \sum_{n \in E} \frac{\chi_E(n)}{3^n},$$

i.e., ϕ maps each $E \subset \mathbb{N}$ to a ternary number $0.a_1a_2a_3\cdots$ such that $a_n = 1$ if $n \in E$ and $a_n = 0$ if $n \notin E$. Then ϕ is one-to-one. So card $\mathbb{N}$ < card $\mathscr{P}(\mathbb{N}) \leq$ card $\mathbb{R}$. □

Remark 0.0.1 In above proof, we intentionally choose ternary numbers instead of binary numbers because the map defined on binary numbers is not one-to-one. For example

$$\phi(\{1\}) = 0.1 = 0.0111111\cdots = \phi(\{2, 3, 4, \dots\}).$$

The extended real number system, denoted by $\overline{\mathbb{R}}$, is $\mathbb{R} \cup \{\infty, -\infty\}$.

For a bounded above set $E \subset \mathbb{R}$, the least upper bound of a set E, denoted by $\sup E$, is a real number α such that

i) $x \leq \alpha$ for all $x \in E$.
ii) For any $\epsilon > 0$, there exists an $x_0 \in E$ such that $\alpha - \epsilon < x_0$.

If E is not bounded above, then $\sup E = \infty$.

For a bounded below set E, the greatest lower bound of a set E, denoted by $\inf E$, is a real number β such that

i) $\beta \leq x$ for all $x \in E$.
ii) For any $\epsilon > 0$, there exists an $x_0 \in E$ such that $x_0 < \beta + \epsilon$.

If E is not bounded below, then $\inf E = -\infty$.

Let $\{x_n\}$ be a sequence. The upper limit of $\{x_n\}$, denoted by $\limsup_{n\to\infty} x_n$, or $\overline{\lim}_{n\to\infty}$, is defined by

$$\limsup_{n\to\infty} x_n = \inf_{k\geq 1}(\sup_{n\geq k}\{x_n\}) = \lim_{k\to\infty}\sup_{n\geq k}\{x_n\}.$$

$\limsup_{n\to\infty} x_n$ is the largest subsequential limit point of $\{x_n\}$. The lower limit of $\{x_n\}$, denoted by $\liminf_{n\to\infty} x_n$, or $\underline{\lim}_{n\to\infty} x_n$, is defined by

$$\liminf_{n\to\infty} x_n = \sup_{k\geq 1}(\inf_{n\geq k}\{x_n\}) = \lim_{k\to\infty}\inf_{n\geq k}\{x_n\}.$$

$\liminf_{n\to\infty} x_n$ is the smallest subsequential limit point of $\{x_n\}$.

Proposition 0.0.7 *Every open set in $\mathbb{R}$ is a countable disjoint union of open intervals.*

Proof

Let U be an open set of $\mathbb{R}$. For any $x \in U$, define

$$J_x = \bigcup_{\substack{x \in I \subset U \\ I \text{ is an open interval}}} I.$$

Clearly J_x is an open interval. For any $x \neq y$, if $J_x \cap J_y \neq \emptyset$, then $J_x \cup J_y$ is an open interval containing both x and y, and therefore $J_x = J_x \cup J_y = J_y$. So

$$U = \bigcup_{x \in U} J_x$$

is a disjoint union of open intervals. Since one can always find a rational number in an open interval, there are at most countable many disjoint open intervals.

□

Measures

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1.1 Introduction: Riemann Integral vs Lebesgue Integral

To understand why we need a theory of measures and a new way to define integrals (Lebesgue integrals), we need to look what is wrong with Riemann's theory of integrals.

Recall the definition of Riemann integrals (Figure 1.1). For a bounded function f defined on $[a, b]$, and for any partition $P : a = x_0 < x_1 < x_2 < \cdots < x_n = b$ of $[a, b]$, we form Riemann upper sum

$$U_P = \sum_{i=1}^{n} M_i(x_i - x_{i-1}), \qquad M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\}$$

and Riemann lower sum

$$L_P = \sum_{i=1}^{n} m_i(x_i - x_{i-1}), \qquad m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}.$$

If $P_1 \subset P_2$, then $L_{P_1} \le L_{P_2} \le U_{P_2} \le U_{P_1}$. So we can define the Riemann upper integral

$$\overline{\int_a^b} f(x)\,dx = \inf_P U_P$$

and Riemann lower integral

$$\underline{\int_a^b} f(x)\,dx = \sup_P U_P.$$

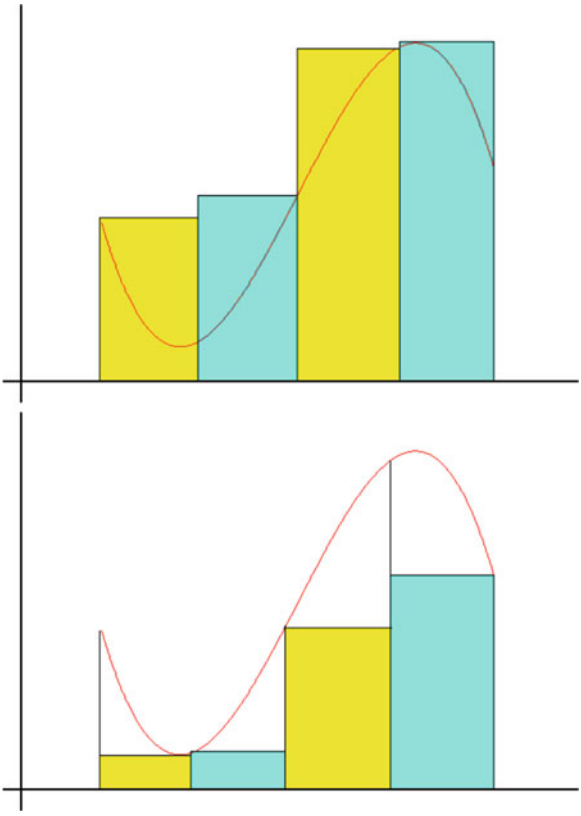


Fig. 1.1 Riemann upper sum and lower sum

f is called Riemann integrable over $[a, b]$ if $\bar{\int}_a^b f(x)\,dx = \underline{\int}_a^b f(x)\,dx$ and their common value is defined as $\int_a^b f(x)\,dx$.

Riemann integral is a terrible theory of integrals, in the sense that it does not have some expected properties, and some quite simple functions are not Riemann integrable. A typical example is the Dirichlet function defined by

$$d(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Obviously $d(x)$ is not Riemann integrable over any $[a, b]$ because $\bar{\int}_a^b d(x)\,dx = b - a$ and $\underline{\int}_a^b d(x)\,dx = 0$.

For nonnegative Riemann integrable functions $\{f_n\}$, $\int_a^b f_n(x)\,dx \geq 0$, so $\sum_{n=1}^{\infty} \int_a^b f_n(x)\,dx$ always exists (maybe ∞). It is natural to expect

$$\int_a^b \sum_{n=1}^{\infty} f_n(x)\,dx = \sum_{n=1}^{\infty} \int_a^b f_n(x)\,dx$$

(it is called "countable additivity"). However, it is not true, as one can see that $d(x) = \sum_{n=1}^{\infty} f_n(x)$, where

$$f_n(x) = \begin{cases} 1, & x = r_n, \\ 0, & \text{otherwise}, \end{cases} \quad \text{for an enumeration } \mathbb{Q} = \{r_1, r_2, r_3, \dots, \}$$

Each f_n is clearly Riemann integrable over any $[a, b]$ and $\int_a^b f_n(x)\,dx = 0$.

Also, for an increasing sequence of nonnegative Riemann integrable functions $0 \le g_1 \le g_2 \le \cdots$ such that $\lim_{n\to\infty} g_n(x) = g(x)$, it is natural to expect the areas of the region bounded by g_n approach the area of the region bounded by g. However, this is not true for Riemann integrals, as one can see that $d(x) = \lim_{n\to\infty} g_n(x)$, where

$$g_n(x) = \begin{cases} 1, & x = r_1, \dots, r_n, \\ 0, & \text{otherwise}, \end{cases} \quad \text{for an enumeration } \mathbb{Q} = \{r_1, r_2, r_3, \dots, \}$$

Each g_n is clearly Riemann integrable over any $[a, b]$ and $\int_a^b g_n(x)\,dx = 0$.

Instead of taking a partition of the x-axis, we can take a partition of the y-axis. For each subinterval $(y_{i-1}, y_i]$, we consider the set $E_i = \{x \in [a, b] : y_{i-1} < f(x) \le y_i\}$ (Figure 1.2). Since on E_i, $y_{i-1} \le f(x) \le y_i$, the "area" of the region under $f(x)$ is bounded below by $y_{i-1} l(E_i)$ and above $y_i l(E_i)$, where $l(E_i)$ is the "length" of E_i, provided that it can be defined. Similar to the Riemann integrals, one can form upper "Lebesgue" sum

$$\sum_{i=1}^{n} y_i l(E_i)$$

and lower "Lebesgue" sum

$$\sum_{i=1}^{n} y_{i-1} l(E_i)$$

(Assume the function values are between y_0 and y_n).

Since the difference between the upper and lower "Lebesgue" sums is at most $(b - a) \max_i (y_i - y_{i-1})$, and goes to zero as $\max_i (y_i - y_{i-1}) \to 0$, such a definition of integral works for any bounded function, provided that one can properly define "length" of E_i. Thus comes the terminology of a "measure" of a set. Precisely, the Lebesgue measure of a subset E of $\mathbb{R}$, denoted by $m(E)$, is a generalization of the length of intervals, i.e., for any interval I bounded by $a \le b$, $m(I) = b - a$.

Now let's see what properties the Lebesgue measure needs to have.

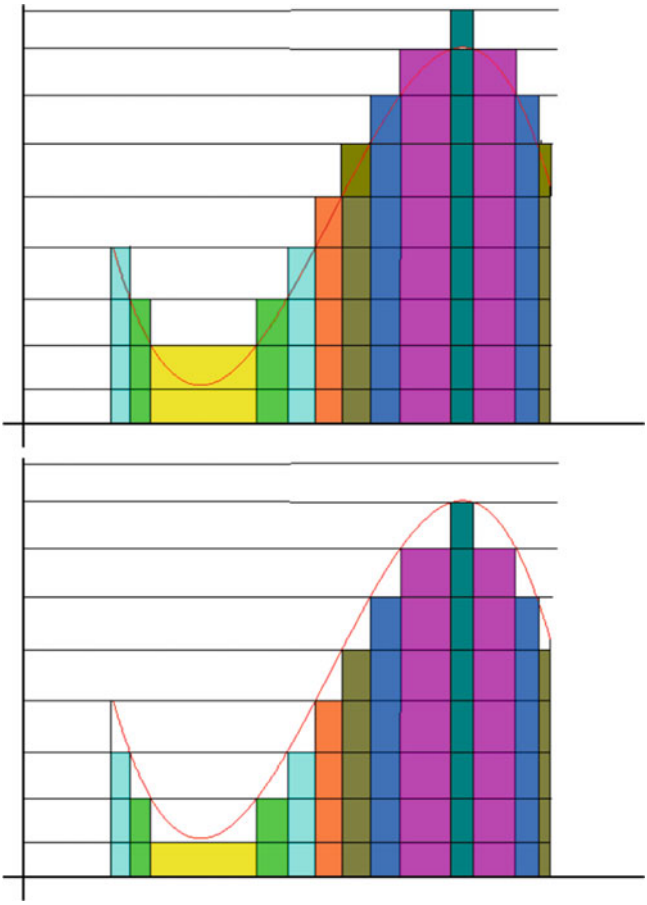


Fig. 1.2 "Lebesgue" upper sum and lower sum

Since we need the integral to have the countable additivity, we want the Lebesgue measure to have the countable additivity, i.e., for countable disjoint sets $\{E_n, n = 1, 2, \dots\}$,

$$m\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} m(E_n).$$

Since the length of an interval is translation invariant, we want the Lebesgue measure to be translation invariant, i.e., for any real number s,

$$m(E) = m(E_s), \quad \text{where} \quad E_s = \{x + s : x \in E\}.$$

Also, we want larger set to have larger measure, i.e., we want monotonicity:

if $E \subset F$, then $m(E) \leq m(F)$.

Unfortunately, such a measure cannot be defined for all subsets of $\mathbb{R}$, as can be seen from the following example.

Example 1.1
Define an equivalence relation on $[0, 1]$ by

$$x \sim y \quad \text{if and only if} \quad x - y \in \mathbb{Q},$$

and let N be the set consisting of exactly one element from each equivalence class. For any $r \in [-1, 1] \cap \mathbb{Q}$, define

$$N_r = \{x + r : x \in N\}.$$

Then $\{N_r : r \in [-1, 1] \cap \mathbb{Q}\}$ are disjoint, because only one element of an equivalence class is in N, and

$$[0, 1] \subset \bigcup_{r \in [0,1] \cap \mathbb{Q}} N_r \subset [-1, 2].$$

If we can define the Lebesgue measure on N, then we must have

$$1 = m([0, 1]) \leq m\left(\bigcup_{r \in [0,1] \cap \mathbb{Q}} N_r\right) = \sum_{r \in [-1,1] \cap \mathbb{Q}} m(N) \leq m([-1, 2]) = 3,$$

which is impossible, because if $m(N) = 0$, then $\sum_{r \in [0,1] \cap \mathbb{Q}} m(N) = 0$, and if $m(N) > 0$, then $\sum_{r \in [0,1] \cap \mathbb{Q}} m(N) = \infty$. ■

So we can only expect to define the Lebesgue measure on a collection of "nice sets." Such a collection needs to have certain structure, and it needs to be a "σ-algebra."

1.2 σ-Algebras

Definition 1.2.1
Let X be a nonempty set. A σ-algebra (algebra) $\mathscr{A}$ on X is a nonempty collection of subsets of X which is closed under complements and countable unions (finite unions); i.e., a subset $\mathscr{A} \subset \mathscr{P}(X)$ is a σ-algebra (algebra) if

i) for any $E \in \mathscr{A}$, $E^c \in \mathscr{A}$,

ii) for any countable (finite) sets $\{E_1, E_2, \dots\} \subset \mathscr{A}$, $\bigcup_n E_n \in \mathscr{A}$.

1

Remark 1.1

a. A σ-algebra $\mathscr{A}$ is closed under countable intersections, because

$$\bigcap_n E_n = \left(\bigcup_n E_n^c\right)^c.$$

b. $\emptyset \in \mathscr{A}$ because $\emptyset = E \cap E^c$.

c. $X \in \mathscr{A}$ because $X = \emptyset^c$.

In fact, a σ-algebra is closed under any countably many set operations. Also from the definition we can see that any intersection of σ-algebras over X is a σ-algebra.

Example 1.2

1. $\mathscr{P}(X)$ is a σ-algebra, and it is the largest σ-algebra over X.
2. $\{\emptyset, X\}$ is a σ-algebra, and it is the smallest σ-algebra over X.
3. The collection of all countable subsets and their complements is a σ-algebra. It is called the σ-algebra of countable and co-countable sets.

■

Definition 1.2.2

Let $\mathscr{E} \subset \mathscr{P}(X)$. The σ-algebra generated by $\mathscr{E}$, denoted by $\mathscr{M}(\mathscr{E})$, is the intersection of all σ-algebras containing $\mathscr{E}$.

Remark 1.2 $\mathscr{M}(\mathscr{E})$ is the smallest σ-algebra containing $\mathscr{E}$. It can be considered as a collection of sets generated from the sets in $\mathscr{E}$ by countable many set operations. For example, the σ-algebra of countable and co-countable sets is generated by the collection $\{\{x\} : x \in X\}$ of all singleton subsets of X.

Following results are oblivious.

Lemma 1.2.3

a) If $\mathscr{E} \subset \mathscr{M}(\mathscr{F})$, then $\mathscr{M}(\mathscr{E}) \subset \mathscr{M}(\mathscr{F})$.

b) If $\mathscr{F} \subset \mathscr{E} \subset \mathscr{M}(\mathscr{F})$, then $\mathscr{M}(\mathscr{E}) = \mathscr{M}(\mathscr{F})$.

Definition 1.2.4

Let X be a metric space. The σ-algebra generated by open sets of X is called the Borel σ-algebra on X, and is denoted by $\mathscr{B}_X$. The sets in $\mathscr{B}_X$ are called Borel sets.

Proposition 1.2.5 *$\mathscr{B}_{\mathbb{R}}$ is generated by any one of the following:*

a) $\mathscr{E}_1 = \{(a, b) : a, b \in \mathbb{R}\}$.
b) $\mathscr{E}_2 = \{[a, b] : a, b \in \mathbb{R}\}$.
c) $\mathscr{E}_3 = \{(a, b] : a, b \in \mathbb{R}\}$.
d) $\mathscr{E}_4 = \{[a, b) : a, b \in \mathbb{R}\}$.
e) $\mathscr{E}_5 = \{(a, \infty) : a \in \mathbb{R}\}$.
f) $\mathscr{E}_6 = \{(-\infty, a) : a \in \mathbb{R}\}$.
g) $\mathscr{E}_5 = \{[a, \infty) : a \in \mathbb{R}\}$.
h) $\mathscr{E}_6 = \{(-\infty, a] : a \in \mathbb{R}\}$.

Proof

Let $\mathscr{U}$ be the collection of all open sets of $\mathbb{R}$. Then $\mathscr{E}_1 \subset \mathscr{U} \subset \mathscr{M}(\mathscr{E}_1)$. So $\mathscr{M}(\mathscr{E}_1) = \mathscr{M}(\mathscr{U}) = \mathscr{B}_{\mathbb{R}}$.

For the other $\mathscr{E}_i$'s, one can easily show that

$$\mathscr{E}_i \subset \mathscr{M}(\mathscr{E}_1), \quad \mathscr{E}_1 \subset \mathscr{M}(\mathscr{E}_i).$$

For example

$$(a, b) = [a, b] \cap [a, a]^c \cap [b, b]^c, \quad [a, b] = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b + \frac{1}{n}\right).$$

□

Definition 1.2.6

The Borel σ-algebra over the extended real number system $\overline{\mathbb{R}}$ is defined as the collection of subset $E \subset \overline{\mathbb{R}}$ such that $E \cap \mathbb{R} \in \mathscr{B}_{\mathbb{R}}$, i.e.

$$\mathscr{B}_{\overline{\mathbb{R}}} = \{E, E \cup \{\infty\}, E \cup \{-\infty\}, E \cup \{\infty, -\infty\} : E \in \mathscr{B}_{\mathbb{R}}\}$$

Exercises

1.1 Let $\mathscr{E} = \{E_1, \ldots, E_n\}$ be a finite partition of X, i.e., $\{E_i\}$ are nonempty and disjoint, and $\bigcup_{i=1}^{n} E_i = X$. Prove that $\mathscr{M}(\mathscr{E})$ has 2^n elements.

1.2 Let $\mathscr{E} = \{E_i\}_1^{\infty}$ be a countably infinite partition of X. Prove that $\text{card}(\mathscr{M}(\mathscr{E})) = c$. (Hint: $\text{card}(\mathscr{P}(\mathbb{N})) = c$.)

1.3 Let $\mathscr{M}$ be an infinite σ-algebra. Prove that $\mathscr{M}$ contains a countably infinite partition of X and $\text{card}(\mathscr{M}(\mathscr{E})) \geq c$.

1.4 Let $\mathscr{E} \subset \mathscr{P}(X)$. Show that

$$\mathscr{M} := \bigcup_{\mathscr{F}\subset\mathscr{E},\operatorname{card}(\mathscr{F})\le c_0} \mathscr{M}(\mathscr{F})$$

is a σ-algebra, and $\mathscr{M} = \mathscr{M}(\mathscr{E})$.

1.5 Let X be an uncountable set. Show that the collection of all countable and co-countable subsets of X forms a σ-algebra.

1.6 Prove that an algebra $\mathscr{A}$ is a σ-algebra if and only if $\mathscr{A}$ is closed under countable decreasing intersection, i.e., if $E_1 \supset E_2 \supset \cdots$ in $\mathscr{A}$, then $\bigcap_{n=1}^{\infty} E_n \in \mathscr{A}$.

1.3 Measures

Now we formally define a measure.

Definition 1.3.1

Let $\mathscr{M}$ be a σ-algebra on X. A measure μ is a function $\mu : \mathscr{M} \to [0, \infty]$ such that

i) $\mu(\emptyset) = 0$
ii) μ is countably additive, i.e., for any **disjoint** $\{E_n, n = 1, \ldots, \infty\} \subset \mathscr{M}$

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n).$$

Remark 1.3 If μ is nontrivial, in the sense that there exists at least one $E \in \mathscr{M}$ such that $\mu(E) \neq \infty$, then the 2nd condition implies the 1st, because

$$\mu(E) = \mu(E \cup \emptyset) = \mu(E) + \mu(\emptyset).$$

Definition 1.3.2

a) $(X, \mathscr{M})$ is called a measurable space.
b) $(X, \mathscr{M}, \mu)$ is called a measure space.
c) Any set in $\mathscr{M}$ is called a measurable set.
d) $(X, \mathscr{M}, \mu)$ is called finite if $\mu(X) < \infty$.
f) $(X, \mathscr{M}, \mu)$ is called σ-finite if $X = \bigcup_{n=1}^{\infty} E_n$ for some $\{E_n\} \subset \mathscr{M}$ such that $\mu(E_n) < \infty$ for all n.
g) $(X, \mathscr{M}, \mu)$ is called semifinite if for all $E \in \mathscr{M}$ with $\mu(E) = \infty$, there exists a $F \in \mathscr{M}$ with $F \subset E$ and $0 < \mu(F) < \infty$.

Example 1.3

Following are examples of measure.

a) Let $\mathscr{M} = \mathscr{P}(X)$ and define $\mu(E)$ to be the number of elements in E. This measure is called the "counting measure" on X.

b) Let $\mathscr{M} = \mathscr{P}(X)$ and fix an α in X. Define

$$\mu_\alpha(E) = \begin{cases} 1, \text{ if } \alpha \in E, \\ 0, \text{ if } \alpha \notin E. \end{cases}$$

μ_α is called point mass at α.

c) Let $\mathscr{M}$ be the σ-algebra of countable and co-countable sets, and define

$$\mu(E) = \begin{cases} 1, \text{ if } E \text{ is co-countable}, \\ 0, \text{ if } E \text{ is countable}. \end{cases}$$

(Can you prove it?)

■

Theorem 1.3.3

Let $(X, \mathscr{M}, \mu)$ be a measure space.

a) (Monotonicity) If $E, F \in \mathbb{M}$ with $E \subset F$, then

$$\mu(E) \leq \mu(F).$$

b) (Countable sub-additivity) If $\{E_n, n = 1, \ldots, \infty\} \subset \mathscr{M}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \mu(E_n).$$

c) (Continuity from below) If $\{E_n, n = 1, \ldots, \infty\} \subset \mathscr{M}$ with $E_1 \subset E_2 \subset \cdots$, then

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} \mu(E_n).$$

d) (Continuity from above) If $\{E_n, n = 1, \ldots, \infty\} \subset \mathscr{M}$ with $E_1 \supset E_2 \supset \cdots$ and at least one $\mu(E_k) < \infty$, then

$$\mu\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} \mu(E_n).$$

Proof

a) $\mu(F) = \mu(E) + \mu(F \setminus E) \geq \mu(E)$.

b) Let $F_n = E_n \setminus \left(\bigcup_{k=1}^{n-1} E_k\right)$. Then $\{F_n\}$ are disjoint. So

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \mu\left(\bigcup_{n=1}^{\infty} F_n\right) = \sum_{n=1}^{\infty} \mu(F_n) \leq \sum_{n=1}^{\infty} \mu(E_n).$$

c)

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{k=1}^{\infty} \mu(E_k \setminus E_{k-1}) = \lim_{n\to\infty} \sum_{k=1}^{n} \mu(E_k \setminus E_{k-1}) = \lim_{n\to\infty} \mu(E_n).$$

d) For any measurable subset A of E_k, since $\mu(E_k) = \mu(A) + \mu(E_k \setminus A)$ and $\mu(E_k) < \infty$, we have

$$\mu(E_k \setminus A) = \mu(E_k) - \mu(A) \text{ and } \mu(A) = \mu(E_k) - \mu(E_k \setminus A).$$

Since the sets $\{E_k \setminus E_n : n = k, k+1, \dots\}$ are increasing, by c),

$$\mu\left(\bigcap_{n=1}^{\infty} E_n\right) = \mu\left(\bigcap_{n=k}^{\infty} E_n\right) = \mu(E_k) - \mu\left(E_k \setminus \bigcap_{n=k}^{\infty} E_n\right)$$

$$= \mu(E_k) - \mu\left(\bigcup_{n=k}^{\infty} (E_k \setminus E_n)\right) = \mu(E_k) - \lim_{n\to\infty} \mu(E_k \setminus E_n)$$

$$= \mu(E_k) - (\mu(E_k) - \lim_{n\to\infty} \mu(E_n)) = \lim_{n\to\infty} \mu(E_n)$$

□

Definition 1.3.4

Let $(X, \mathscr{M}, \mu)$ be a measure space.

a) A null set is a measurable set with measure zero.

b) A property is said true almost everywhere (a.e.) if the exception set is contained in a null set.

c) $(X, \mathscr{M}, \mu)$ is called complete if all the subsets of null sets are measurable.

Theorem 1.3.5

Any measure space $(X, \mathscr{M}, \mu)$ can be expanded to a complete measure space $(X, \overline{\mathscr{M}}, \overline{\mu})$, where

$$\overline{\mathscr{M}} = \{E \cup N : E \in \mathscr{M}, N \text{ is a subset of a } \mu\text{-null set}\}$$

and $\overline{\mu}(E \cup N) = \mu(E)$.

Proof

For any $E \cup N \in \overline{\mathscr{M}}$, where $N \subset F$ for a μ-null set F, since $E \cup N \subset E \cup F$, we have $(E \cup N)^c \supset (E \cup F)^c$, and

$$\begin{aligned}(E \cup N)^c &= ((E \cup N)^c \cap (E \cup F)^c) \cup ((E \cup N)^c \cap (E \cup F)) \\ &= (E \cup F)^c \cup (E^c \cap N^c \cap (E \cup F)) \\ &= (E \cup F)^c \cup (E^c \cap N^c \cap F) \in \overline{\mathscr{M}}\end{aligned}$$

because $E^c \cap N^c \cap F$ is a subset of the μ-null set F (Figure 1.3).

For any $\{E_n \cup N_n\} \subset \overline{\mathscr{M}}$, where $N_n \subset F_n$ for a μ-null set F_n, we have

$$\bigcup_{n=1}^{\infty}(E_n \cup N_n) = \left(\bigcup_{n=1}^{\infty} E_n\right) \cup \left(\bigcup_{n=1}^{\infty} N_n\right) \in \overline{\mathscr{M}}$$

because $\bigcup_{n=1}^{\infty} N_n$ is a subset of the μ-null set $\bigcup_{n=1}^{\infty} F_n$. So $\overline{\mathscr{M}}$ is a σ-algebra.

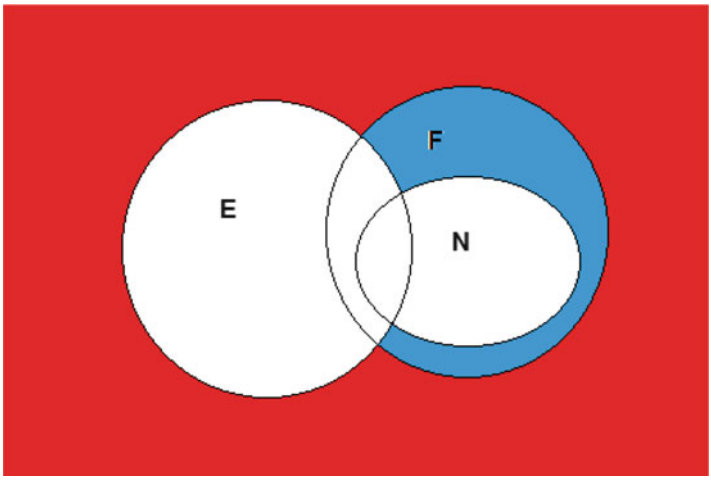


Fig. 1.3 $(E \cup N)^c = (E \cup F)^c \cup (E^c \cap N^c \cap F)$

For any disjoint $\{E_n \cup N_n\} \subset \overline{\mathscr{M}}$,

$$\overline{\mu}\left(\bigcup_{n=1}^{\infty} E_n \cup N_n\right) = \mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n) = \sum_{n=1}^{\infty} \overline{\mu}(E_n \cup N_n).$$

So $\overline{\mu}$ is a measure on $\overline{\mathscr{M}}$. □

Exercises

1.7 Let $(X, \mathscr{M}, \mu)$ be a measure space and $E, F \in \mathscr{M}$ with $F \subset E$. Show that if $\mu(F) < \infty$, then

$$\mu(E \setminus F) = \mu(E) - \mu(F).$$

Explain why it is not true without the condition $\mu(F) < \infty$.

1.8 Let $(X, \mathscr{M})$ be a measurable space and $\mu : \mathscr{M} \to [0, \infty]$ be a set function on $\mathscr{M}$ such that $\mu(E) < \infty$ for at least one $E \in \mathscr{M}$. Prove that μ is a measure on $\mathscr{M}$ if and only if

a. $\mu(E \cup F) = \mu(E) + \mu(F)$ for all disjoint E, F in $\mathscr{M}$, and
b. $\mu\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \mu(E_n)$ for all $\{E_n\}_0^{\infty} \subset \mathscr{M}$.

1.9 Let $\mathscr{M}$ be the σ-algebra of countable and co-countable sets, and define

$$\mu(E) = \begin{cases} 1, & \text{if } E \text{ is co-countable,} \\ 0, & \text{if } E \text{ is countable.} \end{cases}$$

Prove that μ is a measure on $\mathscr{M}$.

1.10 Let $(X, \mathscr{M}, \mu)$ be a non-σ-finite measure space. Define

$$\mathscr{N} = \{E \in \mathscr{M} : \text{either } E \text{ is } \sigma\text{-finite or } E^c \text{ is } \sigma\text{-finite for } \mu\}$$

and a set function ν on $\mathscr{N}$ by

$$\nu(E) = \begin{cases} 0, & \text{if } E \text{ is } \sigma\text{-finite for } \mu, \\ 1, & \text{if } E^c \text{ is } \sigma\text{-finite for } \mu. \end{cases}$$

Prove that $\mathscr{N}$ is a σ-algebra and ν is a measure on $\mathscr{N}$.

1.11 Let $(X, \mathscr{M}, \mu)$ be a measure space and $E \in \mathscr{M}$. Define

$$\mathscr{M}_E = \{E \cap A : A \in \mathscr{M}\}.$$

Show that $(E, \mathscr{M}_E, \mu)$ is also a measure space.

1.12 Let $\mathscr{M}$ be a σ-algebra over X and $\mu : \mathscr{M} \to [0, \infty]$ be a set function such that $\mu(E \cup F) = \mu(E) + \mu(F)$ for any disjoint $E, F \in \mathscr{M}$.

a. Prove that μ is a measure on $\mathscr{M}$ if and only if $\mu(\emptyset) = 0$ and μ is continuous from blow as in Theorem 1.3.3 c).

b. If $\mu(X) < \infty$, prove that μ is a measure on $\mathscr{M}$ if and only if μ is continuous from above as in Theorem 1.3.3 d).

1.13 Let $(X, \mathscr{M}, \mu)$ be a measure space and $E, F \in \mathscr{M}$ such that $\mu(E \triangle F) = 0$. Show that $\mu(E) = \mu(F)$.

1.14 Let $(X, \mathscr{M}, \mu)$ be a measure space, $E \in \mathscr{M}$, and

$$S = \{\mu(F) : F \subset E, \mu(F) < \infty\}$$

be bounded. Prove that

a. there exists an $A \subset E$ such that $\mu(A) = \sup S$,

b. for any $B \subset E \setminus A$, either $\mu(B) = 0$ or $\mu(B) = \infty$, and

c. for any $B \subset E$, $\mu(A \cap B) = \sup\{\mu(F) : F \subset B, \mu(F) < \infty\}$.

1.15 Let $(X, \mathscr{M}, \mu)$ be a semifinite measure space. Prove that

$$\sup\{\mu(F) : F \in \mathscr{M}, F \subset E, \mu(F) < \infty\} = \infty$$

for any $E \in \mathscr{M}$ with $\mu(E) = \infty$.

1.16 Let $(X, \mathscr{M}, \mu)$ be a measure space. Define

$$\nu(E) = \sup\{\mu(F) : F \subset E \text{ and } \mu(F) < \infty\}$$

and

$$\rho(E) = \begin{cases} 0, & \text{if } E \text{ is } \sigma\text{-finite for } \mu, \\ \infty, & \text{if } E \text{ is not } \sigma\text{-finite for } \mu, \end{cases}$$

on $\mathscr{M}$. Show that both ν and ρ are measures on $\mathscr{M}$, ν is semifinite, and $\mu = \nu + \rho$.

1.4 Outer Measures

Next question is: how do we extend the length of intervals to the Lebesgue measure? One familiar technique is used. Imagine we have some measuring rods of various known lengths. To measure the length of a wire, we can lay measuring rods on top of the wire to cover the whole wire. If the measuring rods overlap, the total length will be larger. Otherwise if they are placed end to end, their total length will be closer to the length of the wire (Figure 1.4).



Fig. 1.4 The length of the wire equals the minimum total length of the measuring rods covering the wire

Using such an idea, we can define an outer measure (it cannot be a measure generally as we have shown before) on $\mathscr{P}(X)$ from a collection of sets of known "lengths."

We first define outer measures.

Definition 1.4.1

An outer measure on X is a function $\mu^* : \mathscr{P}(X) \to [0, \infty]$ such that

i) $\mu^*(\emptyset) = 0$;

ii) (Monotonicity) for any $A \subset B$, $\mu^*(A) \leq \mu^*(B)$;

iii) (Countable sub-additivity) $\mu^*\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$.

Proposition 1.4.2 *Let $\mathscr{E}$ be a collection of subsets of X such that $\emptyset \in \mathscr{E}$ and $X = \bigcup_{n=1}^{\infty} E_n$ for some $\{E_n\} \subset \mathscr{E}$, and $\rho : \mathscr{E} \to [0, \infty]$ be a function on $\mathscr{E}$ such that $\rho(\emptyset) = 0$. Define*

$$\mu^*(A) = \inf\left\{\sum_{n=1}^{\infty} \rho(E_n) : E_n \in \mathscr{E} \text{ and } A \subset \bigcup_{n=1}^{\infty} E_n\right\}.$$

Then μ^* is an outer measure.

Roughly speaking, $\mu^*(E)$ is the infimum of total ρ value of countable covering sets in $\mathscr{E}$. The proof of the result is simply verifying each condition of the definition, and is trivial.

Next question is: Under what condition an outer measure becomes a measure? It turns out we just need finite additivity.

Proposition 1.4.3 *Let $\mathscr{M}$ be a σ-algebra and μ^* is an outer measure on $\mathscr{M}$. Then μ^* is a measure if and only if*

$$\mu^*(E \cup F) = \mu^*(E) + \mu^*(F)$$

for any disjoint E and F in $\mathscr{M}$.

Proof

We only need to prove sufficiency. If $\mu^*(E \cup F) = \mu^*(E) + \mu^*(F)$ for any disjoint E, F in $\mathscr{M}$, then for any disjoint $\{E_n\} \subset \mathscr{M}$, by using induction we can prove that

$$\mu^*\left(\bigcup_{n=1}^{k} E_n\right) = \sum_{n=1}^{k} \mu^*(E_n)$$

for all k. Then

$$\mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) \geq \mu^*\left(\bigcup_{n=1}^{k} E_n\right) = \sum_{n=1}^{k} \mu^*(E_n)$$

for all k. So

$$\mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) \geq \sum_{n=1}^{\infty} \mu^*(E_n).$$

Also by the countable subadditivity

$$\mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \mu^*(E_n).$$

So we have countable additivity and μ^* is a measure. □

So for an outer measure to become a measure, we need to find sets such that μ^* is finite additive. Note that any set A defines a partition on all other sets E of $\mathscr{P}(X)$:

$$E = (E \cap A) \cup (E \cap A^c),$$

We can think that each set A is a "cookie cutter"; this "cookie cutter" cuts any other set into 2 halves. A "cookie cutter" is a good "cookie cutter," we call a measurable "cookie cutter," if the sum of the μ^* values of the 2 halves equals the μ^* value of whole for any set E, i.e., if it defines a μ^*-additive partition on all sets in $\mathscr{P}(X)$. We then have the finite additivity. Now we formally define a μ^*-measurable set (Figure 1.5).

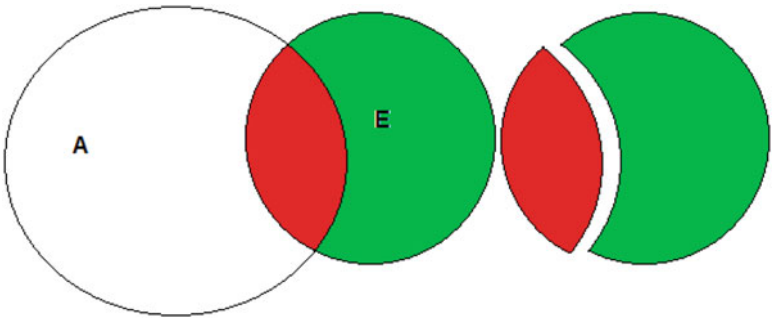


Fig. 1.5 Any set A defines a partition

Definition 1.4.4 (Carathéodory Condition)

Let μ^* be an outer measure on X. A set $A \subset X$ is called μ^*-measurable if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \quad \text{for all subsets } E \text{ of } X.$$

Remark 1.4 Because of the countable sub-additivity, A is μ^*-measurable if and only if

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c) \quad \text{for all subsets } E \text{ of } X.$$

Immediately by using induction we know that any finitely many such measurable "cookie cutters" cut arbitrary E into finitely many μ^*-additive pieces, and thanks to the countable subadditivity, if countably many of them cut E into countably many pieces, they again are μ^*-additive. In particular by restricting to themselves, we have a measure.

Proposition 1.4.5

a) For any μ^*-measurable sets $\{A_1, A_2, \ldots, A_n\}$ and for any subset E of X,

$$\mu^*(E) = \sum_{B_i \in \{A_i, A_i^c\}} \mu^*(E \cap B_1 \cap \cdots \cap B_n).$$

b) For any disjoint μ^*-measurable sets $\{A_1, A_2, \ldots, A_n, \ldots\}$

$$\mu^*(E) = \sum_{i=1}^{\infty} \mu^*(E \cap A_i) + \mu^*\left(E \cap \bigcap_{i=1}^{\infty} A_i^c\right).$$

Proof

a) It is true for $n = 1$ by definition. Assume it is true for $n - 1$,

$$\mu^*(E) = \sum_{B_i \in \{A_i, A_i^c\}} \mu^*(E \cap B_1 \cap \cdots \cap B_{n-1}).$$

Then

$$\begin{aligned}\mu^*(E) &= \sum_{B_i \in \{A_i, A_i^c\}} (\mu^*(E \cap B_1 \cap \cdots \cap B_{n-1} \cap A_n) + \mu^*(E \cap B_1 \cap \cdots \cap B_{n-1} \cap A_n^c)) \\ &= \sum_{B_i \in \{A_i, A_i^c\}} \mu^*(E \cap B_1 \cap \cdots \cap B_n).\end{aligned}$$

So it is true for all n.

b) Homework

□

Now we prove that the collection of all such measurable "cookie cutters" forms a σ-algebra and μ^* becomes a measure among them.

Theorem 1.4.6 (Carathéodory Theorem)
The collection $\mathscr{M}$ of all μ^*-measurable sets is a σ-algebra, and μ^* is a complete measure on $\mathscr{M}$.

Proof
Since A and A^c induce the same partitions, $A \in \mathscr{M}$ if and only if $A^c \in \mathscr{M}$.

Let E be any subset of X. For any $A, B \in \mathscr{M}$, since

$$A^c \cap B^c = (A \cup B)^c, \quad \text{and} \quad (E \cap A \cap B) \cup (E \cap A \cap B^c) \cup (E \cap A^c \cap B) = E \cap (A \cup B),$$

by Proposition 1.4.5 a) and countable sub-additivity,

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c) \\ &\geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c). \end{aligned}$$

So $A \cup B \in \mathscr{M}$, and $\mathscr{M}$ is an algebra.

For any $\{A_1, A_2, \dots, \} \subset \mathscr{M}$, since $\mathscr{M}$ is an algebra, without loss of generality, we can assume they are disjoint. Then by Proposition 1.4.5 b),

$$\begin{aligned} \mu^*(E) &= \sum_{n=1}^{\infty} \mu^*(E \cap A_n) + \mu^* \left(E \cap \bigcap_{n=1}^{\infty} A_n^c \right) \\ &\geq \mu^* \left(E \cap \bigcup_{n=1}^{\infty} A_n \right) + \mu^* \left(E \cap \left(\bigcup_{n=1}^{\infty} A_n \right)^c \right). \end{aligned}$$

So $\mathscr{M}$ is a σ-algebra.

By letting $E = \bigcup_{n=1}^{\infty} A_n$ in Proposition 1.4.5 b) for disjoint $\{A_n\} \subset \mathscr{M}$, we have

$$\mu^* \left(\bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mu^*(A_n).$$

So μ^* is a measure on $\mathscr{M}$.

Finally any μ^*-null set A is measurable because

$$\mu^*(E) \geq \mu^*(E \cap A^c) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

So μ^* is a complete measure on $\mathscr{M}$. □

We will drop the $*$ and use μ instead on $\mathscr{M}$.

Let's summarize what we did so far:

1) Any nonnegative set function ρ with $\rho(\emptyset) = 0$ on a subset $\mathscr{E} \subset \mathscr{P}(X)$ containing $\emptyset$ and covering X induces an outer measure μ^* on $\mathscr{P}(X)$.
2) Any outer measure μ^* on $\mathscr{P}(X)$ restricted on the σ-algebra of all μ^*-measurable sets becomes a measure.

There are two questions:

a) Are all original sets in $\mathscr{E}$ μ^* measurable?
b) If the answer for a) is yes, is $\mu(E) = \rho(E)$ for all $E \in \mathscr{E}$?

The answers are not always yes.

Example 1.4

a) Let $\mathscr{E} = \mathscr{P}(\mathbb{R})$ and define

$$\rho(E) = \begin{cases} 0, & \text{if } E \cap \mathbb{N} = \emptyset, \\ \min E \cap \mathbb{N}, & \text{if } E \cap \mathbb{N} \neq \emptyset. \end{cases}$$

We first find out what the induced outer measure is. Clearly if $E \cap \mathbb{N} = \emptyset$, then E itself is a covering set with the smallest possible ρ value. So $\mu^*(E) = 0$.

If $E \cap \mathbb{N} \neq \emptyset$, then $\rho(E) \geq 1$. The smallest possible ρ is 1, and it can be reached by using the covering set $\mathbb{R}$.

Therefore

$$\mu^*(E) = \begin{cases} 0, & \text{if } E \cap \mathbb{N} = \emptyset, \\ 1, & \text{if } E \cap \mathbb{N} \neq \emptyset. \end{cases}$$

What are the μ^*-measurable sets? Let's try each "cookie cutter" on $\mathbb{R}$. Clearly only $1 + 0 = 1$, so only one of μ^* values of "inside" and "outside" can be 1. So $\mathbb{N}$ must be completely contained either in E or in E^c. Applying such "nice cookie cutter candidate" to any other set, we can see that they are indeed μ^*-measurable "cookie cutters." Therefore

$$\mathscr{M} = \{E \subset \mathbb{R} : \text{either } \mathbb{N} \subset E \text{ or } \mathbb{N} \cap E = \emptyset\}.$$

This example shows that not all sets in $\mathscr{E}$ are μ^*-measurable.

b) Let

$$\mathscr{E} = \{\mathbb{R} \text{ and all finite subsets of } \mathbb{R}\},$$

and ρ on $\mathscr{E}$ is defined by

$$\rho(E) = \begin{cases} \#(E) - 1, & \text{if } E \neq \emptyset \text{ is finite,} \\ 0, & \text{if } E = \emptyset, \\ 1, & \text{if } E = \mathbb{R}, \end{cases}$$

where $\#(E)$ is the number of elements in E.

Any uncountable subset E cannot be covered by countably many finite sets because the union of such sets is still countable. So the only set in $\mathscr{E}$ that can cover E is $\mathbb{R}$ and therefore $\mu^*(E) = 1$.

Any countable set $E = \{x_1, x_2, \dots, \}$ can be covered, for example, by $\{x_1, x_2\}, \{x_3, x_4\}, \dots$, and it can also be covered by $\{x_1\}, \{x_2\}, \dots$. The later give us the smallest total ρ value 0. Therefore $\mu^*(E) = 0$

What are the μ^*-measurable sets? Again let's try each "cookie cutter" on $\mathbb{R}$ first. Since only $0 + 1 = 1$, only one of "inside" and "outside" sets can be countable. So countable and co-countable sets are candidates for "nice cookie cutters." By trying them on all other sets, we know that the μ^*-measurable σ-algebra $\mathscr{M}$ is the countable/co-countable σ-algebra, and $\mathscr{E} \subset \mathscr{M}$. However, for a set $\{x_1, x_2\} \in \mathscr{E}$,

$$\rho(\{x_1, x_2\}) = 1 \neq 0 = \mu(\{x_1, x_2\}).$$

This example shows even if all sets in $\mathscr{E}$ are μ^*-measurable, μ may not agree with ρ generally. ■

Because we need to extend the length of intervals to the Lebesgue measure, we want all intervals to be measurable and the Lebesgue measures of intervals agree with their lengths. So next question we need to answer is: what requirements for $\mathscr{E}$ and ρ will guarantee $\mathscr{E} \subset \mathscr{M}$ and $\mu = \rho$ on $\mathscr{E}$?

Definition 1.4.7

A collection $\mathscr{E} \subset \mathscr{P}(X)$ is called semi-algebra if

i) $\emptyset \in \mathscr{E}$;

ii) for all $E, F \in \mathscr{E}$, $E \cap F \in \mathscr{E}$;

iii) for all $E \in \mathscr{E}$, E^c is a finite disjoint union of members of $\mathscr{E}$.

Remark 1.5 It can be easily verified that for any semi-algebra $\mathscr{E}$,

$$\mathscr{A} = \left\{ \bigcup_{i=1}^{n} A_i : A_i \in \mathscr{E}, A_i \cap A_j = \emptyset, n \in \mathbb{N} \right\}$$

is an algebra.

Recall a partition of X is a collection of nonempty disjoint subsets of X whose union is X. So a semi-algebra is union of finite partitions of X which is closed under intersection.

Definition 1.4.8

Let $\mathscr{E}$ be a semi-algebra. A set function $\mu_0 : \mathscr{E} \to [0, \infty]$ is called a premeasure if

i) $\mu_0(\emptyset) = 0$;

ii) (finite additivity) If $\bigcup_{n=1}^{k} A_n \in \mathscr{E}$ and $\{A_1, \ldots, A_k\} \subset \mathscr{E}$ are disjoint, then

$$\mu_0\left(\bigcup_{n=1}^{k} A_n\right) = \sum_{n=1}^{k} \mu_0(A_n);$$

iii) (monotonicity/countable sub-additivity) If $A \in \mathscr{E}$ and $A \subset \bigcup_{n=1}^{\infty} A_n$ for some $\{A_n\} \subset \mathscr{E}$, then

$$\mu_0(A) \leq \sum_{n=1}^{\infty} \mu_0(A_n).$$

This definition is different from that in Folland's book. However, they are equivalent. We define this way because countable sub-additivity is easier to verify than the countable additivity.

Proposition 1.4.9 *The conditions ii) and iii) are equivalent to*

ii' If $\bigcup_{n=1}^{\infty} A_n \in \mathscr{E}$ *and* $\{A_1, A_2, \ldots\} \subset \mathscr{E}$ *are disjoint, then*

$$\mu_0\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu_0(A_n).$$

Proof

If i) and ii) are true, then for any disjoint $\{A_1, A_2, \ldots\} \subset \mathscr{E}$ such that $A = \bigcup_{n=1}^{\infty} A_n \in \mathscr{E}$, by ii)

$$\mu_0(A) \leq \sum_{n=1}^{\infty} \mu_0(A_n)$$

and

$$\mu_0(A) \geq \mu_0\left(\bigcup_{n=1}^{k} A_n\right)$$

for any k, and by i)

$$\mu_0(A) \geq \mu_0\left(\bigcup_{n=1}^{k} A_n\right) = \sum_{n=1}^{k} \mu_0(A_n)$$

for any k, i.e.

$$\mu_0(A) \geq \sum_{n=1}^{\infty} \mu_0(A_n).$$

So ii') is true.

Conversely, if ii') is true, then ii) is true. Also for any $A \subset B$ in $\mathscr{E}$, $B = A \cup (B \cap A^c)$ and $B \cap A^c = B_1 \cup \cdots \cup B_k$ for some disjoint $\{B_1, \ldots, B_k\} \subset \mathscr{E}$. So

$$\mu_0(B) = \mu_0(A) + \sum_{i=1}^{k} \mu_0(B_i) \geq \mu_0(A).$$

Let $A, A_n, n \in \mathbb{N}$ be in $\mathscr{E}$ such that $A \subset \bigcup_{n=1}^{\infty} A_n$. Without loss of generality (by replacing A_n with $A_n \cap (A_1^c \cap \cdots \cap A_{n-1}^c$ and relabeling), we can assume that $\{A_n\}$ are disjoint. Then $A = \bigcup_{n=1}^{\infty}(A \cap A_n)$ and by ii')

$$\mu_0(A) = \sum_{n=1}^{\infty} \mu_0(A \cap A_n) \leq \sum_{n=1}^{\infty} \mu_0(A_n).$$

□

Proposition 1.4.10 *Let μ_0 be a premeasure on a semi-algebra $\mathscr{E}$ and μ^* be the induced outer measure. Then*

i) $\mu^* = \mu_0$ on $\mathscr{E}$,

ii) every set in $\mathscr{E}$ is μ^*-measurable.

Proof

i) Any $A \in \mathscr{E}$ can be covered by itself. So

$$\mu_0(A) \geq \inf\left\{\sum_{n=1}^{\infty} \mu_0(A_n) : \{A_n\} \subset \mathscr{E}, A \subset \bigcup_{n=1}^{\infty} A_n\right\} = \mu^*(A).$$

Also by property iii) of a premeasure, for any $\{A_n\} \subset \mathscr{E}$ with $A \subset \bigcup_{n=1}^{\infty} A_n$, $\mu_0(A) \leq \sum_{n=1}^{\infty} \mu_0(A_n)$. So

$$\mu_0(A) \leq \inf\left\{\sum_{n=1}^{\infty} \mu_0(A_n) : \{A_n\} \subset \mathscr{E}, A \subset \bigcup_{n=1}^{\infty} A_n\right\} = \mu^*(A).$$

ii) For any $A \in \mathscr{E}$, any $E \in \mathscr{P}(X)$, and any $\epsilon > 0$, there are $\{A_n\} \subset \mathscr{E}$ such that $E \subset \bigcup_{n=1}^{\infty} A_n$ and

$$\mu^*(E) + \epsilon \geq \sum_{n=1}^{\infty} \mu_0(A_n).$$

Assume

$$A^c = \bigcup_{j=1}^{m} B_j, \quad \{B_j\} \subset \mathscr{E} \text{ are disjoint.}$$

Then since

$$E \cap A \subset \bigcup_{n=1}^{\infty}(A_n \cap A), \quad E \cap A^c \subset \bigcup_{n=1}^{\infty}\bigcup_{j=1}^{m}(A_n \cap B_j),$$

by the property ii) of a premeasure,

$$\begin{aligned}\mu^*(E) + \epsilon &\geq \sum_{n=1}^{\infty} \mu_0(A_n) \\ &= \sum_{n=1}^{\infty} \mu_0(A_n \cap A) + \sum_{n=1}^{\infty}\sum_{j=1}^{m} \mu_0(A_n \cap B_j) \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c)\end{aligned}$$

Since ϵ is arbitrary,

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

and A is μ^*-measurable. □

Theorem 1.4.11

Let $\mathscr{E} \subset \mathscr{P}(X)$ be a semi-algebra, μ_0 be a premeasure on $\mathscr{E}$, $\mathscr{M}(\mathscr{E})$ be the σ-algebra generated by $\mathscr{E}$ (not the σ-algebra of all μ^*-measurable sets), μ^* be the outer measure induced by μ_0, and $\mu = \mu^*\Big|_{\mathscr{M}(\mathscr{E})}$ be the induced measure on $\mathscr{M}(\mathscr{E})$. Then μ is the extension of μ_0 such that for all other extension ν of μ_0 on $\mathscr{M}(\mathscr{E})$, we have

$$\nu(E) \leq \mu(E) \quad \forall E \in \mathscr{M}(\mathscr{E})$$

with equality when $\mu(E) < \infty$. If μ_0 is σ-finite (i.e., X can be covered by countably many sets in $\mathscr{E}$ with finite premeasure), then μ is the unique extension of μ_0.

Proof

For any $E \in \mathscr{M}(\mathscr{E})$ and any countable $\{A_n\} \subset \mathscr{E}$ covering E (i.e., $E \subset \bigcup_{n=1}^{\infty} A_n$), we have

$$\nu(E) \leq \sum_{n=1}^{\infty} \nu(A_n) = \sum_{n=1}^{\infty} \mu_0(A_n).$$

So

$$\nu(E) \le \inf\left\{\sum_{n=1}^{\infty} \mu_0(A_n) : \{A_n\} \subset \mathscr{E},\ E \subset \bigcup_{n=1}^{\infty} A_n\right\} = \mu(E).$$

If $\mu(E) < \infty$, then for any $k \in \mathbb{N}$, there are disjoint (why?) $\{A_{kn}\} \subset \mathscr{E}$, $E \subset \bigcup_{n=1}^{\infty} A_{kn}$ such that

$$\mu\left(\bigcup_{n=1}^{\infty} A_{kn}\right) = \sum_{n=1}^{\infty} \mu_0(A_{kn}) < \mu(E) + \frac{1}{k}.$$

and

$$\bigcup_{n=1}^{\infty} A_{kn} \subset \bigcup_{n=1}^{\infty} A_{(k-1)n} \quad \text{for } k = 2, 3, \ldots$$

(why?) Let $A = \bigcap_{k=1}^{\infty} \bigcup_{n=1}^{\infty} A_{kn}$. Then

$$\begin{aligned} \mu(E) \le \mu(A) &= \lim_{k\to\infty} \mu\left(\bigcup_{n=1}^{\infty} A_{kn}\right) = \lim_{k\to\infty} \nu\left(\bigcup_{n=1}^{\infty} A_{kn}\right) \\ &= \nu(A) \le \mu(E) + \lim_{k\to\infty} \frac{1}{k} = \mu(E) \end{aligned}$$

So

$$\mu(A) = \nu(A) = \mu(E)$$

and

$$\nu(A \setminus E) \le \mu(A \setminus E) = \mu(A) - \mu(E) = 0, \quad \text{i.e. } \nu(A \setminus E) = 0$$

Therefore

$$\nu(E) = \nu(A) - \nu((A \setminus E) = \nu(A) = \mu(A) = \mu(E).$$

If μ_0 is σ-finite, then there are disjoint $\{A_n\} \subset \mathscr{E}$, $\mu_0(A_n) < \infty$, such that $X = \bigcup_{n=1}^{\infty} A_n$. For any $E \in \mathscr{M}(\mathscr{E})$

$$\nu(E) = \sum_{n=1}^{\infty} \nu(E \cap A_n) = \sum_{n=1}^{\infty} \mu(E \cap A_n) = \mu(E),$$

i.e., μ is the unique extension of μ_0 on $\mathscr{M}(\mathscr{E})$. □

Remark 1.6 To summarize, the procedure of Carathéodory's construction of a measure is

1) have a semi-algebra $\mathscr{E} \subset \mathscr{P}(X)$ and a premeasure μ_0 on $\mathscr{E}$;
2) extend μ_0 to the outer measure μ^* on $\mathscr{P}(X)$;
3) restrict μ^* on the σ-algebra $\mathscr{M}^*$ of all μ^* measurable sets, then we have a measure.

Different premeasure μ_0's on $\mathscr{E}$ define different $\mathscr{M}^*$'s, but all the measures are defined on $\mathscr{M}(\mathscr{E})$.

Exercises

1.17 Let X be an uncountable set, and $\rho : \mathscr{P}(X) \to \{0, 1\}$ be the set function defined by

$$\rho(E) = \begin{cases} 0, & \text{if } \operatorname{card}(E) < 2, \\ 1, & \text{otherwise.} \end{cases}$$

Describe the outer measure μ^* induced by ρ, and find the σ-algebra of all μ^*-measurable sets.

1.18 Let μ^* be an outer measure on X and $\{A_n\}_1^\infty$ is a sequence of disjoint μ^*-measurable sets. Prove that for any $E \subset X$,

$$\mu^*(E) = \sum_{i=1}^{\infty} \mu^*(E \cap A_i) + \mu^*\left(E \cap \bigcap_{i=1}^{\infty} A_i^c\right).$$

(Theorem 1.4.6 cannot be used here because this result is used to prove Theorem 1.4.6.)

1.19 Let $\mathscr{A}$ be a semi-algebra, μ_0 be a premeasure on $\mathscr{A}$, and μ^* be the outer measure induced by μ_0. Prove that for any $E \subset X$, there is a μ^*-measurable sat A, $A \supset E$, such that

$$\mu(A) = \mu^*(E),$$

and if $\mu^*(E) < \infty$, then for any μ^*-measurable set $B \subset A \setminus E$, we have

$$\mu(B) = 0.$$

1.20 Let μ^* be an outer measure on X and $\mathscr{M}$ be the σ-algebra of all μ^*-measurable sets. Prove that $E \subset X$ with $\mu^*(E) < \infty$ is μ^*-measurable if and only if there is a μ^*-measurable set $A \subset E$ such that $\mu(A) = \mu^*(E)$.

1.21 Let μ^* be an outer measure on X induced by a premeasure on a semi-algebra.

a. Prove that $B \subset X$ is μ^*-measurable if and only if for any μ^*-measurable A with $\mu^*(A) < \infty$,

$$\mu^*(A) = \mu^*(A \cap B) + \mu^*(A \cap B^c).$$

(Hint: Only need to check the Carathéodory condition for those E with $\mu^*(E) < \infty$. Use Exercise 1.19.)

b. If in addition that the premeasure is finite, prove that $B \subset X$ is μ^*-measurable if and only if

$$\mu^*(X) = \mu^*(B) + \mu^*(B^c).$$

1.22 Let μ be a finite measure on $(X, \mathscr{M})$, μ^* be the outer measure induced by μ, and $E \subset X$ be nonmeasurable with $\mu^*(E) = \mu(X)$. Prove that for any $A, B \in \mathscr{M}$, if $A \cap E = B \cap E$, then $\mu(A) = \mu(B)$. (Hint: See Exercise 1.19.)

1.5 Lebesgue-Stieltjes Measures on $\mathbb{R}$

Now we are ready to construct the Lebesgue measure - a measure generalizing the lengths of intervals. Let

$$\mathscr{E}_h = \{(a, b], (a, \infty) : -\infty \le a \le b < \infty\}.$$

Clearly $\mathscr{E}_h$ is a semi-algebra. It is called the semi-algebra of h-intervals (half-open intervals).

If we define premeasure $m((a, b]) = b - a$ and $m(I) = \infty$ for an infinite h-interval I, then the measure generated is called Lebesgue measure and is denoted by m. The corresponding μ^*-measurable sets are called Lebesgue measurable. Since any other kind of interval is either an increasing union or a decreasing intersection of h-intervals, the Lebesgue measure of any interval is indeed its length.

More general premeasures can be defined on $\mathscr{E}_h$.

Let $F : \mathbb{R} \to \mathbb{R}$ be an increasing and right-continuous function. Define

$$\mu_F((a, b]) = F(b) - F(a), \quad \mu_F((a, \infty)) = F(\infty) - F(a) \tag{1.5.1}$$

where

$$F(\infty) = \lim_{x \to \infty} F(x), \quad F(-\infty) = \lim_{x \to -\infty} F(x).$$

Proposition 1.5.1 *μ_F is a premeasure on $\mathscr{E}_h$.*

Proof

i) $\mu_F(\emptyset) = \mu_F((a, a]) = F(a) - F(a) = 0$.

ii) Let $I = \bigcup_{n=1}^{k} I_n$, where $\{I, I_n\} \subset \mathscr{E}_h$ and $\{I_n\}$ are disjoint.

Let $I = (a, b]$ (for $b = \infty$, define $(a, b] = (a, \infty)$). By relabeling we can assume $I_n = (a_n, b_n]$ where $a_1 = a$, $a_{n+1} = b_n$, for $n = 1, 2, \ldots, k-1$, and $b_k = b$. Then

$$\sum_{n=1}^{k} \mu_F(I_n) = \sum_{n=1}^{k} (F(b_n) - F(a_n)) = F(b) - F(a) = \mu_F(I).$$

iii) Let $\{I, I_1, I_2, \dots, \} \subset \mathscr{E}_h$ and $I \subset \bigcup_{n=1}^{\infty} I_n$.

If $I = (a, b]$ is a finite h-interval, let $J_n = I \cap I_n$. Then $I \subset \bigcup_{n=1}^{\infty} J_n$, all $\{J_n\}$ are finite h-intervals, and

$$\mu_F(J_n) \le \mu_F(I_n).$$

Assume $J_n = (a_n, b_n]$. For any $\epsilon > 0$, since F is right continuous, there exists a δ_n such that

$$F(b_n) \le F(b_n + \delta_n) < F(b_n) + \frac{\epsilon}{2^n}.$$

Then for any $0 < \delta < b - a$, $\{(a_n, b_n + \delta_n), n = 1, 2, \dots\}$ is an open cover of $[a + \delta, b]$. So there exists a finite sub-cover $\{(a_{n_i}, b_{n_i} + \delta_{n_i}), 1 = 1, 2, \dots, k\}$. By relabeling and discarding any extra open intervals, we can assume

$$a_{n_1} < a + \delta,\ a_{n_2} < b_{n_1} + \delta_{n_1}, \cdots, a_{n_k} < b_{n_{k-1}} + \delta_{n_{k-1}}, b < b_{n_k} + \delta_{n_k}.$$

Then since F is increasing

$$\begin{aligned}
&F(b) - F(a + \delta) \le F(b_{n_k} + \delta_{n_k}) - F(a_{n_1}) \\
&\le F(b_{n_k} + \delta_{n_k}) - F(a_{n_k}) + F(b_{n_{k-1}} + \delta_{n_{k-1}}) - \cdots \\
&\qquad\qquad - F(a_{n_2}) + F(b_{n_1} + \delta_{n_1}) - F(a_{n_1}) \\
&= \sum_{i=1}^{k} (F(b_{n_i} + \delta_{n_i}) - F(a_{n_i})) \\
&\le \sum_{i=1}^{k} \left(F(b_{n_i}) - F(a_{n_i}) + \frac{\epsilon}{2^{n_i}} \right) \\
&\le \left(\sum_{n=1}^{\infty} \mu_F(J_n) \right) + \epsilon \\
&\le \left(\sum_{n=1}^{\infty} \mu_F(I_n) \right) + \epsilon
\end{aligned}$$

Let $\delta \to 0, \epsilon \to 0$, we have

$$\mu_F(I) = F(b) - F(a) \le \sum_{n=1}^{\infty} \mu_F(I_n).$$

If I is an infinite h-interval, let

$$J_m = \begin{cases} (a, a+m], & \text{if } I = (a, \infty), \\ (b-m, b], & \text{if } I = (-\infty, b], \\ (-m, m], & \text{if } I = (-\infty, \infty). \end{cases}$$

Then $\lim_{m\to\infty} \mu_F(J_m) = \mu_F(I)$. By what we just proved $\mu_F(J_m) \le \sum_{n=1}^{\infty} \mu_F(I_n)$ for all $m \in \mathbb{N}$. So

$$\mu_F(I) \le \lim_{m\to\infty} \mu_F(J_m) \le \sum_{n=1}^{\infty} \mu_F(I_n).$$

□

Definition 1.5.2

The measure induced by the premeasure defined by (1.5.1) for an increasing and right continuous function F is called the Lebesgue-Stieltjes measure generated by F, and is denoted by μ_F.

The Lebesgue measure is the Lebesgue-Stieltjes measure generated by $F(x) = x$.

Note that $\mathscr{B}_{\mathbb{R}} = \mathscr{M}(\mathscr{E}_h)$. So μ_F is defined on $\mathscr{B}_{\mathbb{R}}$ for any increasing and right continuous function F.

Theorem 1.5.3

If $F : \mathbb{R} \to \mathbb{R}$ is an increasing and right continuous function, then there is a unique measure μ_F on $\mathscr{B}_{\mathbb{R}}$ such that

$$\mu_F((a, b]) = F(b) - F(a) \quad \text{for all } a \le b \text{ in } \overline{\mathbb{R}}$$

with the convention that $(a, a] = \emptyset$.
Conversely every measure μ on $\mathscr{B}_{\mathbb{R}}$ which is finite on bounded intervals is a Lebesgue-Stieltjes measure generated by an increasing and right continuous function $F : \mathbb{R} \to \mathbb{R}$.

Proof

The first part is given by Theorem 1.4.11 because μ_F is σ-finite. For the second part, let μ be a measure on $\mathscr{B}_{\mathbb{R}}$ which is finite on bounded intervals. Define

$$F(x) = \begin{cases} \mu((0, x]), & \text{if } x \ge 0, \\ -\mu((x, 0]), & \text{if } x < 0. \end{cases}$$

Then F is increasing (by the monotonicity) and right continuous (by the continuity of a measure), and for any $a < b$

$$F(b)-F(a)=\begin{cases}\mu((0,b])-\mu((0,a])=\mu((a,b]), & \text{if } a\geq 0,\\ \mu((0,b])+\mu((a,0])=\mu((a,b]), & \text{if } a<0 \text{ and } b\geq 0,\\ -\mu((b,0])+\mu((a,0])=\mu((a,b]), & \text{if } b<0.\end{cases}$$

So μ is the Lebesgue-Stieltjes measure generated by F.

□

Lebesgue measure can also be induced by set function on open sets: first define $m((a,b))=b-a$, then since any open set is a countable disjoint union of open intervals, we then have the Lebesgue measure on open sets. By defining outer measure

$$m^*(E)=\inf\{m(U):U \text{ is open and } E\subset U\},$$

and restricting it to the m^*-measurable sets, we then have the Lebesgue measure. Such a procedure works for any Lebesgue-Stieltjes measure.

Theorem 1.5.4

Let μ_F be a Lebesgue-Stieltjes measure generated by F, μ_F^* be the corresponding outer measure on $\mathscr{P}(\mathbb{R})$, and $\mathscr{M}_F$ be the σ-algebra of all μ_F^*-measurable sets.

i) For any $E\in\mathscr{P}(\mathbb{R})$,

$$\mu_F^*(E)=\inf\{\mu_F(U):U \text{ is open and } E\subset U\};$$

ii) For any $E\in\mathscr{M}_F$

$$\mu_F(E)=\sup\{\mu_F(K):K \text{ is compact and } E\supset K\}.$$

Proof

i) By the monotonicity of μ_F^*,

$$\mu_F^*(E)\leq\inf\{\mu_F(U):U \text{ is open and } E\subset U\}.$$

So we only need to prove

$$\inf\{\mu_F(U):U \text{ is open and } E\subset U\}\leq\mu_F^*(E).$$

It is certainly true if $\mu_F^*(E)=\infty$.

If $\mu_F^*(E)<\infty$, then for any $\epsilon>0$, there are h-intervals $\{(a_n,b_n]\}$, $E\subset\bigcup_{n=1}^{\infty}(a_n,b_n]$, such that

$$\sum_{n=1}^{\infty}(F(b_n)-F(a_n))<\mu_F^*(E)+\epsilon.$$

Since F is right continuous, for each n, there is a $\delta_n > 0$ such that

$$F(b_n + \delta_n) \le F(b_n) + \frac{\epsilon}{2^n}.$$

Let

$$U_0 = \bigcup_{n=1}^{\infty} (a_n, b_n + \delta_n).$$

Then $E \subset U_0$ and

$$\mu_F(U_0) \le \sum_{n=1}^{\infty} \mu_F((a_n, b_n + \delta_n)) \le \sum_{n=1}^{\infty} \mu_F((a_n, b_n + \delta_n]) = \sum_{n=1}^{\infty} (F(b_n + \delta_n) - F(a_n))$$
$$< \sum_{n=1}^{\infty} \left(F(b_n) - F(a_n) + \frac{\epsilon}{2^n} \right) < \mu_F^*(E) + \epsilon + \sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \mu_F^*(E) + 2\epsilon.$$

So

$$\inf\{\mu_F(U) : U \text{ is open and } E \subset U\} \le \mu_F(U_0) < \mu_F^*(E) + 2\epsilon.$$

Let $\epsilon \to 0$, we then have

$$\inf\{\mu_F(U) : U \text{ is open and } E \subset U\} \le \mu_F^*(E).$$

ii) Let $E_n = E \cap [-n, n]$. Then $E = \bigcup_{n=1}^{\infty} E_n$ and by the continuity of measures,

$$\lim_{n \to \infty} \mu_F(E_n) = \mu_F(E).$$

For any $\epsilon > 0$, by i), there exists an open set $U \supset (\overline{E}_n \cap E_n^c)$ such that

$$\mu_F(U) < \mu_F(\overline{E}_n \cap E_n^c) + \epsilon.$$

Let $K = \overline{E}_n \cap U^c$. Then K is compact, and since $U^c \subset (\overline{E}_n \cap E_n^c)^c = \overline{E}_n^c \cup E_n$,

$$K = \overline{E}_n \cap U^c \subset \overline{E}_n \cap (\overline{E}_n^c \cup E_n) = E_n \subset E,$$

and

$$\begin{aligned} \mu_F(K) &= \mu_F(E_n) - \mu_F(E_n \cap K^c) = \mu_F(E_n) - \mu_F(E_n \cap (\overline{E}_n^c \cup U)) \\ &= \mu_F(E_n) - \mu_F(E_n \cap U) \ge \mu_F(E_n) - \mu_F((\overline{E}_n^c \cup E_n) \cap U)) \\ &= \mu_F(E_n) - (\mu_F(U) - \mu_F(\overline{E}_n \cap E_n^c)) \\ &> \mu_F(E_n) - \epsilon. \end{aligned}$$

(Note that $U = \left((\overline{E}_n^c \cup E_n) \cap U\right) \cup \left((\overline{E}_n \cap E_n^c) \cap U\right) = \left((\overline{E}_n^c \cup E_n) \cap U\right) \cup (\overline{E}_n \cap E_n^c)$.) So

$$\mu_F(E_n) - \epsilon < \mu_F(K) \leq \sup\{\mu_F(K) : K \text{ is compact and } E \supset K\} \leq \mu_F(E).$$

Let $\epsilon \to 0$,

$$\mu_F(E_n) \leq \sup\left\{\mu_F(K) : K \text{ is compact and } E \supset K\right\} \leq \mu_F(E).$$

Let $n \to \infty$, we have

$$\mu_F(E) = \sup\left\{\mu_F(K) : K \text{ is compact and } E \supset K\right\}.$$

□

For any Lebesgue-Stieltjes measure μ_F, we always have $\mathscr{M}_F \supset \mathscr{B}_{\mathbb{R}}$. However, the difference is not big.

Definition 1.5.5

A countable intersection of open sets is called a G_δ set. A countable union of closed sets is called a F_σ set.

Clearly both G_δ and F_σ are in $\mathscr{B}_{\mathbb{R}}$.

Theorem 1.5.6

Let μ_F be a Lebesgue-Stieltjes measure and $\mathscr{M}_F$ be the σ-algebra of all μ_F^*-measurable sets. Then the following are equivalent

a) $E \in \mathscr{M}_F$.
b) For any $\epsilon > 0$, there exists an open set $U \supset E$ such that $\mu_F^*(U \setminus E) < \epsilon$.
c) $E = V \setminus N_1$ for some G_δ set V and μ_F-null set N_1.
d) For any $\epsilon > 0$, there exists a closed set $K \subset E$ such that $\mu_F^*(E \setminus K) < \epsilon$.
e) $E = H \cup N_2$ for some F_σ set H and μ_F-null set N_2.

Proof

a) $\Rightarrow$ b): We first consider the case when $\mu_F(E) < \infty$. By Theorem 1.5.4, for any $\epsilon > 0$, there is an open set $U \supset E$ such that $\mu_F(E) + \mu_F(U \setminus E) = \mu_F(U) < \mu_F(E) + \epsilon$. So $\mu_F(U \setminus E) < \epsilon$.

If $\mu_F(E) = \infty$, let $E_n = E \cap (-n, n)$. Then for any $\epsilon > 0$, by what we have just proved, there is an open set U_n such that $\mu_F(U_n \setminus E_n) < \frac{\epsilon}{2^n}$. Let $U = \bigcup_{n=1}^{\infty} U_n$. Then U is open, and

$$\mu_F(U\setminus E)=\mu_F\left(\bigcup_{n=1}^{\infty}(U_n\setminus E)\right)\le\sum_{n=1}^{\infty}\mu_F(U_n\setminus E)\le\sum_{n=1}^{\infty}\mu_F(U_n\setminus E_n)<\sum_{n=1}^{\infty}\frac{\epsilon}{2^n}=\epsilon.$$

b) $\Rightarrow$ c): By the condition, there is an open set $U_n\supset E$ such that $\mu_F^*(U_n\setminus E)<\frac{1}{n}$. Let $V=\bigcap_{n=1}^{\infty}U_n$ and $N_1=V\setminus E$. Then V is a G_δ set, $V\supset E$, $E=V\setminus N_1$, and $0\le\mu_F^*(N_1)\le\mu_F^*(U_n\setminus E)<\frac{1}{n}$ for all n, i.e., N_1 is a μ_F-null set.

c) $\Rightarrow$ a): V and N_1 are both measurable. So $E=V\setminus N_1$ is measurable.

Since a) is equivalent to $E^c\in\mathscr{M}_F$, and b) and c) for E^c are equivalent to d) and e) for E, respectively, a), d), and e) are also equivalent. □

Proposition 1.5.7 *For any $E\in\mathscr{M}_F$ with $\mu_F(E)<\infty$, and any $\epsilon>0$, there is a finite disjoint union of open intervals $A=\bigcup_{i=1}^{n}(a_i,b_i)$ such that*

$$\mu_F(E\triangle A)<\epsilon.$$

The proof is left as a homework.

Definition 1.5.8

The σ-algebra of all Lebesgue measurable sets is denoted by $\mathscr{L}$, and Lebesgue measure is denoted by m.

Theorem 1.5.9

For any $E\in\mathscr{L}$ and for any $r,s\in\mathbb{R}$, let

$$E+s=\{x+s:x\in E\},\quad rE=\{rx:x\in E\}.$$

Then $E+s$ and rE are in $\mathscr{L}$, and

$$m(E+s)=m(E),\quad m(rE)=|r|m(E).$$

Proof

Without loss of generality, assume $r\neq 0$. The results are true for open intervals, so they are true for open sets.

For any E (not necessary measurable), because of Theorem 1.5.4 and the fact that each transformation defines a one-to-one correspondence between the collections of open sets covering E and its image, we have

$$\begin{aligned}m^*(E+s)&=\inf\{m(U+s):E\subset U,\ U\text{ is open}\}\\&=\inf\{m(U):E\subset U,\ U\text{ is open}\}=m^*(E)\end{aligned}$$

and

$$m^*(rE) = \inf\{m(rU) : E \subset U,\ U \text{ is open}\}$$
$$= |r| \inf\{m(U) : E \subset U,\ U \text{ is open}\} = |r| m^*(E).$$

Since the measurability is defined by outer measures, $E + s$ or rE is measurable if and only if E is measurable, and for measurable E

$$m(E+s) = m^*(E+s) = m^*(E) = m(E), \quad m(rE) = m^*(rE) = |r| m^*(E) = |r| m(E).$$

□

Definition 1.5.10

The set

$$C = \left\{ \sum_{i=1}^{\infty} \frac{a_i}{3^i} : a_i \in \{0, 2\} \right\}$$

is called the Cantor set.

The Cantor set is the set of all ternary numbers in $[0, 1]$ with only digits 0 and 2. Some numbers have two ternary expressions. For example

$$\frac{1}{3} = 0.1 = 0.022222\cdots, \quad \frac{2}{3} = 0.2 = 0.122222\cdots.$$

Such a number is in C if one of the ternary expression has no digit of 1. For example, both $\frac{1}{3}$ and $\frac{2}{3}$ are in C (Figure 1.6).

The Cantor set is obtained from $[0, 1]$ by removing disjoint open intervals

$$\left\{ \left(\sum_{i=1}^{n-1} \frac{a_i}{3^i} + \frac{1}{3^n}, \sum_{i=1}^{n-1} \frac{a_i}{3^i} + \frac{2}{3^n} \right) : a_i \in \{0, 2\}, n = 1, 2, \ldots \right\}.$$

Proposition 1.5.11

a) C is compact, nowhere dense, totally disconnected, and has no isolated points.
b) $m(C) = 0$.
c) $card(C) = card(\mathbb{R})$.

Proof

a) Homework.
b) The total length of the open intervals removed from $[0, 1]$ in forming C is



Fig. 1.6 Construction of the Cantor set

$$\frac{1}{3}+\frac{2}{9}+\frac{4}{27}+\cdots=\sum_{i=1}^{\infty}\frac{2^{i-1}}{3^i}=\frac{1}{3}\left(\frac{1}{1-\frac{2}{3}}\right)=1.$$

So $m(C)=1-1=0$.

c) Since C is a subset of $\mathbb{R}$, $\text{card}(C)\le\text{card}(\mathbb{R})$. Define $f:C\to[0,1]$

$$f\left(\sum_{i=1}^{\infty}\frac{a_i}{3^i}\right)=\sum_{i=1}^{\infty}\frac{a_i/2}{2^i}. \tag{1.5.2}$$

Then f is onto. So $\text{card}(C)\ge\text{card}([0,1])=\text{card}(\mathbb{R})$. Therefore $\text{card}(C)=\text{card}(\mathbb{R})$.

□

Definition 1.5.12

The Cantor function $f:[0,1]\to[0,1]$ is defined by

$$f\left(\sum_{i=1}^{\infty}\frac{a_i}{3^i}\right)=\begin{cases}\left(\sum_{i=1}^{n-1}\frac{a_i/2}{2^i}\right)+\frac{1}{2^n}, & \text{if } \exists a_j=1, \text{ where } n=\min\{j:a_j=1\},\\ \sum_{i=1}^{\infty}\frac{a_i/2}{2^i}, & \text{otherwise.}\end{cases}$$

For example

$$f(0)=0,\quad f(1)=f\left(\sum_{i=1}^{\infty}\frac{2}{3^i}\right)=\sum_{i=1}^{\infty}\frac{1}{2^i}=1,\quad f\left(\frac{1}{3}\right)=\frac{1}{2},\quad f\left(\frac{2}{3}\right)=\frac{1}{2},$$

$$f\left(\frac{7}{9}\right)=f\left(\frac{2}{3}+\frac{1}{9}\right)=\frac{1}{2}+\frac{1}{4}=\frac{3}{4},\quad f\left(\frac{8}{9}\right)=f\left(\frac{2}{3}+\frac{2}{9}\right)=\frac{1}{2}+\frac{1}{4}=\frac{3}{4}.$$

Proposition 1.5.13 *The Cantor function is well defined, maps the closure of any open interval $I\subset[0,1]\setminus C$ to a single number, is increasing, and continuous.*

Proof

We first show that the Cantor function value does not depend on particular ternary expansions of a number.

Any number with a finite ternary expansion will have another infinite ternary expansion:

$$\sum_{i=1}^{n} \frac{a_i}{3^i} = \sum_{i=1}^{n-1} \frac{a_i}{3^i} + \frac{a_n - 1}{3^n} + \sum_{i=n+1}^{\infty} \frac{2}{3^i}$$

where $a_n \geq 1$.

Certainly if there is an $i < n$ such that $a_i = 1$, then the function values computed using both expansions are the same, because the function value depends only on the digits up to the first 1 of the ternary expansion.

If $a_n = 1$ is the first digit of 1, then by the definition

$$f\left(\sum_{i=1}^{n-1} \frac{a_i}{3^i} + \frac{1}{3^n}\right) = \sum_{i=1}^{n-1} \frac{a_i/2}{2^i} + \frac{1}{2^n} = \sum_{i=1}^{n-1} \frac{a_i/2}{2^i} + \sum_{i=n+1}^{\infty} \frac{1}{2^i} = f\left(\sum_{i=1}^{n-1} \frac{a_i}{3^i} + \sum_{i=n+1}^{\infty} \frac{2}{3^i}\right).$$

If none of the a_i is 1, then by the definition

$$f\left(\sum_{i=1}^{n-1} \frac{a_i}{3^i} + \frac{2}{3^n}\right) = \sum_{i=1}^{n-1} \frac{a_i/2}{2^i} + \frac{1}{2^n} = f\left(\sum_{i=1}^{n-1} \frac{a_i}{3^i} + \frac{1}{3^n} + \sum_{i=n+1}^{\infty} \frac{2}{3^i}\right).$$

Let

$$I = \left(\sum_{i=1}^{n-1} \frac{a_i}{3^i} + \frac{1}{3^n}, \sum_{i=1}^{n-1} \frac{a_i}{3^i} + \frac{2}{3^n}\right) : a_i \in \{0, 2\}$$

be an open interval in $[0, 1] \setminus C$. Then any $x \in \left[\sum_{i=1}^{n-1} \frac{a_i}{3^i} + \frac{1}{3^n}, \sum_{i=1}^{n-1} \frac{a_i}{3^i} + \frac{2}{3^n}\right]$ has a ternary expansion

$$x = \sum_{i=1}^{n-1} \frac{a_i}{3^i} + \frac{1}{3^n} + \sum_{i=n+1}^{\infty} \frac{b_i}{3^i}$$

and $f(x) = \sum_{i=1}^{n-1} \frac{a_i/2}{2^i} + \frac{1}{2^n}$ for all such x's.

Since $f\Big|_C$ is the map defined in (1.5.2), which is clearly increasing, and f is a constant on any interval contained in $[0, 1] \setminus C$, f is increasing on $[0, 1]$ (Figure 1.7).

Since f is increasing and onto $[0, 1] = [f(0), f(1)]$, f is continuous (note that if f is discontinuous at x, then $f(x^-) < f(x^+)$ and the interval $\left(f(x^-), f(x^+)\right)$ is not contained in the range). □

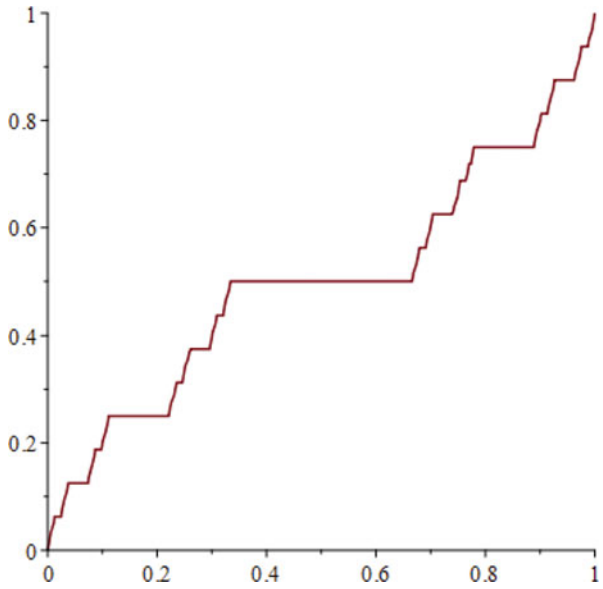


Fig. 1.7 Cantor function (Devil's staircase)

Exercises

1.23 Let $(\mathbb{R}, \mathscr{M}_F, \mu_F)$ be a Lebesgue-Stieltjes measure space, and $E \in \mathscr{M}_F$ with $\mu_F(E) < \infty$. Prove that for any $\epsilon > 0$,

a. there is an open set $U \supset E$ such that

$$\mu_F(U \setminus E) < \epsilon;$$

b. there is an A of disjoint union of finitely many open intervals such that

$$\mu_F(U \setminus A) < \epsilon;$$

c. there is an A of disjoint union of finitely many open intervals such that

$$\mu_F(E \triangle A) < \epsilon.$$

1.24 Let C be the Cantor set. Prove that

a. C is compact,
b. C does not contain any nonempty open set, therefore is nowhere dense and totally disconnect (what would be the measure if it did?)
c. C has no isolated point.

1.25 Let μ_F be a Lebesgue-Stieltjes measure. Prove that $\mu_F([a,b]) = F(b) - F(a^-)$, $\mu_F(\{a\}) = F(a) - F(a^-)$, $\mu_F((a,b)) = F(b^-) - F(a)$, $\mu_F([a,b)) = F(b^-) - F(a^-)$.

1.26 Let E be a bounded subset of $\mathbb{R}$ such that the sets

$$\{E_r = \{x + r : x \in E\} : r \in [0, 1] \cap \mathbb{Q}\}$$

are disjoint. Prove the following statements.

a. If $\bigcup_{r\in[0,1]\cap\mathbb{Q}} E_r$ has nonempty interior, then E is not Lebesgue measurable.
b. If E is Lebesgue measurable, then $m(E) = 0$.

1.27 Let N be the nonmeasure set given in Example 1.1. Prove the following statements.

a. Any Lebesgue measurable set $E \subset N$ must have $m(E) = 0$.
b. Any Lebesgue measurable set with $m(E) > 0$ must contain a Lebesgue nonmeasurable set. (Hint: $\mathbb{R} = \bigcup_{r\in\mathbb{Q}} N_r$.)

1.28 Let f be the Cantor function defined in Definition 1.5.12, C be the Cantor set, and F be the extended Cantor function defined by

$$F(x) = \begin{cases} 0, & x \le 0, \\ f(x), & 0 \le x \le 1, \\ 1, & 1 \le x. \end{cases}$$

Let μ_F be the generated Lebesgue-Stieltjes measure. Prove that $\mu_F(C) = 1$ and $\mu_F(\mathbb{R}\setminus C) = 0$.

1.29 Let $\mathbb{Q} = \{r_n : n = 1, 2, \dots\}$ be an enumeration of all rational numbers.

a. Define

$$G(x) = \sum_{n=1}^{\infty} \frac{1}{2^n} \chi_{[r_n,\infty)}(x).$$

Prove that G is increasing and right continuous.
b. Let μ_G be the Lebesgue-Stieltjes measure generated by G. Prove that $\mu_G(\{r_n\}) = \frac{1}{2^n}$, $\mu_G(\mathbb{R} \setminus \mathbb{Q}) = 0$, and $\mu_G(\mathbb{Q}) = 1$.

1.30 Let E be a Lebesgue measurable set with $m(E) > 0$. Prove that for any $\epsilon > 0$, there is an open interval I such that

$$0 \le m(I) - m(E \cap I) < \epsilon.$$

1.31 (Fat Cantor sets) Let C_α be the set formed by removing middle open intervals of length α^k, $\alpha < \frac{1}{3}$, at the kth step of the construction of the Cantor set. Prove that

$$m(C_\alpha) = \frac{1 - 3\alpha}{1 - 2\alpha} > 0.$$

1.32 Let

$$C_\sigma = \left\{ \sum_{i=1}^{n-1} \frac{a_i}{3^i} + \sum_{i=n}^{\infty} \frac{b_i}{3^i} : a_i \in \{0, 1, 2\}, b_i \in \{0, 2\}, n \in \mathbb{N} \right\}$$

with the convention that $\sum_{i=1}^{0}(\text{anything}) = 0$. Prove that C_σ is uncountable with Lebesgue measure 0. (Hint: C_σ is a countable union of translations of subsets of the Cantor set.)

Integration

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2.1 Measurable Functions

Recall that in order to define the Lebesgue integral for a function f, the set

$$\{x : a < f(x) \leq b\}$$

must be Lebesgue measurable for real numbers a and b. Since $\mathscr{B}_{\mathbb{R}}$ is generated by intervals, one must have that $f^{-1}(E)$ is Lebesgue measurable for all $E \in \mathscr{B}_{\mathbb{R}}$.

Definition 2.1.1

a) Let $(X, \mathscr{M})$ and $(Y, \mathscr{N})$ be measurable spaces. A function $f : X \to Y$ is called $(\mathscr{M}, \mathscr{N})$ measurable if $f^{-1}(E) \in \mathscr{M}$ for every $E \in \mathscr{N}$.

b) Let $(X, \mathscr{M})$ be a measurable spaces. A function $f : X \to \mathbb{R}$ (or $\mathbb{C}$) is called $\mathscr{M}$-measurable if $f^{-1}(E) \in \mathscr{M}$ for every $E \in \mathscr{B}_{\mathbb{R}}$ ($\mathscr{B}_{\mathbb{C}}$).

c) A function $f : \mathbb{R} \to \mathbb{R}$ (or $\mathbb{C}$) is called Lebesgue measurable if it is $\mathscr{L}$-measurable.

d) Let $(X, \mathscr{M})$ be a measurable spaces. A function $f : X \to \overline{\mathbb{R}}$ is called $\mathscr{M}$-measurable if $f^{-1}(E) \in \mathscr{M}$ for every $E \in \mathscr{B}_{\overline{\mathbb{R}}}$.

The following result tells us that to determine whether or not a function is measurable, we only need to check the inverse image of the sets in a generator of the σ-algebra. The proof is trivial.

Proposition 2.1.2 *If $\mathscr{N} = \mathscr{M}(\mathscr{E})$, then $f : X \to Y$ is $(\mathscr{M}, \mathscr{N})$-measurable if and only if*

$$f^{-1}(E) \in \mathscr{M} \text{ for all } E \in \mathscr{E}.$$

For any continuous function, since the inverse image of any open set is open, we have:

Corollary 2.1.3 *Continuous functions $f : X \to Y$ are $(\mathscr{B}_X, \mathscr{B}_Y)$-measurable (in particular, continuous functions $f : \mathbb{R} \to \mathbb{R}$ are Borel, and therefore, Lebesgue measurable).*

Following result is obvious because all the corresponding sets of intervals are generators of open sets.

Proposition 2.1.4 *Let $(X, \mathscr{M})$ be a measurable space. Then $f : X \to \mathbb{R}$ is measurable if and only if one of the following is true.*

a) $\{x : f(x) > \alpha\} \in \mathscr{M}$ *for every* $\alpha \in \mathbb{R}$.
b) $\{x : f(x) \geq \alpha\} \in \mathscr{M}$ *for every* $\alpha \in \mathbb{R}$.
c) $\{x : f(x) < \alpha\} \in \mathscr{M}$ *for every* $\alpha \in \mathbb{R}$.
d) $\{x : f(x) \leq \alpha\} \in \mathscr{M}$ *for every* $\alpha \in \mathbb{R}$.

Clearly by Proposition 2.1.2, Proposition 2.1.4 is still true for $f : X \to \overline{\mathbb{R}}$ except one needs to consider every $\alpha \in \overline{\mathbb{R}}$.

Proposition 2.1.5 *If $f, g : X \to \mathbb{R}$ are $\mathscr{M}$-measurable, then $f + g, cf, fg$ are all $\mathscr{M}$-measurable for any $c \in \mathbb{R}$ (i.e., measurable functions form an algebra over $\mathbb{R}$).*

Proof

i) For any $\alpha \in \mathbb{R}$, $f(x) + g(x) < \alpha$ if and only if there is an $r \in \mathbb{Q}$ such that $f(x) < r < \alpha - g(x)$. Therefore

$$\{x : f(x) + g(x) < \alpha\} = \bigcup_{r \in \mathbb{Q}} (\{x : f(x) < r\} \cap \{x : g(x) < \alpha - r\}) \in \mathscr{M}.$$

So $f + g$ is measurable.

ii) 0 function is certainly measurable.
For nonzero $c \in \mathbb{R}$ and for any $\alpha \in \mathbb{R}$

$$\{x : cf(x) < \alpha\} = \begin{cases} \left\{x : f(x) < \frac{\alpha}{c}\right\} \in \mathscr{M}, & \text{if } c > 0, \\ \left\{x : f(x) > \frac{\alpha}{c}\right\} \in \mathscr{M}, & \text{if } c < 0. \end{cases}$$

So cf is measurable for any $c \in \mathbb{R}$.

iii) For any $\alpha \in \mathbb{R}$, and for any measurable function f

$$\{x : f^2(x) > \alpha\} = \begin{cases} \left\{x : f(x) > \sqrt{\alpha}\right\} \cup \left\{x : f(x) < -\sqrt{\alpha}\right\} \in \mathscr{M}, & \text{if } \alpha \geq 0, \\ X \in \mathscr{M}, & \text{if } \alpha < 0. \end{cases}$$

So f^2 is measurable. Then for any measurable function f, g, $fg = \frac{1}{2}\left((f + g)^2 - f^2 - g^2\right)$ is measurable.

□

Proposition 2.1.6 *Let $\{f_j : j = 1, 2, \dots\}$ be $\overline{\mathbb{R}}$-valued measurable functions on $(X, \mathscr{M})$. Then the functions*

$$g_1 = \sup_j f_j, \quad g_2 = \inf_j f_j, \quad g_3 = \limsup_{j\to\infty} f_j, \quad g_4 = \liminf_{j\to\infty} f_j$$

are all measurable.

Proof
Clearly for any $\alpha \in \overline{\mathbb{R}}$

$$\{x : g_1(x) \le \alpha\} = \bigcap_{j=1}^{\infty} \{x : f_j(x) \le \alpha\} \in \mathscr{M}.$$

So g_1 is measurable.

The other 3 functions are measurable because

$$g_2(x) = -\sup_j(-f_j), \quad g_3 = \inf_k \sup_{j>k} f_j, \quad g_4 = \sup_k \inf_{j>k} f_j.$$

□

Exercises

2.1 Let $f : X \to \overline{\mathbb{R}}$ and $Y = f^{-1}(\mathbb{R})$. Prove that f is $\mathscr{M}$-measurable if and only if f is measurable on Y, and both $f^{-1}(\{-\infty\})$, $f^{-1}(\{\infty\})$ are in $\mathscr{M}$.

2.2 Let $\{f_n\}_1^\infty$ be a sequence of measurable functions on X. Prove that both sets

$$\{x : \lim_{n\to\infty} f_n(x) \text{ exists}\} \quad \text{and} \quad \{x : \lim_{n\to\infty} f_n(x) \text{ does not exists}\}$$

are measurable.

2.3 Prove that

$$\chi_E = \begin{cases} 1, \text{ if } x \in E, \\ 0, \text{ if } x \notin E, \end{cases}$$

is measurable if and only if E is measurable.

2.4 Let $f : X \to \overline{\mathbb{R}}$ be a measurable function. Prove that

$$g(x) = \begin{cases} \frac{1}{f(x)}, & \text{if } f(x) \neq 0, \pm\infty, \\ 0, & \text{if } f(x) = \pm\infty, \\ \infty, & \text{if } f(x) = 0. \end{cases}$$

is measurable.

2.5 Let D be a dense subset of $\mathbb{R}$. Prove that a function $X \to \mathbb{R}$ is measurable if and only if $\{x \in X : f(x) < r\}$ is measurable for every $r \in D$.

2.6 Let $A, B \in \mathscr{M}$. Prove that f is measurable on $A \cup B$ if and only if f is measurable on both A and B, where f is measurable on $E \in \mathscr{M}$ is defined as $f^{-1}(F) \cap E \in \mathscr{M}$ for every $F \in \mathscr{B}_{\mathbb{R}}$.

2.7 Let $g : \mathbb{R} \to \mathbb{R}$ be an increasing function. Describe $g^{-1}((a, \infty))$, where $a \in \mathbb{R}$, and use the result to prove that every monotone function $f : \mathbb{R} \to \mathbb{R}$ is Borel measurable.

2.8 Let $g : [0, 1] \to [0, 2]$ be defined by $g(x) = f(x) + x$, where f is the Cantor function defined in Definition 1.5.12, and $h : [0, 2] \to [0, 1]$ be the inverse function of g.

a. Prove that $m(g(C)) = 1$ where C is the Cantor set.
b. Let $A \subset g(C)$ be a Lebesgue nonmeasurable set (see Problem 1.27b.). Prove that $h(A)$ is Lebesgue measurable but not Borel measurable (note that h is continuous, therefore is a Borel measurable function).
c. Show that $\chi_{h(A)} \circ h$ is not Lebesgue measurable.

2.9 Let $f : \mathbb{R} \to \mathbb{R}$ be continuous and $g : \mathbb{R} \to \mathbb{R}$ be Lebesgue measurable. Prove that $f \circ g$ is Lebesgue measurable, and give an example to show that $g \circ f$ may not be Lebesgue measurable.

2.10 Let $\mathscr{M}$ be the σ-algebra of countable and co-countable sets over $\mathbb{R}$. Prove that an $f : \mathbb{R} \to \mathbb{R}$ is $\mathscr{M}$-measurable if and only if there is a $c \in \mathbb{R}$ such that $f(x) = c$ for all but countably many $x \in \mathbb{R}$.

2.11 Let μ be complete.

a. Prove that if f is measurable and $g = f$ μ-a.e., then g is measurable.
b. Prove that if $\{f_n\}$ are measurable and $f_n \to f$ μ-a.e., then f is measurable.
c. Give an example of $f = g$ μ-a.e., f is measurable, but g is not. (See Exercise 2.8 b.)

2.2 Integrals of Positive Measurable Functions

We will use the term "positive function" for a function f such that $f(x) \geq 0$ for all x, i.e., a nonnegative function. For functions $f(x) > 0$ for all x, we will call them "strictly positive functions."

Recall Figure 1.2: Since the "upper Lebesgue sum" and the "lower Lebesgue sum" always converge to the same values, we can use the limit of the "lower Lebesgue sum" to define the integral. One advantage is that such a definition also works for unbounded functions. From Figure 1.2 we can see the "Lebesgue sum" is basically the area under a function with only finite many values. Such functions are called simple functions.

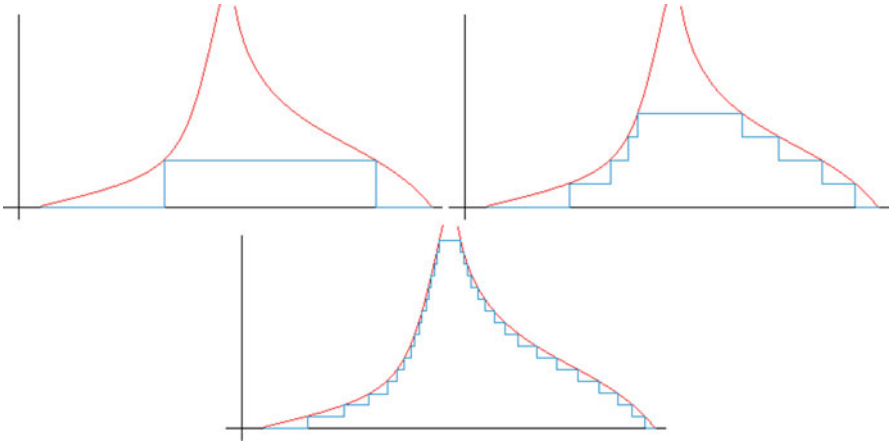


Fig. 2.1 ϕ_0, ϕ_1, and ϕ_2

Definition 2.2.1

A simple function over X is a function defined on the whole X whose range is a finite set.

Definition 2.2.2

Let $E \subset X$. The function defined by

$$\chi_E(x) = \begin{cases} 1, \text{ if } x \in E \\ 0, \text{ if } x \notin E \end{cases}$$

is called a characteristic function.

Clearly a characteristic function is a simple function, and ϕ is a simple function if and only if it is a finite linear combination of characteristic functions.

Definition 2.2.3

For a simple ϕ, if $\text{Range}(\phi) = \{c_1, \ldots, c_k\}$, then

$$\phi = \sum_{j=1}^{k} c_j \chi_{E_j}, \quad \text{where } E_j = \{x \in X : f(x) = c_j\}, \tag{2.2.1}$$

is called the standard representation of ϕ.

2

Theorem 2.2.4

Let $(X, \mathscr{M})$ be a measurable space.

a) If $f : X \to [0, \infty]$ is measurable, then there are simple functions

$$0 \le \phi_1 \le \phi_2 \le \cdots \le f$$

such that $\phi_n \to f$ pointwise, and $\phi_n \to f$ uniformly on any set on which f is bounded.

b) For any measurable function f, there are simple functions

$$0 \le |\phi_1| \le |\phi_2| \le \cdots \le |f|$$

such that $\phi_n \to f$ pointwise, and $\phi_n \to f$ uniformly on any set on which f is bounded.

Proof

a) Define (Figure 2.1)

$$E_n^k = \{x \in X : k2^{-n} < f(x) \le (k+1)2^{-n}\}, \quad n = 0, 1, 2, \ldots, \quad k = 0, 1, 2, \ldots 2^{2n}-1,$$

$$F_n = \{x \in X : 2^n < f(x)\},$$

$$\phi_n = \sum_{k=0}^{2^{2n}-1} k2^{-n}\chi_{E_n^k} + 2^n \chi_{F_n}.$$

Then $0 \le \phi_1 \le \phi_2 \le \cdots \le f$, and for any x such that $f(x) \le 2^n$,

$$0 \le f(x) - \phi_n(x) \le 2^{-n},$$

which means that $\phi_n \to f$ for each x such that $f(x) < \infty$ and uniformly on any set on which f is bounded.

If $f(x) = \infty$, then $x \in F_n$ for all n, and $\phi_n(x) = 2^n \to \infty = f(x)$ as $n \to \infty$. So $\phi_n \to f$ pointwise.

b) Let (Figure 2.2)

$$f^+(x) = \max(0, f(x)), \qquad f^-(x) = \max(0, -f(x)).$$

Then both $f^+ : X \to [0, \infty]$ and $f^- : X \to [0, \infty]$,

$$f(x) = f^+(x) - f^-(x) \quad \text{and} \quad |f(x)| = f^+(x) + f^-(x).$$

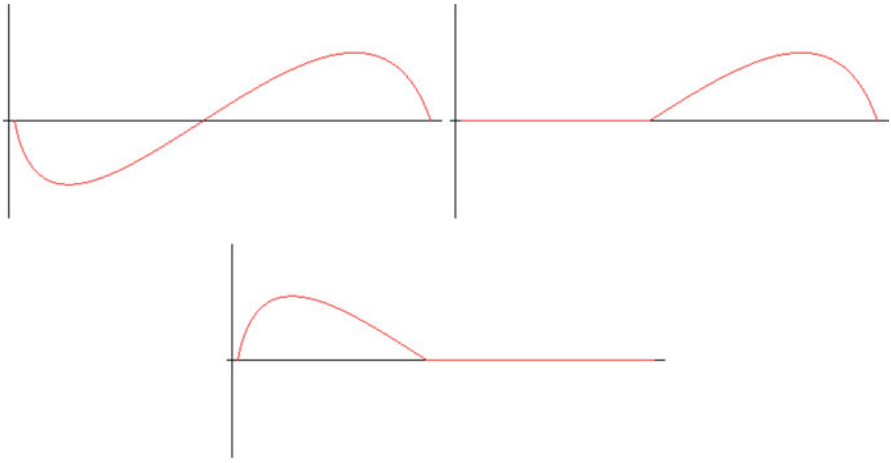


Fig. 2.2 $f(x)$, $f^+(x)$, and $f^-(x)$

By a), there are simple functions

$$0 \le \phi_1^+ \le \phi_2^+ \le \cdots \le f^+, \quad 0 \le \phi_1^- \le \phi_2^- \le \cdots \le f^-$$

$\phi_n^+ \to f^+$, $\phi_n^- \to f^-$ pointwise and uniformly on any set on which f is bounded. Let $\phi_n = \phi_n^+ - \phi_n^-$. Then

$$0 \le |\phi_1| \le |\phi_2| \le \cdots \le |f|$$

$\phi_n \to f$ pointwise, and $\phi_n \to f$ uniformly on any set on which f is bounded.

□

We first define the integrals of positive measurable simple functions.

Definition 2.2.5

Let $(X, \mathscr{M}, \mu)$ be a measure space. For any positive measurable simple functions $\phi(x) = \sum_{j=1}^n y_j \chi_{E_i}$, and for any $E \in \mathscr{M}$, define

$$\int_E \phi \, d\mu = \sum_{j=1}^n y_i \mu(E \cap E_j).$$

Since the representations of a simple function are not unique, before we prove that it does not depend on the representations, we temporarily assume Definition 2.2.5 is only for the standard representation.

2

Lemma 2.2.6 *Assume $\phi(x) = \sum_{i=1}^{n} y_i \chi_{E_i}$ with $\{E_1, \ldots, E_n\}$ disjoint and $y_i \geq 0$, and also $\phi(x) = \sum_{j=1}^{m} z_j \chi_{F_j}$ is the standard representation of ϕ. Then*

$$\sum_{i=1}^{n} y_i \mu(E_i) = \sum_{j=1}^{m} z_j \mu(F_j).$$

Proof

Without loss of generality, assume $y_i > 0$ for each i, and $z_1 = 0$. Then $F_1 = \left(\bigcup_{i=1}^{n} E_i\right)^c$. Since $\{E_i\}$ are disjoint, $F_j = \bigcup_{y_i = z_j} E_i$ for all $j > 1$. Then

$$\sum_{i=1}^{n} y_i \mu(E_i) = \sum_{j=2}^{m} \sum_{y_i = z_j} z_j \mu(E_i) = \sum_{j=2}^{m} z_j \sum_{y_i = z_j} \mu(E_i) = \sum_{j=2}^{m} z_j \mu(F_j) = \sum_{j=1}^{m} z_j \mu(F_j).$$

□

Proposition 2.2.7 *Let ϕ, ψ be positive measurable simple functions, and E be a measurable set. Then*

a) For any real number $c \geq 0$

$$\int_E c\phi \, d\mu = c \int_E \phi \, d\mu.$$

b)

$$\int_E \phi + \psi \, d\mu = \int_E \phi \, d\mu + \int_E \psi \, d\mu.$$

c) If $\phi \leq \psi$, then

$$\int_E \phi \, d\mu \leq \int_E \psi \, d\mu.$$

d) The map $A \mapsto \int_A \phi \, d\mu$ is a measure on $\mathscr{M}$.

Remark 2.1 b) implies that Definition 2.2.5 is well defined.

Proof

Let $\phi = \sum_{i=1}^{n} y_i \chi_{E_i}$ and $\psi = \sum_{j=1}^{m} z_j \chi_{F_j}$ be the standard representations.

a) Without loss of generality, assume $c > 0$. Then $c\phi = \sum_{i=1}^{n} c y_i \chi_{E_i}$ is the standard representation of $c\phi$. So

$$\int_E c\phi \, d\mu = \sum_{i=1}^{n} c y_i \mu(E_i \cap E) = c \sum_{i=1}^{m} y_i \mu(E_i \cap E) = c \int_E \phi \, d\mu.$$

b)

$$\phi + \psi = \sum_{i=1}^{n}\sum_{j=1}^{m}(y_i + z_j)\chi_{E_i \cap F_j}.$$

Since $\{E_i \cap F_j : i = 1, \ldots, n, j = 1, \ldots, m\}$ are disjoint, by Lemma 2.2.6,

$$\begin{aligned}\int_E \phi + \psi \, d\mu &= \sum_{i=1}^{n}\sum_{j=1}^{m}(y_i + z_j)\mu(E_i \cap F_j \cap E) \\ &= \sum_{i=1}^{n} y_i \sum_{j=1}^{m} \mu(E_i \cap F_j \cap E) + \sum_{j=1}^{m} z_j \sum_{i=1}^{n} \mu(E_i \cap F_j \cap E) \\ &= \sum_{i=1}^{n} y_i \mu(E_i \cap E) + \sum_{j=1}^{m} z_j \mu(F_j \cap E) \\ &= \int_E \phi \, d\mu + \int_E \psi \, d\mu.\end{aligned}$$

c) By the given condition, $y_i \le z_j$ if $E_i \cap F_j \neq \emptyset$. So

$$\int_E \phi \, d\mu = \sum_{i=1}^{n}\sum_{j=1}^{m} y_i \mu(E \cap E_i \cap F_j) \le \sum_{j=1}^{m} z_j \mu(E \cap E_i \cap F_j) = \int_E \psi \, d\mu.$$

d) We only need to prove the countable additivity. For any disjoint $\{A_j\} \subset \mathscr{M}$,

$$\begin{aligned}\int_{\bigcup_{j=1}^{\infty} A_j} \phi \, d\mu &= \sum_{i=1}^{n} y_i \mu\left(E_i \cap \bigcup_{j=1}^{\infty} A_j\right) = \sum_{i=1}^{n} y_i \sum_{j=1}^{\infty} \mu(E_i \cap A_j) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{\infty} y_i \mu(E_i \cap A_j) = \sum_{j=1}^{\infty}\sum_{i=1}^{n} y_i \mu(E_i \cap A_j) = \sum_{j=1}^{\infty} \int_{A_j} \phi \, d\mu.\end{aligned}$$

□

Now we are ready to define the integrals of positive measurable functions. Roughly speaking, it is the least upper bound of all "lower Lebesgue sums" (compare with the definition of lower Riemann integrals).

2

Definition 2.2.8

Let $(X, \mathscr{M}, \mu)$ be a measure space, $f : X \to \mathbb{R}$ be a positive measurable function, and $E \in \mathscr{M}$. Define

$$\int_E f \, d\mu = \sup \left\{ \int_E \phi \, d\mu : 0 \le \phi \le f, \text{ and } \phi \text{ is simple} \right\}.$$

Remark 2.2 Clearly by definition, we have $\int_E cf \, d\mu = c \int_E f \, d\mu$ for any $c \ge 0$, and if $f \le g$, then $\int_E f \, d\mu \le \int_E g \, d\mu$.

Since we can always define $X = E$ and the σ-algebra to be

$$\mathscr{M}_E = \{A \cap E : A \in \mathscr{M}\},$$

we will drop E and simply use $\int f \, d\mu$.

The following result shows that unlike the Riemann's theory of integration, the integrals defined this way indeed have an expected property.

Theorem 2.2.9 (Monotone Convergence Theorem)
Let $0 \le f_1(x) \le f_2(x) \le \cdots$ be a sequence of increasing positive measurable functions, and $f(x) = \lim_{n\to\infty} f_n(x)$. Then

$$\int f \, d\mu = \lim_{n\to\infty} \int f_n \, d\mu.$$

Proof

Since $f_n \le f$ for all n,

$$\lim_{n\to\infty} \int f_n \, d\mu \le \int f \, d\mu.$$

For all α with $0 < \alpha < 1$, and for any simple functions ϕ with $0 \le \phi \le f$, define

$$E_n = \{x : f_n(x) \ge \alpha\phi(x)\}.$$

Then

$$E_1 \subset E_2 \subset \cdots \quad \text{and} \quad \bigcup_{n=1}^{\infty} E_n = X.$$

By Propositions 2.2.7

$$\lim_{n\to\infty} \int f_n \, d\mu \ge \lim_{n\to\infty} \int_{E_n} f_n \, d\mu \ge \alpha \lim_{n\to\infty} \int_{E_n} \phi \, d\mu = \alpha \int \phi \, d\mu.$$

Let $\alpha \to 1$, we have

$$\lim_{n\to\infty} \int f_n \, d\mu \geq \int \phi \, d\mu,$$

so

$$\lim_{n\to\infty} \int f_n \, d\mu \geq \sup \left\{ \int \phi \, d\mu : 0 \leq \phi \leq f \right\} = \int f \, d\mu.$$

Therefore

$$\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu.$$

□

Remark 2.3 By the Monotone Convergence Theorem, the integral of a positive measurable function f can be defined as the limit of the integrals of the positive simple functions increasing to f

Corollary 2.2.10

a) For any positive measurable functions f, g and any positive real numbers a, b,

$$\int af + bg \, d\mu = a \int f \, d\mu + b \int g \, d\mu.$$

b) If $0 \leq g \leq f$ and f, g are measurable, then

$$\int f - g \, d\mu = \int f \, d\mu - \int g \, d\mu.$$

Proof

a) Let

$$0 \leq \phi_1 \leq \phi_2 \leq \cdots \leq f, \quad 0 \leq \psi_1 \leq \psi_2 \leq \cdots \leq g$$

be simple functions such that $\phi_n \to f$ and $\psi_n \to g$. Then by the Monotone Convergence Theorem

$$\begin{aligned} \int af + bg \, d\mu &= \lim_{n\to\infty} \int a\phi_n + b\psi_n \, d\mu \\ &= a \lim_{n\to\infty} \int \phi_n \, d\mu + b \lim_{n\to\infty} \int \psi_n \, d\mu \\ &= a \int f \, d\mu + b \int g \, d\mu. \end{aligned}$$

b) By a), $\int g \, d\mu + \int (f - g) \, d\mu = \int f \, d\mu$.

□

2

Following property is another one that the Riemann integrals do not have.

Theorem 2.2.11
Let $\{f_n\}$ be a sequence of positive measurable functions. Then

$$\int \sum_{n=1}^{\infty} f_n \, d\mu = \sum_{n=1}^{\infty} \int f_n \, d\mu.$$

Proof
By the Monotone Convergence Theorem and Corollary 2.2.10,

$$\begin{aligned} \int \sum_{n=1}^{\infty} f_n \, d\mu &= \int \lim_{m\to\infty} \sum_{n=1}^{m} f_n \, d\mu = \lim_{m\to\infty} \int \sum_{n=1}^{m} f_n \, d\mu \\ &= \lim_{m\to\infty} \sum_{n=1}^{m} \int f_n \, d\mu = \sum_{n=1}^{\infty} \int f_n \, d\mu. \end{aligned}$$

□

Proposition 2.2.12 *For any positive measurable function f, $\int f \, d\mu = 0$ if and only if $f = 0$ a.e.*

Proof
"⇒":
Let

$$E_n = \left\{ x : f(x) > \frac{1}{n} \right\}, \quad E = \{x : f(x) > 0\}.$$

Then

$$E_1 \subset E_2 \subset \cdots \quad \text{and} \quad \bigcup_{n=1}^{\infty} E_n = E.$$

Since

$$0 \le \frac{1}{n}\mu(E_n) \le \int_{E_n} f \, d\mu \le \int f \, d\mu = 0$$

we must have $\mu(E_n) = 0$ for all n. Therefore

$$\mu(E) = \lim_{n\to\infty} \mu(E_n) = 0.$$

"⇐":

Let $E = \{x : f(x) > 0\}$. Then $\mu(E) = 0$. So $\int_E \phi \, d\mu = 0$ for any simple function ϕ, $0 \le \phi \le f$. Therefore

$$\begin{aligned}\int f \, d\mu &= \int_E f \, d\mu + \int_{E^c} f \, d\mu \\ &= \sup\left\{\int_E \phi \, d\mu : 0 \le \phi \le f, \ \phi \text{ is simple}\right\} + \int_{E^c} 0 \, d\mu \\ &= 0.\end{aligned}$$

□

Theorem 2.2.13 (Fatou's Lemma)
For any sequence of positive measurable functions $\{f_n\}$.

$$\int \liminf_{n\to\infty} f_n \, d\mu \le \liminf_{n\to\infty} \int f_n \, d\mu.$$

Proof

For any $i \ge k$, $\inf_{n\ge k} f_n \le f_i$. So

$$\int \inf_{n\ge k} f_n \, d\mu \le \int f_i \, d\mu, \quad \text{for all } i \ge k,$$

and therefore

$$\int \inf_{n\ge k} f_n \, d\mu \le \inf_{n\ge k} \int f_n \, d\mu.$$

So by the Monotone Convergence Theorem

$$\begin{aligned}\int \liminf_{n\to\infty} f_n \, d\mu &= \int \lim_{k\to\infty} \inf_{n\ge k} f_n \, d\mu = \lim_{k\to\infty} \int \inf_{n\ge k} f_n \, d\mu \\ &\le \lim_{k\to\infty} \inf_{n\ge k} \int f_n \, d\mu = \liminf_{n\to\infty} \int f_n \, d\mu.\end{aligned}$$

□

Exercises

2.12 Let $f : X \to \overline{\mathbb{R}}$ be a positive measurable function such that $\int f \, d\mu < \infty$. Prove that $f < \infty$ a.e., and $\{x \in X : f(x) > 0\}$ is σ-finite.

2.13 Let $\{f_n : n \in \mathbb{N}\}$ be a sequence of positive measurable functions such that $f_n \to f$ pointwise (therefore $f \ge 0$, and is also measurable) and $\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu < \infty$.

2

a. Prove that for any measurable set E, $\lim_{n\to\infty} \int_E f_n \, d\mu = \int_E f \, d\mu$.
b. Give an example of $\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu = \infty$ and $\lim_{n\to\infty} \int_E f_n \, d\mu \neq \int_E f \, d\mu$.

2.14 Let $(X, \mathscr{M}, \mu)$ be a measure space and $f \geq 0$ be measurable. Prove that $\lambda(E) := \int_E f \, d\mu$ is a measure on $\mathscr{M}$, $\int \phi \, d\lambda = \int \phi f \, d\mu$ for any measurable simple function $\phi \geq 0$, and $\int g \, d\lambda = \int gf \, d\mu$ for any measurable function $g \geq 0$.

2.15 Let $f_1 \geq f_2 \geq \cdots \geq 0$ be measurable functions from X to $\mathbb{R}$ with $\int f_k \, d\mu < \infty$ for some k. Prove that there is a measurable function $f \geq 0$ such that

$$\lim_{n\to\infty} f_n(x) = f(x), \quad \text{and} \quad \lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu.$$

2.16 Let $f \geq 0$ be measurable function such that $\int f \, d\mu < \infty$. Prove that there is a sequence of measurable sets $E_1 \subset E_2 \subset \cdots$ with $\mu(E_n) < \infty$ for all n such that

$$\lim_{n\to\infty} \int_{E_n} f \, d\mu = \int f \, d\mu.$$

2.17 Let μ be the counting measure on $(X, \mathscr{P}(X))$ and f be a positive function on X. Prove that

$$\int f \, d\mu = \sup_{E\subset X \text{ is finite}} \sum_{x\in E} f(x).$$

2.3 Integrals of Integrable Functions

A naive idea is to simply define the integral of f as the integral of f^+ minus the integral of f^-. However, we would encounter situations of $\infty - \infty$. In order to avoid such a situation, we simply define the integral for those functions such that both $\int f^+ \, d\mu$ and $\int f^- \, d\mu$ are finite; i.e., the integrable functions.

Definition 2.3.1

A measurable function f is called integrable on X if

$$\int |f| \, d\mu < \infty.$$

The set of all integrable functions on X is denoted by $L^1(X)$.
For any $f \in L^1(X)$, define

$$\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu.$$

Proposition 2.3.2

a) For all $f, g \in L^1$ and for all $a, b \in \mathbb{R}$, $af + bg \in L^1$ and

$$\int af + bg \, d\mu = a \int f \, d\mu + b \int g \, d\mu.$$

b) Let $f, g \in L^1$. If $f \geq g$ on E, then

$$\int_E f \, d\mu \geq \int_E g \, d\mu.$$

Proof

a) $\int |af+bg| \, d\mu \leq \int |a||f|+|b||g| \, d\mu = |a| \int |f| \, d\mu + |b| \int |g| \, d\mu < \infty$. So $|af+bg| \in L^1$.

To prove the linearity of integrals, we only need to show

$$\int af \, d\mu = a \int f \, d\mu \quad \text{and} \quad \int f + g \, d\mu = \int f \, d\mu + \int g \, d\mu.$$

For any $a \in \mathbb{R}$,

$$\begin{aligned} \int af \, d\mu &= \int (af)^+ \, d\mu - \int (af)^- \, d\mu \\ &= \begin{cases} a \int f^+ \, d\mu - a \int f^- \, d\mu = a \left(\int f^+ \, d\mu - \int f^- \, d\mu \right), & a > 0, \\ |a| \int f^- \, d\mu - |a| \int f^+ \, d\mu = a \left(\int f^+ \, d\mu - \int f^- \, d\mu \right), & a < 0, \\ 0, & a = 0. \end{cases} \\ &= a \int f \, d\mu. \end{aligned}$$

Also since

$$f + g = (f + g)^+ - (f + g)^- = (f^+ - f^-) + (g^+ - g^-)$$

we have

$$(f + g)^+ + f^- + g^- = (f + g)^- + f^+ + g^+$$

and

$$\int (f+g)^+ \, d\mu + \int f^- \, d\mu + \int g^- \, d\mu = \int (f+g)^- \, d\mu + \int f^+ \, d\mu + \int g^+ \, d\mu,$$

which means that

$$\begin{aligned}\int f+g\,d\mu &= \int (f+g)^+\,d\mu - \int (f+g)^-\,d\mu \\ &= \int f^+\,d\mu - \int f^-\,d\mu + \int g^+\,d\mu - \int g^-\,d\mu \\ &= \int f\,d\mu + \int g\,d\mu.\end{aligned}$$

b) Since $f-g \geq 0$ on E,

$$\int_E f-g\,d\mu = \int_E f\,d\mu - \int_E g\,d\mu \geq 0, \quad i.e. \quad \int_E f\,d\mu \geq \int_E g\,d\mu.$$

□

Proposition 2.3.3 *For any $f \in L^1$,*

$$\left|\int f\,d\mu\right| \leq \int |f|\,d\mu.$$

Proof

$$\left|\int f\,d\mu\right| = \left|\int f^+\,d\mu - \int f^-\,d\mu\right| \leq \int f^+\,d\mu + \int f^-\,d\mu = \int |f|\,d\mu.$$

□

Proposition 2.3.4

a) For any $f \in L^1$, $\{x : f(x) \neq 0\}$ is a σ-finite set.

b) Let $f, g \in L^1$. Then

$$\int_E f\,d\mu = \int_E g\,d\mu \quad \text{for all } E \in \mathscr{M}$$

if and only if $f = g$ a.e.

Proof

a) Let

$$E = \{x : f(x) \neq 0\} \quad \text{and} \quad E_n = \left\{x : |f(x)| \geq \frac{1}{n}\right\}.$$

Clearly

$$E = \{x : |f(x)| > 0\} = \bigcup_{n=1}^{\infty} E_n.$$

Since

$$\frac{1}{n}\mu(E_n) \le \int_{E_n} |f|\, d\mu \le \int |f|\, d\mu < \infty,$$

we must have $\mu(E_n) < \infty$ and therefore $E = \bigcup_{n=1}^{\infty} E_n$ is σ-finite.

b) By definition if $f = g$ a.e., then $\mu(F) = 0$ where

$$F = \{x : f \neq g\}.$$

Then for any $E \in \mathscr{M}$

$$\begin{aligned}\int_E f\, d\mu &= \int_{E\cap F} f\, d\mu + \int_{E\cap F^c} f\, d\mu = \int_{E\cap F^c} f\, d\mu \\ &= \int_{E\cap F^c} g\, d\mu = \int_{E\cap F} g\, d\mu + \int_{E\cap F^c} g\, d\mu = \int_E g\, d\mu.\end{aligned}$$

Conversely, $\int_E f\, d\mu = \int_E g\, d\mu$ for all $E \in \mathscr{M}$ implies $\int_E f - g\, d\mu = 0$ for all $E \in \mathscr{M}$. Let

$$E_n = \left\{x : f(x) - g(x) \ge \frac{1}{n}\right\}, \quad F_n = \left\{f(x) - g(x) \le -\frac{1}{n}\right\}.$$

Then

$$\frac{1}{n}\mu(E_n) \le \int_{E_n} f - g\, d\mu = 0 \text{ and } \frac{1}{n}\mu(F_n) \le \int_{F_n} g - f\, d\mu = 0.$$

So $\mu(E_n) = 0$, $\mu(F_n) = 0$, and

$$\mu\left(\{x : f(x) \neq g(x)\}\right) \le \sum_{n=1}^{\infty}(\mu(E_n) + \mu(F_n)) = 0,$$

i.e., $f = g$ a.e.

□

Theorem 2.3.5 (Dominated Convergence Theorem)
Let $\{f_n\}$ be a sequence of measurable functions such that
a) $f_n \to f$ a.e., and
b) there exists a $g \in L^1$ such that $|f_n| \le g$ a.e. for all n.

Then $f_n, f \in L^1$ and

$$\int f\, d\mu = \int \lim_{n\to\infty} f_n\, d\mu = \lim_{n\to\infty} \int f_n\, d\mu.$$

2

Proof

f is measurable after redefining its values on a set of measure 0 containing all the divergent points. $f_n, f \in L^1$ because $|f_n|, |f| \le g$ a.e.

Both $g + f_n$ and $g - f_n$ are positive measurable functions and

$$g + f_n \to g + f, \quad \text{and} \quad g - f_n \to g - f \ \text{ a.e.}$$

By Fatou's Lemma

$$\int g\,d\mu + \int f\,d\mu \le \liminf_{n\to\infty} \int g + f_n\,d\mu = \int g\,d\mu + \liminf_{n\to\infty} \int f_n\,d\mu$$

and

$$\int g\,d\mu - \int f\,d\mu \le \liminf_{n\to\infty} \int g - f_n\,d\mu = \int g\,d\mu - \limsup_{n\to\infty} \int f_n\,d\mu$$

i.e.

$$\limsup_{n\to\infty} \int f_n\,d\mu \le \int f\,d\mu \le \liminf_{n\to\infty} \int f_n\,d\mu.$$

So

$$\limsup_{n\to\infty} \int f_n\,d\mu = \liminf_{n\to\infty} \int f_n\,d\mu = \lim_{n\to\infty} \int f_n\,d\mu = \int f\,d\mu.$$

□

Theorem 2.3.6

Suppose $\{f_n\} \subset L^1$ *and* $\sum_{n=1}^{\infty} \int |f_n|\,d\mu < \infty$. *Then* $\sum_{n=1}^{\infty} f_n$ *converges a.e. to a function in* L^1 *and*

$$\int \sum_{n=1}^{\infty} f_n\,d\mu = \sum_{n=1}^{\infty} \int f_n\,d\mu.$$

Proof

Let $g(x) = \sum_{n=1}^{\infty} |f_n|$. Then by Monotone Convergence Theorem

$$\int g\,d\mu = \sum_{n=1}^{\infty} \int |f_n|\,d\mu < \infty,$$

which means that $\sum_{n=1}^{\infty} f_n$ converges a.e. to a function in L^1 (why?)

Since $\left|\sum_{n=1}^{k} f_n\right| \le g$, by Dominant Convergence Theorem

$$\int \sum_{n=1}^{\infty} f_n \, d\mu = \int \lim_{k\to\infty} \sum_{n=1}^{k} f_n \, d\mu = \lim_{k\to\infty} \int \sum_{n=1}^{k} f_n \, d\mu$$
$$= \lim_{k\to\infty} \sum_{n=1}^{k} \int f_n \, d\mu = \sum_{n=1}^{\infty} \int f_n \, d\mu.$$

□

It can be easily shown that for any function in L^1, the $\|f\|_1 = \int |f| \, d\mu$ is a norm (which is called L_1 norm).

Definition 2.3.7

$f_n \to f$ in L^1 if $\|f_n - f\|_1 \to 0$.

Theorem 2.3.8

a) For any measure space $(X, \mathscr{M}, \mu)$, simple functions are dense in $L^1(\mu)$.

b) For any Lebesgue-Stieltjes measure space $(\mathbb{R}, \mathscr{M}, \mu_F)$, step functions (i.e., simple functions of intervals) are dense in $L^1(\mu_F)$.

c) For any Lebesgue-Stieltjes measure space $(\mathbb{R}, \mathscr{M}, \mu_F)$, continuous functions with compact support are dense in $L^1(\mu_F)$ ($support(f) = \overline{\{x \in \mathbb{R} : f(x) \neq 0\}}$).

Proof

a) For any $f \in L^1(\mu)$, there exist simple functions $\{\phi_n\}$ such that

$$0 \le |\phi_1| \le |\phi_2| \le \cdots \le |f| \quad \text{and} \quad \phi_n \to f \ \text{ pointwise.}$$

So by the Dominated Convergence Theorem (what is the dominant function?)

$$\lim_{n\to\infty} \|\phi_n - f\| = \lim_{n\to\infty} \int |\phi_n - f| \, d\mu = 0,$$

i.e., simple functions are dense in $L^1(\mu)$.

b) By a), we only need to show that a simple function in L^1 can be approximated in L^1 norm by step functions. Since a simple function in L^1 is a finite linear combination of characteristic functions of measurable sets with finite measures, we only need to show this for a characteristic function of measurable set with a finite measure.

Let $E \in \mathscr{M}$ with $\mu_F(E) < \infty$, then by Proposition 1.5.7, for any $\epsilon > 0$, there is A, which is a finite disjoint union of open intervals, such that

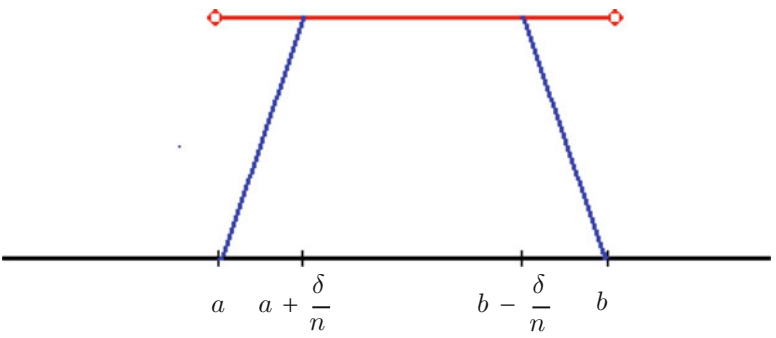


Fig. 2.3 Approximate $\chi_{(a,b)}$ by continuous functions

$$\mu_F(E \triangle A) < \epsilon.$$

Then χ_A is a step function, and

$$\int |\chi_E - \chi_A| \, d\mu_F = \mu_F(E \triangle A) < \epsilon.$$

So step functions are dense in $L^1(\mu_F)$.

c) By b), we only need to show that the characteristic function of a bounded open interval can be approximated in L^1 norm by continuous functions (Figure 2.3). For a given such open interval (a, b), let $\delta = \frac{b-a}{3}$ and define

$$f_n(x) = \begin{cases} 0, & x \le a, \\ \frac{n}{\delta}(x-a), & a \le x \le a + \frac{\delta}{n}, \\ 1, & a + \frac{\delta}{n} \le x \le b - \frac{\delta}{n}, \\ \frac{n}{\delta}(b-x), & b - \frac{\delta}{n} \le x \le b, \\ 0, & b \le x. \end{cases}$$

Then f_n is continuous, $f = \chi_{(a,b)}$ on $(-\infty, a] \cup \left(a + \frac{\delta}{n}, b - \frac{\delta}{n}\right] \cup [b, \infty)$, and on $\left(a, a + \frac{\delta}{n}\right] \cup \left(b - \frac{\delta}{n}, b\right)$, we have $|f_n - \chi_{(a,b)}| \le 1$. Therefore

$$\begin{aligned} \int |f_n - \chi_{(a,b)}| \, d\mu_F &\le \mu_F\left(\left(a, a + \frac{\delta}{n}\right] \cup \left(b - \frac{\delta}{n}, b\right)\right) \\ &= F\left(a + \frac{\delta}{n}\right) - F(a) + F\left(b^-\right) - F\left(b - \frac{\delta}{n}\right) \\ &\to 0 \text{ as } n \to \infty. \end{aligned}$$

So continuous functions are dense in $L^1(\mu_F)$.

□

Theorem 2.3.9
Let $f : X \times [a,b] \to \mathbb{R}$ such that $f(x,t) \in L^1(\mu(x))$ for all $t \in [a,b]$ and

$$F(t) = \int f(x,t)\, d\mu(x).$$

a) If there is a $g \in L^1(\mu(x))$ such that $|f(x,t)| \le g(x)$ for all $(x,t) \in X \times [a,b]$ and $f(x,t)$ is continuous in t for all $x \in X$, then $F(t)$ is continuous on $[a,b]$.
b) If $\frac{\partial f}{\partial t}$ exists and there is a $g \in L^1(\mu(x))$ such that $\left|\frac{\partial f}{\partial t}(x,t)\right| \le g(x)$ for all $(x,t) \in X \times [a,b]$, then

$$\frac{dF}{dt} = \frac{d}{dt}\int f(x,t)\, d\mu(x) = \int \frac{\partial f}{\partial t}(x,t)\, d\mu(x).$$

Proof
The proofs are simple applications of the Dominated Convergence Theorem. The dominant functions are corresponding $g(x)$'s. Note that for b), by Mean Value Theorem,

$$\left|\frac{f(x,t+h) - f(x,t)}{h}\right| = \left|\frac{\partial f}{\partial t}(x,t_1)\right| \le g(x),$$

also $\lim_{t\to t_0} F(t) = L$ if and only if $\lim_{n\to\infty} F(t_n) = L$ for any $t_n \to t_0$. □

Exercises

2.18 Prove that for any integrable function f and for any $x_0 \in X$ such that $\{x_0\}$ is measurable,

$$\int_{\{x_0\}} f\, d\mu = f(x_0)\mu(\{x_0\}).$$

2.19 Let μ be the counting measure on $(X, \mathcal{P}(X))$ and $f \in L^1(\mu)$. Prove that the set $E = \{x \in X : f(x) \neq 0\}$ is countable.

2.20 a. Prove that if $\{f_n\}$ are integrable, $f_n \to f$ uniformly, and $\mu(X) < \infty$, then $f \in L^1(\mu)$ and $\lim_{n\to\infty} \int f_n\, d\mu = \int f\, d\mu$.
b. Show that $\{x^{-\frac{n+1}{n}}\}$ are Lebesgue integrable over $[1,\infty)$ and $x^{-\frac{n+1}{n}} \to x^{-1}$ uniformly, but x^{-1} is not Lebesgue integrable over $[1,\infty)$.
c. Show that $\{n^{-1}\chi_{[-n,n]}\}$ are Lebesgue integrable over $\mathbb{R}$ and $n^{-1}\chi_{[-n,n]} \to 0$ uniformly, but $\int n^{-1}\chi_{[-n,n]}\, dm \not\to 0$.

2.21 Prove the generalized Dominated Convergence Theorem: Let f_n, f be measurable and $g_n, g \in L^1$ such that $|f_n| \leq g_n$, $f_n \to f$ and $g_n \to g$ a.e., and $\int g_n \, d\mu \to \int g \, d\mu$. Then $f_n, f \in L^1$ and

$$\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu.$$

2.22 Let $f_n, f \in L^1$ and $f_n \to f$ a.e. Use the generalized Dominated Convergence Theorem to prove that $\int |f_n - f| \, d\mu \to 0$ if and only if $\int |f_n| \, d\mu \to \int |f| \, d\mu$.

2.23

a. Let $f \in L^1(\mu)$. Prove that for any $\epsilon > 0$, there is an $E \in \mathscr{M}$ such that $\mu(E) < \infty$ and $\left|\int_X f \, d\mu - \int_E f \, d\mu\right| < \epsilon$.

b. Let $f \in L^1(m)$. Prove that the function $F(x) = \int_{-\infty}^{x} f(t) \, dm(t)$ is continuous on $\mathbb{R}$.

2.24 Let g be a continuous increasing function on $[a, b]$, and let $c = g(a)$, $d = g(b)$. Prove that for any Borel set $E \subset [c, d]$, $m(E) = \mu_g(g^{-1}(E))$, and for any Borel measurable and Lebesgue integrable function f, $f \circ g$ is μ_g-integrable, and

$$\int_c^d f(y) \, dy = \int_a^b f(g(x)) \, d\mu_g(x),$$

where μ_g is the Lebesgue Stieltjes measure generated by g.

2.25 Let $M \in \mathbb{R}$, and $\{f_n\}$ be a sequence of measurable functions such that $|f_n(x)| \leq M$ for all x and n, and $f_n \to f$ a.e. Prove that for any measurable set E with $\mu(E) < \infty$,

$$\lim_{n\to\infty} \int_E f_n \, d\mu = \int_E f \, d\mu.$$

2.26 Let $\mathbb{Q} = \{r_n : n = 1, 2, \ldots, \}$ be an enumeration of all rational numbers and μ_F be the Lebesgue-Stieltjes measure in Exercise 1.29 b.,

$$f(x) = \begin{cases} \frac{1}{2^n}, & x = r_n, \\ 0 & \text{otherwise.} \end{cases}$$

Evaluate $\int f \, d\mu_F$.

2.27 Let $(X, \mathscr{M}, \mu)$ be a measure space and $f \in L^1(\mu)$. Prove that if $\int_E |f| \, d\mu \leq \mu(E)$ for all $E \in \mathscr{M}$, then $-1 \leq f \leq 1$ a.e.

2.28

a. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Find

$$\lim_{n\to\infty} \int_0^1 f(x^n) \, dx.$$

b. Let $f : \mathbb{R} \to \mathbb{R}$ be Lebesgue integrable. Find

$$\lim_{n\to\infty} \int_{\mathbb{R}} f(x+n) \frac{x^2}{1+x^2}\, dx.$$

2.29 Compute the following limits and justify the calculation:

a. $\lim_{n\to\infty} \int_0^\infty \frac{\sin(x/n)}{(1+x/n)^n}\, dx.$
b. $\lim_{n\to\infty} \int_0^1 \frac{1+nx^2}{(1+x^2)^n}\, dx.$
c. $\lim_{n\to\infty} \int_0^\infty \frac{n\sin(x/n)}{x(1+x^2)}\, dx.$
d. $\lim_{n\to\infty} \int_0^\infty \frac{1}{x^{1/n}+x^n}\, dx.$
e. $\lim_{n\to\infty} \int_0^\infty \frac{e^{-x}}{1+x^{1/n}}\, dx.$

2.4 Relationship Between Riemann and Lebesgue Integrals

Definition 2.4.1

A partition P of an interval $[a, b]$ is a set of real numbers $\{x_0, x_1, \ldots, x_n\}$ such that

$$a = x_0 < x_1 < \cdots < x_n = b,$$

and $\|P\|$ is defined by

$$\|P\| = \max\{x_i - x_{i-1}, i = 1, 2, \ldots, n\}.$$

Notice that for any bounded function f on $[a, b]$, we can always find a sequence of partitions

$$P_1 \subset P_2 \subset \cdots$$

such that the corresponding Riemann upper and lower sums have the property that

$$\lim_{n\to\infty} U_{P_n} = \overline{\int_a^b} f(x)\, dx, \quad \lim_{n\to\infty} L_{P_n} = \underline{\int_a^b} f(x)\, dx,$$

because by definitions of Riemann upper and lower integrals, we can always find partitions $\{Q_n\}$ and $\{R_n\}$ such that

$$\lim_{n\to\infty} U_{Q_n} = \overline{\int_a^b} f(x)\, dx, \quad \lim_{n\to\infty} L_{R_n} = \underline{\int_a^b} f(x)\, dx.$$

Then simply let

$$P_n = \bigcup_{i=1}^{n} (Q_i \cup R_i).$$

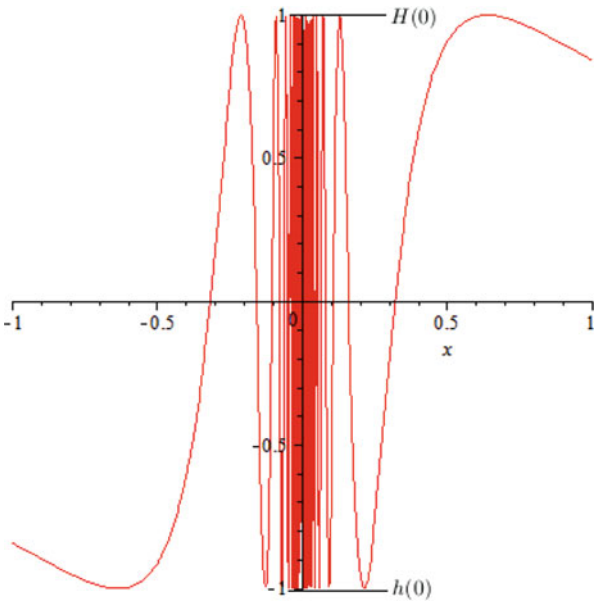


Fig. 2.4 $H(0)$ and $h(0)$ of $\sin\left(\frac{1}{x}\right)$

Definition 2.4.2

For a function $f(x)$ (Figure 2.4), define the upper envelope function to be

$$H(x) = \lim_{\delta \to 0} \sup_{|y-x| \le \delta} f(y)$$

and the lower envelope function to be

$$h(x) = \lim_{\delta \to 0} \inf_{|y-x| \le \delta} f(y)$$

Clearly

1) $h(x) \le f(x) \le H(x)$ and
2) $h(x) = f(x) = H(x)$ if and only if f is continuous at x.

Theorem 2.4.3

Let f be a bounded real function on $[a, b]$.

a) If f is Riemann integrable on $[a, b]$, then it is Lebesgue integrable on $[a, b]$, and

$$\int_a^b f(x)\,dx = \int_{[a,b]} f\,dm.$$

(continued)

Theorem 2.4.3 (continued)

b) *f is Riemann integrable on $[a,b]$ if and only if it is continuous m-a.e. on $[a,b]$, i.e., if and only if*

$$m\left(\{x \in [a,b] : f \text{ is discontinuous at } x\}\right) = 0.$$

Proof

a) Let $P_1 \subset P_2 \subset \cdots$ be a sequence of partitions such that $\|P_n\| \to 0$ and

$$\lim_{n\to\infty} U_{P_n} = \bar{\int_a^b} f(x)\,dx, \quad \lim_{n\to\infty} L_{P_n} = \underline{\int_a^b} f(x)\,dx.$$

For each $P_n = \{a = x_0 < \cdots < x_{k_n} = b\}$, define

$$\phi_n = \sum_{i=1}^{k_n} M_i \chi_{[x_{i-1},x_i]}, \quad \psi_n = \sum_{i=1}^{k_n} m_i \chi_{[x_{i-1},x_i]}, \quad M_i = \sup_{[x_{i-1},x_i]} f(x), \quad m_i = \inf_{[x_{i-1},x_i]} f(x).$$

Then

$$\psi_1 \le \psi_2 \le \cdots \le f \le \cdots \le \phi_2 \le \phi_1 \quad \text{a.e. on } [a,b]$$

and

$$\int_{[a,b]} \phi_n\,dm = U_{P_n} \to \bar{\int_a^b} f(x)\,dx, \quad \int_{[a,b]} \psi_n\,dm = L_{P_n} \to \underline{\int_a^b} f(x)\,dx.$$

For each x, since $\{\phi_n\}$ and $\{\psi_n\}$ are bounded monotone sequences,

$$g = \lim_{n\to\infty} \psi_n, \quad G = \lim_{n\to\infty} \phi_n$$

exist and are measurable, and by the Dominated Convergence Theorem (what is the dominant function?),

$$\int_{[a,b]} g\,dm = \lim_{n\to\infty} \int_{[a,b]} \psi_n\,dm = \int_a^b f(x)\,dx = \lim_{n\to\infty} \int_{[a,b]} \phi_n\,dm = \int_{[a,b]} G\,dm.$$

So

$$\int_{[a,b]} G - g\,dm = 0$$

and since $G - g \ge 0$, $g = f = G$ a.e. on $[a,b]$. Therefore f is measurable and

$$\int_{[a,b]} f\,dm = \int_{[a,b]} g\,dm = \int_a^b f(x)\,dx.$$

b) Homework.

□

Exercises

2.30 Let $f : \mathbb{R} \to \mathbb{R}$ be bounded on $[a, b]$, H and h be defined by Definition 2.4.2, $P_1 \subset P_2 \subset \cdots$ be a sequence of partitions such that $\|P_n\| \to 0$ and

$$\lim_{n\to\infty} U_{P_n} = \bar{\int}_a^b f(x)\,dx, \quad \lim_{n\to\infty} L_{P_n} = \underline{\int}_a^b f(x)\,dx,$$

and G and g be defined as in the proof of Theorem 2.4.3.

a. Prove that for any x which is not a partition point of any P_n, $g(x) = h(x)$ and $G(x) = H(x)$.

b. Prove that

$$\underline{\int}_a^b f(x)\,dx = \int_{[a,b]} h\,dm, \quad \bar{\int}_a^b f(x)\,dx = \int_{[a,b]} H\,dm,$$

where the right-hand sides of both equalities are Lebesgue integrals.

c. Prove that f is Riemann integrable if and only if $H(x) - h(x) = 0$ m-a.e., and therefore if and only if f is continuous m-a.e. on $[a, b]$.

2.5 Convergence in Measure

Definition 2.5.1

A sequence of measurable functions $\{f_n\}$ is called to converge to another measurable function f in measure if for any $\epsilon > 0$

$$\mu(\{x : |f_n(x) - f(x)| \geq \epsilon\}) \to 0 \quad \text{as} \quad n \to \infty.$$

We have encountered several kinds of convergence, for example $f_n \to f$ pointwise; a.e.; uniformly; in L^1; in measure; etc. In this section we study the relations among them.

Example 2.1

i) $f_n = \frac{1}{n}\chi_{[0,n]}$.
$f_n \to 0$ pointwise, uniformly, and in measure. $f_n \not\to 0$ in L^1 (Figure 2.5).

ii) $g_n = \chi_{[n,n+1]}$

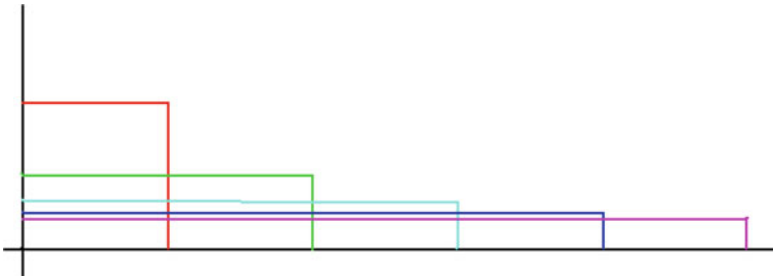


Fig. 2.5 $f_n(x)$

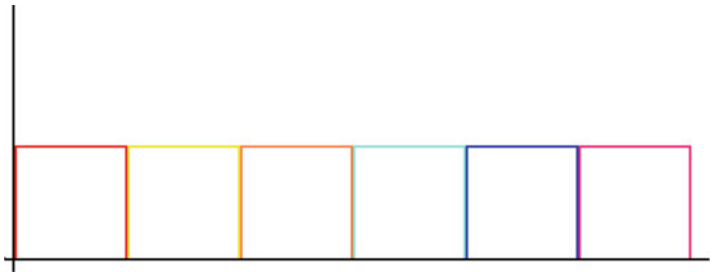


Fig. 2.6 $g_n(x)$

$g_n \to 0$ pointwise. $f_n \not\to 0$ uniformly, in measure, or in L^1 (Figure 2.6).

iii) $h_n = n\chi_{\left[0,\frac{1}{n}\right]}$
$h_n \to 0$ a.e., and in measure. $h_n \not\to 0$ uniformly, or in L^1 (Figure 2.7).

iv) $l_n = \chi_{[j2^{-k},(j+1)2^{-k}]}$, where $0 \le j < 2^k$, and $n = 2^k + j$ (Figure 2.8).
$l_n \to 0$ in L^1, and in measure. $h_n \not\to 0$ anywhere (so certainly not uniformly)

■

Proposition 2.5.2 *If $f_n \to f$ in L^1, then $f_n \to f$ in measure.*

Proof

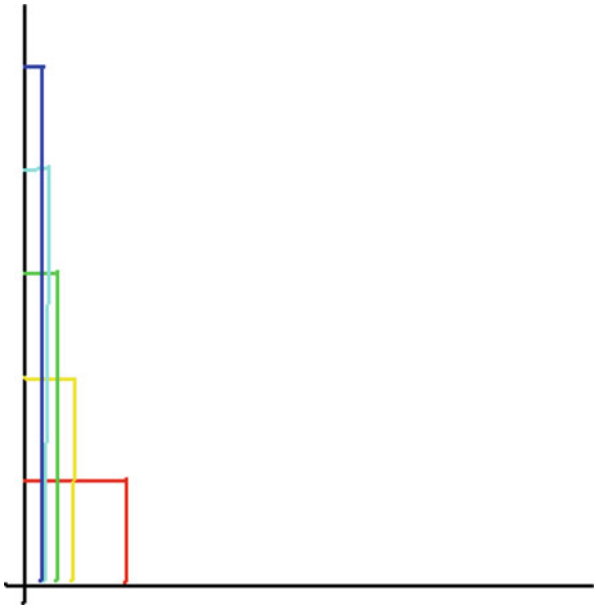
For any $\epsilon > 0$, let

$$E_{n,\epsilon} = \{x : |f_n(x) - f(x)| > \epsilon\}.$$

Then

$$\int |f_n - f|\, d\mu \ge \int_{E_{n,\epsilon}} |f_n - f|\, d\mu \ge \epsilon\mu(E_{n,\epsilon}).$$

Fig. 2.7 $h_n(x)$



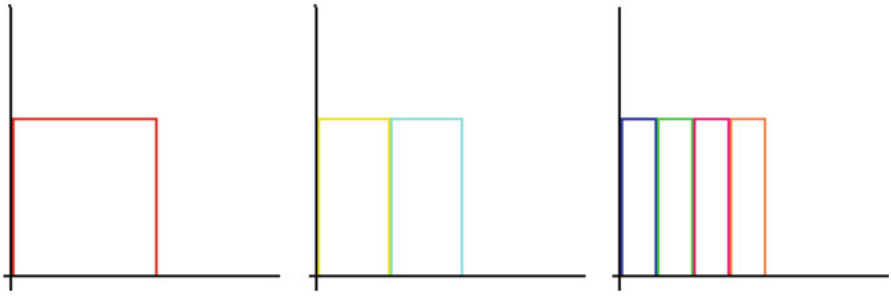


Fig. 2.8 $l_n(x)$

Therefore $f_n \to f$ in L^1 implies that $f_n \to f$ in measure. □

The opposite is not true, for example, i) and iii) of Example 2.1.

Theorem 2.5.3
If $f_n \to f$ in measure, then there is a subsequence $\{f_{n_k}\} \subset \{f_n\}$ such that $f_{n_k} \to f$ a.e.

Proof
There are $n_1 < n_2 < \cdots$ such that

$$\mu\left(\left\{x : |f_n(x) - f(x)| \geq \frac{1}{k}\right\}\right) < \frac{1}{2^k} \quad \text{for all } n \geq n_k.$$

We show that $f_{n_k} \to f$ a.e. Let

$$E_k = \left\{x : |f_{n_k}(x) - f(x)| \geq \frac{1}{k}\right\} \quad \text{and} \quad A = \bigcap_{m=1}^{\infty} \bigcup_{k=m}^{\infty} E_k.$$

Then

$$\mu(A) = \lim_{m\to\infty} \mu\left(\bigcup_{k=m}^{\infty} E_k\right) \leq \lim_{m\to\infty} \sum_{k=m}^{\infty} \mu(E_k) = 0.$$

For all $x \in A^c = \bigcup_{m=1}^{\infty} \bigcap_{k=m}^{\infty} E_k^c$, $x \in \bigcap_{k=m}^{\infty} E_k^c$ for some m. Therefore for all $k \geq m$

$$0 \leq |f_{n_k}(x) - f(x)| < \frac{1}{k} \to 0,$$

i.e., $f_{n_k} \to f$ on A^c as $k \to \infty$. □

As we can see in Example 2.1, $f_n \to f$ a.e. does not imply $f_n \to f$ in measure. However, this is true if $\mu(X) < \infty$.

Theorem 2.5.4
If $f_n \to f$ a.e. and $\mu(X) < \infty$, then $f_n \to f$ in measure.

Proof
For any $\epsilon > 0$, let

$$E_n(\epsilon) = \{x : |f_m(x) - f(x)| \geq \epsilon \text{ for some } m \geq n\}. \tag{2.5.2}$$

Then $E_1 \supset E_2 \supset \cdots$ and $\{x : |f_n(x) - f(x)| \geq \epsilon\} \subset E_n$. So

$$\lim_{n\to\infty} \mu\left(\{x : |f_n(x) - f(x)| \geq \epsilon\}\right) \leq \lim_{n\to\infty} \mu(E_n)$$
$$= \mu\left(\bigcap_{n=1}^{\infty} E_n\right) = \mu\left(\{x : f_n(x) \not\to f(x)\}\right) = 0,$$

i.e., $f_n \to f$ in measure.

□

Theorem 2.5.5 (Egoroff's Theorem)
If $f_n \to f$ a.e. and $\mu(X) < \infty$, then for any $\epsilon > 0$, there is a measurable $E \subset X$ with $\mu(E) < \epsilon$ such that $f_n \to f$ uniformly on E^c.

Proof
For any $k \in \mathbb{N}$, let $E_n(1/k)$ be the set defined by (2.5.2). Then for any $\epsilon > 0$, since $\mu(E_n(1/k)) \to 0$ as $n \to \infty$, there is an n_k such that $\mu(E_{n_k}(1/k)) < \frac{\epsilon}{2^k}$. Let

$$E = \bigcup_{k=1}^{\infty} E_{n_k}(1/k), \quad \text{then} \quad \mu(E) \le \sum_{k=1}^{\infty} \mu(E_{n_k}(1/k)) < \sum_{k=1}^{\infty} \frac{\epsilon}{2^k} = \epsilon.$$

We claim that $\{f_n\}$ converges uniformly on E^c. For any $\delta > 0$, choose $k_0 > \frac{1}{\delta}$. Then for all

$$x \in E^c = \bigcap_{k=1}^{\infty} E^c_{n_k}(1/k) \subset E^c_{n_{k_0}}(1/k_0)$$

we have $|f_n(x) - f(x)| < \frac{1}{k_0} < \delta$ for all $n \ge n_{k_0}$; i.e., $f_n \to f$ uniformly on E^c.

□

Exercises

2.31 Prove that $f_n \to f$ in measure if and only if for any $\epsilon > 0$, there is a $N > 0$ such that

$$\mu(\{x : |f_n - f| \ge \epsilon\}) < \epsilon$$

for all $n \ge N$.

2.32 Prove that if $f_n \to f$ and $g_n \to g$ in measure, then $af_n + bf_n \to af + bg$ in measure for any $a, b \in \mathbb{R}$.

2.33 Prove the FatouŠs Lemma for convergence in measure: Let f_n, f be positive measurable functions such that $f_n \to f$ in measure. Then

$$\int f \, d\mu \le \liminf_{n\to\infty} \int f_n \, d\mu.$$

(Hint: there is a subsequence $\{f_{n_k}\} \subset \{f_n\}$ such that $\lim_{k\to\infty} \int f_{n_k} \, d\mu = \liminf_{n\to\infty} \int f_n \, d\mu$.)

2.34 Prove the Dominated Convergence Theorem for convergence in measure: Let f_n, f be measurable functions such that $f_n \to f$ in measure, and there is a $g \in L^1$ such that $|f_n| \leq g$. Then

$$\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu.$$

2.35 Prove that if $f_n \to f$ in measure and there is a $g \in L^1$ such that $|f_n| \leq g$, then $f_n \to f$ in L^1.

2.36

a. Prove that $f_n \to f$ in measure if and only if $\frac{|f_n - f|}{1+|f_n - f|} \to 0$ in measure.

b. Prove that if $\mu(X) < \infty$, then $f_n \to f$ in measure if and only if

$$\int \frac{|f_n - f|}{1 + |f_n - f|} \, d\mu \to 0.$$

2.37 Prove that if $f_n \to f$ in measure and there is an $M > 0$ such that $|f_n| \leq M$ for all x and for all n, then $|f| \leq M$.

2.6 Product σ-Algebra

Let $(X_i, \mathscr{M}_i), i = 1, \ldots, n$ be measure spaces.

Definition 2.6.1

For sets $E_i \in \mathscr{M}_i$, the set

$$E_1 \times E_2 \times \cdots \times E_n$$

is called a measurable cuboid, and each E_i is called a side of the measurable cuboid $E_1 \times E_2 \times \cdots \times E_n$.

The σ-algebra of subsets of $X_1 \times X_2 \times \cdots \times X_n$ generated by all measurable cuboids is called the product σ-algebra, and is denoted by

$$\mathscr{M}_1 \otimes \mathscr{M}_2 \otimes \cdots \otimes \mathscr{M}_n.$$

A measurable cuboid $E_1 \times E_2$ of two sides is called a measurable rectangle.

In fact, we are going to show that the Borel σ-algebra of $\mathbb{R}^n$ (i.e., the σ-algebra generated by open sets of $\mathbb{R}^n$) equals the product of Borel σ-algebras of $\mathbb{R}$. We'll use a collection of special Borel measurable cuboid in $\mathbb{R}^n$.

2

Definition 2.6.2

A Borel measurable cuboid in $\mathbb{R}^n$ whose sides are closed intervals of equal lengths h is called a cube of size h.

Define $\mathscr{Q}_k$ to be the collection of all cubes of size 2^{-k} with vertices at $2^{-k}\mathbb{Z}^n$, i.e.

$$\mathscr{Q}_k = \left\{ \left[\frac{m_1}{2^k}, \frac{m_1+1}{2^k}\right] \times \left[\frac{m_2}{2^k}, \frac{m_2+1}{2^k}\right] \times \cdots \times \left[\frac{m_n}{2^k}, \frac{m_n+1}{2^k}\right] : m_i \in \mathbb{Z} \right\},$$

and let

$$\mathscr{Q} = \bigcup_{k=1}^{\infty} \mathscr{Q}_k.$$

Remark 2.4 $\mathscr{Q}_k$ is countable (so is $\mathscr{Q}$) and all cubes in $\mathscr{Q}_k$ have disjoint interiors. Furthermore, for any cubes $Q_1 \in \mathscr{Q}_{k_1}$, $Q_2 \in \mathscr{Q}_{k_2}$ with $k_1 < k_2$, either Q_1, Q_2 have disjoint interiors or $Q_1 \supset Q_2$.

Lemma 2.6.3 *For any open set $U \subset \mathbb{R}^n$ and any $k_0 \in \mathbb{N}$, U is a union of cubes in $\bigcup_{k=k_0}^{\infty} \mathscr{Q}_k$ with disjoint interiors.*

Proof

We first show that

$$U = \bigcup_{Q \in \bigcup_{k=k_0}^{\infty} \mathscr{Q}_k,\ Q \subset U} Q.$$

Clearly

$$\bigcup_{Q \in \bigcup_{k=k_0}^{\infty} \mathscr{Q}_k,\ Q \subset U} Q \subset U.$$

Conversely, for any $x \in U$, there is a $k \geq k_0$, such that

$$N_{\frac{\sqrt{n}}{2^k}}(x) \subset U,$$

where $N_{\frac{\sqrt{n}}{2^k}}(x)$ is the open ball centered at x with radius $\frac{\sqrt{n}}{2^k}$. Since $\mathbb{R}^n = \bigcup_{Q \in \mathscr{Q}_{k+1}} Q$, there is a $Q \in \mathscr{Q}_{k+1}$ such that $x \in Q$, and since the distance between any 2 points in Q is at most $\frac{\sqrt{n}}{2^{k+1}} < \frac{\sqrt{n}}{2^k}$ (when 2 points are located at opposite vertices),

$$Q \subset N_{\frac{\sqrt{n}}{2^k}}(x) \subset U.$$

so

$$x \in \bigcup_{Q \in \bigcup_{k=k_0}^{\infty} \mathscr{Q}_k,\ Q \subset U} Q.$$

Next we show that we can make such Q's with disjoint interiors. Label the cubes in

$$\{Q \in \mathscr{Q}_{k_0} : Q \subset U\} \quad \text{as} \quad \{Q_{k_0 j} : j = 1, 2, \dots\}.$$

Generally for $k > k_0$, we label the cubes in

$$\left\{ Q \in \mathscr{Q}_k : Q \subset U \text{ and } Q \not\subset \bigcup_{i=k_0}^{k-1} \bigcup_j Q_{ij} \right\} = \{Q_{kj}, j = 1, 2, \dots\}.$$

Then

$$U = \bigcup_{k=k_0}^{\infty} \bigcup_j Q_{kj}$$

is countable union of cubes with disjoint interiors (Figure 2.9). □

Proposition 2.6.4

$$\mathscr{B}_{\mathbb{R}^n} = \bigotimes_1^n \mathscr{B}_{\mathbb{R}}.$$

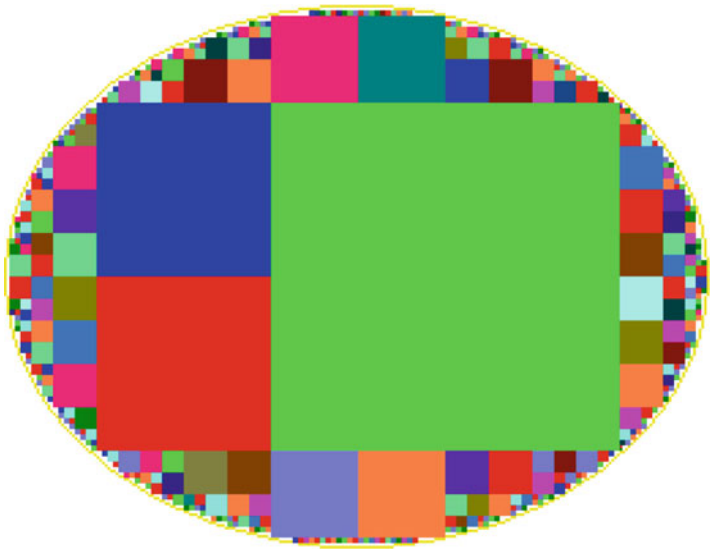


Fig. 2.9 Every open set in $\mathbb{R}^n$ is a union of cubs with disjoint interiors

2

Proof

Let $\mathscr{E}$ be the collection of all Borel measurable cuboids. We certainly have $\mathscr{Q} \subset \mathscr{E}$. By Lemma 2.6.3, any open set $U \subset \mathbb{R}^n$ is a countable union of cubes. So $\mathscr{B}_{\mathbb{R}^n} = \mathscr{M}(\mathscr{Q}) \subset \mathscr{M}(\mathscr{E}) = \bigotimes_1^n \mathscr{B}_{\mathbb{R}}$. According to Lemma 1.2.3 b), we only need to show that $\mathscr{E} \subset \mathscr{B}_{\mathbb{R}^n}$. Since

$$E_1 \times E_2 \times \cdots \times E_n = \bigcap_{i=1}^{n} \mathbb{R} \times \cdots \times E_i \times \cdots \times \mathbb{R},$$

we only need to show $\mathbb{R} \times \cdots \times E_i \times \cdots \times \mathbb{R} \in \mathscr{B}_{\mathbb{R}^n}$. By symmetry, it is sufficient to show $E \times \mathbb{R}^{n-1} \in \mathscr{B}_{\mathbb{R}^n}$ for any $E \in \mathscr{B}_{\mathscr{R}}$.

Let

$$\mathscr{M} = \{E \in \mathbb{R} : E \times \mathbb{R}^{n-1} \in \mathscr{B}_{\mathbb{R}^n}\}.$$

Clearly $\mathscr{M}$ is a σ-algebra which contains all the open sets of $\mathbb{R}$, therefore $\mathscr{B}_{\mathscr{R}} \subset \mathscr{M}$. So $E \times \mathbb{R}^{n-1} \in \mathscr{B}_{\mathbb{R}^n}$ for all $E \in \mathscr{B}_{\mathscr{R}}$.

□

To define the product measure, we start from a semi-algebra and a premeasure. Let $(X, \mathscr{M}, \mu)$ and $(Y, \mathscr{N}, \nu)$ be measure spaces.

Definition 2.6.5

Let

$$\mathscr{E} = \{E \times F : E \in \mathscr{M}, F \in \mathscr{N}\}$$

be the collection of all measurable rectangles and define the set function $\rho : \mathscr{E} \to [0, \infty]$ by

$$\rho(E \times F) = \mu(E)\nu(F),$$

with the convention that $\infty \cdot 0 = 0$.

$\mathscr{E}$ clearly is a semi-algebra.

Lemma 2.6.6 *ρ is a premeasure on the semi-algebra $\mathscr{E}$.*

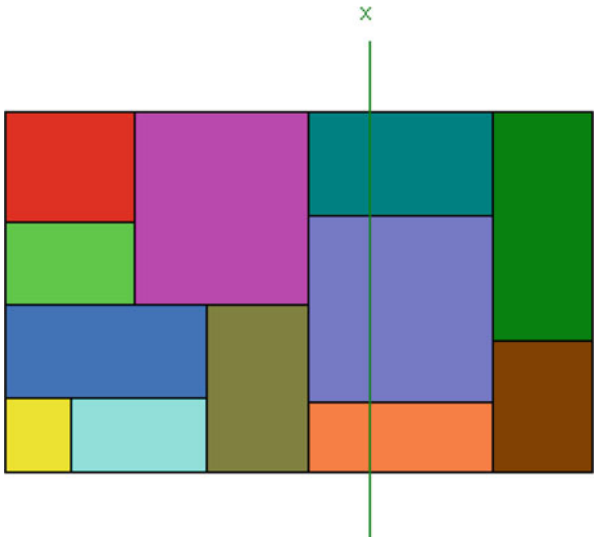
Proof

It is sufficient to prove that if

$$A \times B = \bigcup_{i=1}^{\infty} (A_i \times B_i)$$

in $\mathscr{E}$ and the union is disjoint union, then

Fig. 2.10 $B = \bigcup_{x \in A_i} B_i$



$$\rho(A \times B) = \sum_{i=1}^{\infty} \rho(A_i \times B_i).$$

Note that for each fixed $x \in A$ (Figure 2.10)

$$B = \left\{ y : (x, y) \in \bigcup_{i=1}^{\infty} (A_i \times B_i) \right\} = \bigcup_{x \in A_i} B_i.$$

So

$$\chi_A(x)\nu(B) = \nu(B) = \sum_{x \in A_i} \nu(B_i) = \sum_{i=1}^{\infty} \chi_{A_i}(x)\nu(B_i).$$

Therefore

$$\mu(A)\nu(B) = \int \chi_A(x)\nu(B)\, d\mu(x) = \sum_{i=1}^{\infty} \int \chi_{A_i}(x)\nu(B_i)\, d\mu(x) = \sum_{i=1}^{\infty} \mu(A_i)\nu(B_i).$$

□

Definition 2.6.7

The product measure, denoted by $\mu \times \nu$, is the measure on $\mathscr{M} \otimes \mathscr{N}$ induced by the premeasure $\rho(E \times F) = \mu(E)\nu(F)$ on the semi-algebra $\mathscr{E} = \{E \times F : E \in \mathscr{M}, F \in \mathscr{N}\}$ of measurable rectangles.

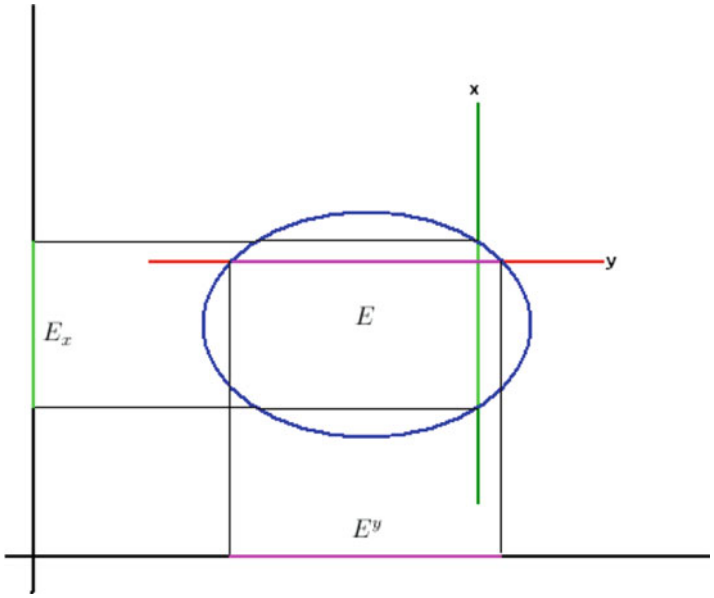


Fig. 2.11 E_x and E^y

Definition 2.6.8

Let $E \subset X \times Y$. For each fixed x, the x-section E_x of E is a subset of Y defined by

$$E_x = \{y \in Y : (x, y) \in E\}$$

and for each fixed y, the y-section E^y of E is a subset of X defined by (Figure 2.11)

$$E^y = \{x \in X : (x, y) \in E\}.$$

Let $f(x, y) : X \times Y \to \mathbb{R}$. For each fixed x, denote $f_x(y) = f(x, y)$ to be the function $Y \to \mathbb{R}$, and for each fixed y, denote $f^y(x) = f(x, y)$ to be the function $X \to \mathbb{R}$.

Proposition 2.6.9

a) Let $E \in \mathscr{M} \otimes \mathscr{N}$. Then for any $x \in X$, $E_x \in \mathscr{N}$, and for any $y \in Y$, $E^y \in \mathscr{M}$.

b) Let $f(x, y)$ be a $\mathscr{M} \otimes \mathscr{N}$-measurable function. Then for any fixed $x \in X$, $f_x(y)$ is a $\mathscr{N}$-measurable, and for any fixed $y \in Y$, $f^y(x)$ is a $\mathscr{M}$-measurable function.

Proof

a) Let $\mathscr{R}$ be the collection of subsets of $X \times Y$ such that a) is true. Then clearly $\mathscr{R}$ contains $\mathscr{E}$, is closed under countable union, and is closed under complement. So $\mathscr{R}$ is a σ-algebra containing $\mathscr{E}$. Therefore

$$\mathscr{M} \otimes \mathscr{N} \subset \mathscr{R}.$$

b) It follows from a) because for each Borel $B \subset \mathbb{R}$,

$$(f_x)^{-1}(B) = (f^{-1}(B))_x, \quad (f^y)^{-1}(B) = (f^{-1}(B))^y.$$

□

We'll need a new terminology in order to prove the Tonelli/Fubini theorem.

Definition 2.6.10

A monotone class is a collection of subsets of X which is closed under countable increasing union and decreasing intersection.
For a collection $\mathscr{E}$ of subsets, the monotone class generated by $\mathscr{E}$, denoted by $\mathscr{C}(\mathscr{E})$, is the intersection of all monotone classes containing $\mathscr{E}$.

Lemma 2.6.11 *For any algebra $\mathscr{A}$ of subsets of X,*

$$\mathscr{C}(\mathscr{A}) = \mathscr{M}(\mathscr{A}).$$

Proof

Clearly $\mathscr{C}(\mathscr{A}) \subset \mathscr{M}(\mathscr{A})$ because a σ-algebra is a monotone class. So we only need to prove the other inclusion.

For each $E \in \mathscr{C}(\mathscr{A})$, let

$$\mathscr{C}(E) = \{F \in \mathscr{C}(\mathscr{A}) : \{E \setminus F, F \setminus E, E \cap F\} \subset \mathscr{C}(\mathscr{A})\}.$$

Then because of symmetry,

$$F \in \mathscr{C}(E) \quad \text{if and only if} \quad E \in \mathscr{C}(F). \tag{2.6.3}$$

Also $\mathscr{C}(E)$ is closed under countable increasing union and decreasing intersection because

$$E \setminus \left(\bigcup F_i\right) = \bigcap (E \setminus F_i), \quad E \setminus \left(\bigcap F_i\right) = \bigcup (E \setminus F_i), \quad \left(\bigcup F_i\right) \setminus E = \bigcup (F_i \setminus E),$$

$$\left(\bigcap F_i\right) \setminus E = \bigcap (F_i \setminus E), \quad E \cap \left(\bigcup F_i\right) = \bigcup (E \cap F_i), \quad E \cap \left(\bigcap F_i\right) = \bigcap (E \cap F_i)$$

are either increasing union or decreasing intersection in $\mathscr{C}(\mathscr{A})$ for any increasing union or decreasing intersection of $\{F_i\}$ in $\mathscr{C}(E)$.

If $E \in \mathscr{A}$, then $\mathscr{A} \subset \mathscr{C}(E)$ because $\mathscr{A}$ is closed under the operations defining $\mathscr{C}(E)$. So

$$\mathscr{C}(\mathscr{A}) \subset \mathscr{C}(E) \quad \text{for all } E \in \mathscr{A},$$

which in turn implies that

$$\mathscr{A} \subset \mathscr{C}(F) \quad \text{for all } F \in \mathscr{C}(\mathscr{A})$$

because of (2.6.3). So

$$\mathscr{C}(\mathscr{A}) \subset \mathscr{C}(F) \quad \text{for all } F \in \mathscr{C}(\mathscr{A}),$$

which implies that $\mathscr{C}(\mathscr{A})$ is closed under set subtraction and intersection. Since it contains X, it is closed under complement, and since $E \cup F = (E^c \cap F^c)^c$, it is closed under union. Therefore $\mathscr{C}(\mathscr{A})$ is an algebra, and since it is closed under increasing union, it is a σ-algebra. Therefore

$$\mathscr{M}(\mathscr{A}) \subset \mathscr{C}(\mathscr{A}).$$

□

Theorem 2.6.12

Let $(X, \mathscr{M}, \mu)$ and $(Y, \mathscr{N}, \nu)$ be σ-finite measure spaces. For any $E \in \mathscr{M} \otimes \mathscr{N}$, the function $f_E(x) = \nu(E_x)$ is $\mathscr{M}$-measurable, the function $g_E(y) = \mu(E^y)$ is $\mathscr{N}$-measurable, and

$$\mu \times \nu(E) = \int f_E(x)\, d\mu(x) = \iint \chi_E\, d\nu(y) d\mu(x),$$

$$\mu \times \nu(E) = \int g_E(y)\, d\nu(y) = \iint \chi_E\, d\mu(x) d\nu(y).$$

Proof

We first assume μ and ν are finite.

Let $\mathscr{C}$ be the collection of sets in $\mathscr{M} \otimes \mathscr{N}$ such that

i) $f_E(x) = \nu(E_x)$ is $\mathscr{M}$-measurable,

ii) $g_E(y) = \mu(E^y)$ is $\mathscr{N}$-measurable,

iii) $\mu \times \nu(E) = \iint \chi_E\, d\nu(y) d\mu(x) = \iint \chi_E\, d\mu(x) d\nu(y)$,

and let $\mathscr{A}$ be the algebra of all finite disjoint union of measurable rectangles (see Remark 1.5). Clearly $\mathscr{A} \subset \mathscr{C}$ because for a measurable rectangle $A \times B$, $f_{A\times B}(x) = \nu(B)\chi_A(x)$ and $g_{A\times B}(y) = \mu(A)\chi_B(y)$, and for a finite disjoint union of measurable rectangles $E = \bigcup_{i=1}^n E_i$, $f_E = \sum_{i=1}^n f_{E_i}$ and $g_E = \sum_{i=1}^n g_{E_i}$.

By Lemma 2.6.11, we only need to show that $\mathscr{C}$ is a monotone class. For any

$$E_1 \subset E_2 \subset \cdots, \quad \text{and} \quad F_1 \supset F_2 \supset \cdots \text{ in } \mathscr{C},$$

let $E = \bigcup_{i=1}^\infty E_i$ and $F = \bigcap_{i=1}^\infty F_i$. Then by the continuity of a measure

$$f_E = \lim_{i\to\infty} f_{E_i}, \quad g_E = \lim_{i\to\infty} g_{E_i}, \quad f_F = \lim_{i\to\infty} f_{F_i}, \quad g_F = \lim_{i\to\infty} g_{F_i},$$

are measurable, and by Monotone Convergence Theorem

$$\mu\times\nu(E) = \lim_{i\to\infty} \mu\times\nu(E_i) = \lim_{i\to\infty} \iint \chi_{E_i}\, d\nu(y)d\mu(x) = \iint \chi_E\, d\nu(y)d\mu(x)$$

and also

$$\mu\times\nu(E) = \lim_{i\to\infty} \mu\times\nu(E_i) = \lim_{i\to\infty} \iint \chi_{E_i}\, d\mu(x)d\nu(y) = \iint \chi_E\, d\mu(x)d\nu(y),$$

and by the Dominated Convergence Theorem (what is the dominant function?)

$$\mu\times\nu(F) = \lim_{i\to\infty} \mu\times\nu(F_i) = \lim_{i\to\infty} \iint \chi_{F_i}\, d\nu(y)d\mu(x) = \iint \chi_F\, d\nu(y)d\mu(x)$$

and also

$$\mu\times\nu(F) = \lim_{i\to\infty} \mu\times\nu(F_i) = \lim_{i\to\infty} \iint \chi_{F_i}\, d\mu(x)d\nu(y) = \iint \chi_F\, d\mu(x)d\nu(y).$$

So $\mathscr{C}$ is a monotone class containing $\mathscr{A}$. Therefore

$$\mathscr{M}\otimes\mathscr{N} = \mathscr{M}(\mathscr{A}) = \mathscr{C}(\mathscr{A}) \subset \mathscr{C} \subset \mathscr{M}\otimes\mathscr{N},$$

i.e.

$$\mathscr{C} = \mathscr{M}\otimes\mathscr{N}.$$

Next we prove it for σ-finite μ and ν. There exist

$$X_1 \subset X_2 \subset \cdots \subset X, \quad \mu(X_i) < \infty\ \forall i, \quad \bigcup_{i=1}^\infty X_i = X$$

$$Y_1 \subset Y_2 \subset \cdots \subset Y, \quad \mu(Y_i) < \infty\ \forall i, \quad \bigcup_{i=1}^\infty Y_i = Y.$$

2

For each $E \in \mathscr{M} \otimes \mathscr{N}$

$$f_E = \lim_{i\to\infty} f_{E\cap(X_i\times Y_i)}, \quad g_E = \lim_{i\to\infty} g_{E\cap(X_i\times Y_i)}$$

are measurable, and by the Monotone Convergence Theorem

$$\mu\times\nu(E) = \lim_{i\to\infty} \mu\times\nu(E\cap(X_i\times Y_i))$$
$$= \lim_{i\to\infty} \iint \chi_{E\cap(X_i\times Y_i)}\, d\nu(y)d\mu(x) = \iint \chi_E\, d\nu(y)d\mu(x)$$

$$\mu\times\nu(E) = \lim_{i\to\infty} \mu\times\nu(E\cap(X_i\times Y_i))$$
$$= \lim_{i\to\infty} \iint \chi_{E\cap(X_i\times Y_i)}\, d\mu(x)d\nu(y) = \iint \chi_E\, d\mu(x)d\nu(y).$$

□

Now we are ready to prove Tonelli's Theorem.

Theorem 2.6.13 (Tonelli's Theorem)
Let $(X, \mathscr{M}, \mu)$ and $(Y, \mathscr{N}, \nu)$ be σ-finite measure spaces. For any positive $\mu\times\nu$-measurable function f, the function

$$g(x) = \int f_x\, d\nu(y)$$

is μ-measurable, and the function

$$h(y) = \int f^y\, d\mu(x)$$

is ν-measurable, and

$$\int f\, d(\mu\times\nu) = \iint f(x,y)\, d\nu(y)d\mu(x) = \iint f(x,y)\, d\mu(x)d\nu(y).$$

Proof
The result is true for simple functions by Theorem 2.6.12, and since every positive measurable function is the limit of an increasing sequence of positive simple functions, the result is true for any positive measurable functions by the Monotone Convergence Theorem. □

By using the Tonelli's theorem, we have the Fubini's theorem.

Theorem 2.6.14 (Fubini's Theorem)
Let $(X, \mathscr{M}, \mu)$ and $(Y, \mathscr{N}, \nu)$ be σ-finite measure spaces. For any $f \in L^1(\mu \times \nu)$, the functions

$$f^y \in L^1(\mu) \text{ for a.e. } y \in Y, \quad f_x \in L^1(\nu) \text{ for a.e. } x \in X$$

and the a.e. defined functions

$$g(x) = \int f_x \, d\nu(y) \in L^1(\mu), \quad h(y) = \int f^y \, d\mu(x) \in L^1(\nu),$$

and

$$\int f \, d(\mu \times \nu) = \iint f(x, y) \, d\nu(y) d\mu(x) = \iint f(x, y) \, d\mu(x) d\nu(y).$$

Proof
By the Tonelli's theorem

$$\iint |f(x, y)| d\mu(x) d\nu(y) = \iint |f(x, y)| d\nu(y) d\mu(x) = \int |f(x, y)| \, d(\mu \times \nu) < \infty.$$

Therefore $\int |f(x, y)| d\nu(y) \in L^1(\mu)$ and $\int |f(x, y)| d\mu(x) \in L^1(\nu)$, which implies that $g(x) \in L^1(\mu)$ and $h(y) \in L^1(\mu)$. They also imply $\int |f(x, y)| d\nu(y) < \infty$ μ-a.e. and $\int |f(x, y)| d\mu(x) < \infty$ ν-a.e., which implies that $f^y \in L^1(\mu)$ and $f_x \in L^1(\nu)$ a.e. Then by applying the Tonelli's theorem to both f^+ and f^-, we have

$$\int f \, d(\mu \times \nu) = \iint f(x, y) \, d\nu(y) d\mu(x) = \iint f(x, y) \, d\mu(x) d\nu(y).$$

□

Exercises

2.38 Let $(X, \mathscr{M}, \mu)$ and $(Y, \mathscr{N}, \nu)$ be measure spaces, f and g be $\mathscr{M}$- and $\mathscr{N}$-measurable, respectively. Prove that both $f(x) + g(y)$ and $f(x)g(x)$ are $\mathscr{M} \otimes \mathscr{N}$ measurable.

2.39 Let $(X_i, \mathscr{M}_i, \mu_i)$ be σ-finite measure spaces. Prove that $\mathscr{M}_1 \otimes \mathscr{M}_2 \otimes \mathscr{M}_3 = (\mathscr{M}_1 \otimes \mathscr{M}_2) \otimes \mathscr{M}_3$ and $\mu_1 \times \mu_2 \times \mu_3 = (\mu_1 \times \mu_2) \times \mu_3$.

2

2.40 This exercise shows that the Tonelli's theorem is not true if one of the space is not σ-finite. Let $X = Y = [0, 1]$, $\mathscr{M} = \mathscr{N} = \mathscr{B}_{[0,1]}$, μ is the Lebesgue, ν is the counting measure, and $D = \{(x, x) : x \in [0, 1]\}$ be the diagonal. Compute

$$\iint \chi_D \, d\mu d\nu, \quad \iint \chi_D \, d\nu d\mu, \quad \int \chi_D \, d(\mu \times \nu).$$

2.41 This exercise shows that the FubiniŠs theorem is not true if the function is not integrable over the product measure. Let $X = Y = \mathbb{N}$, $\mathscr{M} = \mathscr{N} = \mathscr{P}(\mathbb{N})$, $\mu = \nu$ be the counting measure, and

$$f(m, n) = \begin{cases} 1, & m = n, \\ -1, & m = n + 1, \\ 0, & \text{otherwise.} \end{cases}$$

Compute

$$\iint f(m, n) \, d\mu(m) d\nu(n), \quad \iint f(m, n) \, d\nu(n) d\mu(m), \quad \int |f| \, d(\mu \times \nu).$$

2.42 Let $\{a_{ij} : i = 1, 2, \ldots, j = 1, 2, \ldots\} \subset \mathbb{R}^2$ such that $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |a_{ij}| < \infty$. Prove that for any partition $\{E_k : k = 1, 2, \ldots, \}$ of $\mathbb{N}^2$ (i.e., $\{E_k\}$ are disjoint and $\bigcup_{k=1}^{\infty} E_k = \mathbb{N}^2$)

$$\sum_{k=1}^{\infty} \sum_{(i,j) \in E_k} a_{ij} = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}.$$

2.43 Let f, g be Lebesgue integrable on $[a, b]$, $F(x) = \int_a^x f(t) \, dt$, and $G(x) = \int_a^x g(t) \, dt$. Prove that

$$\int_a^b F(x) g(x) \, dx = F(b)G(b) - \int_a^b f(x) G(x) \, dx.$$

2.44 Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function and $\Gamma = \{(x, f(x)) : x \in \mathbb{R}\}$ be the graph of f in $\mathbb{R}^2$. Prove that $m \times m(\Gamma) = 0$.

2.45 This exercise shows that if one of the space is the counting measure space over $\mathbb{N}$, the other space does not have to be σ-finite for both Tonelli's and FubiniŠs theorems.

a. For any sequence of positive measurable functions $\{f_n\}$ over a measure space $(X, \mathscr{M}, \mu)$, prove that

$$\int \sum_{n=1}^{\infty} f_n(x) \, d\mu(x) = \sum_{n=1}^{\infty} \int f_n(x) \, d\mu(x),$$

b. For any sequence of measurable functions $\{f_n\}$ over a measure space $(X, \mathscr{M}, \mu)$, if $\sum_{n=1}^{\infty} |f_n(x)|$ is integrable, then each $f_n(x)$ is integrable, $\sum_{n=1}^{\infty} f_n(x)$ converges a.e. and is integrable, and

$$\int \sum_{n=1}^{\infty} f_n(x)\, d\mu(x) = \sum_{n=1}^{\infty} \int f_n(x)\, d\mu(x).$$

2.46 Let f be Lebesgue integrable on $(0, a)$ and $g(x) = \int_x^a \frac{f(t)}{t}\, dt$. Prove that g is integrable on $(0, a)$ and

$$\int_0^a g(x)\, dx = \int_0^a f(x)\, dx.$$

2.7 Change of Variables for n-Dimensional Lebesgue Integrals

Let $m^n = m \times m \times \cdots \times m$ be the n-dimensional Lebesgue measure.

Theorem 2.7.1
Lebesgue outer measure on $\mathbb{R}^n$ is translation invariant. More precisely, for any $a \in \mathbb{R}^n$, define $\tau_a : \mathbb{R}^n \to \mathbb{R}^n$ by $\tau_a(x) = x - a$. Then
a) For any $E \subset \mathbb{R}^n$,

$$(m^n)^*(\tau_a(E)) = (m^n)^*(E).$$

In particular, E is Lebesgue measurable if and only if $\tau_a(E)$ is, and

$$m^n(\tau_a(E)) = m^n(E).$$

b) $f(y)$ is a Lebesgue measurable on $\mathbb{R}^n$ if and only if $f(\tau_a(x))$ is, and if f is Lebesgue measurable, then

$$\int f(y)\, dy = \int f(\tau_a(x))\, dx \tag{2.7.4}$$

provided any one side is defined.

Proof

a) Let $\mathscr{E}$ be the semi-algebra of all measurable cuboids. By Theorem 1.5.9, $m^n(A) = m^n(\tau_a(A))$ for any measurable cuboids A. For any $E \subset \mathbb{R}^n$, since τ_a defines a one-to-one correspondence between measurable cuboids covering E and $\tau_a(E)$, we have

$$
\begin{aligned}
(m^n)^*(E) &= \inf\left\{\sum_{i=1}^{\infty} m^n(A_i),\, E \subset \bigcup_{i=1}^{\infty} A_i,\ \{A_i\} \subset \mathscr{E}\right\} \\
&= \inf\left\{\sum_{i=1}^{\infty} m^n(\tau_a(A_i)),\, \tau_a(E) \subset \bigcup_{i=1}^{\infty} \tau_a(A_i),\ \{A_i\} \subset \mathscr{E}\right\} \\
&= (m^n)^*(\tau_a(E))
\end{aligned}
$$

Since the measurability is defined by an equality of outer measures, E is measurable if and only if $\tau_a(E)$ is.

b) By a), the result is true for simple functions. By taking the limit of an increasing sequence of simple functions and applying the Monotone Convergence Theorem, we conclude that the result is true for nonnegative measurable functions. Then by applying it to f^+ and f^- we conclude that the result is true for all integrable functions.

□

Theorem 2.7.2

The n-dimensional Lebesgue outer measure $(m^n)^*$ on $\mathscr{P}(\mathbb{R}^n)$ can also be defined as

$$(m^n)^*(E) = \inf\{m^n(U) : E \subset U \text{ and } U \text{ is open}\}.$$

Proof

By the monotonicity of an outer measure, we certainly have

$$(m^n)^* E) \leq \inf\{m^n(U) : E \subset U \text{ and } U \text{ is open}\}.$$

So we only need to prove

$$\inf\{m^n(U) : E \subset U \text{ and } U \text{ is open}\} \leq (m^n)^*(E).$$

It is certainly true if $(m^n)^*(E) = \infty$. So we only consider the case when $(m^n)^*(E) < \infty$.

By the definition of product measures, for any given $\epsilon > 0$, there exist measurable cuboids $\{A_1, A_2, \dots\}$ such that

$$E \subset \bigcup_{k=1}^{\infty} A_k, \quad \text{and} \quad \sum_{k=1}^{\infty} m^n(A_k) < (m^n)^*(E) + \frac{\epsilon}{2}.$$

For each A_k, by applying Theorem 1.5.4 to each side of A_k, we have an open cuboids $U_k \supset A_k$ such that

$$m^n(U_k) < m^n(A_k) + \frac{\epsilon}{2^{k+1}}.$$

Let

$$U = \bigcup_{k=1}^{\infty} U_k.$$

Then $E \subset U$ and

$$m^n(U) \leq \sum_{k=1}^{\infty} m^n(U_k) \leq \sum_{k=1}^{\infty} \left(m^n(A_k) + \frac{\epsilon}{2^{k+1}} \right) = \sum_{k=1}^{\infty} m^n(A_k) + \frac{\epsilon}{2} < (m^n)^*(E) + \epsilon.$$

So

$$\inf\{m^n(U) : E \subset U \text{ and } U \text{ is open}\} \leq m^n(U) < (m^n)^*(E) + \epsilon.$$

Let $\epsilon \to 0$, we have

$$\inf\{m^n(U) : E \subset U \text{ and } U \text{ is open}\} \leq (m^n)^*(E).$$

□

Theorem 2.7.3
Let $\mathscr{L}^n$ be the σ-algebra of all Lebesgue measurable sets in $\mathbb{R}^n$. Then the following are equivalent
a) $E \in \mathscr{L}^n$.
b) *For any $\epsilon > 0$, there exists an open set $U \supset E$ such that $(m^n)^*(U \setminus E) < \epsilon$.*
c) $E = V \setminus N_1$ *for some G_δ set V and m^n-null set N_1.*
d) *For any $\epsilon > 0$, there exists a closed set $K \subset E$ such that $(m^n)^*(E \setminus K) < \epsilon$.*
e) $E = H \cup N_2$ *for some F_σ set H and m^n-null set N_2.*

The proof is identical to Theorem 1.5.6.

Proposition 2.7.4 *For any Lebesgue measurable set $E \subset \mathbb{R}^n$ with $m^n(E) < \infty$, and any $\epsilon > 0$, there is a set A, which is a finite union of cubes with disjoint interior, such that*

$$m^n(E \triangle A) < \epsilon.$$

Proof
By Theorem 2.7.2, there is an open set $U \supset E$ such that

$$m^n(U \setminus E) < \frac{\epsilon}{2}.$$

2

By Lemma 2.6.3, $U = \bigcup_{i=1}^{\infty} Q_i$ for some cubes $\{Q_i\}$ with disjoint interior. Since the Lebesgue measure of the boundary of any cube is 0, we have $m^n(U) = \sum_{i=1}^{\infty} m^n(Q_i) < \infty$. So there is an n such that

$$\sum_{i=n+1}^{\infty} m^n(Q_i) < \frac{\epsilon}{2}.$$

Let $A = \bigcup_{i=1}^{n} Q_i$. Then

$$m^n(U \setminus A) = \sum_{i=n+1}^{\infty} m^n(Q_i) < \frac{\epsilon}{2}$$

and

$$m^n(E \triangle A) \leq m^n(E \setminus A) + m^n(A \setminus E) \leq m^n(U \setminus A) + m^n(U \setminus E) < \epsilon.$$

□

Theorem 2.7.5

Let A be an $n \times n$ nonsingular matrix and let $T_A : \mathbb{R}^n \to \mathbb{R}^n$ be defined as

$$y = T_A(x) = Ax,$$

where the points in $\mathbb{R}^n$ are represented as $n \times 1$ vectors. Then

a) For any $F \subset \mathbb{R}^n$,

$$(m^n)^*(T_A(F)) = |\det(A)|(m^n)^*(F).$$

In particular F is Lebesgue measurable if and only if $T_A(F)$ is, and

$$m^n(T_A(F)) = |\det(A)|m^n(F).$$

b) $f(y)$ is a Lebesgue measurable on $\mathbb{R}^n$ if and only if $f(T_A(x))$ is, and if f is Lebesgue measurable, then

$$\int f(y)\, dy = |\det(A)| \int f(T_A(x))\, dx \tag{2.7.5}$$

provides any one side is defined.

Proof

a) Because of Theorem 2.7.2 and the fact that T_A induces a bijection on the collections of all open sets covering a set E and its image $T_A(E)$, we only need to prove it for open sets (see the proof of Theorem 1.5.9), and because of Lemma 2.6.3 and the fact that the boundary of a cube has Lebesgue measure 0, we only need to prove it for a cube. Assume F is a cube.

Since $A = E_1 \cdots E_k$ and $\det(A) = \det(E_1) \cdots \det(E_k)$ for some elementary matrices $E_1 \ldots, E_k$, without loss of generality, we can assume A is an elementary matrix.

i) A is a type I elementary matrix. Then there is an $i \in \{1, \ldots, n\}$ and a $c \neq 0$ such that $y = T_A(x)$ is given by $y_i = cx_i$ and $y_j = x_j$ for all $j \neq i$, and $\det(A) = c$. Because of Theorem 1.5.9 and the fact that $T_A(F)$ is a measurable cuboid,

$$m^n(T_A(F)) = |c| m^n(F) = |\det(A)| m^n(F)$$

ii) A is a type II elementary matrix. Then there are $i \neq j$ in $\{1, \ldots, n\}$ such that $y = T_A(x)$ is given by $y_i = x_j$, $y_j = x_i$, and $y_l = x_l$ for all $l \neq i, j$, and $\det(A) = -1$. Then $T_A(F)$ is a cube of the same size. So

$$m^n(T_A(F)) = m^n(F) = |\det(A)| m^n(F)$$

iii) A is a type III elementary matrix. Then there are $i \neq j$ in $\{1, \ldots, n\}$ and a $k \in \mathbb{R}$, such that $y = T_A(x)$ is given by $y_i = x_i$, $y_j = x_j + kx_i$, and $y_l = x_l$ for all $l \neq i, j$, and $\det(A) = 1$. Because translation invariance of the Lebesgue measure, without loss of generality, assume

$$F = [0, h] \times \cdots \times [0, h].$$

Then by Tonelli's Theorem

$$m^n(T_A(F)) = \int_0^h \cdots \int_0^h \int_{ky_i}^{h+ky_i} dy_j dy_i \cdots dy_l = h^n = m^n(F) = |\det(A)| m^n(F).$$

b) Since

$$\chi_{T_A(E)}(y) = \chi_{T_A(E)}(T_A(x)) = \chi_E(x),$$

the result is true for simple function. Then the result follows.

□

Since a rotation is a linear map induced by an orthogonal matrix, and $|\det(A)| = 1$ for any orthogonal matrix A, we have

2

Corollary 2.7.6 *Lebesgue measure is invariant under rotations.*

Definition 2.7.7

Let $\Omega \subset \mathbb{R}^n$ be open. A map $G : \Omega \to \mathbb{R}^n$ is called a diffeomorphism if it is one-to-one and differentiable, and its inverse is also differentiable.
G is called a C^1 diffeomorphism if it is a diffeomorphism with continuous and nonsingular Jacobian matrix $DG(x)$ for all $x \in \Omega$.

Definition 2.7.8

Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$ (see ▶ Chapter 5 of the book). For any $n \times n$ matrix A, the induced operator norm $\|A\|$ is defined as

$$\|A\| = \sup_{\|x\|>0} \frac{\|Ax\|}{\|x\|}.$$

Clearly we have $\|Ax\| \leq \|A\|\|x\|$, and for any two $n \times n$ matrices A and B, $\|AB\| \leq \|A\|\|B\|$, and $\|I\| = 1$. It can be proved that

$$\|x\|_\infty = \max_{1\leq i\leq n} |x_i|$$

is a norm on $\mathbb{R}^n$ (which is called the ∞-norm) and the induced operator norm for $n \times n$ matrix $A = [a_{ij}]$ is given by

$$\|A\|_\infty = \max_{1\leq i\leq n} \sum_{j=1}^{n} |a_{ij}|.$$

Theorem 2.7.9

Let $\Omega \subset \mathbb{R}^n$ be open and $G : \Omega \to \mathbb{R}^n$ be a C^1 diffeomorphism.

a) If $f(y)$ is Lebesgue measurable on $G(\Omega)$, then $f(G(x))$ is Lebesgue measurable on Ω and

$$\int_{G(\Omega)} f(y)\, dy = \int_\Omega |\det(DG(x))| f(G(x))\, dx$$

whenever the left-hand side is defined.

b) If $E \subset \Omega$ is Lebesgue measurable, then $G(E)$ is Lebesgue measurable and

$$m^n(G(E)) = \int_E |\det(DG(x))|\, dx.$$

Proof

We first show that if E is measurable, then $G(E)$ is measurable, and

$$m^n(G(E)) \le \int_E |\det(DG(x))|\, dx. \tag{2.7.6}$$

For any cube $Q \subset \Omega$ of size h and for any $\epsilon > 0$, since $DG(x)$ is uniformly continuous on Q and $DG^{-1}(y)DG(x) \to I$ as $x \to y$, there is a $j_0 \in \mathbb{N}$, such that $\|DG^{-1}(y)DG(x)\|_\infty < 1+\epsilon$ for all $y \in Q$ and for all x such that $\|x-y\|_\infty < \frac{h}{2^{j_0}}$.

For all $j \ge j_0$, we can write $Q = \bigcup_{i=1}^{2^{jn}} Q_i$, where $\{Q_i\}$ have sizes $\frac{h}{2^j}$ and disjoint interior. Consider one of the Q_i. Let the center of Q_i be y_i. Then each Q_i is defined by $Q_i = \left\{x \in \mathbb{R}^n : \|x-y_i\|_\infty \le \frac{h}{2^{j+1}}\right\}$.

Let $F : Q_i \to \mathbb{R}^n$ be defined by $F(x) = DG^{-1}(y_i)G(x)$. Then for any $x \in Q_i$

$$\begin{aligned}
\|F(x)-F(y_i)\|_\infty &= \left\| \int_0^1 \frac{d}{dt} F(y_i + t(x-y_i))\, dt \right\|_\infty \\
&= \left\| \left(\int_0^1 DF(y_i + t(x-y_i))\, dt \right)(x-y_i) \right\|_\infty \\
&\le \int_0^1 \|DG^{-1}(y_i)DG(y_i+t(x-y_i))\|_\infty\, dt \|x-y_i\|_\infty \\
&\le (1+\epsilon)\frac{h}{2^{j+1}},
\end{aligned}$$

i.e., $F(Q_i) \subset R$ where R is the cube defined by $R = \left\{x \in \mathbb{R}^n : \|x-F(y_i)\|_\infty \le (1+\epsilon)\frac{h}{2^{j+1}}\right\}$. So by Theorem 2.7.5

$$\begin{aligned}
m^n(G(Q_i)) &= m^n(DG(y_i)F(Q_i)) = |\det(DG(y_i))| m^n(F(Q_i) \le |\det(DG(y_i))| m^n(R) \\
&= (1+\epsilon)^n |\det(DG(y_i))| m^n(Q_i)
\end{aligned}$$

and

$$m^n(G(Q)) \le \sum_{i=1}^{2^{jn}} m^n(G(Q_i)) \le (1+\epsilon)^n \sum_{i=1}^{2^{jn}} |\det(DG(y_i))| m^n(Q_i) = (1+\epsilon)^n \int_Q \phi_j(x)\, dx$$

where $\phi_j(x) = \sum_{i=1}^{2^{jn}} |\det(DG(y_i))| \chi_{Q_i}(x)$. Let $j \to \infty$. Since $\phi_j(x) \to |\det(DG(x))|$ a.e., by the Dominated Convergence Theorem

$$m^n(G(Q)) \le (1+\epsilon)^n \int_Q |\det(DG(x))|\, dx.$$

Let $\epsilon \to 0$, we have

$$m^n(G(Q)) \le \int_Q |\det(DG(x))|\, dx.$$

By Lemma 2.6.3, for all open set $U \subset \Omega$,

$$m^n(G(U)) \le \int_U |\det(DG(x))|\, dx.$$

For any bounded G_δ set V in Ω, there are open sets $U_1 \supset U_2 \supset \cdots$ in Ω such that U_1 is bounded and $\bigcap_{i=1}^\infty U_i = V$. Then by continuity of m^n and the Dominated Convergence Theorem

$$m^n(G(V) = \lim_{i\to\infty} m^n(G(U_i)) \le \lim_{i\to\infty} \int_{U_i} |\det(DG(x))|\, dx = \int_V |\det(DG(x))|\, dx.$$

For any unbounded bounded G_δ set V in Ω, there are bounded G_δ sets $V_1 \subset V_2 \subset \cdots \subset V$ such that $\bigcup_{i=1}^\infty V_i = V$. Then by continuity of m^n and the Monotone Convergence Theorem

$$m^n(G(V) = \lim_{i\to\infty} m^n(G(V_i)) \le \lim_{i\to\infty} \int_{V_i} |\det(DG(x))|\, dx = \int_V |\det(DG(x))|\, dx.$$

For any null set N in Ω, there is a G_δ set $V \supset N$ such that $m^n(V) = 0$. Then

$$m^n(G(N)) \le m^n(G(V)) \le \int_V |\det(DG(x))|\, dx = 0,$$

so $G(N)$ is a null set.

Finally for any measurable set E, there is a null set N and a G_δ set V such that $E = V \setminus N$. Note that $G(V)$ is a G_δ set because G^{-1} is continuous, so $G(E) = G(V) \setminus G(N)$ is measurable, and

$$m^n(G(E)) \le m^n(G(V)) \le \int_V |\det(DG(x))|\, dx = \int_E |\det(DG(x))|\, dx.$$

Now we prove a). By applying (2.7.6) to simple functions and taking limit, we know that for any positive measurable function f,

$$\int_{G(\Omega)} f(y)\, dy \le \int_\Omega |\det(DG(x))| f(G(x))\, dx.$$

Conversely, since $G(G^{-1}(y))$ is the identity map on $G(\Omega)$, by the chain rule we have

$$I = DG(G^{-1}(y))DG^{-1}(y).$$

By applying the same reasoning to function $|\det(DG(x))| f(G(x))$ and transformation $x = G^{-1}(y)$, we have

$$\begin{aligned}
&\int_{\Omega} |\det(DG(x))| f(G(x))\, dx \\
&= \int_{G^{-1}(G(\Omega))} |\det(DG(x))| f(G(x))\, dx \qquad (let x = G^{-1}(y)) \\
&\leq \int_{G(\Omega)} |\det(DG^{-1}(y))|\, |\det(DG(G^{-1}(y))| f(y)\, dy \\
&= \int_{G(\Omega)} f(y)\, dy,
\end{aligned}$$

So

$$\int_{G(\Omega)} f(y)\, dy = \int_{\Omega} |\det(DG(x))| f(G(x))\, dx$$

for any positive measurable function f.

By considering f^+ and f^-, we know that a) is also true for all L^1 function f on $G(\Omega)$. Finally by letting $f = \chi_{G(E)}$ we have b).

□

Exercises

2.47 Show that Theorem 2.7.5 a) remains valid if A is singular. However, b) is no longer true for singular A.

2.48 Let V be a proper linear subspace of $\mathbb{R}^n$ and $a \in \mathbb{R}^n$. Prove that the set

$$\{x + a : x \in V\}$$

has Lebesgue measure 0.

2.49 Let $\mathbb{R}^2 \ni (a, b) \neq (0, 0)$ and $s : \mathbb{R}^n \times \mathbb{R}^n$ be defined by $s(x, y) = ax + by$. Prove that for any Lebesgue measurable set E in $\mathbb{R}^n$, $s^{-1}(E)$ is Lebesgue measurable in $\mathbb{R}^n \times \mathbb{R}^n$.

Signed Measures and Differentiation

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3.1 Signed Measures

Definition 3.1.1
Let $(X, \mathscr{M})$ be a measurable space. A signed measure of $(X, \mathscr{M})$ is a countably additive set function $\nu : \mathscr{M} \to [-\infty, \infty)$ or $(-\infty, \infty]$ such that $\nu(\emptyset) = 0$.

Example 3.1

1) Let μ_1, μ_2 be two finite measures. Then $\mu_1 - \mu_2$ is a signed measure.
2) Let $f \in L^1(\mu)$. Then $\nu(E) = \int_E f \, d\mu$ is a signed measure.

■

Definition 3.1.2
Let ν be a signed measure on $\mathscr{M}$. A set $E \in \mathscr{M}$ is called a positive, a negative, or a null set if $\nu(A) \geq 0$, $\nu(A) \leq 0$, or $\nu(A) = 0$, respectively, for all subset $A \subset E$.

Lemma 3.1.3 *Let ν be a signed measure on $(X, \mathscr{M})$ and $\nu(E) < \infty$ for all $E \in \mathscr{M}$. For any $E \in \mathscr{M}$ with $0 < \nu(E) < \infty$, if E is not a positive set, then E contains a positive set $A \subset E$ such that*

$$\nu(A) > \nu(E) > 0.$$

Proof
Since E is not a positive set, there is a $C \subset E$ such that $\nu(C) < 0$. Then

$$\nu(E \setminus C) = \nu(E) - \nu(C) > \nu(E).$$

Let

$$n_1 = \min\left\{n \in \mathbb{N} : \exists B \subset E, \nu(B) > \nu(E) + \frac{1}{n}\right\},$$

and A_1 be the corresponding set such that

$$\nu(A_1) > \nu(E) + \frac{1}{n_1}$$

Generally if A_{j-1} is not positive, then let

$$n_j = \min\left\{n \in \mathbb{N} : \exists B \subset A_{j-1}, \nu(B) > \nu(A_{j-1}) + \frac{1}{n}\right\}, \tag{3.1.1}$$

and A_j be the corresponding set such that

$$\nu(A_j) > \nu(A_{j-1}) + \frac{1}{n_j}.$$

If A_j is positive for some j, then let $A = A_j$, and we are done. Otherwise let $A = \bigcap_{j=1}^{\infty} A_j$. We have

$$\left(\sum_{j=1}^{\infty} \frac{1}{n_j}\right) + \nu(E) \leq \lim_{j\to\infty} \nu(A_j) = \nu(A) < \infty$$

which means that $\nu(A) > \nu(E) > 0$ and $n_j \to \infty$ as $j \to \infty$. We claim that A is a positive set. If not, there is a $B \subset A$ such that $\nu(B) > \nu(A) + \frac{1}{n}$ for some $n \in \mathbb{N}$. Choose j large enough such that $n_j > n$. Then $B \subset A_j$ and

$$\nu(B) > \nu(A) + \frac{1}{n} \geq \nu(A_{j-1}) + \frac{1}{n}$$

which contradicts the choice of n_j (3.1.1). Therefore A is positive.

□

Theorem 3.1.4 (Hahn Decomposition Theorem)
For any signed measure ν on $(X, \mathscr{M})$, there exists a partition $\{P, N\}$ of positive-negative sets of X (i.e., $P \cup N = X$ and $P \cap N = \emptyset$).
If $\{P', N'\}$ is another such a partition, then $P \triangle P'$ (which equals $N \triangle N'$) is a ν-null set.

Proof

Without loss of generality, assume $\nu : \mathscr{M} \mapsto [-\infty, \infty)$

If $\nu(E) \leq 0$ for all $E \in \mathscr{M}$, then take $N = X$, $P = \emptyset$. Otherwise by Lemma 3.1.3, there is a positive set $A \subset X$ such that $\nu(A) > 0$.

Let

$$\lambda = \sup\{\nu(A) : A \text{ is positive}\}.$$

Then $\lambda > 0$ and there are positive sets $\{A_n\}$ such that $\nu(A_n) \to \lambda$. Let

$$P = \bigcup_{n=1}^{\infty} A_n.$$

Since P is positive,

$$\lambda \geq \nu(P) \geq \nu(A_n).$$

Let $n \to \infty$. Then $\nu(P) = \lambda$.

$N = X \setminus P$ must be a negative set, because otherwise by Lemma 3.1.3, there is a positive set $A \subset X \setminus P$ such that $\nu(A) > 0$, then $P \cup A$ is positive and $\nu(P \cup A) = \nu(P) + \nu(A) > \lambda$ contradicts the definition of λ.

If $\{P', N'\}$ is another such a partition, then $P' \setminus P = P' \cap N$ and $P \setminus P' = P \cap N'$ are subsets of both positive and negative sets, so they must be null sets. □

Definition 3.1.5

A positive-negative partition $\{P, N\}$ of X is called a Hahn decomposition for ν.

Definition 3.1.6

Two signed measures μ and ν are called mutually singular, denoted by

$$\mu \perp \nu,$$

if there is a partition $\{E, F\}$ of X such that E is μ-null and F is ν-null.

Theorem 3.1.7 (Jordan Decomposition Theorem)

For any signed measure ν, there are unique mutually singular positive measures ν^+ and ν^- such that

$$\nu = \nu^+ - \nu^-.$$

Proof

Let $\{P, N\}$ be a Hahn decomposition for ν and define

$$\nu^+(E) = \nu(E \cap P), \quad \nu^-(E) = -\nu(E \cap N).$$

Then both ν^+, ν^- are positive measures,

$$\nu^+ \perp \nu^-, \quad \nu = \nu^+ - \nu^-.$$

To prove the uniqueness, assume there are positive measures μ^+, μ^- and partition $\{P', N'\}$ of X,

$$\nu = \mu^+ - \mu^-, \quad \mu^+ \perp \mu^-, \quad P' \text{ is } \mu^-\text{-null and } N' \text{ is } \mu^+\text{-null.}$$

Then (P', N') is another Hahn decomposition for ν. So $P \text{`}\triangle P = N \text{`}\triangle N$ is ν-null.

For any $E \in \mathscr{M}$,

$$\nu^+(E) = \nu(P \cap E) = \nu(P \cap E \cap P') = \nu(P' \cap E) = \mu^+(E)$$

$$-\nu^-(E) = \nu(N \cap E) = \nu(N \cap E \cap N') = \nu(N' \cap E) = -\mu^-(E)$$

i.e., $\nu^+ = \mu^+$ and $\nu^- = \mu^-$ □

Definition 3.1.8

a) The decomposition in Theorem 3.1.7 is called the Jordan decomposition of ν.

b) For a Jordan decomposition $\nu = \nu^+ - \nu^-$, the measure $\nu^+ + \nu^-$ is called total variation of ν and is denoted by

$$|\nu| = \nu^+ + \nu^-.$$

c) $L^1(\nu)$ for a signed measure ν is defined as

$$L^1(\nu) = L^1(\nu^+) \cap L^1(\nu^-).$$

For any $f \in L^1(\nu)$, the ν-integral of f is defined by

$$\int f \, d\nu = \int f \, d\nu^+ - \int f \, d\nu^-.$$

d) A signed measure is called finite (resp. σ-finite) if $|\nu|$ is finite (resp. σ-finite).

e) Let ν be a signed measure and μ be a positive measure. ν is called absolutely continuous with respect to μ, denoted by

$$\nu \ll \mu,$$

if for any $E \in \mathscr{M}$, $\mu(E) = 0$ implies that $\nu(E) = 0$.

Remark 3.1 It can be easily proved that $\nu \ll \mu$ iff $\nu^+ \ll \mu$, $\nu^- \ll \mu$,, and iff $|\nu| \ll \mu$.

Theorem 3.1.9
Let ν be a finite signed measure and μ be a positive measure. Then $\nu \ll \mu$ if and only if for any $\epsilon > 0$, there is a $\delta > 0$ such that $\mu(E) < \delta$ implies that $|\nu(E)| < \epsilon$.

Proof
Without loss of generality, assume ν is positive.

$\Leftarrow$:

For all E with $\mu(E) = 0$, the condition implies that $\nu(E) < \epsilon$ for any $\epsilon > 0$, which means that $\nu(E) = 0$.

$\Rightarrow$:

If not, there exists an $\epsilon_0 > 0$ such that for all $n \in \mathbb{N}$, there exists an $E_n \in \mathscr{M}$, with $\mu(E_n) \leq \frac{1}{2^n}$ and $\nu(E_n) \geq \epsilon_0$. Let

$$F_m = \bigcup_{n=m}^{\infty} E_n, \quad A = \bigcap_{m=1}^{\infty} F_m.$$

Then (why?)

$$\mu(A) = \lim_{m\to\infty} \mu(F_m) \leq \lim_{m\to\infty} \sum_{n=m}^{\infty} \frac{1}{2^n} = \lim_{m\to\infty} \frac{1}{2^{m-1}} = 0,$$

and

$$\nu(A) = \lim_{m\to\infty} \nu(F_m) \geq \liminf_{m\to\infty} \nu(E_m) \geq \epsilon_0 > 0,$$

a contradiction.

□

Since for any $f \in L^1(\mu)$, $\nu(E) = \int_E f \, d\mu$ is a finite signed measure absolutely continuous with respect to μ, we have

Corollary 3.1.10 *If $f \in L^1(\mu)$, then for any $\epsilon > 0$, there is a $\delta > 0$ such that $\mu(E) < \delta$ implies that $\left|\int_E f \, d\mu\right| < \epsilon$.*

Exercises

3.1 Let ν be a signed measure on $(X, \mathscr{M})$. Prove that $E \in \mathscr{M}$ is ν-null if and only if $|\nu|(E) = 0$

3.2 Let μ_F, μ_G be the Lebesgue-Stieltjes measures on $(\mathbb{R}, \mathscr{B})$ defined in Exercises 1.28 and1.29, and m be the Lebesgue measure on $(\mathbb{R}, \mathscr{B})$. Prove that $\mu_F \perp m$, $\mu_G \perp m$, and $\mu_F \perp \mu_G$.

3.3 Let ν, μ be signed measures on $(X, \mathscr{M})$. Prove that the following conditions are equivalent:

i) $\nu \perp \mu$.
ii) $|\nu| \perp \mu$.
iii) $\nu^+ \perp \mu$ and $\nu^- \perp \mu$.

3.4 Let ν be a signed measure on $(X, \mathscr{M})$. Prove that

a. $L^1(\nu) = L^1(|\nu|)$,
b. $\left|\int f \, d\nu\right| \leq \int |f| \, d|\nu|$ for any $f \in L^1(\nu)$.

3.5 Let ν be a signed measure on $(X, \mathscr{M})$. Prove that for any $E \in \mathscr{M}$

a.

$$|\nu|(E) = \sup\left\{\sum_{i=1}^{\infty} |\nu(E_i)| : \{E_i\} \subset \mathscr{M},\ E_i \cap E_j = \emptyset,\ \bigcup_{i=1}^{\infty} E_i = E\right\},$$

b.

$$|\nu|(E) = \sup\left\{\left|\int_E f \, d\nu\right| : f \in L^1(\nu), |f| \leq 1\right\}.$$

3.6 Let ν be a signed measure on $(X, \mathscr{M})$. Prove that for any $E \in \mathscr{M}$

$$\nu^+(E) = \sup\{\nu(F) : F \in \mathscr{M}, F \subset E\},$$

and

$$\nu^-(E) = -\inf\{\nu(F) : F \in \mathscr{M}, F \subset E\}.$$

3.7 Let ν be a signed measure on $(X, \mathscr{M})$. Prove that for any $E \in \mathscr{M}$, the following conditions are equivalent:

i) E is ν-null.
ii) $\nu^+(E) = 0$ and $\nu^-(E) = 0$.
iii) $|\nu|(E) = 0$.

3.8 Let ν be a signed measure, and λ, ρ be positive measures on $(X, \mathscr{M})$ such that $\nu = \lambda - \rho$. Prove that $\nu^+ \leq \lambda$ and $\nu^- \leq \rho$.

3.9 Prove that $|\nu_1 + \nu_2| \leq |\nu_1| + |\nu_2|$ for any signed measures on $(X, \mathscr{M})$ such that the left-hand side is defined.

3.10 Let μ be a positive measure on $(X, \mathscr{M})$ and $f \in L^1(\mu)$. Prove that

a. $\nu : \mathscr{M} \to (-\infty, \infty)$ defined by $\nu(E) = \int_E f \, d\mu$ is a finite signed measure on $(X, \mathscr{M})$,
b. $\nu^+(E) = \int_E f^+ \, d\mu$, $\nu^-(E) = \int_E f^- \, d\mu$, and $|\nu|(E) = \int_E |f| \, d\mu$.

3.2 The Lebesgue-Radon-Nikodym Theorem

Lemma 3.2.1 *Suppose that ν and μ are positive finite measures on $(X, \mathscr{M})$. Then either $\nu \perp \mu$ or there is an $\epsilon > 0$ and an $E \in \mathscr{M}$ such that $\mu(E) > 0$ and E is a positive set of $\nu - \epsilon\mu$.*

Proof

Let $\{P_n, N_n\}$ be Hahn decomposition of $\nu - \frac{1}{n}\mu$. If $\mu(P_n) > 0$ for some n, then let $E = P_n$. Otherwise $\mu(P_n) = 0$ for all n, which means that

$$\mu\left(\bigcup_{n=1}^{\infty} P_n\right) = 0.$$

Since $\left(\nu - \frac{1}{n}\mu\right)\left(\bigcap_{k=1}^{\infty} N_k\right) \leq 0$, we must have $\nu\left(\bigcap_{k=1}^{\infty} N_k\right) \leq \frac{1}{n}\mu\left(\bigcap_{k=1}^{\infty} N_k\right)$ for all n, which means that

$$\nu\left(\bigcap_{k=1}^{\infty} N_k\right) = 0.$$

So $\mu \perp \nu$. □

Theorem 3.2.2 (Lebesgue-Radon-Nikodym Theorem)
Let ν be a σ-finite signed measure and μ be a σ-finite positive measure on $(X, \mathscr{M})$. Then exists unique σ-finite signed measures λ and ρ such that

$$\lambda \perp \mu, \quad \rho \ll \mu, \quad \text{and} \quad \nu = \lambda + \rho.$$

Moreover, there is a measurable function f with at least one of $\int f^+ \, d\mu$, $\int f^- \, d\mu$ finite, such that for all $E \in \mathscr{M}$

$$\rho(E) = \int_E f \, d\mu.$$

Any two such functions are equal μ-a.e.

Proof

Case I: ν and μ are finite positive measures.

Let

$$\mathscr{F} = \left\{ f : X \mapsto [0, \infty] \, : \int_E f \, d\mu \leq \nu(E), \forall E \in \mathscr{M} \right\}.$$

$\mathscr{F}$ is nonempty since the zero function is in $\mathscr{F}$. We claim that if $f, g \in \mathscr{F}$, then $h = \max(f, g) \in \mathscr{F}$. Let $A = \{x : f(x) > g(x)\}$. Then h is given by

$$h(x) = \begin{cases} f(x), & x \in A, \\ g(x), & x \in A^c, \end{cases}$$

and

$$\int_E h \, d\mu = \int_{E\cap A} f \, d\mu + \int_{E\cap A^c} g \, d\mu \le \nu(E \cap A) + \nu(E \cap A^c) = \nu(E).$$

Let

$$\alpha = \sup \left\{ \int f \, d\mu : f \in \mathscr{F} \right\}.$$

Then $\alpha \le \nu(X) < \infty$, and there exist $\{f_n\} \subset \mathscr{F}$ such that $\int f_n \, d\mu \to \alpha$. Let $g_n = \max(f_1, \ldots, f_n)$. Then $g_n \in \mathscr{F}$ and $g_1 \le g_2 \le \cdots$. Let $f = \lim_{n\to\infty} g_n$. Then (why?)

$$\alpha = \lim_{n\to\infty} \int f_n \, d\mu \le \lim_{n\to\infty} \int g_n \, d\mu = \int f \, d\mu \le \alpha$$

i.e., $\int f \, d\mu = \alpha$.

We claim that μ and the measure λ defined by

$$\lambda(E) = \nu(E) - \int_E f \, d\mu$$

are mutually singular. If not, by Lemma 3.2.1, there is an $\epsilon_0 > 0$ and an $A \in \mathscr{M}$, such that

$$\mu(A) > 0 \quad \text{and} \quad \lambda(A \cap E) \ge \epsilon_0 \mu(A \cap E), \quad \forall E \in \mathscr{M},$$

which means that (note that λ is a positive measure)

$$\epsilon_0 \mu(A \cap E) \le \lambda(A \cap E) \le \lambda(E) = \nu(E) - \int_E f \, d\mu \quad \forall E \in \mathscr{M}$$

or

$$\int_E f + \epsilon_0 \chi_A \, d\mu \le \nu(E).$$

So $f + \epsilon_0 \chi_A \in \mathscr{F}$. However

$$\int f + \epsilon_0 \chi_A \, d\mu = \int f \, d\mu + \epsilon_0 \mu(A) > \alpha,$$

a contradiction.

Therefore $\lambda \perp \mu$. Let

$$\rho(E) = \int_E f \, d\mu.$$

Then $\nu = \lambda + \rho$, $\lambda \perp \mu$, $\rho \ll \mu$.

If there is another $\hat{f}$ and $\hat{\lambda}$ such that

$$\nu(E) = \hat{\lambda}(E) + \int_E \hat{f} \, d\mu = \lambda(E) + \int_E f, \, d\mu,$$

then

$$(\hat{\lambda} - \lambda)(E) = \int_E f - \hat{f} \, d\mu, \quad \forall E \in \mathscr{M}.$$

The measure on the left-hand side is mutually singular to μ, and the measure on the right-hand side is absolutely continuous with respect to μ. Therefore they both have to be zero measures, and $f = \hat{f}$ μ-a.e.

Case II: ν and μ are σ-finite positive measures.

Then there are partitions $\{A_n\}$ and $\{B_n\}$ of X such that $\mu(A_n) < \infty$ and $\nu(B_n) < \infty$. Let

$$C_{ij} = A_i \cap B_j, \quad \mu_{ij}(E) = \mu(E \cap C_{ij}), \quad \nu_{ij}(E) = \nu(E \cap C_{ij}).$$

Then by what we just proved, there are positive measures $\lambda_{ij} \perp \mu_{ij}$ and functions f_{ij} such that

$$\nu_{ij}(E) = \lambda_{ij}(E) + \int_E f_{ij} \, d\mu_{ij}.$$

Since $\nu_{ij}(C_{ij}^c) = 0$, $\mu_{ij}(C_{ij}^c) = 0$, we must have $\lambda_{ij}(C_{ij}^c) = 0$, and we can also assume $f_{ij} = f_{ij}\chi_{C_{ij}}$. Define

$$\lambda(E) = \sum_{i,j} \lambda_{ij}(E \cap C_{ij}), \quad \text{and} \quad f = \sum_{i,j} f_{ij}.$$

Then $\lambda \perp \mu$ and $\nu(E) = \lambda(E) + \int_E f \, d\mu$.

Case III: ν is a σ-finite signed measure.

Let $\nu = \nu^+ - \nu^-$ be the Jordan decomposition of ν. Then by what we just proved, there are positive measures $\lambda^+ \perp \mu$, $\lambda^- \perp \mu$, functions f^+, f^-, and partitions $\{A^+, B^+\}$ and $\{A^-, B^-\}$ of X, such that

$$\nu^+(E) = \lambda^+(E) + \int_E f^+ \, d\mu, \quad \nu^-(E) = \lambda^-(E) + \int_E f^- \, d\mu,$$

and A^+ is λ^+-null, A^- is λ^--null, and both B^+, B^- are μ-null. Let $\lambda = \lambda^+ - \lambda^-$ and $f = f^+ - f^-$. Then $\lambda \perp \mu$ because $A^+ \cap A^-$ is λ-null and $B^+ \cup B^-$ is μ-null, and we have

$$\nu(E) = \lambda(E) + \int_E f \, d\mu.$$

□

Definition 3.2.3

a) The decomposition $\nu = \lambda + \rho$ with $\lambda \perp \mu$ and $\rho \ll \mu$ is called the Lebesgue decomposition of ν with respect to μ.
b) For $\nu \ll \mu$, we must have $\nu(E) = \int_E f \, d\mu$ for some function f. Such f is called the Radon-Nikodym derivative of ν with respect to μ, and is denoted by

$$\frac{d\nu}{d\mu} = f.$$

Proposition 3.2.4 *Suppose ν is a σ-finite signed measure and μ, λ are σ-finite positive measures such that*

$$\nu \ll \mu \ll \lambda.$$

a) If $g \in L^1(\nu)$, then $g\frac{d\nu}{d\mu} \in L^1(\mu)$ and

$$\int g \, d\nu = \int g \frac{d\nu}{d\mu} \, d\mu.$$

b) $\nu \ll \lambda$ and $\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu}\frac{d\mu}{d\lambda}$ λ-a.e.

Proof

a) It is true for simple functions. Then by the Monotone Convergence Theorem it is true for positive measurable functions for both ν^+ and ν^-. So it is true for all functions in L^1.
b) By a)

$$\nu(E) = \int_E \frac{d\nu}{d\mu} \, d\mu = \int_E \frac{d\nu}{d\mu}\frac{d\mu}{d\lambda} \, d\lambda.$$

□

Corollary 3.2.5 *If $\mu \ll \lambda$ and $\lambda \ll \mu$, then $\frac{d\lambda}{d\mu}\frac{d\mu}{d\lambda} = 1$.*

Proposition 3.2.6 *For positive measures $\mu_1, \ldots, \mu_n$,*

$$\mu_j \ll \mu_1 + \cdots + \mu_n, \qquad \forall j = 1, \ldots, n.$$

Exercises

3.11 Let ν be a signed measure and μ be a positive measure on $(X, \mathscr{M})$. Prove that the following conditions are equivalent

i) $\nu \ll \mu$.
ii) $|\nu| \ll \mu$.
iii) $\nu^+ \ll \mu$ and $\nu^- \ll \mu$.

3.12 Let ν, μ be σ-finite positive measures on $(X, \mathscr{M})$ such that $\nu \ll \mu$. Prove that $\frac{d\nu}{d\mu} \geq 0$ μ-a.e.

3.13 Let ν be a signed measure and μ be a positive measure on $(X, \mathscr{M})$. Prove that $\frac{d|\nu|}{d\mu} = \left|\frac{d\nu}{d\mu}\right|$ μ-a.e.

3.14 Let $\mu, \nu_i, i \in \mathbb{N}$ be signed measures on $(X, \mathscr{M})$ such that $\nu_i(E) > -\infty$ for all i. Prove that if $\mu \perp \nu_i$ for all i, then $\mu \perp \sum_{i=1}^{\infty} \nu_i$.

3.15 Let $\lfloor x \rfloor$ be the largest integer less than or equal to x (the floor function).

a. Prove that $\mu_{\lfloor x \rfloor} \ll \mu_{\lfloor 2x \rfloor}$.
b. Find $\frac{d\mu_{\lfloor x \rfloor}}{d\mu_{\lfloor 2x \rfloor}}$.
c. Find the Lebesgue decomposition of $\mu_{\lfloor 2x \rfloor}$ with respect to $\mu_{\lfloor x \rfloor}$.

3.16 a. Let μ_i, ν_i be σ-finite positive measures on $(X_i, \mathscr{M}_i)$ such that $\nu_i \ll \mu_i$, $i = 1, 2$. Prove that $\nu_1 \times \nu_2 \ll \mu_1 \times \mu_2$ and

$$\frac{d(\nu_1 \times \nu_2)}{d(\mu_1 \times \mu_2)}(x_1, x_2) = \frac{d\nu_1}{d\mu_1}(x_1)\frac{d\nu_2}{d\mu_2}(x_2).$$

b. Prove the same conclusion if ν_1, ν_2 are signed measures and μ_1, μ_2 are finite positive measures.

3.17 Let μ, ν be σ-finite positive measures on $(X, \mathscr{M})$ with $\nu \ll \mu$ and let $\lambda = \mu + \nu$. Prove that

$$\frac{d\mu}{d\lambda} = 1 - \frac{d\nu}{d\lambda}, \quad 0 \leq \frac{d\nu}{d\lambda} < 1, \quad \text{and} \quad \frac{d\nu}{d\mu} = \frac{\frac{d\nu}{d\lambda}}{1 - \frac{d\nu}{d\lambda}}.$$

3.3 Monotone Functions

Definition 3.3.1

A collection $\mathscr{C}$ of intervals is called a Vitali cover of a set $E \subset \mathbb{R}$ if every point $x \in E$ can be covered by an interval in $\mathscr{C}$ of arbitrarily small positive length, i.e.

$$\forall x \in E, \ \forall \epsilon > 0, \ \exists I \in \mathscr{C} \text{ such that } x \in I \text{ and } 0 < m(I) < \epsilon.$$

Lemma 3.3.2 (Vitali's Lemma) *Let E be a subset of $\mathbb{R}$ with $m^*(E) < \infty$ and $\mathscr{C}$ be a Vitali cover of E. Then for any $\epsilon > 0$, there are finitely many disjoint $\{I_1, \ldots, I_n\} \subset \mathscr{C}$ such that*

$$m^*\left(E \setminus \left(\bigcup_{i=1}^{n} I_i\right)\right) < \epsilon.$$

Proof

Without loss of generality, assume all the intervals in $\mathscr{C}$ are closed intervals.

Since $m^*(E) < \infty$, there is an open set $U \supset E$ with $m(U) < \infty$. Since each point in E is covered by intervals in $\mathscr{C}$ of arbitrarily small lengths, we can only consider those intervals contained in U and assume all intervals in $\mathscr{C}$ are contained in U.

We will choose $\{I_1, \ldots, I_n\}$ inductively. Pick any $I_1 = [a_1, b_1] \in \mathscr{C}$ with $m(I_i) > 0$. If $E \subset I_1$, then we are done. Otherwise let

$$k_1 = \sup\{m(I) : I \in \mathscr{C}, I \cap I_1 = \emptyset\}$$

(why does the sup exist?). Then $0 < k_1 \leq m(U) < \infty$, So we can find $I_2 = [a_2, b_2] \in \mathscr{C}$ with $I_1 \cap I_2 = \emptyset$ and $m(I_2) > \frac{1}{2}k_1$. Generally, assume disjoint $\{I_1, \ldots, I_m\}$ are chosen. If $E \subset \bigcup_{i=1}^{m} I_i$, then we are done. Otherwise let

$$k_m = \sup\{m(I) : I \in \mathscr{C}, I \cap I_i = \emptyset \text{ for } i = 1, \ldots, m\}$$

and pick $I_{m+1} = [a_{m+1}, b_{m+1}] \in \mathscr{C}$ such that $I_{m+1} \cap I_i = \emptyset$ and $m(I_{m+1}) > \frac{1}{2}k_m$.

By induction, if $E \not\subset \bigcup_{i=1}^{n} I_i$ for any n, then we have $\{I_1, I_2, \ldots\} \subset \mathscr{C}$ such that

$$m(I_{n+1}) > \frac{1}{2}\sup\{m(I) : I \in \mathscr{C}, I \cap I_i = \emptyset \text{ for } i = 1, \ldots, n\} \quad \forall n.$$

Since $\{I_i, i = 1, \ldots, \}$ are disjoint and contained in U, $\sum_{i=1}^{\infty} m(I_n)$ converges, and for any $\epsilon > 0$, there exists an n such that

$$\sum_{i=n+1}^{\infty} m(I_n) < \frac{\epsilon}{5}.$$

Let (Figure 3.1)

$$J_i = [a_i - 2m(I_i), b_i + 2m(I_i)], \quad i = 1, 2, \ldots.$$

We claim that

$$E \setminus \left(\bigcup_{i=1}^{n} I_i\right) \subset \bigcup_{i=n+1}^{\infty} J_i.$$

For any $x \in E \setminus \left(\bigcup_{i=1}^{n} I_i\right)$, we can always find an $I \in \mathscr{C}$ such that

$$x \in I, \quad m(I) > 0, \quad \text{and} \quad I \cap I_i = \emptyset, \quad \text{for } i = 1, \ldots, n.$$

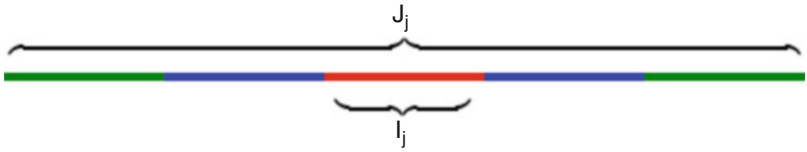


Fig. 3.1 I_j and J_j

If $I \cap I_i = \emptyset$ for all $i > n$ too, then $m(I) \leq k_i < 2m(I_{i+1})$ for all i, which contradicts the facts that $m(I_{i+1}) \to 0$ as $i \to \infty$ and $m(I) > 0$.

So there must exist a $j > n$ such that

$$I \cap I_i = \emptyset \text{ for all } i = 1, \ldots, j-1 \text{ and } I \cap I_j \neq \emptyset.$$

Then

$$m(I) \leq k_{j-1} < 2m(I_j)$$

which implies that

$$I \subset J_j \quad \text{and} \quad x \in I \subset J_j \subset \bigcup_{i=n+1}^{\infty} J_i.$$

Therefore

$$m^*\left(E \setminus \left(\bigcup_{i=1}^{n} I_i\right)\right) \leq \sum_{i=n+1}^{\infty} m(J_i) = 5 \sum_{i=n+1}^{\infty} m(I_i) < \epsilon.$$

□

We now are ready to prove properties of monotonic functions on $\mathbb{R}$. The results are given for increasing functions and can be easily adapted to decreasing functions f because $-f$ is increasing.

Theorem 3.3.3

Let $f : \mathbb{R} \to \mathbb{R}$ be increasing and $g(x) = f(x^+)$. Then

a) The set of points of discontinuity of f is countable.

b) f, g are differentiable m-a.e.

c) $f' = g'$ at each x where both $f'(x)$ and $g'(x)$ are defined.

d) $f' \in L^1[a, b]$ for all bounded $[a, b]$ and

$$\int_a^b f' \, dm \leq f(b) - f(a).$$

Proof

a) At each point of discontinuity x of f, both one-side limits $f(x^+)$, $f(x^-)$ exist (why?), and $f(x^-) < f(x^+)$. So we can always find a rational number $r \in (f(x^-), f(x^+))$. The map $x \mapsto r$ is one-to-one. Therefore the set of points of discontinuity of f is countable.

b) Define

$$D^{\pm} f = \limsup_{h \to 0^{\pm}} \frac{f(x+h) - f(x)}{h}, \quad D_{\pm} f = \liminf_{h \to 0^{\pm}} \frac{f(x+h) - f(x)}{h}.$$

We will show that the sets where any two of them are unequal have Lebesgue measure 0. We only prove for one pair. The proofs for other pairs are similar.

For any bounded interval $[a, b]$ with $a < b$ and any $v < u$ in $\mathbb{Q}$, define

$$E_{v,u} = \{x \in [a, b] : D_- f(x) < v < u < D^+ f(x)\}.$$

We show that $m^*(E_{v,u}) = 0$.

For any $\epsilon > 0$, there is an open set $U \supset E_{v,u}$ such that

$$m(U) < m^*(E_{v,u}) + \epsilon.$$

For all $x \in E_{v,u}$, since $D_- f(x) < v$, there exists arbitrarily small $h > 0$ such that

$$\frac{f(x) - f(x-h)}{h} < v, \quad \text{i.e.} \quad f(x) - f(x-h) < vh.$$

All such intervals $\{[x-h, x]\}$ form a Vitali cover of $E_{v,u}$ and by Lemma 3.3.2, there are disjoint $\{I_1, \ldots, I_m\}$, $I_i = (x_i - h_i, x_i) \subset U$ (note that we change the closed intervals to open one), such that

$$f(x_i) - f(x_i - h_i) < vh_i \quad \text{and} \quad m^*\left(E_{v,u} \setminus \left(\bigcup_{i=1}^{m} I_i\right)\right) < \epsilon,$$

which means that

$$m^*(E_{v,u}) \le m^*\left(E_{v,u} \cap \left(\bigcup_{i=1}^{m} I_i\right)\right) + m^*\left(E_{v,u} \setminus \left(\bigcup_{i=1}^{m} I_i\right)\right)$$

$$< m^*\left(E_{v,u} \cap \left(\bigcup_{i=1}^{m} I_i\right)\right) + \epsilon.$$

So we have

$$m^*\left(E_{v,u}\cap\left(\bigcup_{i=1}^{m}I_i\right)\right)>m^*(E_{v,u})-\epsilon,$$

and

$$\sum_{i=1}^{m}(f(x_i)-f(x_i-h_i))<v\sum_{i=1}^{m}h_i<vm(U)<v(m^*(E_{v,u})+\epsilon).$$

Next, for all $y\in E_{v,u}\cap\left(\bigcup_{i=1}^{m}I_i\right)$, since $D^+f(y)>u$, there is arbitrarily small $k>0$ such that

$$[y,y+k]\subset\left(\bigcup_{i=1}^{m}I_i\right),\quad\frac{f(y+k)-f(y)}{k}>u,\quad\text{i.e.}\quad f(y+k)-f(y)<uk.$$

All such intervals form a Vitali cover of $E_{v,u}\cap\left(\bigcup_{i=1}^{m}I_i\right)$, so there are disjoint intervals $\{J_1,\ldots,J_p\}$, $J_i=[y_i,y_i+k_i]$, such that

$$m^*\left(\left(E_{v,u}\cap\left(\bigcup_{i=1}^{m}I_i\right)\right)\setminus\left(\bigcup_{j=1}^{p}J_j\right)\right)<\epsilon,$$

which means that

$$m^*\left(E_{v,u}\cap\left(\bigcup_{i=1}^{m}I_i\right)\right)\le m^*\left(E_{v,u}\cap\left(\bigcup_{i=1}^{m}I_i\right)\cap\left(\bigcup_{j=1}^{p}J_j\right)\right)$$

$$+m^*\left(\left(E_{v,u}\cap\left(\bigcup_{i=1}^{m}I_i\right)\right)\setminus\left(\bigcup_{j=1}^{p}J_j\right)\right)<m^*\left(E_{v,u}\cap\left(\bigcup_{j=1}^{p}J_j\right)\right)+\epsilon.$$

So we have

$$m^*\left(E_{v,u}\cap\left(\bigcup_{j=1}^{p}J_j\right)\right)>m^*\left(E_{v,u}\cap\left(\bigcup_{i=1}^{m}I_i\right)\right)-\epsilon>m^*(E_{v,u})-2\epsilon,$$

and

$$\sum_{j=1}^{p}(f(y_j+k_j)-f(y_j))>u\sum_{j=1}^{p}k_j=um\left(\bigcup_{j=1}^{p}J_j\right)>u(m^*(E_{v,u})-2\epsilon).$$

Therefore we have

$$\begin{aligned} u(m^*(E_{v,u}) - 2\epsilon) &< \sum_{j=1}^{p} (f(y_j + k_j) - f(y_j)) \\ &= \sum_{i=1}^{n} \sum_{J_j \subset I_i} (f(y_j + k_j) - f(y_j)) \\ &\leq \sum_{i=1}^{m} (f(x_i) - f(x_i - h_i)) \\ &< v(m^*(E_{v,u}) + \epsilon). \end{aligned}$$

Let $\epsilon \to 0$, we have

$$um^*(E_{v,u}) \leq vm^*(E_{v,u}).$$

Since $u > v$, it is possible only if $m^*(E_{v,u}) = 0$.

Since

$$\{x : D_- f(x) < D^+ f(x)\}$$

is a countable union of such $E_{v,u}$,

$$m^* \left(\{x : D_- f(x) < D^+ f(x)\} \right) = 0.$$

Similarly

$$m^* \left(\{x : D_+ f(x) < D^- f(x)\} \right) = 0,$$

$$m^* \left(\{x : D_+ f(x) < D^+ f(x)\} \right) = 0,$$

$$m^* \left(\{x : D_- f(x) < D^- f(x)\} \right) = 0,$$

and f' exists m-a.e.

c) Note that for each point of continuity x of f, we have $f(x) = g(x)$. At each x where both $f'(x)$ and $g'(x)$ are defined, f is continuous, and since the set of points of discontinuity of f is countable, there are $h_n \to 0$ such that f is continuous at all $x + h_n$. Then since both $f'(x)$ and $g'(x)$ are defined,

$$f'(x) = \lim_{n \to \infty} \frac{f(x + h_n) - f(x)}{h_n} = \lim_{n \to \infty} \frac{g(x + h_n) - g(x)}{h_n} = g'(x).$$

d) Homework.

□

Exercises

3.18 Let f be an increasing Lebesgue measurable function on $\mathbb{R}$. Prove that for any $a, b \in \mathbb{R}$, $f' \in L^1([a,b])$ and

$$\int_a^b f'(x)\,dx \le f(b) - f(a).$$

(Hint: Redefine $f(x) \equiv f(b)$ for $x \ge b$ and apply the Fatou's lemma to $g_n = n\left(f\left(x+\frac{1}{n}\right) - f(x)\right)$.)

3.19 Prove that an increasing function $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ if and only if it is onto $[f(a), f(b)]$.

3.20 Prove that the function $G(x)$ defined in Exercise 1.29 is strictly increasing, continuous at every irrational points, and discontinuous at every rational points.

3.4 Functions of Bounded Variation

Let $a < b$ be real numbers, f be a function defined on $[a,b]$, and

$$\delta = \{x_0, \ldots, x_n\}, a = x_0 < x_1 < \cdots < x_n = b$$

be a partition of $[a,b]$. Define

$$t_\delta = \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})|.$$

Definition 3.4.1

A function f defined on $[a,b]$ is called of bounded variation on $[a,b]$, denoted by $f \in BV[a,b]$, if

$$\sup_\delta t_\delta < \infty.$$

Remark 3.2 Functions of bounded variation can also be defined on an infinite interval. If one or both of $a, b = \pm\infty$, then the sup can be taken over all $\delta = \{x_0, \ldots, x_n\}$ with $a < x_0 < x_1 < \cdots < x_n < b$. However, if $f \in BV[a,b]$, then for any sequence $\{t_n\} \subset (a,b)$ goes to the boundaries, $\{f(t_n)\}$ is a Cauchy sequence, and by the Cauchy criterion $\lim_{x\to b^-} f(x)$ and $\lim_{x\to a^+} f(x)$ exist. So if we define $f(-\infty) = \lim_{x\to-\infty} f(x)$ and/or $f(\infty) = \lim_{x\to\infty} f(x)$, and call a function "defined on $[a,b]$" for an infinite interval if the limit(s) to ∞ and/or $-\infty$ exist, then Definition 3.4.1 is also valid for infinite intervals.

For $f \in BV[a,b]$, define

$$p_\delta = \sum_{i=1}^{n} (f(x_i) - f(x_{i-1}))^+, \quad n_\delta = \sum_{i=1}^{n} (f(x_i) - f(x_{i-1}))^-,$$

where

$$y^+ = \max(y, 0), \quad \text{and} \quad y^- = \max(-y, 0).$$

Then

$$f(b) - f(a) = p_\delta - n_\delta, \quad \text{and} \quad t_\delta = p_\delta + n_\delta.$$

Definition 3.4.2

Let $f \in BV[a,b]$. Then

$$P_a^b = \sup_\delta p_\delta, \quad N_a^b = \sup_\delta n_\delta, \quad T_a^b = \sup_\delta t_\delta$$

are called the positive, negative, and total variations of f over $[a,b]$.
We will use $P_a(x), N_a(x), T_a(x)$ for P_a^x, N_a^x, T_a^x and $P^b(x), N^b(x), T^b(x)$ for P_x^b, N_x^b, T_x^b if x is a variable.
If $a = -\infty$, we'll drop the subscript and use $P(x), N(x), T(x)$ for $P_{-\infty}^x, N_{-\infty}^x, T_{-\infty}^x$, respectively.

Example of functions in $BV[a,b]$ are monotone functions, functions with bounded derivatives, and $\int_a^x g\, dm$ for any $g \in L^1[a,b]$

Lemma 3.4.3 *For any $f \in BV[a,b]$*
a) $P_a(x), N_a(x), T_a(x)$ are increasing functions on $[a,b]$
b) $f(x) - f(a) = P_a(x) - N_a(x)$ and $T_a(x) = P_a(x) + N_a(x)$.

Proof

We only need to prove b). For any partition δ of $[a,x]$, we have $p_\delta = n_\delta + f(x) - f(a)$ and $t_\delta = p_\delta + n_\delta = 2p_\delta - (f(x) - f(a))$. By taking sup over all partitions we have

$$P_a(x) = N_a(x) + f(x) - f(a) \quad \text{or} \quad f(x) - f(a) = P_a^b - N_a^b,$$

and

$$T_a(x) = 2P_a(x) - (f(x) - f(a)) \quad \text{or} \quad T_a(x) = P_a(x) + N_a(x).$$

□

Lemma 3.4.4 *$T_a(x) - f(x)$ and $T_a(x) + f(x)$ are increasing functions on $[a,b]$.*

Proof
$T_a(x) - f(x) = 2N_a(x) - f(a)$ and $T_a(x) + f(x) = 2P_a(x) + f(a)$. □

Theorem 3.4.5
a) $f \in BV[a,b]$ *if and only if* f *is a difference of two increasing functions on* $[a,b]$ *(if* $[a,b]$ *is an infinite interval, function on* $[a,b]$ *means that the limits of* $f(x)$ *as* x *goes to the boundaries exist.)*
b) If $f \in BV[a,b]$*, then the one side limit of* $f(x)$ *exists for any* $x \in [a,b]$.
c) If $f \in BV[a,b]$*, then the set of points of discontinuity of* f *on* $[a,b]$ *is countable.*
d) If $f \in BV[a,b]$ *and* $g(x) = f(x^+)$*, then* f' *exists and* $f' = g'$ *a.e. on* $[a,b]$.

Proof
Because of Theorem 3.3.3, we only need to prove a). Any monotone function on $[a,b]$ is of bounded variation because for any partition δ

$$t_\delta = |f(b) - f(a)|.$$

Conversely if $f \in BV[a,b]$, then $f(x) = P_a(x) - N_a(x) + f(a)$ is a difference of two increasing functions. □

Definition 3.4.6
A function on $\mathbb{R}$ is called a function of locally bounded variation, denoted by $f \in LBV$ if $f \in BV[a,b]$ for any bounded interval $[a,b]$.

Corollary 3.4.7 *$f \in LBV$ if and only if f is a difference of two increasing functions on $\mathbb{R}$.*

Proof
Clearly any monotone function is in LBV. Conversely, if $f \in LBV$, then

$$f(x)-f(0) = P_0(x)-N_0(x), \text{ for } x \geq 0, \text{ and } f(0)-f(x) = P^0(x)-N^0(x), \text{ for } x < 0$$

So if we define

$$P(x) = \begin{cases} P_0(x), & x \geq 0, \\ -P^0(x), & x < 0, \end{cases} \quad N(x) = \begin{cases} N_0(x), & x \geq 0, \\ -N^0(x), & x < 0. \end{cases}$$

then both $P(x)$ and $N(x)$ are increasing and

$$f(x) = P(x) - N(x) + f(0)$$

is a difference of two increasing functions. □

Lemma 3.4.8 *Let $a < b < c$ and $f \in BV[a, c]$. Then*

$$P_a^c = P_a^b + P_b^c, \quad N_a^c = N_a^b + N_b^c, \quad T_a^c = T_a^b + T_b^c.$$

Proof

There are sequences of partitions $\{\alpha_n\}$ of $[a, b]$ and $\{\beta_n\}$ of $[b, c]$ such that $t_{\alpha_n} \to T_a^b$, $t_{\beta_n} \to T_b^c$. For any n

$$T_a^c \geq t_{\alpha_n \cup \beta_n} = t_{\alpha_n} + t_{\beta_n}.$$

Let $n \to \infty$, then we have $T_a^c \geq T_a^b + T_b^c$.

Conversely, there is a sequence of partitions $\{\gamma_n\}$ of $[a, c]$ such that $t_{\gamma_n} \to T_a^c$. Let

$$\hat{\gamma}_n = \gamma_n \cup \{b\}, \quad \alpha_n = \hat{\gamma}_n \cap [a, b], \quad \beta_n = \hat{\gamma}_n \cap [b, c]$$

Then

$$T_a^b + T_b^c \geq t_{\alpha_n} + t_{\beta_n} = t_{\hat{\gamma}_n} \geq t_{\gamma_n}.$$

Let $n \to \infty$, then we have $T_a^b + T_b^c \geq T_a^c$.

The proofs for positive and negative variations are similar. □

Lemma 3.4.9 *Let $f \in BV[a, b]$ and f is right continuous, then $T_a(x)$ is also right continuous. In particular, if $a = -\infty$, then $\lim_{x \to -\infty} T(x) = 0$.*

Proof

Let $x \in [a, b]$ be given such that $f(x^+) = f(x)$. Let $d = T_a(x^+) - T_a(x)$. Then since $T_a(x)$ is increasing,

$$T_a(z) - T_a(x) = T_x^z \geq d \quad \forall z > x. \tag{3.4.2}$$

By the given condition, for any $\epsilon > 0$, there is a $\delta > 0$ such that for finite x

$$|f(\hat{x}) - f(x)| < \epsilon \quad \text{and} \quad T_a(\hat{x}) - T_a(x^+) < \epsilon$$

for all $x < \hat{x} < x + \delta$ (for all $\hat{x} < -\delta$ if $x = a = -\infty$).

By (3.4.2),

$$d \leq T_a(\hat{x}) - T_a(x) = T_a(\hat{x}) - T_a(x^+) + T_a(x^+) - T_a(x) < \epsilon + d.$$

For any such $\hat{x}$, there is a partition $\alpha : x = x_0 < x_1 < \cdots < x_n = \hat{x}$ such that

$$t_\alpha > \frac{3}{4}(T_x^{\hat{x}}) = \frac{3}{4}(T_a(\hat{x}) - T_a(x)) \geq \frac{3}{4}d.$$

So

$$t_\alpha - |f(x_1) - f(x)| \geq \frac{3}{4}d - \epsilon$$

where x_1 is the second partition point of α. Then there is a partition β of $[x, x_1]$ such that

$$t_\beta \geq \frac{3}{4}T_x^{x_1} \geq \frac{3}{4}d.$$

So we have

$$d + \epsilon > T_x^{\hat{x}} \geq t_{\alpha\cup\beta} = t_\alpha - |f(x_1) - f(x)| + t_\beta \geq \frac{3}{4}d - \epsilon + \frac{3}{4}d = \frac{3}{2}d - \epsilon.$$

Let $\epsilon \to 0$, we have $d \geq \frac{3}{2}d$, so we must have $d = 0$. Since by our definition $f(-\infty) = \lim_{x\to-\infty} f(x)$ is right continuous, $\lim_{x\to-\infty} T(x) = T_{-\infty}^{-\infty} = 0$. □

If $f \in BV[-\infty, \infty]$, then $f(-\infty) = \lim_{x\to-\infty} f(x)$ exists, and we can always replace $f(x)$ with $g(x) = f(x) - f(-\infty)$. Then $g(-\infty) = 0$

Definition 3.4.10

A function f is called a normalized function of bounded variation on $\mathbb{R}$, denoted by $f \in NBV$ if $f \in BV[-\infty, \infty]$, $f(-\infty) = 0$, and $f(x^+) = f(x)$ for all $x \in \mathbb{R}$.

Theorem 3.4.11

If μ is a finite signed Borel measure on $\mathbb{R}$ and let $f(x) = \mu((-\infty, x])$, then $f \in NBV$. Conversely if $f \in NBV$, then there is a unique finite signed Borel measure μ_f such that

$$f(x) = \mu_f((-\infty, x]).$$

Moreover

$$|\mu_f| = \mu_T.$$

Proof

If μ is a finite signed measure, define

$$f^\pm(x) = \mu^\pm((-\infty, x]).$$

Then $f^{\pm}$ are increasing and right continuous and $f^{\pm}(-\infty) = 0$, $f^{\pm}(\infty) < \infty$. So $f(x) = f^{+} - f^{-} \in NBV$.

Conversely if $f \in NBV$, then

$$f(x) = P(x) - N(x).$$

Let

$$\mu_f = \mu_P - \mu_N.$$

Then

$$f(x) = P(x) - N(x) = \mu_P((-\infty, x]) - \mu_N((-\infty, x]) = \mu_f((-\infty, x]).$$

The uniqueness follows the uniqueness of Lebesgue-Stieltjes measures.

Next we show $|\mu_f| = \mu_T$. For any partition $\delta : -\infty = x_0 < x_1 < \cdots < x_n = x$,

$$|\mu_f|((-\infty, x]) = \sum_{i=1}^{n} |\mu_f|((x_{i-1}, x_i]) \geq \sum_{i=1}^{n} |\mu_f((x_{i-1}, x_i])| = \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| = t_\delta.$$

So

$$T(x) = \sup_{\delta} t_\delta \leq |\mu_f|((-\infty, x]).$$

Conversely for any E, since $\mu_f = \mu_P - \mu_N$,

$$\mu_f^{+}(E) = \mu_f(E \cap A_P) = \mu_P(E \cap A_P) - \mu_N(E \cap A_P) \leq \mu_P(E)$$

$$\mu_f^{-}(E) = -\mu_f(E \cap A_n) = \mu_N(E \cap A_n) - \mu_P(E \cap A_n) \leq \mu_N(E).$$

where $\{A_P, A_n\}$ is a Hahn decomposition for μ_f. So

$$\mu_P + \mu_N \geq \mu_f^{+} + \mu_f^{-} = |\mu_f|,$$

which means that

$$T(x) = P(x) + N(x) = \mu_P((-\infty, x]) + \mu_N((-\infty, x]) \geq |\mu_f|((-\infty, x]).$$

Therefore

$$T(x) = |\mu_f|((-\infty, x]) \quad \text{and} \quad |\mu_f| = \mu_T.$$

□

Corollary 3.4.12 *$\mu_f^+ = \mu_P$ and $\mu_f^- = \mu_N$.*

Proof
Homework □

Exercises

3.21 Let f be a continuous function of bounded variation on $[a, b]$ and $\delta : x_0 = a < x_1 < \cdots < x_{n-1} < x_{n_\delta} = b$ be partitions of $[a, b]$. Prove that

$$\lim_{\|\delta\|\to 0} \sum_{i=1}^{n_\delta} |f(x_i) - f(x_{i-1})|^2 = 0.$$

3.22 Let $\{f_i : i \in \mathbb{N}\}$ be functions $\mathbb{R} \to \mathbb{R}$ and $f_i \to f$ pointwise. Prove that for any $a, b \in \mathbb{R}$, and any partition δ of $[a, b]$, we have $t_\delta(f) \le \liminf_{i\to\infty} T_a^b(f_i)$, so

$$T_a^b(f) \le \liminf_{i\to\infty} T_a^b(f_i).$$

3.23 Let $f, g \in NBV$. Prove that for any $a < b$ in $\mathbb{R}$,

$$\int_{[a,b]} f(x)\, d\mu_g + \int_{[a,b]} g(x^-)\, d\mu_f = f(b)g(b) - f(a^-)g(a^-).$$

3.5 Absolutely Continuous Functions

Definition 3.5.1
A function $f : [a, b] \to \mathbb{R}$ is called absolutely continuous on $[a, b]$ if for any $\epsilon > 0$, there exists a $\delta > 0$ such that for any disjoint intervals $\{(a_i, b_i)\}_1^n$ in $[a, b]$ with $\sum_{i=1}^n (b_i - a_i) < \delta$, we have

$$\sum_{i=1}^{n} |f(b_i) - f(a_i)| < \epsilon.$$

Example 3.2
$F(x) = \int_a^x f(t)\, dt$ is absolutely continuous for any $f \in L^1[a, b]$ because of Corollary 3.1.10. ■

Proposition 3.5.2 *If f is absolutely continuous on $[a, b]$, then $f \in BV[a, b]$.*

Proof
Let δ be the one in the definition of absolute continuity for $\epsilon = 1$, N be the natural number such that $d = \frac{b-a}{N} < \delta$, and $x_i = a + id, i = 0, 1, \ldots, N$. Then for any partition $\alpha : z_0 =$

$x_{i-1} < z_1 < \cdots < z_n = x_i$ of $[x_{i-1}, x_i]$, since $\sum_{i=1}^{n}(z_i - z_{i-1}) = x_i - x_{i-1} < \delta$, we have $t_\alpha < 1$. Therefore $T_{x_{i-1}}^{x_i} \le 1$, and by Lemma 3.4.8

$$T_a^b = \sum_{i=1}^{N} T_{x_{i-1}}^{x_i} \le N.$$

□

Lemma 3.5.3 *Let $f \in L^1[a,b]$. Then if $\int_a^x f\,dm = 0$ for all $x \in [a,b]$, then $f = 0$ a.e. in $[a,b]$.*

Proof

By replacing f by $f\chi_{[a,b]}$ we may assume $f \in L^1(\mathbb{R})$. By the given condition $\int_{-\infty}^{x} f\,dm = 0$ for all x. So

$$\int_{(c,d)} f\,dm = \int_{-\infty}^{d} f\,dm - \int_{-\infty}^{c} f\,dm = 0$$

for all open intervals (c,d). So

$$\int_U f\,dm = 0$$

for all open sets U and

$$\int_K f\,dm = \int_{\mathbb{R}} f\,dm - \int_{K^c} f\,dm = 0$$

for all closed sets K.

Let

$$E^+ = \{x : f(x) > 0\}.$$

For any closed set $K \subset E^+$, since $f\chi_K \ge 0$ and

$$\int_{\mathbb{R}} f\chi_K\,dm = \int_K f\,dm = 0,$$

by Proposition 2.2.12, $f\chi_K = 0$ a.e., which means that $m(K) = 0$. Therefore by Theorem 1.5.4, $m(E^+) = \sup\{m(K) : K \subset E^+, \quad K \text{ is compact}\} = 0$. Similarly

$$m(\{x : f(x) < 0\}) = 0.$$

So $f = 0$ a.e.

□

Theorem 3.5.4
Let $-\infty < a < b < \infty$. For any $f \in L^1[a,b]$,

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x) \quad \text{a.e. on } [a,b].$$

Proof
$F(x) = \int_a^x f(t)\,dt$ is absolutely continuous, therefore in $BV[a,b]$ and F' exists a.e. on $[a,b]$.

Case 1: $f(x)$ is bounded on $[a,b]$.

There exists an $M > 0$ such that $|f(x)| \le M$. Let

$$f_n(x) = \frac{F\left(x+\frac{1}{n}\right) - F(x)}{\frac{1}{n}} = n\int_x^{x+\frac{1}{n}} f(t)\,dt.$$

Then

$$|f_n| \le n\int_x^{x+\frac{1}{n}} M\,dt = M.$$

Since $f_n \to F'$ a.e., for any $c \in (a,b]$, by the Dominated Convergence Theorem and the fact that F is continuous,

$$\begin{aligned}
\int_a^c F'(x)dx &= \lim_{n\to\infty}\int_a^c f_n(x)\,dx \\
&= \lim_{n\to\infty} n\int_a^c F\left(x+\frac{1}{n}\right) - F(x)\,dx \\
&= \lim_{n\to\infty}\left(n\int_{a+\frac{1}{n}}^{c+\frac{1}{n}} F(x)\,dx - n\int_a^c F(x)\,dx\right) \\
&= \lim_{n\to\infty}\left(n\int_c^{c+\frac{1}{n}} F(x)\,dx - n\int_a^{a+\frac{1}{n}} F(x)\,dx\right) \\
&= F(c) - F(a) \\
&= \int_a^c f(x)\,dx
\end{aligned}$$

i.e.

$$\int_a^c F'(x) - f(x)\,dx = 0.$$

By Lemma 3.5.3, $F' = f$ a.e. on $[a, b]$.

Case 2, $f \in L^1[a, b]$.

Without loss of generality (by considering f^+ and f^-), we can assume $f \geq 0$.

Let

$$f_n(x) = \min(n, f(x)) \quad \text{and} \quad G_n(x) = \int_a^x (f(x) - f_n(x))\, dx.$$

Then for any n, since $G_n(x)$ is increasing, so G_n' exists and $G_n' \geq 0$ a.e. Since f_n is bounded,

$$\frac{d}{dx} \int_a^x f_n(t)\, dt = f_n(x) \text{ and } F'(x) = G_n'(x) + f_n(x) \geq f_n(x) \text{ a.e.}$$

Let $n \to \infty$, we have

$$F'(x) \geq f(x) \text{ a.e. and } \int_a^b F'(x)\, dx \geq \int_a^b f(x)\, dx = F(b) - F(a).$$

Also since F is increasing, by Theorem 3.3.3 d)

$$\int_a^b F'(x)\, dx \leq F(b) - F(a).$$

Therefore

$$\int_a^b F'(x)\, dx = F(b) - F(a) = \int_a^b f(x)\, dx, \quad \text{or} \quad \int_a^b F'(x) - f(x)\, dx = 0,$$

and by Proposition 2.2.12, $F' = f$ a.e. on $[a, b]$.

□

Lemma 3.5.5 *If f is absolutely continuous on $[a, b]$ and $f'(x) = 0$ a.e., then f is a constant.*

Proof

We will show $f(a) = f(c)$ for any $c \in [a, b]$. For any $\epsilon > 0$, let δ be the one in the definition of absolute continuity, and

$$E = \{x \in (a, c) : f'(x) = 0\}.$$

Then $m(E) = c - a$, and for any $x \in E$, there is arbitrarily small $h > 0$ such that $[x, x + h] \subset [a, c]$ and

$$\left| \frac{f(x + h) - f(x)}{h} \right| < \epsilon.$$

All such intervals $[x, x+h]$ form a Vitali cover of E. By the Vitali Lemma, there are finite disjoint $\{[x_i, y_i], i = 1, \ldots, k\}$ such that

$$m\left(E \setminus \bigcup_{i=1}^{k} [x_i, y_i]\right) = m\left([a, c] \setminus \bigcup_{i=1}^{k} [x_i, y_i]\right) < \delta.$$

By relabeling, we can also assume $y_i < x_{i+1}$ for all $i = 1, \ldots, k-1$. Let $a = y_0$ and $c = x_{k+1}$. Then

$$\begin{aligned} |f(c) - f(a)| &\leq \sum_{i=0}^{k} |f(x_{i+1}) - f(y_i)| + \sum_{i=1}^{k} |f(y_i) - f(x_i)| \\ &< \epsilon + \epsilon \sum_{i=1}^{k} |y_i - x_i| \leq \epsilon + \epsilon(c - a). \end{aligned}$$

Let $\epsilon \to 0$, we then have $f(c) = f(a)$, and f is a constant on $[a, b]$. □

Theorem 3.5.6 (The Fundamental Theorem of Calculus for Lebesgue Integrals)
Let $-\infty < a < b < \infty$ and $f : [a, b] \to \mathbb{R}$. Then the following are equivalent:
a) f is absolutely continuous on $[a, b]$.
b) $f(x) = \int_a^x g(t)\, dt + f(a)$ for some $g \in L^1[a, b]$.
c) f is differentiable a.e. on $[a, b]$, $f' \in L^1[a, b]$, and

$$f(x) = \int_a^x f'(t)\, dt + f(a).$$

Proof

a)⇒c):

By Proposition 3.5.2, $f \in BV[a, b]$, so $f' \in L^1[a, b]$ by Theorem 3.3.3. Let

$$h(x) = f(x) - \int_a^x f'(t)\, dt.$$

Then h is absolutely continuous, and by Theorem 3.5.4, $h' = 0$ a.e. Therefore by Lemma 3.5.5, $h(x) = h(a) = f(a)$, or $f(x) = \int_a^x f'(t)\, dt + f(a)$.

c)⇒b):

Trivial.

b)⇒a):

Corollary 3.1.10.

□

Definition 3.5.7

A function f is called singular if $f' = 0$ m-a.e.

For example, the Cantor function is a singular function.

Proposition 3.5.8 *Any increasing function f can be written as $f = g + h$, where g is a singular increasing function, and h is an absolutely continuous increasing function.*

Proof

Let

$$g(x) = f(x) - \int_0^x f'(t)\,dt, \quad \text{and} \quad h(x) = \int_0^x f'(t)\,dt$$

with the conversion that for any integrable function $p(x)$

$$\int_0^x p(t)\,dt = \begin{cases} \int_{[0,x]} p\,dm, & x \geq 0, \\ -\int_{[x,0]} p\,dm, & x < 0, \end{cases}$$

Then $f(x) = g(x) + h(x)$. g is singular functions because of Theorem 3.3.3 d) and Theorem 3.5.4, and h is absolutely continuous because of Corollary 3.1.10.

h is increasing because $f' \geq 0$ a.e. For any $x < y$, $g(y) - g(x) = f(y) - f(x) - \int_x^y f'(t)\,dt \geq 0$ by Theorem 3.3.3 d). So g is increasing too. □

Theorem 3.5.9

Let F be an increasing and right continuous function on $\mathbb{R}$, μ_F be the Lebesgue-Stieltjes measure generated by F, and $\mu_F = \lambda + \rho$ with $\lambda \perp m$, $\rho \ll m$, be the Lebesgue decomposition of μ_F with respect to m. Then

$$F' = \frac{d\rho}{dm} \quad m\text{-a.e.}$$

Proof

Without loss of generality (by applying a translation), assume $F(0) = 0$. Define

$$G(x) = \begin{cases} \lambda((0,x]), & x \geq 0, \\ -\lambda((x,0]), & x < 0 \end{cases}$$

$$H(x) = \begin{cases} \rho((0,x]) = \int_0^x \frac{d\rho}{dm}\,dm, & x \geq 0, \\ -\rho((x,0]) = -\int_x^0 \frac{d\rho}{dm}\,dm, & x < 0. \end{cases}$$

Then $F(x) = G(x) + H(x)$ and by Theorem 3.5.4 $H' = \frac{d\rho}{dm}$ and $F' = G' + H' = G' + \frac{d\rho}{dm}$ m-a.e.

We only need to prove that $G' = 0$ m-a.e. Let (A, B) be a partition of $\mathbb{R}$ such that A is λ-null and B is m-null. Since G' exists m-a.e., without loss of generality (by combining an m-null set with B), assume G' exists everywhere in A.

Let

$$E = \{x \in A : G'(x) > 0\}, \quad E_k = \left\{x \in A \cap [-k, k] : G'(x) > \frac{1}{k}\right\}.$$

We claim that $m(E_k) = 0$ for all $k \in \mathbb{N}$. If not, $m(E_k) > 0$ for some k. Then by Theorem 1.5.4, there is an open set $U \supset A$ such that $\lambda(U) < \frac{1}{2k} m(E_k)$. For any $x \in E_k$, there are arbitrarily small positive h's such that $I = (x - h, x] \subset U$ and

$$\frac{G(x) - G(x-h)}{h} > \frac{1}{k}, \quad \text{or} \quad \lambda(I) = G(x) - G(x-h) > \frac{1}{k} m(I).$$

The collection of all such intervals forms a Vitali cover of E_k. By Vitali's Lemma, there are disjoint $\{I_1, \ldots, I_j\}$ such that

$$m\left(E_k \setminus \bigcup_{i=1}^{j} I_i\right) < \frac{1}{2} m(E_k).$$

Then

$$m(E_k) \le m\left(\bigcup_{i=1}^{j} I_i\right) + m\left(E_k \setminus \bigcup_{i=1}^{j} I_i\right) < \sum_{i=1}^{j} m(I_i) + \frac{1}{2} m(E_k),$$

or

$$2 \sum_{i=1}^{j} m(I_i) > m(E_k).$$

So

$$m(E_k) < 2 \sum_{i=1}^{j} m(I_i) < 2k \sum_{i=1}^{j} \lambda(I_i) \le 2k\lambda(U) < m(E_k),$$

a contradiction. So $m(E_k) = 0$ for all k and $m(E) = \lim_{k \to \infty} m(E_k) = 0$, i.e., $G' = 0$ m-a.e. □

Next result shows that the singular-absolutely-continuous decomposition of an increasing right continuous function coincides with the Lebesgue decomposition with respect to m of its Lebesgue-Stieltjes measure.

Proposition 3.5.10 *Let F be increasing and right continuous on $\mathbb{R}$, and $F = G + H$ where $H(x) = \int_a^x F'(t)\,dt$ for some a is absolutely continuous, and $G = F - H$ is increasing and singular. Then G is right continuous, and*

$$\mu_F = \mu_G + \mu_H$$

is the Lebesgue decomposition of μ_F with respect to the Lebesgue measure m on $\mathscr{B}_{\mathbb{R}}$, i.e.

$$\mu_G \perp m, \quad \mu_H \ll m.$$

Proof

G is right continuous because $G = F - H$ and H is continuous, and we certainly have $\mu_F = \mu_G + \mu_H$ on $\mathscr{B}_{\mathbb{R}}$ because of Theorem 1.4.13. Let $\mu_F = \lambda + \rho$ be the Lebesgue decomposition of μ_F with respect to the Lebesgue measure m. Then since $F' = \frac{d\rho}{dm}$ m-a.e., $\rho = \mu_H$. For any $E \in \mathscr{B}_{\mathbb{R}}$ and for any $n \in \mathbb{N}$

$$\begin{aligned}\lambda(E \cap [-n, n]) &= \mu_F(E \cap [-n, n]) - \rho(E \cap [-n, n]) \\ &= \mu_F(E \cap [-n, n]) - \mu_H(E \cap [-n, n]) \\ &= \mu_G(E \cap [-n, n])\end{aligned}$$

Therefore

$$\lambda(E) = \lim_{n\to\infty} \lambda(E \cap [-n, n]) = \lim_{n\to\infty} \mu_G(E \cap [-n, n]) = \mu_G(E)$$

and $\lambda = \mu_G$. □

Corollary 3.5.11 *If $F \in NBV$, then $\mu_F \perp m$ if and only if $F' = 0$ m-a.e., and $\mu_F \ll m$ if and only if $F(x) = \int_{-\infty}^x F'(t)\,dt$.*

Proof

Write $F = G + H$ where $H(x) = \int_{-\infty}^x F'(t)\,dt$ is absolutely continuous and $G = F - H$ is increasing and singular. Since $\mu_F = \mu_G + \mu_H$ is the Lebesgue decomposition of μ_F with respect to m, $\mu_F \perp m$ if and only if $H = 0$ and $\mu_F \ll m$ if and only if $G = 0$ m-a.e. □

Exercises

3.24 Show that a singular function is absolutely continuous if and only if it is a constant.

3.25 A function $f : \mathbb{R} \to \mathbb{R}$ is called a Lipschitz function on $\mathbb{R}$ if there is a $M \in \mathbb{R}$ such that $|f(x) - f(y)| \le M|x - y|$ for all x, y in $\mathbb{R}$. Prove that f is a Lipschitz on $\mathbb{R}$ if and only if f is absolutely continuous and f' is bounded a.e. on $\mathbb{R}$.

3.26

a. Let $\{F_j : j \in \mathbb{N}\}$ be a sequence of increasing right continuous functions on $\mathbb{R}$ such that $F(x) = \sum_{j=1}^{\infty} F_j(x) < \infty$ for all $x \in \mathbb{R}$, and

$$\lambda_{F_j} = \lambda_j + \rho_j, \quad \lambda_j \perp m, \quad \rho_j \ll m$$

be the Lebesgue decomposition of μ_{F_j} with respect to the Lebesgue measure m. Prove that

$$\lambda = \sum_{j=1}^{\infty} \lambda_j, \quad \rho = \sum_{j=1}^{\infty} \rho_j$$

are measures on $\mathscr{B}(\mathbb{R})$, and $\mu_F = \lambda + \rho$ is the Lebesgue decomposition of μ_F with respect to m.

b. Let $\{F_j : j \in \mathbb{N}\}$ be a sequence of increasing functions on $\mathbb{R}$ such that $F(x) = \sum_{j=1}^{\infty} F_j(x) < \infty$ for all $x \in \mathbb{R}$. Prove that

$$F'(x) = \sum_{j=1}^{\infty} F_j'(x) \quad \text{a.e.}$$

Topology: A Generalization of Open Sets

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4.1 Topology and Continuity

Definition 4.1.1

A collection $\mathscr{T}$ of subsets of X is called a topology on X if

i) $\emptyset \in \mathscr{T}$, $X \in \mathscr{T}$,

ii) for all $\{U_\alpha : \alpha \in A\} \subset \mathscr{T}$,

$$\bigcup_{\alpha \in A} U_\alpha \in \mathscr{T},$$

iii) for all $\{U_i : i = 1, \ldots, n\} \subset \mathscr{T}$,

$$\bigcap_{i=1}^{n} U_i \in \mathscr{T}.$$

The sets in a topology $\mathscr{T}$ are called open sets. The $(X, \mathscr{T})$ is called a topological space.

Example 4.1

a) $\mathscr{P}(X)$ is a topology, which is called the discrete topology.

b) $\{\emptyset, X\}$ is a topology, which is called the trivial topology.

c) $\{\emptyset, U : U^c$ is a finite set$\}$ is a topology, which is called the cofinite topology.

d) Let $\mathscr{P}$ be the set of all polynomials over $\mathbb{R}^n$,

$$\left\{\{x \in \mathbb{R}^n : \exists f \in S \text{ such that } f(x) \neq 0\} : S \subset \mathscr{P}\right\}$$

is a topology, which is called the Zariski topology.

e) Let $x_0 \in X$. $\{E \subset X : x_0 \notin E\} \cup \{X\}$ is a topology, and $\{E \subset X : x_0 \in E\} \cup \{\emptyset\}$ is also a topology.

■

4

Definition 4.1.2

Let $(X, \mathscr{T})$ be a topological space. For any subset $Y \subset X$,

$$\mathscr{T}_Y = \{U \cap Y : U \in \mathscr{T}\}$$

is called the relative topology induced by Y.

Definition 4.1.3

Let $A \subset X$ and $\mathscr{T}$ be topological on X

a) For any $U \in \mathscr{T}$, U^c is called a closed set.
b) A^o is the largest open set contained in A and is called the interior of A.
c) $\overline{A}$ is the smallest closed set containing in A and is called the closure of A.
d) $\partial A = \overline{A} \setminus A^o$ is called the boundary of A.
e) A is called dense in X if $\overline{A} = X$.
f) A is called nowhere dense if $(\overline{A})^o = \emptyset$.
g) A is called a neighborhood of a point $x \in X$ if $x \in A^o$.
h) $x \in X$ is called an accumulation point of a set A if $A \cap (U \setminus \{x\}) \neq \emptyset$ for any neighborhood U of x. The set of all accumulation points is denoted by $\text{acc}(A)$.
i) A topology $\mathscr{T}_1$ is called weaker than another topology $\mathscr{T}_2$ ($\mathscr{T}_2$ is stronger than $\mathscr{T}_1$) if $\mathscr{T}_1 \subset \mathscr{T}_2$.
j) $\mathscr{N} \subset \mathscr{T}$ is called a neighborhood base at x if
   1) $x \in V$ for all $V \in \mathscr{N}$
   2) if $x \in U$ for a $U \in \mathscr{T}$, then there is $V \in \mathscr{N}$ such that $V \subset U$.

k) $\mathscr{B} \subset \mathscr{T}$ is called a base of $\mathscr{T}$ if it contains neighborhood bases of all $x \in X$.
l) $(X, \mathscr{T})$ is called first countable if every point $x \in X$ has a countable neighborhood base. $(X, \mathscr{T})$ is called second countable if $\mathscr{T}$ has a countable base.
m) $(X, \mathscr{T})$ is called separable if X has a countable dense subset.
n) A sequence $\{x_j\}$ is called to converge to x if for all neighborhood U of x, there exists a $J \in \mathbb{N}$ such that $x_j \in U$ for all $j \geq J$.

Definition 4.1.4 (Separation Axioms)

a) A topological space $(X, \mathscr{T})$ is T_0 if for any $x \neq y$ in X, there is an open set V contains one but does not contain the other.
b) A topological space $(X, \mathscr{T})$ is T_1 if for any $x \neq y$ in X, there are open set U, $x \in U$ and $y \notin U$. (and therefore $\exists$ open set V, $y \in V$, and $x \notin V$.)
c) A topological space $(X, \mathscr{T})$ is T_2 (Hausdorff space) if for any $x \neq y$ in X, there are open sets V, U, $V \cap U = \emptyset$, $x \in V$, and $y \in U$.
d) A topological space $(X, \mathscr{T})$ is T_3 (Regular space) if it is T_1, and for all closed set A and for all $x \notin A$, there are open sets V, U, $V \cap U = \emptyset$, $x \in V$, and $A \subset U$.
e) A topological space $(X, \mathscr{T})$ is T_4 (Normal space) if it is T_1, and for all disjoint closed sets A, B, there are open sets V, U, $V \cap U = \emptyset$, $A \subset V$, and $B \subset U$.

In Example 4.1, the trivial topology is not T_0. The topologies in e) are T_0, but not T_1. The Zariski topology is T_1, but not Hausdorff.

Definition 4.1.5

Let X, Y be topological spaces. $f : X \to Y$ is called continuous if $f^{-1}(V)$ is open in X for any open set V in Y.
f is called continuous at an $x \in X$ if for all neighborhood V of $f(x)$, $f^{-1}(V)$ is a neighborhood of x.

Remark 4.1 Since $f^{-1}(V^c) = (f^{-1}(V))^c$, $f : X \to Y$ is continuous if and only if $f^{-1}(F)$ is closed in X for any closed set F in Y.

Definition 4.1.6

Let $E \subset \mathbb{R}$
a) $B(X, E)$ is a set of all bounded functions $f : X \to E$, and $B(X) = B(X, \mathbb{R})$.
b) $C(X, E)$ is a set of all continuous functions $f : X \to E$, and $C(X) = C(X, \mathbb{R})$
c) $BC(X, E) = B(X, E) \cap C(X, E)$.

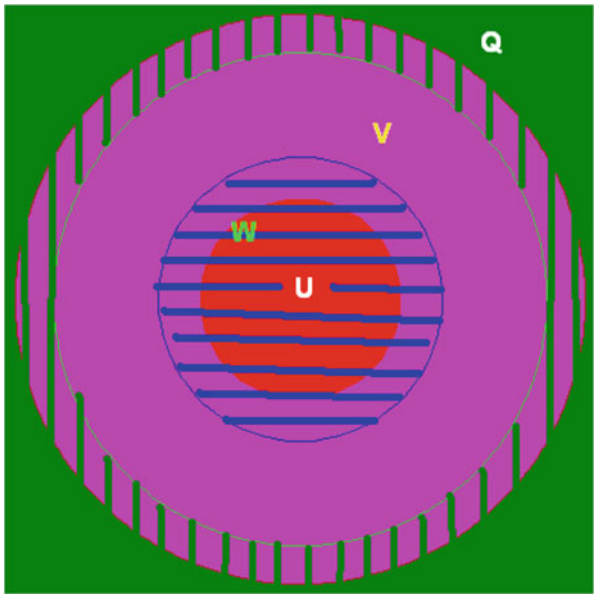
Definition 4.1.7

The uniform norm $\| \cdot \|_u$ on $B(X)$ is defined by

$$\|f\|_u = \sup\{|f(x)| : x \in X\}.$$

Clearly $B(X)$ is complete (i.e., every Cauchy sequence converges) under the uniform norm.

Fig. 4.1
$\overline{U} \subset W \subset \overline{W} \subset Q^c \subset V$



Lemma 4.1.8 *Let X be a normal topological space. Then for any open sets U, V with $\overline{U} \subset V$, there exists an open set W such that*

$$\overline{U} \subset W \subset \overline{W} \subset V.$$

Proof
$\overline{U}$ and V^c are disjoint closed sets. Since X is normal, there are disjoint open sets W, Q such that $W \supset \overline{U}$ and $Q \supset V^c$. Therefore (Figure 4.1)

$$\overline{U} \subset W \subset \overline{W} \subset Q^c \subset V.$$

□

Lemma 4.1.9 *Suppose that A and B are disjoint closed subsets of a normal space X. Let*

$$\Delta = \left\{ \frac{k}{2^n} : n \geq 1,\ 0 < k < 2^n \right\}.$$

Then there is a family of open sets

$$\{U_r : r \in \Delta\}$$

such that

$$A \subset U_r \subset B^c \quad \text{and} \quad \overline{U}_r \subset U_s \ \text{for any } r < s.$$

Proof

Let $U_0 = A^o$ and $U_1 = B^c$. Generally for any open set $U_{\frac{k}{2^n}}$, $U_{\frac{k+1}{2^n}}$, $0 \le n$ and $0 \le k < 2^n$, such that

$$U_{\frac{k}{2^n}} \subset \overline{U}_{\frac{k}{2^n}} \subset U_{\frac{k+1}{2^n}},$$

by Lemma 4.1.8, there is an open set $U_{\frac{2k+1}{2^{n+1}}}$ such that

$$U_{\frac{k}{2^n}} \subset \overline{U}_{\frac{k}{2^n}} = \overline{U}_{\frac{2k}{2^{n+1}}} \subset U_{\frac{2k+1}{2^{n+1}}} \subset \overline{U}_{\frac{2k+1}{2^{n+1}}} \subset U_{\frac{2k+2}{2^{n+1}}} = U_{\frac{k+1}{2^n}}.$$

□

Theorem 4.1.10 (Urysohn's Lemma)

Let X be a normal space. If A and B are disjoint closed sets in X, then there is a $f \in C(X, [0, 1])$ such that $f(x) = 0$ for all $x \in A$, and $f(x) = 1$ for all $x \in B$.

Proof

Let $\{U_r : r \in \Delta\}$ be as in Lemma 4.1.9, and let $U_1 = X$ (not B^c as in the proof of Lemma 4.1.9). Define

$$f(x) = \inf\{r : x \in U_r\}.$$

Clearly $0 \le f(x) \le 1$. For all $x \in A$, $f(x) = \inf\left\{\frac{k}{2^n}, n = 1, 2, \ldots, k = 1, 2, \ldots, 2^n\right\} = 0$, and for all $x \in B$, $x \in U_1$ and $x \notin U_r$ for all $r < 1$. So $f(x) = 1$.

Now we show that f is continuous. Since an open set in $\mathbb{R}$ is a union of disjoint intervals, and since each interval $(\alpha, \beta) = (-\infty, \beta) \cap (\alpha, \infty)$, we only need to show $f^{-1}((\alpha, \infty))$ and $f^{-1}((-\infty, \beta))$ are open for any α, β in $\mathbb{R}$. Clearly for $\alpha < 0$ or $\beta > 1$, the sets are all open set X, so we only need to show them for $\alpha \ge 0$ and $\beta \le 1$.

Let $\beta \le 1$. For any $x \in f^{-1}((-\infty, \beta))$, since $f(x) = \inf\{r : x \in U_r\} < \beta$, there exists an $r_0 < \beta$ such that $x \in U_{r_0} \subset \bigcup_{r<\beta} U_r$. Conversely, for any $x \in \bigcup_{r<\beta} U_r$, $x \in U_{r_0}$ for some $r_0 < \beta$. So $f(x) = \inf\{r : x \in U_r\} \le r_0 < \beta$ and $x \in f^{-1}((-\infty, \beta))$. Therefore

$$f^{-1}((-\infty, \beta)) = \bigcup_{r<\beta} U_r$$

is open.

Let $\alpha \ge 0$. For any $x \in f^{-1}((\alpha, \infty))$, since $f(x) = \inf\{r : x \in U_r\} > \alpha$, there is an r_0, $\alpha < r_0 < f(x)$ such that $x \notin U_{r_0}$, or $x \in U_{r_0}^c$. Pick an $s \in \Delta$, $\alpha < s < r_0$. Then $\overline{U}_s \subset U_{r_0}$. So $x \in U_{r_0}^c \subset \overline{U}_s^c \subset \bigcup_{\alpha<r} \overline{U}_r^c$. Conversely, for any $x \in \bigcup_{\alpha<r} \overline{U}_r^c$, there is an $s > \alpha$ such that $x \notin U_s \supset U_r$ for all $r \le s$. So $f(x) = \inf\{r : x \in U_r\} \ge s > \alpha$, i.e., $x \in f^{-1}((\alpha, \infty))$. Therefore

$$f^{-1}((\alpha, \infty)) = \bigcup_{\alpha < r} \overline{U}_r^c$$

is also open, and f is continuous.

□

4

Theorem 4.1.11 (Tietze Extension Theorem)
Let X be a normal space. If A is a closed subset of X and $f \in C(A, [a, b])$, then there is an $F \in C(X, [a, b])$ such that $F\Big|_A = f$.

Proof
Let $g(x) = \frac{f(x)-a}{b-a}$. Then $g \in C(A, [0, 1])$.

Let

$$B_1 = g^{-1}\left(\left[0, \frac{1}{3}\right]\right) \cap A, \qquad C_1 = g^{-1}\left(\left[\frac{2}{3}, 1\right]\right) \cap A.$$

Then by the UrysohnŠs lemma, there is a $g_1 \in C\left(X, \left[0, \frac{1}{3}\right]\right)$ such that $g_1 = 0$ on B_1 and $g_1 = \frac{1}{3}$ on C_1. We have $0 \le g - g_1 \le \frac{1}{3}$ on B_1, $\frac{1}{3} \le g - g_1 \le \frac{2}{3}$ on C_1, and $0 \le g - g_1 \le \frac{2}{3}$ on $A \setminus (B_1 \cup C_1)$, so

$$0 \le g - g_1 \le \frac{2}{3}$$

on A.

Let

$$B_2 = (g - g_1)^{-1}\left(\left[0, \frac{2}{3^2}\right]\right) \cap A, \qquad C_1 = (g - g_1)^{-1}\left(\left[\frac{2^2}{3^2}, \frac{2}{3}\right]\right) \cap A.$$

Then there is a $g_2 \in C\left(X, \left[0, \frac{2}{3^2}\right]\right)$ such that $g_2 = 0$ on B_2 and $g_2 = \frac{2}{3^2}$ on C_2. Similarly, we have

$$0 \le g - g_1 - g_2 \le \frac{2^2}{3^2}$$

on A.

Continuing this process, we then have $g_1, g_2, \ldots$ such that

$$0 \le g - \sum_{n=1}^{k} g_n \le \left(\frac{2}{3}\right)^k \quad \text{on } A \text{ and } |g_n| \le \frac{2^{n-1}}{3^n}.$$

Let

$$G(x) = \sum_{n=1}^{\infty} g_n.$$

Then G converges uniformly, so $G \in C(X, [0, 1])$, and $g = G$ on A. Finally let

$$F(x) = G(x)(b - a) + a.$$

Then on A, $F(x) = g(x)(b - a) + a = f(x)$ and $F \in C(X, [a, b])$. □

Corollary 4.1.12 *If X is normal, $A \subset X$ is closed, and $f \in C(A)$, then there exists an $F \in C(X)$ such that $F\Big|_A = f$.*

Proof

Let $g = \frac{f}{1+|f|}$. Then $g \in C(A, [-1, 1])$ and $g \neq \pm 1$ on A. So there is $G \in C(X, [-1, 1])$, $G\Big|_A = g$. Note that g and f have the same signs, which means that

$$g|f| = |g|f.$$

From the definition of g we derive $f = g + g|f| = g + f|g|$. So f can be solved

$$f = \frac{g}{1 - |g|}.$$

Since G is continuous, $G^{-1}(\{\pm 1\})$ is closed and $G^{-1}(\{\pm 1\}) \cap A = \emptyset$. By Urysohn's Lemma, there is an $h \in C(X, [0, 1])$, $h = 1$ on A and $h = 0$ on $G^{-1}(\{\pm 1\})$. Then $hG \in C(X, (-1, 1))$ and $hG\Big|_A = g$. Let

$$F(x) = \frac{h(x)G(x)}{1 - |h(x)G(x)|}.$$

Then $F \in C(X)$ and $F\Big|_A = f$. □

Exercises

4.1 Show that $\mathscr{T} = \{(a, \infty) : a \in \overline{\mathbb{R}}\}$, with the convention that $(\infty, \infty) = \emptyset$ is a topology on $\mathbb{R}$ which is T_0 but not T_1

4.2 Let $\mathscr{T}$ be the collection of subsets of $\mathbb{R}^2$ containing $\emptyset$, $\mathbb{R}$, and all the complements of finite number of lines and points. Prove that $\mathscr{T}$ is a topology on $\mathbb{R}^2$ which is T_2 but not Hausdorff.

4.3 Let X be a metric space with the metric ρ and $F \subset X$ be closed. For any $x \in X$, define

$$\rho(x, F) = inf\{\rho(x, y) : y \in F\}.$$

a. Prove that $\rho(x, F) = 0$ if and only if $x \in F$.
b. Prove that $U = \{x \in X : \rho(x, F) < c\}$ is an open set containing F for any $c > 0$.
c. Prove that X is normal.

4.4 Prove that in a Hausdorff space, every singleton set $\{x\}$ is closed.

4.5 Let

$$\mathscr{T} = \{U \cap (V \cap \mathbb{Q}) : U, V \subset \mathbb{R} \text{ are open sets in Euclidean topology}\}.$$

Prove that $\mathscr{T}$ is a topology on $\mathbb{R}$ which is Hausdorff but not regular.

4.6
a. Prove that $\overline{\bigcup_{i=1}^{n} E_i} = \bigcup_{i=1}^{n} \overline{E}_i$ in any topological space.
b. Show that $\overline{\bigcap_{\alpha \in A} E_\alpha} \neq \bigcap_{\alpha \in A} \overline{E}_\alpha$ generally.

4.7 Let A be a dense subset of a topological space $(X, \mathscr{T})$. Prove that for any open set U,

$$A \subset U^c \cup (\overline{U \cap A}), \quad U \subset \overline{U \cap A}, \quad \text{and} \quad \overline{U} = \overline{U \cup A}.$$

4.8 Let X be a topological space, $A \subset X$ be closed, and $g \in C(A)$ with $g = 0$ on ∂A. Prove that

$$G(x) = \begin{cases} g(x), & x \in A \\ 0, & x \in A^c, \end{cases}$$

is continuous on X.

4.9 Let X be a topological space, Y be a Hausdorff space, and $f, g \in C(X, Y)$. Prove that
a. $\{x : f(x) = g(x)\}$ is closed, and
b. if $f = g$ on a dense subset of X, then $f = g$ on X.

4.10 Let $(\mathbb{R}, \mathscr{T}_c)$ and $(\mathbb{R}, \mathscr{T}_e)$ be the cofinite and Euclidean topological spaces on $\mathbb{R}$, respectively. Prove that every continuous function $f : (\mathbb{R}, \mathscr{T}_c) \to (\mathbb{R}, \mathscr{T}_e)$ is a constant function.

4.11 Let X be a Hausdorff space. Prove that the following statements are equivalent.
i) X is normal.
ii) For any disjoint closed sets A, B, there exists a $f \in C(X, [0, 1])$ such that $f = 0$ on A and $f = 1$ on B.
iii) Any $f \in C(A, [a, b])$ with A closed can be extended to a function $F \in C(X, [a, b])$ such that $F\Big|_A = f$.

4.2 Nets: A Generalization of Sequences

Definition 4.2.1

S is a directed set if there is a binary relation $\preceq$ such that

i) for all $\alpha \in S$, $\alpha \preceq \alpha$;
ii) if $\alpha \preceq \beta$ and $\beta \preceq \gamma$, then $\alpha \preceq \gamma$;
iii) for any $\alpha, \beta \in S$, there is a $\gamma \in S$ such that $\alpha \preceq \gamma$ and $\beta \preceq \gamma$.

Example 4.1

a) $\mathbb{N}$ with the ordinary order.
b) $\mathbb{R} \setminus \{a\}$ with $x \preceq y$ if $|x - a| \geq |y - a|$.
c) The set of all partitions on $[a, b]$ with $P \preceq Q$ if $\|P\| \geq \|Q\|$
d) The set of all neighborhoods of a point with $U \preceq V$ if $U \supset V$.
e) The collection of all finite subsets of an infinite set with $E \preceq F$ if $E \subset F$.
f) $A \times B$ for any directed sets A and B, with $(\alpha, \beta) \preceq (\alpha', \beta')$ if $\alpha \preceq \alpha'$ and $\beta \preceq \beta'$.

■

Definition 4.2.2

A net in X is a mapping $\alpha \to x_\alpha$, $\alpha \in A$, $x_\alpha \in X$, from a directed set A into X (Figure 4.2).

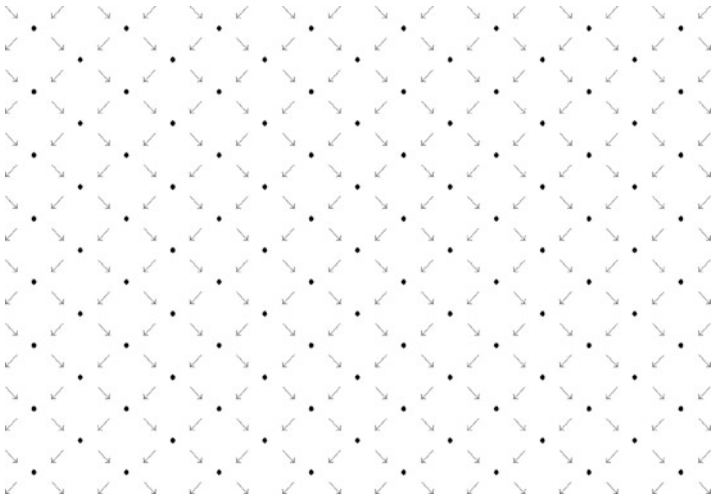


Fig. 4.2 A net

Definition 4.2.3

a) A net $\langle x_\alpha \rangle_{\alpha \in A}$ is called eventually in E if there is an $\alpha_0 \in A$ such that $x_\alpha \in E$ for all $\alpha \succeq \alpha_0$.
b) A net $\langle x_\alpha \rangle_{\alpha \in A}$ is called frequently in E if for all $\alpha \in A$, there is a $\beta \succeq \alpha$ such that $x_\beta \in E$.
c) A point $x \in X$ is called a limit point of a net $\langle x_\alpha \rangle_{\alpha \in A}$ if for all neighborhood U of x, $\langle x_\alpha \rangle$ is eventually in U. $\langle x_\alpha \rangle$ is called to converge to x and is denoted by $x_\alpha \to x$.
d) A point $x \in X$ is called a cluster point of a net $\langle x_\alpha \rangle_{\alpha \in A}$ if for all neighborhood U of x, $\langle x_\alpha \rangle$ is frequently in U.

We first prove a result about the closure of a set, which will be needed to prove some results about nets next.

Proposition 4.2.4

i) $\overline{A} = A \cup acc(A)$.
ii) *A is closed if and only if* $acc(A) \subset A$.

Proof

i) We will show $\overline{A}^c = (A \cup \text{acc}(A))^c$.

If $x \in \overline{A}^c = (A^c)^o$ (why are they equal?), then A^c is a neighborhood of x which does not contain any point of A. So $x \notin A \cup \text{acc}(A)$, i.e., $x \in (A \cup \text{acc}(A))^c$.

Conversely, if $x \in (A \cup \text{acc}(A))^c$, then there is an open neighborhood U of x such that $U \cap A = \emptyset$, which means that $A \subset U^c$. So $\overline{A} \subset U^c$ and $x \in U \subset \overline{A}^c$.

ii) By i), $\text{acc}(A) \subset \overline{A}$. So if A is closed $\text{acc}(A) \subset \overline{A} = A$, and if $\text{acc}(A) \subset A$, then $A = A \cup \text{acc}(A) = \overline{A}$.

□

Following results are similar to that of sequences on a metric space.

Proposition 4.2.5 *Let X be a topological space, $E \subset X$ and $x \in X$*

i) *x is an accumulation point of E if and only if there is a net $\langle x_\alpha \rangle \subset E \setminus \{x\}$ that converges to x.*

ii) *$x \in \overline{E}$ if and only if there is a net $\langle x_\alpha \rangle$ in E that converges to x.*

Proof

i) "⇒"

Let A be the directed set of all neighborhoods of x with $U_1 \preceq U_2$ if $U_2 \subset U_1$. For any $\alpha \in A$, let x_α be any point in $(\alpha \setminus \{x\}) \cap E$. Then $x_\alpha \to x$.

"⇐"

Assume $x_\alpha \to x$ and $\langle x_\alpha \rangle \subset E \setminus \{x\}$. Then for any neighborhood U of x, there is a $x_\alpha \in U$ by the definition of convergence, and $x_\alpha \neq x$. So x is an accumulation point of E.

ii) If $x \in \overline{E} = \text{acc}(E) \cup E$, then either $x \in \text{acc}(E)$ or $x \in E$. If $x \in \text{acc}(E)$, then by i), there is a net $\langle x_\alpha \rangle \subset E \setminus \{x\} \subset E$ such that $x_\alpha \to x$. On the other hand, if $x \in E$, then let $x_n = x$ for all $n \in \mathbb{N}$, and $x_n \to x$.

Conversely, let $x_\alpha \to x$ be a net $\langle x_\alpha \rangle \subset E$. If $x_\alpha = x$ for some α, then $x = x_\alpha \in E \subset \overline{E}$. Otherwise $\langle x_\alpha \rangle \subset E \setminus \{x\}$, and by i), $x \in \text{acc}(E) \subset \overline{E}$.

□

Proposition 4.2.6 *Let X and Y be topological spaces and $f : X \to Y$. Then f is continuous at $x \in X$ if and only if for every net $\langle x_\alpha \rangle$ converges to x, $f(x_\alpha) \to f(x)$.*

Proof

Assume f is continuous at x. For any neighborhood V of $f(x)$, $f^{-1}(V)$ is a neighborhood of x. So for any $x_\alpha \to x$, $\langle x_\alpha \rangle$ is eventually in $f^{-1}(V)$, which means that $\langle f(x_\alpha) \rangle$ is eventually in V, i.e., $f(x_\alpha) \to f(x)$.

Conversely, if f is not continuous at x, then there is a neighborhood V of $f(x)$ such that $f^{-1}(V)$ is not a neighborhood of x, which means that $x \notin (f^{-1}(V))^o$, or $x \in \left((f^{-1}(V))^o\right)^c = \overline{(f^{-1}(V))^c} = \overline{f^{-1}(V^c)}$. By Proposition 4.2.5, there is a $\langle x_\alpha \rangle \subset f^{-1}(V^c)$, $x_\alpha \to x$. Then $f(x_\alpha) \not\to f(x)$ because $\langle f(x_\alpha) \rangle \subset V^c$ and $f(x) \in V^o$. □

Recall the definition of a subsequence: $\{x_{n_k}\} \subset \{x_n\}$ is a subsequence if $n_1 < n_2 < \cdots$, which implies that for any n_0 there is a k_0 such that $n_0 \le n_{k_0} < n_{k_0+1} < n_{k_n+2} \cdots$. The concept of subnet is defined similarly, i.e., a subnet is a subset of the net as a set, with some regularity property of the indices. Here is the formal definition.

Definition 4.2.7

Let $\langle x_\alpha \rangle_{\alpha \in A}$ be a net. A net $\langle y_\alpha \rangle_{\alpha \in B}$ together with a map $\beta \mapsto \alpha_\beta$ from B to A is called a subnet of $\langle x_\alpha \rangle$ if for any $\alpha_0 \in A$, there is a $\beta_0 \in B$ such that

i) for any $\beta \succeq \beta_0$, we have $\alpha_\beta \succeq \alpha_0$, and

ii) $y_\beta = x_{\alpha_\beta}$.

We have a result similar to that of sequences (note that a cluster point of a sequence is a subsequential limit point of the sequence).

Proposition 4.2.8 *Let $\langle x_\alpha \rangle$ be a net in a topological space X. Then $x \in X$ is a cluster point of $\langle x_\alpha \rangle$ if and only if $\langle x_\alpha \rangle$ has a subnet that converges to x.*

Proof

Assume x is a cluster point of $\langle x_\alpha \rangle_{\alpha \in A}$. Let $\mathcal{N}$ be the directed set of all neighborhoods of x with $U_1 \preceq U_2$ if $U_2 \subset U_1$, and let $B = \mathcal{N} \times A$. For each $\beta = (U, \alpha) \in B$, define α_β be the one such that

$$\alpha_\beta \succeq \alpha \quad \text{and} \quad x_{\alpha_\beta} \in U.$$

Then $\langle x_{\alpha_\beta} \rangle_{\beta \in B}$ is a subnet that converges to x.

Conversely, if $\langle x_{\alpha_\beta} \rangle$ is a subnet that converges to x, then for any neighborhood U of x, there is a β_1 such that $x_{\alpha_\beta} \in U$ for all $\beta \succeq \beta_1$. Since $\langle x_{\alpha_\beta} \rangle \subset \langle x_\alpha \rangle$, for all α, there is a β_2 such that $\alpha_\beta \succeq \alpha$ for all $\beta \succeq \beta_2$. Then for all $\beta \succeq \beta_1$ and $\beta \succeq \beta_2$, $\alpha_\beta \succeq \alpha$ and $x_{\alpha_\beta} \in U$. So $\langle x_\alpha \rangle$ is frequently in U and x is a cluster point of $\langle x_\alpha \rangle$. □

4

Exercises

4.12 Prove that a topological space X is Hausdorff if and only if the limit of any convergent net is unique.

4.13 Let $\langle x_\alpha \rangle_{\alpha \in A}$ and

$$E_\alpha = \{x_\beta : \beta \succeq \alpha\}.$$

Prove that x is a cluster point of $\langle x_\alpha \rangle$ if and only if $x \in \bigcap_{\alpha \in A} \overline{E}_\alpha$.

4.3 Compact Spaces

Definition 4.3.1

A collection $\{U_\alpha : \alpha \in A\}$ of open sets is called an open cover of a set E if

$$E \subset \bigcup_{\alpha \in A} U_\alpha.$$

Definition 4.3.2

A set E is called compact if every open cover of E has a finite subcover.
E is called precompact if $\overline{E}$ is compact.
A topological space is called compact if X is compact.

Note that $\{U_\alpha : \alpha \in A\}$ covers X if and only if $\bigcap_{\alpha \in A} U_\alpha^c = \emptyset$, and $\{U_\alpha : \alpha \in A\}$ has a finite subcover if and only if $\bigcap_{i=1}^n U_{\alpha_i}^c = \emptyset$ for some $\{\alpha_1, \ldots, \alpha_n\} \subset A$. Therefore the contrapositive of the definition gives us an equivalent condition for compactness.

We first define a terminology.

Definition 4.3.3

A collection of closed sets $\{F_\alpha : \alpha \in A\}$ is said to have the finite intersection property if $\bigcap_{i=1}^n F_{\alpha_i} \neq \emptyset$ for all finite subset $\{\alpha_1, \ldots, \alpha_n\} \subset A$.

Proposition 4.3.4 *A topological space X is compact if and only if every family of closed sets $\{F_\alpha : \alpha \in A\}$ with the finite intersection property has nonempty intersection, i.e.*

$$\bigcap_{\alpha \in A} F_\alpha \neq \emptyset.$$

Proposition 4.3.5 *A closed subset of a compact space is compact.*

Proof
Let K be a closed subset of a compact space X. Then for any open cover $\{U_\alpha : \alpha \in A\}$ of K,

$$\{U_\alpha : \alpha \in A\} \cup \{K^c\}$$

is an open cover of X, so there is a subcover $\{U_{\alpha_1}, \ldots, U_{\alpha_n}, K^c\}$ of X. Then $\{U_{\alpha_1}, \ldots, U_{\alpha_n}\}$ is a finite subcover of K. □

Proposition 4.3.6 *Let F be a compact subset of a Hausdorff space X and $x \notin F$. Then there are disjoint open sets U, V such that*

$$F \subset U \quad \textit{and} \quad x \in V.$$

Proof
For any $y \in F$, there are disjoint open sets U_y, V_y such that $y \in U_y$ and $x \in V_y$. $\{U_y : y \in F\}$ is an open cover of F. So there is a finite subcover $\{U_{y_1}, \ldots, U_{y_n}\}$. Let $U = \bigcup_{i=1}^n U_{y_i}$ and $V = \bigcap_{i=1}^n V_{y_i}$. Then U and V are disjoint, and $U \supset F$ and $x \in V$.

□

Proposition 4.3.7 *Every compact subset of a Hausdorff space is closed.*

Proof
Let F be a compact set in a Hausdorff space X. If $F = X$, then it is closed. Otherwise, for all $x \notin F$, by Proposition 4.3.6, there are disjoint open sets U_x, V_x such that $U_x \supset F$ and $x \in V_x$. Then

$$F^c = \bigcup_{x \in F^c} V_x$$

is open. So F is closed. □

Proposition 4.3.8 *Every compact Hausdorff space is normal.*

Proof
Let E and F be disjoint closed sets. Then they are compact. For all $x \in F$, by Proposition 4.3.6, there are disjoint open sets U_x, V_x such that $U_x \supset E$ and $x \in V_x$. $\{V_x : x \in F\}$ is an open cover of F. So there is a finite subcover $\{V_{x_1}, \ldots, V_{x_m}\}$. Let

$$U = \bigcap_{i=1}^m U_{x_i}, \quad \text{and} \quad V = \bigcup_{i=1}^m V_{x_i}.$$

Then $U \cap V = \emptyset$, $E \subset U$, and $F \subset V$. So X is normal. □

Proposition 4.3.9 *A continuous function maps a compact set to a compact set.*

Proof
Let $f : X \to Y$ and $K \subset X$ be compact. Then for any open cover $\{U_\alpha : \alpha \in A\}$ of $f(K)$, $\{f^{-1}(U_\alpha) : \alpha \in A\}$ is an open cover of K. So there is a finite subcover $K \subset \bigcup_{i=1}^{n} f^{-1}(U_{\alpha_i})$. Then $f(K) \subset \bigcup_{i=1}^{n} U_{\alpha_i}$, and therefore $f(K)$ is compact. □

Corollary 4.3.10 *Every continuous function $f : X \to \mathbb{R}$ is bounded on a compact set.*

Proof
Let $f : X \to \mathbb{R}$ be continuous and $K \subset X$ be compact. Then $f(K) \subset \mathbb{R}$ is compact, and therefore is bounded. □

Definition 4.3.11
Let X, Y be topological spaces. $f : X \to Y$ is called a homeomorphism if it is bijection and continuous, and its inverse is also continuous.

Proposition 4.3.12 *Let X be compact and Y be Hausdorff. Then any continuous bijection $f : X \to Y$ is a homeomorphism.*

Proof
By Proposition 4.3.9, for any closed (compact) set K of X, $f(K) = (f^{-1})^{-1}(K)$ is closed (compact). So f^{-1} is continuous. □

Theorem 4.3.13
Let X be a topological space. Then the following are equivalent.
a) X is compact.
b) Every net in X has a cluster point.
c) Every net in X has a convergent subnet.

Proof
"b) $\Leftrightarrow$ c)"
Proposition 4.2.8.
"a) $\Rightarrow$ b)"
Let $\langle x_\alpha \rangle_{\alpha \in A}$ be a net. Define $E_\alpha = \{x_\beta : \beta \succeq \alpha\}$. For any $\alpha, \beta \in A$, there is a $\gamma \in A$ such that $\gamma \succeq \alpha$ and $\gamma \succeq \beta$. So $E_\gamma \subset E_\alpha \cap E_\beta$, and $\{E_\alpha : \alpha \in A\}$ has the finite intersection property.

By Proposition 4.3.4,

$$\bigcap_{\alpha\in A} \overline{E}_\alpha \neq \emptyset.$$

For any $x \in \bigcap_{\alpha\in A} \overline{E}_\alpha$, we claim that it is a cluster point of $\langle x_\alpha\rangle$.

For any neighborhood U of x and for any $\alpha \in A$, since $x \in \overline{E}_\alpha$, we have either $x \in E_\alpha$ or $x \in \mathrm{acc}(E_\alpha)$.

If $x \in E_\alpha$, then $x = x_\beta$ for some $\beta \succeq \alpha$ by the definition of E_α. So $x_\beta = x \in U$. Otherwise, $x \in \mathrm{acc}(E_\alpha)$ and $U \cap (E_\alpha \setminus \{x\}) \neq \emptyset$, and again there is a $\beta \succeq \alpha$ such that $x_\beta \in U \cap (E_\alpha \setminus \{x\}) \subset U$. Therefore $\langle x_\alpha\rangle$ is frequently in U and x is a cluster point of $\langle x_\alpha\rangle$.

"b) $\Rightarrow$ a)"

If X is not compact, then there is an open cover $\{U_a : a \in A\}$ which does not have any finite subcover. Let S be the directed set of all finite subsets of $\{U_a : a \in A\}$ with $\alpha \preceq \beta$ if $\alpha \subset \beta$.

For each

$$\alpha = \{U_{a_1}, \ldots, U_{a_n}\}$$

pick $x_\alpha \in \left(\bigcup_{i=1}^n U_{a_i}\right)^c$. Then $\langle x_\alpha\rangle_{\alpha\in S}$ is a net.

We show that $\langle x_\alpha\rangle_{\alpha\in S}$ does not have any cluster point. For any $x \in X$, x must belong to a U_{a_0} for some $a_0 \in A$. Let $\beta = \{U_{a_0}\}$. Then for any $\alpha = \{U_{a_0}, U_{a_1}, \ldots, U_{a_n}\}$,

$$x_\alpha \notin \bigcup_{i=0}^{n} U_{a_i} \supset U_{a_0},$$

i.e., $\langle x_\alpha\rangle$ is not frequently in U_{a_0}. Therefore x is not a cluster point of $\langle x_\alpha\rangle$. □

Definition 4.3.14

A topological space X is called countably compact if every countable open cover of X has a finite subcover.

A topological space X is called sequentially compact if every sequence in X has a convergent subsequence.

Exercises

4.14 Prove that every sequentially compact space is countably compact.

4.15 Let X be a topological space and $E \subset X$. Prove that a set $K \subset E$ is relatively compact in E if and only if it is compact in X.

4.16 Prove that if X is countably compact, then every sequence in X has a cluster point. If X is also first countable, then X is sequentially compact.

4.4 Locally Compact Hausdorff Spaces

Definition 4.4.1
A topological space is called locally compact if every point has a compact neighborhood.

Proposition 4.4.2 *Let X be a locally compact Hausdorff space, $U \subset X$ is open, and $x \in U$. Then U contains a compact neighborhood K of x, i.e., there is a compact K such that*

$$x \in K^o \subset K \subset U.$$

Proof
By replacing U with $U \cap F^o$ for any compact neighborhood F of x, we may assume $\overline{U}$ is compact. By Proposition 4.3.6, there are disjoint open sets V, W such that $x \in V$ and $U^c \cap \overline{U} \subset W$. $V \subset W^c$ because V, W are disjoint. Let $K = \overline{U \cap V}$. Since

$$U \cap V \subset \overline{U} \cap W^c \subset \overline{U} \cap (U^c \cap \overline{U})^c = \overline{U} \cap (U \cup \overline{U}^c) = U$$

and $\overline{U} \cap W^c$ is closed, we have that K is a compact neighborhood of x contained in U. □

Proposition 4.4.3 *Let X be a locally compact Hausdorff space, K be compact, U be open, and $K \subset U \subset X$. Then there exists a precompact open set V such that*

$$K \subset V \subset \overline{V} \subset U.$$

Proof
By Proposition 4.4.2, every $x \in K$ has a compact neighborhood N_x. $\{N_x^o : x \in K\}$ is an open cover of K. So there are $x_1, \ldots, x_n$ such that

$$K \subset \bigcup_{i=1}^{n} N_{x_i}^o.$$

Let $V = \bigcup_{i=1}^n N_{x_i}^o$. Then $\overline{V} \subset \bigcup_{i=1}^n N_{x_i}$ is compact and

$$K \subset V \subset \overline{V} \subset U.$$

□

Theorem 4.4.4 (LCH Urysohn's Lemma)
Let X be a locally compact Hausdorff space, K be compact, U be open, and $K \subset U \subset X$. Then there exists an $f \in C(X, [0, 1])$ such that $f = 1$ on K and $f = 0$ outside a compact subset of U.

Proof
Let V be the one in Proposition 4.4.3. Then $\overline{V}$ is a normal space under the relative topology. So there is a $\hat{f} \in C(\overline{V}, [0,1])$ such that $\hat{f} = 1$ on K and $\hat{f} = 0$ on $\overline{V} \cap V^c$. Define

$$f(x) = \begin{cases} \hat{f}(x), & x \in \overline{V} \\ 0, & x \in \overline{V}^c. \end{cases}$$

We show that f is continuous. For any closed $E \subset \mathbb{R}$, since $\hat{f}$ is continuous on $\overline{V}$, $\hat{f}^{-1}(E)$ is relatively closed in $\overline{V}$, i.e., $\hat{f}^{-1}(E) = F \cap \overline{V}$ for some closed set $F \subset X$. So

$$f^{-1}(E) = \begin{cases} \hat{f}^{-1}(E) = F \cap \overline{V}, & \text{if } 0 \notin E, \\ \hat{f}^{-1}(E) \cup V^c = (F \cap \overline{V}) \cup V^c, & \text{if } 0 \in E \end{cases}$$

is always closed, and therefore f is continuous. □

Theorem 4.4.5 (LCH Tietze Extension Theorem)
Let X be a locally compact Hausdorff space and $K \subset X$ be compact. Then every $f \in C(K)$ can be extended to a function $F \in C(X)$ such that $F = 0$ outside a compact set.

Proof
Homework. □

Definition 4.4.6
a) Let $f \in C(X)$. The support of f, denoted by $\operatorname{supp}(f)$, is defined by

$$\operatorname{supp}(f) = \overline{\{x : f(x) \neq 0\}}.$$

b) $C_c(X) = \{f \in C(X) : \operatorname{supp}(f) \text{ is compact}\}$.
c) $f \in C(X)$ is called to vanish at infinity if the set

$$\{x : |f(x)| \geq \epsilon\}$$

is compact for any $\epsilon > 0$.
d) $C_0(X) = \{f \in C(X) : f \text{ vanishes at infinity}\}$.

Remark 4.2

Clearly $C_c(X) \subset C_0(X) \subset BC(X)$.

Proposition 4.4.7 *Let X be a locally compact Hausdorff space. Then*

$$C_0(X) = \overline{C_c(X)}$$

under the uniform norm.

Proof

For any $f \in \overline{C_c(X)}$, there is a sequence $\{f_n\}_1^\infty \subset C_c(X)$ such that $\|f_n - f\|_u \to 0$ as $n \to \infty$. For any $\epsilon > 0$, there is an N such that $\|f_n - f\|_u < \epsilon$ for all $n \geq N$. For any $x \in (\text{supp}\,(f_N))^c$, since $f_N(x) = 0$, we have $|f(x)| < \epsilon$. So $\{x : |f(x)| \geq \epsilon\} \subset \text{supp}\,(f_N))$ is compact, and $f \in C_0(X)$.

Conversely, for any $f \in C_0(X)$, let

$$K_n = \left\{x : |f(x)| \geq \frac{1}{n}\right\}$$

Then K_n is compact. By Theorem 4.4.4, there is $g_n \in C_c(X, [0, 1])$ such that $g_n = 1$ on K_n and $g_n = 0$ outside of a compact set. Let $f_n = g_n f$. Then $f_n \in C_c(X)$, and

$$|f_n - f| = |f||g_n - 1| = 0$$

if $x \in K_n$ and

$$|f_n - f| = |f||g_n - 1| \leq |f| < \frac{1}{n}$$

if $x \notin K_n$. So $\|f_n - f\|_u \leq \frac{1}{n} \to 0$ as $n \to \infty$, and $f \in \overline{C_c(X)}$. □

Definition 4.4.8 (One-Point Compactification)

Let $(X, \mathcal{T})$ be a locally compact Hausdorff space. Define

$$X^* = X \cup \{\infty\}, \quad \mathcal{T}^* = \mathcal{T} \cup \left\{K^c \cup \{\infty\} : K \subset X \text{ is compact}\right\}.$$

It can be proved (Homework) that $(X^*, \mathcal{T}^*)$ is a compact topological space, and it is called the one-point compactification of $(X, \mathcal{T})$.

Exercises

4.17 Let X be a locally compact Hausdorff space and $K \subset X$ be compact, and $U \supset K$ be a precompact open set.

a. Prove that every $f \in C(K)$ can be extended to a function $g \in C(\overline{U})$ such that $g = 0$ on $\overline{U} \setminus U$.

b. Prove that every $f \in C(K)$ can be extended to a function $F \in C(X)$ such that $F = 0$ on U^c. (See Exercise 4.8.)

4.18 (One-point compactification.) Let $(X, \mathscr{T})$ be a non-compact topological space. Pick any element not in X and denote it by ∞. Define

$$X^* = X \cup \{\infty\}, \quad \mathscr{T}^* = \mathscr{T} \cup \left\{K^c \cup \{\infty\} : K \subset X \text{ is compact in } (X, \mathscr{T})\right\}.$$

Prove that

a. $\mathscr{T}^*$ is a topology on X^* such that $(X, \mathscr{T})$ is the relative topological space of $(X^*, \mathscr{T}^*)$, and

b. $(X^*, \mathscr{T}^*)$ is a compact space.

4.19 Prove that the one-point compactification $(X^*, \mathscr{T}^*)$ of a Hausdorff space $(X, \mathscr{T})$ is Hausdorff if and only if $(X, \mathscr{T})$ is locally compact.

4.20 Let $(X^*, \mathscr{T}^*)$ be the one-point compactification of a locally compact Hausdorff space $(X, \mathscr{T})$. Prove that a function $f \in C(X)$ can be extended to a function $F \in C(X^*)$ if and only if $f = g +$ constant for a $g \in C_0(X)$.

4.21 Prove that in Euclidean topology, the one-point compactification of $\mathbb{R}$ is homeomorphic to the unit circle S^1 in $\mathbb{R}^2$.

4.22 Let U be an open subset of a compact Hausdorff space X and U^* be the one-point compactification of U. Prove that the function $f : X \to U^*$ defined by

$$f(x) = \begin{cases} x, & x \in U, \\ \infty, & x \in U^c, \end{cases}$$

is continuous.

4.5 The Stone-Weierstrass Theorem

Let X be a compact Hausdorff space (so it is normal). Then $C(X)$ is a normed space under the uniform norm

$$\|f\|_u = \max\{|f(x)| : x \in X\}.$$

Definition 4.5.1

a) A subset $\mathscr{A} \subset C(X)$ is called to separate points if for any $x \neq y$ in X, there exists an $f \in \mathscr{A}$ such that

$$f(x) \neq f(y).$$

b) An algebra $\mathscr{A}$ over $\mathbb{R}$ is a vector space over $\mathbb{R}$ with a multiplication defined.

c) A subset $\mathscr{A} \subset C(X)$ is called a lattice if for all $f, g \in \mathscr{A}$, both $\max(f(x), g(x)) \in \mathscr{A}$ and $\min(f(x), g(x)) \in \mathscr{A}$.

Lemma 4.5.2 *Let $\mathbb{R}^2$ be the algebra with operations defined by*

$$a(x_1, y_1) + b(x_2, y_2) = (ax_1 + bx_2, ay_1 + by_2),$$

$$(x_1, y_1)(x_2, y_2) = (x_1x_2, y_1y_2).$$

Then the only subalgebras are

$$\mathbb{R}^2, \quad \{(0, 0)\}, \quad \{(x, 0) : x \in \mathbb{R}\}, \quad \{(0, y) : y \in \mathbb{R}\}, \quad \{(x, x) : x \in \mathbb{R}\}.$$

Proof
Let $\mathscr{A}$ be a subalgebra. If there is a $(a, b) \in \mathscr{A}$ such that

$$0 \neq a \neq b \neq 0,$$

then $(a^2, b^2) \in \mathscr{A}$ and $\{(a, b), (a^2, b^2)\}$ form a basis of $\mathbb{R}^2$. So $\mathscr{A} = \mathbb{R}^2$. Otherwise for all $(x, y) \in \mathscr{A}$, either $x = y$, or $y = 0$ and there exists an $x \neq 0$, or $x = 0$ and there exists a $y \neq 0$, or both $x = 0, y = 0$. So $\mathscr{A}$ is either $\{(x, x) : x \in \mathbb{R}\}$, or $\{(x, 0) : x \in \mathbb{R}\}$, or $\{(0, y) : y \in \mathbb{R}\}$, or $\{(0, 0)\}$, respectively. □

Lemma 4.5.3 *For all $\epsilon > 0$, there exists a polynomial $p(x)$ such that*

$$p(0) = 0, \quad \textit{and} \quad \Big||x| - p(x)\Big| < \epsilon \ \textit{for all} \ x \in [-1, 1].$$

Proof
Let

$$(1 - t)^{1/2} = 1 + c_1t + c_2t^2 + \cdots$$

be the Maclaurin Series expansion for $(1 - t)^{1/2}$. Then the series converges uniformly on $[-1, 1]$. So for any $\epsilon > 0$, there is a polynomial $q(t)$ such that

$$|(1 - t)^{1/2} - q(t)| < \frac{\epsilon}{2} \ \text{ for all } \ t \in [-1, 1].$$

Let $t = 1 - x^2$. Then

$$\Big||x| - q(1 - x^2)\Big| < \frac{\epsilon}{2} \ \text{ for all } \ x \in [-1, 1].$$

Let $x = 0$, we have $|q(1)| < \frac{\epsilon}{2}$. Define

$$p(x) = q(1 - x^2) - q(1).$$

Then $p(x)$ is a polynomial, $p(0) = 0$, and

$$\Big||x| - p(x)\Big| \leq \Big||x| - q(1-x^2)\Big| + |q(1)| < \epsilon \quad \text{for all} \quad x \in [-1, 1].$$

□

Note that $C(X)$ is an algebra under function addition, function multiplication, and multiplication be real numbers.

Lemma 4.5.4 *Let $\mathscr{A}$ be a closed (under the uniform norm) subalgebra of $C(X)$. Then for every $f \in \mathscr{A}$, we have $|f| \in \mathscr{A}$ and $\mathscr{A}$ is a lattice.*

Proof

If $f \equiv 0$, then $|f| = f \in \mathscr{A}$. Otherwise let

$$h(x) = \frac{1}{\|f\|_u} f(x).$$

Then $-1 \leq h(x) \leq 1$ and by Lemma 4.5.3, for any $\epsilon > 0$, there is a polynomial $p(x)$ such that

$$\Big||h(x)| - p(h(x))\Big| < \epsilon \quad \text{for all} \quad x \in X.$$

Since $\mathscr{A}$ is an algebra, $p(h(x)) \in \mathscr{A}$. So $|h(x)| \in \overline{\mathscr{A}} = \mathscr{A}$ and therefore $|f| = \|f\|_u |h| \in \mathscr{A}$.

Finally, for any $f, g \in \mathscr{A}$,

$$\max(f, g) = \frac{1}{2}(f + g + |f - g|) \in \mathscr{A}, \quad \min(f, g) = \frac{1}{2}(f + g - |f - g|) \in \mathscr{A}.$$

So $\mathscr{A}$ is a lattice.

□

Lemma 4.5.5 *Let $\mathscr{A}$ be a closed lattice in $C(X)$.and $f \in C(X)$. If for every pair x, y in X, there is a function $g_{xy} \in \mathscr{A}$ such that*

$$g_{xy}(x) = f(x), \quad g_{xy}(y) = f(y),$$

then $f \in \mathscr{A}$.

Proof

For any $\epsilon > 0$ and for any $x, y \in X$, define

$$U_{xy} = \{z \in X : f(z) < g_{xy}(z) + \epsilon\}, \quad V_{xy} = \{z \in X : f(z) > g_{xy}(z) - \epsilon\}.$$

Then U_{xy} and V_{xy} are open and contain both x and y. $\{U_{xy} : x \in X\}$ is an open cover of X. So there are $x_1, x_2, \ldots, x_n$ such that

$$X = \bigcup_{i=1}^{n} U_{x_i y}.$$

Define

$$g_y(z) = \max(g_{x_1 y}(z), \ldots, g_{x_n y}(z).$$

Then $g_y \in \mathscr{A}$ and

$$f < g_y + \epsilon \text{ on } X, \text{ and } g_y - \epsilon < f < g_y + \epsilon \text{ on } V_y := \bigcap_{i=1}^{n} V_{x_1 y}.$$

$\{V_y : y \in X\}$ is an open cover of X. So there are $y_1, \ldots, y_m$ such that

$$X = \bigcup_{i=1}^{m} V_{y_i}.$$

Define

$$g(z) = \min(g_{y_1}(z), \ldots, g_{y_m}(z)).$$

Then $g \in \mathscr{A}$ and (Figure 4.3)

$$g - \epsilon < f < g + \epsilon, \text{ or } \|f - g\|_\infty < \epsilon.$$

Since $\mathscr{A}$ is closed, $f \in \mathscr{A}$. □

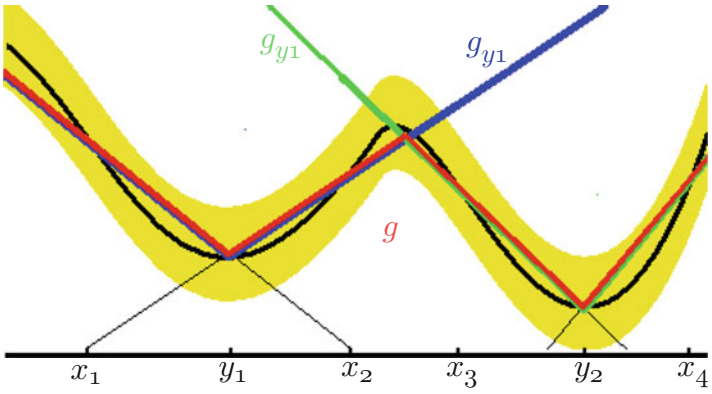


Fig. 4.3 Construction of g

Theorem 4.5.6 (The Stone-Weierstrass Theorem)
Let X be a compact Hausdorff space and $\mathscr{A}$ be a closed subalgebra of $C(X)$ that separates points. Then either

$$\mathscr{A} = C(X)$$

or there exists a $x_0 \in X$ such that

$$\mathscr{A} = \{f \in C(X) : f(x_0) = 0\}.$$

In particular, if the constant function $1 \in \mathscr{A}$, then $\mathscr{A} = C(X)$

Proof
Since $\mathscr{A}$ is closed, it is a lattice by Lemma 4.5.4. For each fixed pair $x \neq y$ in X, let

$$\mathscr{A}_{xy} = \{(f(x), f(y)) : f \in \mathscr{A}\} \subset \mathbb{R}^2.$$

Then by Lemma 4.5.2, $\mathscr{A}_{xy}$ is one of

$$\mathbb{R}^2, \quad \{(0,0)\}, \quad \{(u,0) : u \in \mathbb{R}\}, \quad \{(0,v) : u \in \mathbb{R}\}, \quad \{(u,u) : u \in \mathbb{R}\}.$$

Since $\mathscr{A}$ separates points, $\mathscr{A}_{xy}$ cannot be $\{(0,0)\}$ or $\{(u,u) : u \in \mathbb{R}\}$.

If for all $x \in X$, there exists a $g \in \mathscr{A}$ such that $g(x) \neq 0$. Then $\mathscr{A}_{xy} = \mathbb{R}^2$ for all x, y in X. For any $f \in C(X)$ and for any x, y in X, since $(f(x), f(y)) \in \mathbb{R}^2 = \mathscr{A}_{xy}$, there is a $g_{xy} \in \mathscr{A}$ such that

$$g_{xy}(x) = f(x), \quad g_{xy}(y) = f(y).$$

Then by Lemma 4.5.5, $f \in \mathscr{A}$, i.e., $\mathscr{A} = C(X)$.

Otherwise, there is an $x_0 \in X$ such that $g(x_0) = 0$ for all $g \in \mathscr{A}$. Since $\mathscr{A}$ separates points, such x_0 is unique. For any $f \in C(X)$ with $f(x_0) = 0$, and for any x, y in X,

$$(f(x), f(y)) \in \begin{cases} \{(u,0) : u \in \mathbb{R}\} = \mathscr{A}_{xy}, & \text{if } y = x_0, \\ \{(0,v) : v \in \mathbb{R}\} = \mathscr{A}_{xy}, & \text{if } x = x_0, \\ \mathbb{R}^2 = \mathscr{A}_{xy}, & \text{if } x \neq x_0, y \neq x_0. \end{cases}$$

So there is a $g_{xy} \in \mathscr{A}$ such that

$$g_{xy}(x) = f(x), \quad g_{xy}(y) = f(y).$$

Again by Lemma 4.5.5, $f \in \mathscr{A}$, i.e., $\mathscr{A} = \{f \in C(X) : f(x_0) = 0\}$. □

Corollary 4.5.7 *Let $\mathscr{B}$ be a subalgebra of $C(X)$ that separates points. Then either*

$$\overline{\mathscr{B}} = C(X)$$

or there is an $x_0 \in X$ such that

$$\overline{\mathscr{B}} = \{f \in C(X) : f(x_0) = 0\}.$$

In particular, if $1 \in \mathscr{B}$, then $\overline{\mathscr{B}} = C(X)$.

Example 4.3

a) Any bounded closed interval $[a, b]$ is compact. The set $\mathscr{B}$ of all polynomials is an algebra that separates points (by the polynomial $x \in \mathscr{B}$) and contains 1. Therefore $\overline{\mathscr{B}} = C([a, b])$, i.e., any continuous function can be uniformly approached by a sequence of polynomials on $[a, b]$.

b) All continuous periodic functions of period 2π can be considered as functions on the unit circle, which is compact. The set of all finite linear combinations of

$$\{1, \cos x, \sin x, \cos 2x, \sin 2x, \dots\}$$

is an algebra that separates points and contains 1. Therefore any continuous periodic functions of period 2π can be uniformly approached by a sequence of finite linear combinations of $\{1, \cos x, \sin x, \cos 2x, \sin 2x, \dots\}$.

■

Exercises

4.23 (The Stone-Weierstrass Theorem for noncompact spaces.) Let X be a noncompact locally compact Hausdorff space, and $\mathscr{A}$ be a subalgebra of $C_0(X, \mathbb{R})$ that separates points. Prove that either $\overline{\mathscr{A}} = C_0(X, \mathbb{R})$ or $\overline{\mathscr{A}} = \{f \in C_0(X, \mathbb{R}) : f(x_0) = 0\}$ for some $x_0 \in X$. (Hint: If there is an x_0 such that $f(x_0) = 0$ for all $f \in \mathscr{A}$, then consider the one-point compactification of $X \setminus \{x_0\}$ with $\infty = x_0$. Otherwise consider the one-point compactification of X.)

4.24 Let X and Y be compact Hausdorff spaces. Prove that

$$\mathscr{A} = \left\{ \sum_{i=1}^{n} g_i(x) h_i(y) : g_i \in C(X), h \in C(Y), n \in \mathbb{N} \right\}$$

is dense in $C(X \times Y)$ in the Euclidean topology.

4.25 Let $g \in C([a, b])$ be strictly increasing, $a < b$ in $\mathbb{R}$. Prove that for any $f \in C([a, b])$ and for any $\epsilon > 0$, there are $a_0, a_1, \dots, a_n \in \mathbb{R}$ such that

$$\max_{x \in [a,b]} |f(x) - (a_0 + a_1 g(x) + a_2 g^2(x) + \cdots + a_n g^n(x))| < \epsilon.$$

Elements of Functional Analysis

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5.1 Banach Spaces

Definition 5.1.1

Let X be a vector space over $\mathbb{R}$. A norm $\|\cdot\|$ on X is a function $X \to [0, \infty)$ such that

i) $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = 0$;
ii) (Triangle Inequality) $\|x + y\| \leq \|x\| + \|y\|$ for any $x, y \in X$;
iii) $\|cx\| = |c|\|x\|$ for any $c \in \mathbb{R}$ and any $x \in X$.

$(X, \|\cdot\|)$ is called a normed space.

Example 5.1

a) Let X be a finite dimensional vector space and $\{u_1, \ldots, u_n\}$ be a basis. Then any vector $x \in X$ has a unique expression $x = a_1u_1 + \cdots + a_nu_n$. Then

$$\|x\| = |a_1| + \cdots + |a_n|$$

is a norm on X.

b) For the vector space in a),

$$\|x\| = \max_i |a_i|$$

is another norm.

c) For the space $C[a, b]$ of all continuous real value functions on $[a, b]$,

$$\max_{x\in[a,b]} |f(x)|$$

is a norm on $C[a, b]$.

d)

$$\|f\| = \int |f|\, d\mu$$

is a norm on $L^1(\mu)$ if we identify all functions equal each other a.e.

■

5

A normed space is a metric space with the metric

$$\rho(x, y) = \|x - y\|,$$

so is a topological space.

Definition 5.1.2

A complete (i.e., every Cauchy sequence converges) normed space is called a Banach space.

Definition 5.1.3

Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on X are called equivalent if there are positive numbers c_1, c_2 such that

$$c_1\|x\|_1 \le \|x\|_2 \le c_2\|x\|_1 \quad \text{for all } x \in X.$$

Equivalent norms define the same topology.

Definition 5.1.4

A series $\sum_{n=1}^{\infty} x_n$ in a normed space X is called to converge absolutely if

$$\sum_{n=1}^{\infty} \|x_n\|$$

converges.

Theorem 5.1.5

A normed space X is complete if and only if every absolutely convergent series converges.

Proof

Let X be complete. Then for any absolutely convergent series $\sum_{n=1}^{\infty} x_n$,

$$\left\| \sum_{n=N+1}^{N+m} x_n \right\| \leq \sum_{n=N+1}^{N+m} \|x_n\| \leq \sum_{n=N+1}^{\infty} \|x_n\| \to 0$$

for any m as $N \to \infty$. So $\left\{ \sum_{n=1}^{N} x_n : N = 1, 2, \dots \right\}$ is a Cauchy sequence, and

$$\lim_{N\to\infty} \sum_{n=1}^{N} x_n = \sum_{n=1}^{\infty} x_n$$

exists.

Conversely, assume every absolutely convergent series converges. Then for any Cauchy sequence $\{x_n\} \subset X$, and for any j, there is an n_j such that

$$\|x_k - x_m\| < \frac{1}{2^j} \quad \text{for all } k, m \geq n_j.$$

Let $y_1 = x_{n_1}$ and $y_j = x_{n_j} - x_{n_{j-1}}$ for $j > 1$. Then

$$\sum_{j=1}^{k} y_j = x_{n_k} \quad \text{and} \quad \sum_{j=1}^{\infty} \|y_j\| \leq \|y_1\| + \sum_{j=1}^{\infty} \frac{1}{2^j} = \|y_1\| + 1 < \infty.$$

So

$$\lim_{k\to\infty} x_{n_k} = \lim_{j\to\infty} \sum_{j=1}^{k} y_j = x$$

exists.

We show that $x_n \to x$ as $n \to \infty$. For any $\epsilon > 0$, there is an N such that

$$\|x_n - x_m\| < \frac{\epsilon}{2} \quad \text{for all } n, m \geq N.$$

For all $n \geq N$, we can always choose an $n_k \geq N$ such that $\|x_{n_k} - x\| < \frac{\epsilon}{2}$. Then

$$\|x_n - x\| \leq \|x_n - x_{n_k}\| + \|x_{n_k} - x\| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

□

Definition 5.1.6

Let X be a normed space and $M \subset X$ be a linear subspace. Define an equivalence relation

$$x \sim y \quad \text{if and only if} \quad x - y \in M.$$

Then the set of all equivalence classes is called the quotient space of X over M, and is denoted by X/M. Clearly the set of all equivalence class of x is $x + M$. So

$$X/M = \{x + M : x \in X\}.$$

X/M is a vector space with the operations defined by

$$a(x + M) + b(y + M) = (ax + by) + M$$

for any $a, b \in \mathbb{R}$. Clearly

$$x + M + M = x + M.$$

So M is the zero element of X/M.

X/M is also a normed space with the quotient norm

$$\|x + M\| = \inf_{y \in M} \|x + y\|.$$

Exercises

5.1 Let X be a normed space over $\mathbb{R}$. Prove that the vector addition, $X \times X \to X$: $(x, y) \mapsto x + y$, the multiplication by a scalar, $\mathbb{R} \times X \to X$: $(c, x) \mapsto cx$, and the norm, $X \to \mathbb{R}$: $x \mapsto \|x\|$, are all continuous.

5.2 Let X be a normed space. Prove that the closure of a subspace of X is a subspace.

5.2 Bounded Linear Operators

Definition 5.2.1

Let X and Y be normed spaces. A linear map $T : X \to Y$ is called bounded if there is a $c > 0$ such that

$$\|T(x)\| \leq c\|x\| \quad \text{for all } x \in X.$$

Note that the definition of bounded linear operators is different from the definition of bounded functions. It does not mean $T(X)$ is bounded.

Proposition 5.2.2 *Let X and Y be normed vector spaces, and $T : X \to Y$ be a linear map. Then the following are equivalent.*

a) T *is continuous.*

b) T *is continuous at* 0.

c) T *is bounded.*

Proof

"a) ⇒ b)":

Trivial.

"b) ⇒ c)":

There is a $\delta > 0$ such that $\|T(x)\| \leq 1$ whenever $\|x\| \leq \delta$. For any nonzero $w \in X$, let

$$z = \frac{\delta}{\|w\|} w.$$

then $\|z\| = \delta$. By the linearity

$$\frac{\delta}{\|w\|}\|T(w)\| = \|T(z)\| \leq 1, \quad \text{or} \quad \|T(w)\| \leq \frac{1}{\delta}\|w\|,$$

i.e., T is bounded.

"c) ⇒ a)": Since T is bounded, there exists a $c \in \mathbb{R}$ such that $\|T(x)\| \leq c\|x\|$ for all $x \in X$. Then

$$\lim_{w \to x} \|T(w) - T(x)\| = \lim_{w \to x} \|T(w - x)\| \leq c \lim_{w \to x} \|w - x\| = 0,$$

i.e., T is continuous at any x. □

Definition 5.2.3

Let $T : X \to Y$ be a bounded linear operator. Then

$$\|T\| = \sup_{x \neq 0} \frac{\|T(x)\|}{\|x\|} = \sup_{x \neq 0} \left\| T\left(\frac{1}{\|x\|} x\right) \right\| = \sup_{\|x\|=1} \|T(x)\|$$

is called the operator norm induced by the norms of X and Y.

Example 5.2

Let $C^\infty([0, 1])$ be the space of all C^∞ functions on $[0, 1]$, and define

$$\|f\| = \max_{x \in [0,1]} |f(x)|.$$

Then $(C^\infty([0,1]), \|\cdot\|)$ is a normed space. Both $\frac{d}{dx}$ and $\int_0^x$ are linear operators on $C^\infty([0,1])$. However, $\frac{d}{dx}$ is unbounded, because

$$\left\| \frac{d}{dx} x^n \right\| = \|n x^{n-1}\| = n = n\|x^n\|$$

but $\int_0^x$ is bounded with $\left\|\int_0^x\right\| = 1$, because

$$\left\| \int_0^x f(t)\, dt \right\| \le \int_0^1 \|f\|\, dt = \|f\|, \text{ for all } f \in C^\infty([0,1]), \text{ and } \left\| \int_0^x 1\, dt \right\| = 1 = \|1\|.$$

■

Example 5.3

Consider $\mathbb{R}^n$. For any $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

$$\|x\|_1 := |x_1| + \cdots + |x_n|, \quad \|x\|_2 := \sqrt{x_1^2 + \cdots + x_n^2}, \quad \|x\|_\infty = \max(|x_1|, \ldots, |x_n|)$$

are all norms. An $n \times n$ matrix

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$$

defines a linear operator $A : x \mapsto Ax$. We have

$$\begin{aligned} \|Ax\|_1 &= |a_{11}x_1 + \cdots + a_{1n}x_n| + \cdots + |a_{n1}x_1 + \cdots + a_{nn}x_n| \\ &\le (|a_{11}| + \cdots + |a_{n1}|)|x_1| + \cdots + (|a_{1n}| + \cdots + |a_{nn}|)|x_n| \\ &\le \|x\|_1 \max_j (|a_{1j}| + \cdots + |a_{nj}|), \\ \|Ax\|_2 &= \sqrt{x^t A^t A x} \\ &\le \sqrt{x^t Q^t \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix} Q x} \\ &\le \|x\|_2 \max_i \sqrt{\lambda_i} \end{aligned}$$

where the columns of Q are orthonormal eigenvectors and $\lambda_1, \ldots, \lambda_n$ are eigenvalues of A^tA, and

$$\begin{aligned}\|Ax\|_\infty &= \max_i(|a_{i1}x_1 + \cdots + a_{in}x_n|)\\ &\le \|x\|_\infty \max_i(|a_{i1}| + \cdots + |a_{in}|)\end{aligned}$$

So

$$\|A\|_1 \le \max_j(|a_{1j}| + \cdots + |a_{nj}|), \quad \|A\|_2 \le \max_i \sqrt{\lambda_i}, \quad \|A\|_\infty \le \max_i(|a_{i1}| + \cdots + |a_{in}|).$$

Actually they are equal. To show these, we only need to find particular x's such that inequalities become equalities.

For the 1-norm, let $x_j = 1$ and $x_i = 0$ for all $i \ne j$, where jth column has the largest sum of absolute values. Then

$$\|Ax\|_1 = |a_{1j}| + \cdots + |a_{nj}| = (|a_{1j}| + \cdots + |a_{nj}|)\|x\|_1.$$

For the 2-norm, let x be the unit eigenvector of λ_i, where λ_i is the largest eigenvalue of A^tA. Then

$$\|Ax\|_2 = \sqrt{\lambda_i} = \sqrt{\lambda_i}\|x\|_2.$$

For the ∞-norm, assume ith row has the largest sum of absolute values, and let

$$x = \begin{bmatrix} \text{sign}\,(a_{i1}) \\ \vdots \\ \text{sign}\,(a_{in}) \end{bmatrix}$$

Then

$$\|Ax\|_\infty = |a_{i1}| + \cdots + |a_{in}| = (|a_{i1}| + \cdots + |a_{in}|)\|x\|_\infty.$$

Therefore, $\|A\|$ equals the largest column sum of absolute values, $\|A\|_2$ equals the largest singular value (the square roots of the eigenvalues of A^tA are called the singular values of A), and $\|A\|_\infty$ equals the largest column sum of absolute values.

■

Let $L(X, Y)$ be the set of all bounded linear operators for X to Y. Then $L(X, Y)$ is a vector space with operations defined by $(aT + bS)(x) = aT(x) + bS(x)$ for any $T, S \in L(X, Y)$ and any $a, b \in \mathbb{R}$.

Proposition 5.2.4 *$L(X, Y)$ is a normed space under the induced operator norm.*

Proof

We only need to verify the triangle inequality, which comes directly from the definition

$$\begin{aligned}\|T\| + \|S\| &= \sup_{\|x\|=1} \|T(x)\| + \sup_{\|x\|=1} \|S(x)\| \\ &\geq \sup_{\|x\|=1} (\|T(x)\| + \|S(x)\|) \\ &\geq \sup_{\|x\|=1} \|T(x) + S(x)\| \\ &= \|T + S\|.\end{aligned}$$

□

Proposition 5.2.5 *If Y is complete (i.e., a Banach space), then $L(X, Y)$ is complete (i.e., a Banach space).*

Proof

Let $\{T_n\} \subset L(X, Y)$ be a Cauchy sequence. Then for any $x \in X$, $\{T_n(x)\}$ is a Cauchy sequence in Y because $\|T_n(x) - T_m(x)\| = \|(T_n - T_m)(x)\| \leq \|T_n - T_m\|\|x\|$. Define

$$T(x) = \lim_{n\to\infty} T_n(x).$$

T is certainly linear. So we only need to show it is bounded and $\|T_n - T\| \to 0$ as $n \to \infty$.

Since $\{T_n\}$ is Cauchy, for any $\epsilon > 0$, there exists a N such that $\|T_n - T_m\| \leq \epsilon$ for all $m, n \geq N$. So for any $x \in X$ and any $n \geq N$

$$\|T_n(x)\| \leq \|T_n(x) - T_N(x)\| + \|T_N(x)\| \leq \epsilon\|x\| + \|T_N\|\|x\| = (\|T_N\| + \epsilon)\|x\|.$$

So $\|T(x)\| = \lim_{n\to\infty} \|T_n(x)\| \leq (\|T_N\| + \epsilon)\|x\|$, i.e., T is bounded. Also

$$\|(T_n - T)(x)\| = \lim_{m\to\infty} \|T_n(x) - T_m(x)\| \leq \limsup_{m\to\infty} \|T_n - T_m\|\|x\| \leq \epsilon\|x\|,$$

i.e., $\|T_n - T\| \leq \epsilon$ for all $n \geq N$, i.e., $\|T_n - T\| \to 0$ as $n \to \infty$. □

Remark 5.1 If $T \in L(X, Y)$ and $S \in L(Y, Z)$, then

$$S(T(x)) \leq \|S\|\|T(x)\| \leq \|S\|\|T\|\|x\|,$$

i.e., $\|ST\| \leq \|S\|\|T\|$.

Definition 5.2.6

$T \in L(X, Y)$ is called invertible, or an isomorphism, if T is one-to-one and onto, and T^{-1} is bounded.
$T \in L(X, Y)$ is called an isometry if

$$\|T(x)\| = \|x\|$$

for all $x \in X$.

Definition 5.2.7

a) A linear functional on a normed space X is a linear map from X to $\mathbb{R}$.
b) The space $L(X, \mathbb{R})$ of all bounded linear functionals is called the dual space of X, and is denoted by X^*.

For any bounded linear functional f on X, we have $|f(x)| \leq \|f\|\|x\|$. If we define $\rho(x) = \|f\|\|x\|$, then we have $\rho(x + y) \leq \rho(x) + \rho(y)$, $\rho(cx) = c\rho(x)$ for any $c \geq 0$ in R, and $f(x) \leq \rho(x)$.

Definition 5.2.8

A sublinear functional on X is a map $\rho : X \to \mathbb{R}$ such that
i) $\rho(x + y) \leq \rho(x) + \rho(y)$,
ii) $\rho(cx) = c\rho(x)$ for any $c \geq 0$ in $\mathbb{R}$.

Theorem 5.2.9 (Hahn-Banach Theorem)

Let X be a real vector space, ρ be a sublinear functional on X, $M \subset X$ be a subspace, and f be a linear functional on M such that

$$f(x) \leq \rho(x) \quad \textit{for all} \quad x \in M.$$

Then there is a linear functional F on X such that $F(x) \leq \rho(x)$ for all $x \in X$ and $F\Big|_M = f$.

Proof

We show first that f can be extended to a linear functional g on $M_g := M + \text{span}\,(x)$ for any $x \neq 0$ in $X \setminus M$. The only thing we need to do is to properly define $g(x)$. If such $g(x)$ can be defined, then $g(x) = g(x + y) - g(y) \leq \rho(x + y) - f(y)$ and $g(x) = g(y) - g(y - x) \geq f(y) - \rho(y - x)$ for any $y \in M$.

Such a number indeed exists, because for any $y_1, y_2 \in M$, $f(y_1)+f(y_2) = f(y_1+y_2) \le \rho(y_1+y_2) \le \rho(y_1-x)+\rho(y_2+x)$, which means that $f(y_1)-\rho(y_1-x) \le \rho(y_2+x)-f(y_2)$. By taking $y_1 = y_2 = 0$ we have

$$-\infty < -\rho(-x) \le \sup_{y\in M}(f(y) - \rho(y-x)) \le \inf_{y\in M}(\rho(y+x) - f(y)) \le \rho(x) < \infty.$$

Take any $\alpha \in \left[\sup_{y\in M}(f(y) - \rho(y-x)), \inf_{y\in M}(\rho(y+x) - f(y))\right]$ and define

$$g(y+cx) = f(y) + c\alpha, \quad \text{for } y + cx \in M_g.$$

g is clearly linear. We show that $g(x) \le \rho(z)$ for any $z \in M_g$.

For $c > 0$,

$$g(y+cx) = c\left(f\left(\frac{1}{c}y\right) + \alpha!\right) \le c\left(f\left(\frac{1}{c}y\right) + \rho\left(x + \frac{1}{c}y\right) - f\left(\frac{1}{c}y\right)\right) = \rho\left(y + cx\right).$$

For $c < 0$,

$$\begin{aligned} g(y+cx) &= |c|\left(f\left(\frac{1}{|c|}y\right) - \alpha\right) \le |c|\left(f\left(\frac{1}{|c|}y\right) + \rho\left(\frac{1}{|c|}y - x\right) - f\left(\frac{1}{|c|}y\right)\right) \\ &= \rho\left(y + cx\right). \end{aligned}$$

So $g(z) \le \rho(z)$ for all $z \in M_g$.

Let $\mathscr{F} = \{(g, M_g)\}$ be the set of all such extensions of f and define a partial order

$$(g_1, M_{g_1}) \le (g_2, M_{g_2}) \quad \text{if and only if } M_{g_1} \subset M_{g_2} \text{ and } g_2\Big|_{M_{g_1}} = g_1.$$

For a linearly ordered subset $\{(g_\alpha, M_{g_\alpha}) : \alpha \in A\}$, define

$$\mathscr{M} = \bigcup_{\alpha\in A} M_{g_\alpha}, \quad g(x) = g_\alpha(x) \text{ if } x \in M_{g_\alpha},$$

then $(g, \mathscr{M})$ is an upper bound. So by Zorn's Lemma, $\mathscr{F}$ has a maximum element (F, X) (Why is whole X? Why is not a subspace of X?). □

We can memorize the Hahn-Banach theorem roughly as "*local functionals can be expended*."

Remark 5.2 The null space of any nonzero bounded linear functional f is huge. It has codimension 1, in the sense that $X = N(f) + \text{span}(z)$ for a z such that $f(z) \neq 0$, because any $x \in X$ can be written as $x = \left(x - \frac{f(x)}{f(z)}z\right) + \frac{f(x)}{f(z)}z$, and $x - \frac{f(x)}{f(z)z} \in N(f)$.

The following theorem is used more often than Hahn-Banach theorem itself.

Theorem 5.2.10

Let X be a normed vector space.

a) If M is a closed subspace of X, $x \notin M$, then there is an $g \in X^*$, $g(x) \neq 0$ and $g\Big|_M = 0$. Furthermore such g can be taken to satisfy $\|g\| = 1$ and $g(x) = \inf_{y\in M} \|x - y\|$.

b) If $x \in X$ is nonzero, then there is an $g \in X^*$ such that $\|g\| = 1$ and $g(x) = \|x\|$.

c) If $x \neq y$ in X, then there is an $g \in X^*$ such that $g(x) \neq g(y)$.

d) For $x \in X$, define $\hat{x} : X^* \to \mathbb{R}$ by

$$\hat{x}(f) = f(x),$$

then $x \mapsto \hat{x}$ is a linear isometry from X to X^{**}.

Proof

a) Take the g in the first half of the proof of Theorem 5.2.9 with $f = 0$, $\rho(x) = \|x\|$, and $\alpha = g(x) = \inf_{y\in M}(\rho(y + x) - f(y)) = \inf_{y\in M} \|x - y\|$ ($\alpha > 0$ because M is closed), and extend it to the whole of X. The only thing that needs to be proved is $\|g\| = 1$. Since $\pm g(x) = g(\pm x) \leq \|x\|$, we have $\|g\| = \sup_{\|x\|=1} |g(x)| \leq 1$. Conversely, there are $\{y_n\} \subset M$ such that $\|x - y_n\| \to g(x)$ as $n \to \infty$. So

$$\|g\| = \sup_{z\neq 0} \frac{|g(z)|}{\|z\|} \geq \frac{|g(x - y_n)|}{\|x - y_n\|} = \frac{g(x)}{\|x - y_n\|}.$$

Let $n \to \infty$, we have $\frac{g(x)}{\|x-y_n\|} \to \frac{g(x)}{g(x)} = 1$ and $\|g\| \geq 1$. So $\|g\| = 1$.

b) Take $M = \{0\}$ in a), so $g(x) = \inf_{y\in M} \|x - y\| = \|x\|$.

c) Replace x by $x - y$ in b).

d) The map $x \mapsto \hat{x}$ is clearly linear, and by c), it is one-to-one. So we only need to show $\|\hat{x}\| = \|x\|$ for all nonzero $x \in X$.

$$\|\hat{x}\| = \sup_{\|f\|=1} |\hat{x}(f)| = \sup_{\|f\|=1} |f(x)| \leq \|f\|\|x\| = \|x\|.$$

Conversely, take f to be the g in b), then

$$\|\hat{x}\| = \sup_{\|f\|=1} |\hat{x}(f)| \geq |\hat{x}(g)| = \|x\|.$$

So $\|\hat{x}\| = \|x\|$ for all $x \in X$.

□

Property d) basically says that we can consider X as a subspace of X^{**}.

Definition 5.2.11

X is called reflexive if $X = X^{**}$.

Lemma 5.2.12

a) A subset $E \subset X$ is dense in X if and only if $E \cap U \neq \emptyset$ for all nonempty open set $U \subset X$.

b) A subset $E \subset X$ is nowhere dense if and only if $\overline{E}^c$ is open and dense in X.

Proof

a) If $E \cap U = \emptyset$ for a nonempty open set U, then $\overline{E} \subset U^c \neq X$ and E is not dense in X. Conversely if E is not dense in X, then $U = \overline{E}^c$ is a nonempty open set disjoint with E.

b) Since $(\overline{E}^o)^c = \overline{E^c}$ by DeMorgan's law, $\overline{E}^o = \emptyset$ if and only if $\overline{E^c} = X$.

□

Theorem 5.2.13 (The Baire Category Theorem)

Let X be a complete metric space.

a) If $\{U_n\}_1^\infty$ is a sequence of open dense subsets of X, then $\bigcap_{n=1}^\infty U_n$ is dense in X.

b) X is not a countable union of nowhere dense sets.

Proof

a) Let $N_r(x) = \{y \in X : d(y, x) < r\}$. We only need to show that $N_r(x) \cap \bigcap_{n=1}^\infty U_n \neq \emptyset$ for any $x \in X$ and for any real number $r > 0$.

Let such $N_r(x)$ be given. Since $N_r(x) \cap U_1$ is nonempty open, there is a $\overline{N_{r_1}(x_1)} \subset N_r(x) \cap U_1$ with $0 < r_1 < \frac{1}{2}$. Generally for $k \geq 1$, since $N_{r_k}(x_k) \cap U_{k+1}$ is nonempty open, there is a $\overline{N_{r_{k+1}}(x_{k+1})} \subset N_{r_k}(x_k) \cap U_{k+1}$ with $0 < r_{k+1} < \frac{1}{2^{k+1}}$.

For all n, k, since $x_{n+k} \in N_{r_{n+k}}(x_{n+k}) \subset N_{r_n}(x_n)$, we have $d(x_{n+k}, x_n) < \frac{1}{2^n} \to 0$ as $n \to \infty$, and therefore $\{x_n\}$ is a Cauchy sequence. Let

$$y = \lim_{n\to\infty} x_n.$$

Then $y \in \overline{N_{r_n}(x_n)} \subset U_n$ for all n. So $y \in N_r(x) \cap \bigcap_{n=1}^\infty U_n$.

b) For any nowhere dense sets $\{E_n\}_1^\infty$, $\{\overline{E_n}^c\}$ are open dense. Since $\bigcap_{n=1}^\infty \overline{E_n}^c = \left(\bigcup_{n=1}^\infty \overline{E}\right)^c$ is dense by a), it is not empty. So $\bigcup_{n=1}^\infty E \subset \bigcup_{n=1}^\infty \overline{E} \subsetneq X$.

□

We can memorize the 2nd statement of the Baire category theorem (which is the one most often used) roughly as "*a complete metric space is not countable union of thin sets.*"

Definition 5.2.14

A function $f : X \to Y$ is called open if $f(U)$ is open for any open set $U \subset X$.

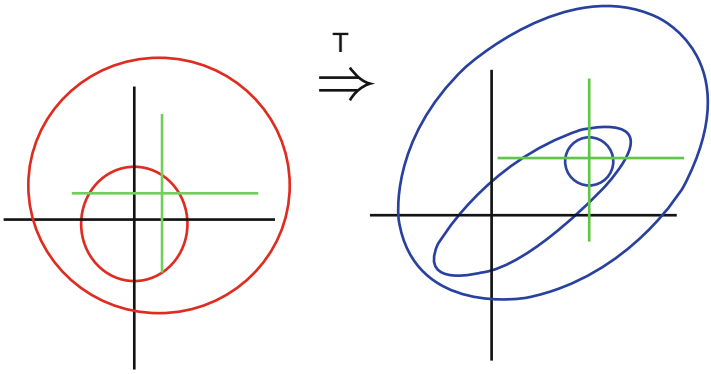


Fig. 5.1 $N_\eta(0) \subset \overline{T(N_1(0))}$

We first prove a lemma.

Lemma 5.2.15 *Let X and Y be Banach spaces and $T \in L(X, Y)$ be surjective. Then there exists an $\epsilon > 0$ such that $N_\epsilon(0) \subset T(N_1(0))$.*

Proof

Since $X = \bigcup_{n=1}^{\infty} nN_{\frac{1}{2}}(0)$, where

$$nN_{\frac{1}{2}}(0) = \{nx : x \in N_{\frac{1}{2}}(0)\},$$

we have $Y = T(X) = \bigcup_{n=1}^{\infty} nT(N_{\frac{1}{2}}(0))$. So by the Baire category theorem, $T(N_{\frac{1}{2}}(0))$ is not nowhere dense. So there is a $y \in T(N_{\frac{1}{2}}(0))$ and $\eta > 0$ such that

$$N_\eta(y) \subset \overline{T(N_{\frac{1}{2}}(0))}.$$

Let $y = T(x)$. Then $\|x\| < \frac{1}{2}$. We have

$$N_\eta(0) = N_\eta(y) - y \subset \overline{T(N_{\frac{1}{2}}(0))} - y = \overline{T(N_{\frac{1}{2}}(0)) - y} = \overline{T(N_{\frac{1}{2}}(0) - x)} \subset \overline{T(N_1(0))},$$

where for any set E, $E - y$ is defined as (Figure 5.1)

$$E - y = \{x - y : x \in E\}.$$

Since T is linear, we have $N_{\frac{\eta}{a}}(0) \subset \overline{T(N_{\frac{1}{a}}(0))}$ for any $a > 0$. We use this property to show that

$$N_{\frac{\eta}{2}}(0) \subset T(N_1(0)).$$

For any $y \in N_{\frac{\eta}{2}}(0) \subset \overline{T(N_{\frac{1}{2}}(0))}$, there is an $x_1 \in N_{\frac{1}{2}}(0)$ such that $\|y - T(x_1)\| < \frac{\eta}{4}$. Then $y - T(x_1) \in N_{\frac{\eta}{2^2}}(0) \subset \overline{T(N_{\frac{1}{2^2}}(0))}$, so there is an $x_2 \in N_{\frac{1}{2^2}}(0)$ such that $\|y - T(x_1) - T(x_2)\| < \frac{\eta}{2^3}$. Continuing this process, we then have a sequence $\{x_n\}$ such that $\|x_n\| < \frac{1}{2^n}$ and $\left\|y - T\left(\sum_{i=1}^n x_i\right)\right\| < \frac{\eta}{2^{n+1}}$. Since $\sum_{i=1}^\infty \|x_i\| < \sum_{i=1}^\infty \frac{1}{2^i} < \infty$, $x = \sum_{i=1}^\infty x_i$ exists, $\|x\| \leq \sum_{i=1}^\infty \|x_i\| < \sum_{i=1}^\infty \frac{1}{2^i} = 1$, and $T(x) = \lim_{n\to\infty} \sum_{i=1}^n T(x_i) = y$, i.e., $N_{\frac{\eta}{2}}(0) \subset T(N_1(0))$. □

5

Theorem 5.2.16 (The Open Mapping Theorem)
Let X and Y be Banach spaces. If $T \in L(X, Y)$ is surjective, then T is open.

Proof
Let $U \subset X$ be open. We show $T(U)$ is open.

For any $y \in T(U)$, $y = T(x)$ for some $x \in U$. Since U is open, there is an $\epsilon > 0$ such that $N_\epsilon(x) \subset U$, which means that $\epsilon N_1(0) + x \subset U$. So $\epsilon T(N_1(0)) + y \subset T(U)$. By Lemma 5.2.15, there exists a $\delta > 0$ such that $N_\delta(0) \subset T(N_1(0))$. Therefore

$$N_{\epsilon\delta}(y) = N_{\epsilon\delta}(0) + y = \epsilon N_\delta(0) + y \subset \epsilon T(N_1(0)) + y \subset T(U),$$

and therefore $T(U)$ is open. □

Under the conditions that the spaces are Banach and the map is linear and bounded, we can memorize the open mapping theorem roughly as "*onto implies open*."

There are some interesting corollaries of the open mapping theorem.

Corollary 5.2.17 *If X and Y are Banach spaces and $T \in L(X, Y)$ is bijective, then T is an isomorphism.*

Proof
Since T is onto, T is open. Since T is one-to-one, $S = T^{-1}$ exists. For any open set $U \subset X$, $S^{-1}(U) = T(U)$ is open. So S is continuous, and therefore is bounded. □

Corollary 5.2.18 *Let $(X, \|\cdot\|_1)$ and $(X, \|\cdot\|_2)$ both be Banach spaces. If there exists a $c > 0$ such that $\|x\|_1 \leq c\|x\|_2$ for all $x \in X$, then $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent.*

Proof
Define $T : (X, \|\cdot\|_2) \to (X, \|\cdot\|_1)$ by $T(x) = x$. Then since $\|T(x)\| = \|x\|_1 \leq c\|x\|_2$, T is bounded linear operator. Since T is bijective, by Corollary 5.2.17, T^{-1} is bounded. So $\|x\|_2 = \|T^{-1}(x)\| \leq \|T^{-1}\|\|x\|_1$, i.e.

$$\frac{1}{\|T^{-1}\|}\|x\|_2 \leq \|x\|_1 \leq c\|x\|_2.$$

Therefore $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent. □

Definition 5.2.19

Let X and Y be normed spaces. The product ∞-norm is defined by

$$\|(x, y)\|_\infty = \max(\|x\|, \|y\|), \quad x \in X,\ y \in Y.$$

For any $p \geq 1$, the product p-norm, denoted by $\|\cdot\|_p$, is defined by

$$\|(x, y)\|_p = \sqrt[p]{\|x\|^p + \|y\|^p} \quad x \in X,\ y \in Y.$$

Clearly the product ∞-norm and the product 1-norm are norms on $X \times Y$. It can be proved (using the discrete form of the Minkowski inequality, Theorem 6.1.5) that the product p-norms, $p \in [1, \infty]$, are all norms on $X \times Y$. In fact, all of them are equivalent.

Proposition 5.2.20

a) All the product p-norms, $p \in [1, \infty]$, are equivalent.
b) In $X \times Y$, a sequence $(x_n, y_n) \to (x, y)$ if and only if $x_n \to x$ and $y_n \to y$.
c) If both X and Y are Banach spaces, then $X \times Y$ is a Banach space under any of the product p-norm.

Proof

The first statement follows from the inequality

$$\begin{aligned} \max(\|x\|, \|y\|) &\leq \sqrt[p]{\|x\|^p + \|y\|^p} \\ &\leq \sqrt[p]{(\max(\|x\|, \|y\|))^p + (\max(\|x\|, \|y\|))^p} \\ &= \sqrt[p]{2}\max(\|x\|, \|y\|). \end{aligned}$$

The second statement is obvious if we use product ∞-norm. Then the third statement follows. □

From now on without specific mention, the topology on $X \times Y$ is always the topology induced by any of the product p-norms.

Definition 5.2.21

A map $T : X \to Y$ is called closed if the graph of T,

$$\Gamma(T) = \{(x, y) \in X \times Y : y = T(x)\},$$

is a closed subset of $X \times Y$.

Remark 5.3 A continuous map is always closed, because for any $(x, y) \in \overline{\Gamma(T)}$, there is $\{(x_n, T(x_n))\} \subset \Gamma(T)$ such that $(x_n, T(x_n)) \to (x, y)$. Since T is continuous

$$y = \lim_{n\to\infty} T(x_n) = T(x),$$

i.e., $(x, y) \in \Gamma(T)$.

The closed graph theorem tells us that the converse is also true if T is linear and both X and Y are Banach spaces.

Theorem 5.2.22 (The Closed Graph Theorem)
If X and Y are Banach spaces and $T : X \to Y$ be a closed linear map, then T is bounded.

Proof
Since T is linear, $\Gamma(T)$ is a subspace of $X \times Y$. So if $\Gamma(T)$ is closed, then it is a closed subspace of the Banach space $(X \times Y, \|\cdot\|_\infty)$, therefore is a Banach space itself. Let $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ be defined by

$$\pi_1((x, y)) = x, \quad \pi_2((x, y)) = y.$$

Then since

$$\|\pi_1((x, y))\| = \|x\| \le \|(x, y)\|_\infty, \quad \|\pi_2((x, y))\| = \|y\| \le \|(x, y)\|_\infty,$$

both π_1 and π_2 are bounded.

Since $\pi_1 : \Gamma(T) \to X$ is bijection, by Corollary 5.2.17, π_1^{-1} is bounded. Therefore $T = \pi_2 \circ \pi_1^{-1}$ is bounded.

□

Under the conditions that the spaces are Banach and the map is linear, we can memorize the closed graph theorem roughly as "*closed implies bounded*."

Theorem 5.2.23 (Uniform Boundedness Principle)
Let X be a Banach space, Y be a normed space, and $\mathscr{A} \subset L(X, Y)$. If for each $x \in X$, $\sup_{T\in\mathscr{A}} \|T(x)\| < \infty$, then $\sup_{T\in\mathscr{A}} \|T\| < \infty$.

Proof
Since $\|T\| = \sup_{\|x\|=1} \|T(x)\|$, to show $\sup_{T\in\mathscr{A}} \|T\| < \infty$, we only need to show that

$$\sup_{T\in\mathscr{A},\|x\|=1} \|T(x)\| < \infty,$$

and since

$$\sup_{T\in\mathscr{A},\|x\|=1} \|T(x)\| = \frac{1}{r} \sup_{T\in\mathscr{A},\|x\|=r} \|T(x)\|$$

for any $r > 0$, we only need to show that

$$\overline{N_r(0)} \subset \{x \in X : \sup_{T\in\mathscr{A}} \|T(x)\| \le n\}$$

for some $r > 0$ and some $n \in \mathbb{N}$.

Let

$$E_n = \{x \in X : \sup_{T\in\mathscr{A}} \|T(x)\| \le n\}.$$

Then since $X = \bigcup_{n=1}^{\infty} E_n$, by the Baire category theorem, there exists an n such that E_n is not nowhere dense, which means that $\overline{N_r(z)} \subset \overline{E_n}$ for some z and some $r > 0$. Then for all $x \in \overline{N_r(0)}$, both $x + z, z \in \overline{N_r(z)}$, so

$$\|T(x)\| \le \|T(x+z)\| + \|T(-z)\| = \|T(x+z)\| + \|T(z)\| \le 2n$$

i.e., $\overline{N_r(0)} \subset E_{2n}$.

□

Under conditions that spaces are Banach (actually, only X needs to be Banach) and maps are bounded linear, we can memorize the uniform boundedness principle roughly as "*pointwise bounded implies uniformly bounded*."

Finally in this section, we define some terminologies.

Definition 5.2.24

Let X be a normed space. A sequence $\{x_n\} \subset X$ is said to converge weakly to an x in X if $f(x_n) \to f(x)$ for every bounded linear functional $f \in X^*$.

A sequence of bounded linear functionals $\{f_n\} \subset X^*$ is said to converge to an $f \in X^*$ in the weak* topology if $f_n(x) \to f(x)$ for every $x \in X$, i.e., converges pointwise.

Note that $f_n \to f$ weakly if $\phi(f_n) \to \phi(f)$ for every $\phi \in X^{**}$. The difference between $f_n \to f$ in the weak topology and in the weak* topology is that the former requires for every ϕ in X^{**} and the latter requires only for every ϕ_x defined by $\phi_x(f) = f(x)$ in $X \subset X^{**}$.

Exercises

5.3 Let T be an linear operator from a normed space X to a normed space Y. Prove that

$$\sup\{\|T(x)\| : \|x\| = 1\} = \sup\left\{\frac{\|T(x)\|}{\|x\|}\right\} = \inf\{c \in R : \|T(x)\| \leq c\|x\| \ \forall x \in X\}.$$

5.4 Let X be a n-dimensional vector space over $\mathbb{R}$ and $\{e_1, \ldots, e_n\}$ be a basis for X. For any $x = a_1e_1 + \cdots + a_ne_n \in X$, defined

$$\|x\|_\infty = \max_i |a_i|.$$

Prove the following statements.

a. $\|\cdot\|_\infty$ is a norm on X.

b. The map $\phi : (\mathbb{R}^n, \|\cdot\|_\infty) \to (X, \|\cdot\|_\infty)$ defined by

$$\phi((a_1, \ldots, a_n)) = a_1e_1 + \cdots + a_ne_n$$

is continuous, where $\|\cdot\|_\infty$ on $\mathbb{R}^n$ is defined by Example 5.3.

c. The set $K = \{x \in X : \|x\|_\infty = 1\}$ is compact in $(X, \|\cdot\|_\infty)$. (Hint: $\phi^{-1}(K)$ is compact in $\mathbb{R}^n$.)

d. All norms on a finite dimensional vector space over $\mathbb{R}$ are equivalent.

5.5 Let X, Y be vector spaces and X be finite dimensional. Prove that any linear operator $T : X \to Y$ is bounded. (Hint: Show T is continuous at 0. See Exercise 5.4.)

5.6 Prove that any finite dimensional normed vector space is a Banach space. (Hint: Exercise 5.4.)

5.7 Let X be a Banach space.

a. Prove that for any $R \in L(X, X)$ with $\|R\| < 1$, $\sum_{i=0}^{\infty} R^i$ converges in $L(X, X)$ and $I - R$ is invertible with $(I - R)^{-1} = \sum_{i=0}^{\infty} R^i$.

b. Prove that the set of all invertible operators in $L(X, X)$ is open by showing that if T is invertible and

$$\|S - T\| < \frac{1}{\|T^{-1}\|},$$

then S is invertible. (Hint: $ST^{-1} = I - (I - ST^{-1})$ and the given condition implies that $\|I - ST^{-1}\| < 1$.)

5.8 Let X be a normed vector space and M be a proper closed subspace of X. Prove the following statements

a.

$$\|x + M\| = \inf\{\|x + y\| : y \in M\}$$

is a norm on X/M.

b. For any $\epsilon > 0$, there exists an $x \in X$ such that $\|x\| = 1$ and $\|x + M\| \geq 1 - \epsilon$.
c. The projection $\pi : X \to X/M$ defined by $\pi(x) = x + M$ has norm 1.
d. X/M is complete if X is complete.

5.9 Let X and Y be normed vector spaces and $T \in L(X, Y)$. Prove that

$$N(T) = \{x \in X : T(x) = 0\}$$

is a closed subspace of X.

5.10 Let X and Y be normed vector spaces and $T \in L(X, Y)$. Prove that $S : X/N(T) \to Y$ defined by $S(x + N(T)) = T(x)$ is a well-defined one-to-one bounded linear operator and $\|S\| = \|T\|$. (See Exercise 5.8 c.)

5.11 Prove that a linear functional f on a normed vector space X is bounded if and only if $f^{-1}(\{0\})$ is closed. (Hint: If $|f(x_n)| \to \infty$ for a sequence of unit vectors $\{x_n\}$, then $y_n = \frac{1}{f(x_n)} x_n \to 0$, but $f(y_n) \not\to 0$.)

5.12 Let X be a normed vector space.
a. Prove that if M is a closed subspace and $x \notin M$, then $M + \mathbb{C}x$ is closed. (Hint: Theorem 5.2.10 a).)
b. Prove that every finite-dimensional subspace of X is closed.

5.13 Let X be an infinite-dimensional normed vector space.
a. Prove that there is a sequence of unit vectors $\{x_j\}$ such that $\|x_j - x_k\| \geq \frac{1}{2}$ for $j \neq k$. (See Exercise 5.8 b.)
b. Prove that X is not locally compact by showing that $B_\epsilon = \{x : \|x\| \leq \epsilon\}$ is not compact for any $\epsilon > 0$. (Hint: Consider $\{\epsilon x_j\}$ for the $\{x_j\}$ in a.)

5.14 Let M be an finite-dimensional subspace of a normed space X. Prove that there is a closed subspace N such that $X = M \oplus N$, i.e., $X = M + N$ and $M \cap N = \{0\}$. (Hint: Theorem 5.2.10 a).)

5.15 Let X and Y be normed vector spaces, $X^* \times Y^*$ be the normed space with $\|(f, g)\| = \max(\|f\|, \|g\|)$, and $X \times Y$ be the normed space with $\|(x, y)\| = \|x\| + \|y\|$. Prove that $\alpha : X^* \times Y^* \to (X \times Y)^*$ defined by $\alpha(f, g)(x, y) = f(x) + g(y)$ is an isometric isomorphism (the norm on $(X \times Y)^*$ is the induced operator norm).

5.16 Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be norms on a vector space X, and both $(X, \|\cdot\|_1)$ and $(X, \|\cdot\|_2)$ are Banach spaces. Prove that if there is a $c > 0$ such that $\|\cdot\|_1 \leq c\|\cdot\|_2$, then the norms are equivalent. (Hint: Corollary 5.2.17.)

5.17 Let X and Y be Banach spaces, $T \in L(X, Y)$, $N(T) = \{x \in X : T(x) = 0$, and $M = \text{range}(T)$. Prove that $X/N(T)$ and M are isomorphic if and only if M is closed. (See Exercises 5.8 and 5.9.)

5.18 Let X be a Banach space and $S \subset X$ such that $\sup_{x \in S} f(x) < \infty$ for all $f \in X^*$. Prove that $\sup_{x \in S} \|x\| < \infty$.

5.19 Let X and Y be Banach spaces and $T : X \to Y$ be a linear operator. Prove that if $f \circ T \in X^*$ for every $f \in Y^*$, then T is bounded.

5.20 Let X and Y be Banach spaces and $\{T_n\} \subset L(X, Y)$ such that $\lim_{n\to\infty} T_n(x)$ exists for every $x \in X$. Define $T(x) = \lim_{n\to\infty} T_n(x)$. Prove that T is a bounded linear operator from X to Y.

5.3 Hilbert Spaces

Definition 5.3.1

Let H be a vector space over $\mathbb{C}$ (or $\mathbb{R}$). A function $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{C}$ (or $\mathbb{R}$) is called an inner product if

i) $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$ for all $x, y, z \in H$ and all $a, b \in \mathbb{C}$ (or $\mathbb{R}$).
ii) $\langle x, y \rangle = \overline{\langle y, x \rangle}$ for all $x, y \in H$.
iii) $\langle x, x \rangle \geq 0$ (which means that it is real) for all $x \in H$, and $\langle x, x \rangle = 0$ if and only if $x = 0$.

$(H, \langle \cdot, \cdot \rangle)$ is called an inner product space.
For any x in an inner product space, define

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

Example 5.4

a) The dot product in $\mathbb{C}^3$ defined by

$$(x_1, y_1, z_1) \cdot (x_2, y_2, z_2) = x_1\overline{x}_2 + y_1\overline{y}_2 + x_3\overline{x}_3$$

is an inner product.

b) For the space of all $m \times n$ real matrices

$$\langle A, B \rangle = \text{trace} A^t B$$

is an inner product.

c) For the space P^2 of all polynomials of degree at most 2 over $\mathbb{R}$,

$$\langle p, q \rangle = \int_{-1}^{1} p(x)q(x)\, dx$$

is an inner product.

d) For the same space P^2 as in c),

$$\langle p, q\rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1)$$

is also an inner product.

e) Let $(X, \mathscr{M}, \mu)$ be a measure space and $L^2(\mu)$ be the space of all equivalent classes of measurable functions such that $\int f^2\, d\mu < \infty$ with the equivalence relation $f \sim g$ if $f = g$ a.e. Then

$$\langle f, g\rangle = \int fg\, d\mu$$

is an inner product.

■

We will show later that $\|x\| = \sqrt{\langle x, x\rangle}$ is a norm on H. We first prove the Schwarz inequality.

Theorem 5.3.2 (Schwarz Inequality)
For any x, y in an inner product space H

$$|\langle x, y\rangle| \le \|x\|\|y\|$$

with equality if and only if x and y are linearly dependent.

Proof
By the definition of inner product we have

$$\begin{aligned}\big\|\|y\|^2 x - \langle x, y\rangle y\big\|^2 &= \|y\|^4\|x\|^2 - 2\|y\|^2|\langle x, y\rangle|^2 + |\langle x, y\rangle|^2\|y\|^2 \\ &= \|y\|^2(\|y\|^2\|x\|^2 - |\langle x, y\rangle|^2).\end{aligned} \tag{5.3.1}$$

So $\|y\|^2\|x\|^2 - |\langle x, y\rangle|^2 \ge 0$, or $\|x\|\|y\| \ge |\langle x, y\rangle|$. Also the right-hand side of (5.3.1) is zero if and only if $\|y\|^2 x - \langle x, y\rangle y = 0$, i.e., if and only if x, y are linearly dependent. □

Proposition 5.3.3 *$\|x\| = \sqrt{\langle x, x\rangle}$ is a norm in H.*

Proof
We only need to show the triangle inequality. By the Schwarz inequality

$$\begin{aligned}\|x + y\|^2 = \langle x + y, x + y\rangle &= \|x\|^2 + 2\mathrm{Re}\langle x, y\rangle + \|y\|^2 \\ &\le \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2,\end{aligned}$$

i.e., $\|x + y\| \le \|x\| + \|y\|$. □

Definition 5.3.4

The norm $\|x\| = \sqrt{\langle x, x\rangle}$ is called the induced norm by the inner product.
A complete inner product space under the induced norm is called a Hilbert space.

Proposition 5.3.5 *If $x_n \to x$, $y_n \to y$, then $\langle x_n, y_n\rangle \to \langle x, y\rangle$.*

Proof

$$\begin{aligned}
|\langle x_n, y_n\rangle - \langle x, y\rangle| &= |\langle x_n, y_n\rangle - \langle x, y_n\rangle + \langle x, y_n\rangle - \langle x, y\rangle| \\
&\le |\langle x_n, y_n\rangle - \langle x, y_n\rangle| + |\langle x, y_n\rangle - \langle x, y\rangle| \\
&\le \|x_n - x\|\|y_n\| + \|x\|\|y_n - y\| \\
&\le \|x_n - x\|(\|y_n - y\| + \|y\|) + \|x\|\|y_n - y\| \\
&\to 0
\end{aligned}$$

as $n \to \infty$. □

Theorem 5.3.6 (The Parallelogram Law)
For all x, y in an inner product space H,

$$\left\|x + y\|^2 + \|x - y\right\|^2 = 2(\|x\|^2 + \|y\|^2).$$

Proof

$$\begin{aligned}
\|x + y\|^2 + \|x - y\|^2 &= \|x\|^2 + 2\mathrm{Re}\langle x, y\rangle + \|y\|^2 + \|x\|^2 - 2\mathrm{Re}\langle x, y\rangle + \|y\|^2 \\
&= 2(\|x\|^2 + \|y\|^2).
\end{aligned}$$

□

The reason it is called the parallelogram law is that in $\mathbb{R}^2$, $\|x+y\|$ and $\|x-y\|$ are the lengths of the diagonals of the parallelogram formed by x and y. So the parallelogram law says that the sum of the length squares of the sides of a parallelogram equals the sum of length squares of its diagonals.

Analogizing to the dot product in $\mathbb{R}^3$, we can define angle between two nonzero elements of an inner product space by

$$\cos\theta = \frac{\langle x, y\rangle}{\|x\|\|y\|}.$$

Therefore we have a concept of orthogonality.

Definition 5.3.7

Two nonzero vectors x, y in an inner product space H are called orthogonal, denoted by $x \perp y$, if $\langle x, y\rangle = 0$.
For any subset $E \subset H$, $E^{\perp}$ is defined by

$$E^{\perp} = \{x \in H : \langle x, y\rangle = 0 \text{ for all } y \in E\}.$$

Remark 5.4 $E^{\perp}$ is closed because

$$E^{\perp} = \bigcap_{y \in E} f_y^{-1}(\{0\})$$

where f_y is the bounded linear functional defined by $f_y(x) = \langle x, y\rangle$.

Theorem 5.3.8 (The Pythagorean Theorem)
If $\{x_1, \ldots, x_n\} \subset H$, and $x_i \perp x_j$ for all $i \neq j$, then

$$\left\| \sum_{i=1}^{n} x_i \right\|^2 = \sum_{i=1}^{n} \|x_i\|^2.$$

Proof

$$\left\| \sum_{i=1}^{n} x_i \right\|^2 = \left\langle \sum_{i=1}^{n} x_i, \sum_{i=1}^{n} x_i \right\rangle = \sum_{i,j=1}^{n} \langle x_i, x_j\rangle = \sum_{i=1}^{n} \langle x_i, x_i\rangle = \sum_{i=1}^{n} \|x_i\|^2.$$

□

Recall for two subspaces $M, N \subset X$,

$$M + N = \{x + y : x \in M, y \in N\}$$

and we write $M \oplus N$ for $M + N$ if $N \cap N = \{0\}$.

Theorem 5.3.9
Let M be a closed subspace of a Hilbert space H. Then

$$H = M \oplus M^{\perp}.$$

Proof

Clearly $M \cap M^{\perp} = \{0\}$. So we only need to show that any $x \in H$ can be written as $x = y + z$, for some $y \in M$ and $z \in M^{\perp}$.

For any $x \in H$, let

$$\delta = \rho(x, M) = \inf\{\|x - y\| : y \in M\}.$$

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Then there is a sequence $\{y_n\} \subset M$ such that $\|x - y_n\| \to \delta$. By the parallelogram law,

$$\begin{aligned} 2\|x - y_n\|^2 + 2\|x - y_m\|^2 &= \|2x - y_n - y_m\|^2 + \|y_n - y_m\|^2 \\ &= 4\left\|x - \frac{1}{2}(y_n + y_m)\right\|^2 + \|y_n - y_m\|^2, \end{aligned}$$

or

$$\begin{aligned} \|y_n - y_m\|^2 &= 2\|x - y_n\|^2 + 2\|x - y_m\|^2 - 4\left\|x - \frac{1}{2}(y_n + y_m)\right\|^2 \\ &\le 2\|x - y_n\|^2 + 2\|x - y_m\|^2 - 4\delta^2 \to 0 \end{aligned}$$

as $n, m \to \infty$. So $\{y_n\}$ is a Cauchy sequence. Let $y = \lim_{n\to\infty} y_n$ and $z = x - y$. Then $\|z\| = \delta$.

We show that $z \in M^{\perp}$. If not, there exists a $u \in M$ such that $\langle z, u\rangle \neq 0$. Let

$$y_0 = y + \frac{\langle z, u\rangle}{\|u\|^2} u \in M.$$

Then

$$\|x - y_0\|^2 = \left\|z - \frac{\langle z, u\rangle}{\|u\|^2} u\right\|^2 = \|z\|^2 - 2\frac{|\langle z, u\rangle|^2}{\|u\|^2} + \frac{|\langle z, u\rangle|^2}{\|u\|^2} = \delta^2 - \frac{|\langle z, u\rangle|^2}{\|u\|^2} < \delta^2,$$

which contradicts the definition of δ. So $z \in M^{\perp}$.

□

Theorem 5.3.10 (Riesz Representation Theorem)

Let H be a Hilbert space. For all $f \in H^*$, there is a unique $y \in H$ such that f is defined by

$$f(x) = \langle x, y\rangle.$$

Proof

If $f \equiv 0$, then take $y = 0$. Otherwise $M = f^{-1}(\{0\})$ is a proper closed subspace of H. Pick a $z \in M^{\perp}$ with $\|z\| = 1$ and define $y = \overline{f(z)}z$. For any $x \in H$, since $f(x)z - f(z)x \in M$

$$0 = \langle f(x)z - f(z)x, z\rangle = f(x) - \langle x, \overline{f(z)}z\rangle = f(x) - \langle x, y\rangle,$$

i.e., $f(x) = \langle x, y\rangle$.

If $f(x) = \langle x, y\rangle = \langle x, y_1\rangle$, then $x \mapsto \langle x, y - y_1\rangle$ is a zero functional. In particular $\langle y - y_1, y - y_1\rangle = 0$. So $y = y_1$ and such a y is unique. □

Definition 5.3.11

A subset $\{u_\alpha : \alpha \in A\}$ is called orthonormal if $\langle x_\alpha, x_\beta\rangle = 0$ for all $\alpha \neq \beta$, and $\|x_\alpha\| = 1$ for all α.

Remark 5.5 For a set $\{a_\alpha : \alpha \in A\}$ of positive real numbers, $\sum_{\alpha\in A} a_\alpha$ is defined as

$$\sum_{\alpha\in A} a_\alpha = \sup_{F\subset A \text{ is finite}} \sum_{\alpha\in F} a_\alpha.$$

Theorem 5.3.12 (Bessel's Inequality)

If $\{u_\alpha : \alpha \in A\}$ is an orthonormal set in an inner product space H, then for all $x \in H$,

$$\sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 \leq \|x\|^2.$$

In particular, $\{\alpha : \langle x, u_\alpha\rangle \neq 0\}$ is countable.

Proof

For any finite subset $F \subset A$

$$\begin{aligned}
0 &\leq \left\| x - \sum_{\alpha\in F} \langle x, u_\alpha\rangle \right\|^2 \\
&= \|x\|^2 - 2\mathrm{Re}\left\langle x, \sum_{\alpha\in F} \langle x, u_\alpha\rangle u_\alpha \right\rangle + \sum_{\alpha\in F} |\langle x, u_\alpha\rangle|^2 \\
&= \|x\|^2 - 2\sum_{\alpha\in F} |\langle x, u_\alpha\rangle|^2 + \sum_{\alpha\in F} |\langle x, u_\alpha\rangle|^2 \\
&= \|x\|^2 - \sum_{\alpha\in F} |\langle x, u_\alpha\rangle|^2.
\end{aligned}$$

For the second part, since $\sum_{\alpha\in A}|\langle x,u_\alpha\rangle|^2$ converges, $E_n = \left\{\alpha : |\langle x,u_\alpha\rangle| > \frac{1}{n}\right\}$ is finite. So $\{\alpha : \langle x,u_\alpha\rangle \neq 0\} = \bigcup_{n=1}^{\infty} E_n$ is countable. □

Theorem 5.3.13

Let $\{u_\alpha : \alpha \in A\}$ be an orthonormal set in a Hilbert space H. Then the following are equivalent.

a) ***(Completeness)*** *If $\langle x,u_\alpha\rangle = 0$ for all α, then $x = 0$.*

b) ***(Parseval's identity)*** $\sum_{\alpha\in A}|\langle x,u_\alpha\rangle|^2 = \|x\|^2$.

c) $x = \sum_{\alpha\in A}\langle x,u_\alpha\rangle u_\alpha$ *for any* $x \in H$.

Proof

a) ⇒ c):

For any $x \in H$, let $\{\langle x,u_{\alpha_n}\rangle : n = 1,2,\dots\}$ be all nonzero $\langle x,u_{\alpha_n}\rangle$'s.

By Bessel's inequality,

$$\left\|\sum_{i=n}^{m}\langle x,u_{\alpha_n}\rangle u_{\alpha_n}\right\|^2 = \sum_{i=n}^{m}|\langle x,u_{\alpha_n}\rangle|^2 \to 0$$

as $n,m \to \infty$. So $\left\{\sum_{i=1}^{n}\langle x,u_{\alpha_i}\rangle u_{\alpha_i} : n = 1,2,\dots\right\}$ is a Cauchy sequence, and $z = \sum_{i=1}^{\infty}\langle x,u_{\alpha_i}\rangle u_{\alpha_i}$ exists. Let $y = x - z$. Then $\langle y,u_\alpha\rangle = 0$ for all α. So $y = 0$, i.e.

$$x = z = \sum_{i=1}^{\infty}\langle x,u_{\alpha_i}\rangle u_{\alpha_i} = \sum_{\alpha\in A}\langle x,u_\alpha\rangle u_\alpha.$$

c) ⇒ b): Let $\{\alpha : \langle x,u_\alpha\rangle \neq 0\} = \{\alpha_1,\alpha_2,\dots\}$. Then

$$\|x\|^2 = \lim_{n\to\infty}\left\|\sum_{i=1}^{n}\langle x,u_{\alpha_i}\rangle u_{\alpha_i}\right\|^2 = \lim_{n\to\infty}\sum_{i=1}^{n}|\langle x,u_{\alpha_i}\rangle|^2 = \sum_{i=1}^{\infty}|\langle x,u_{\alpha_i}\rangle|^2 = \sum_{\alpha\in A}|\langle x,u_\alpha\rangle|^2.$$

b) ⇒ a):

Trivial. □

Definition 5.3.14

An orthonormal set in Theorem 5.3.13 is called an orthonormal basis of H.

Proposition 5.3.15 *Every Hilbert space has an orthonormal basis.*

Proof

Let $\mathscr{B}$ be the set of all orthonormal sets. Define a partial order on $\mathscr{B}$

$$F_1 \leq F_2 \quad \text{if and only if} \quad F_1 \subset F_2.$$

For a linearly ordered subset $\{F_\alpha : \alpha \in A\}$, define

$$F = \bigcup_{\alpha \in A} F_\alpha.$$

then F is an upper bound. So by Zorn's Lemma, $\mathscr{B}$ has a maximum element B.

Assume $B = \{u_\beta : \beta \in D\}$. Since B is maximum, if $\langle x, u_\beta \rangle = 0$ for all $\beta \in D$, we must have $x = 0$. So by Theorem 5.3.13, B is an orthonormal basis. □

Proposition 5.3.16 *A Hilbert space H is separable if and only if it has a countable orthonormal basis.*

Proof

Assume H is separable and $\{x_n : n = 1, 2, \dots\}$ is a countable dense subset. Inspecting x_n's one-by-one and discarding any x_n which is linearly dependent with $x_1, \dots, x_{n-1}$, we then have a linearly independent $\{y_n\}_1^\infty$ whose linear combinations are dense in H. By applying the Gram-Schmidt process we have an orthonormal $\{u_n\}_1^\infty$ whose span is dense in H. We claim that $\{u_n\}_1^\infty$ is an orthonormal basis. By Theorem 5.3.13, we only need to prove that if $\langle x, u_n \rangle = 0$ for all n, then $x = 0$. If not, then for any $z \in \text{span}\{u_n\}$, $\|x - z\| = \sqrt{\langle x - z, x - z \rangle} = \sqrt{\|x\|^2 + \|z\|^2} \geq \|x\|$, which contradicts the assumption that $\text{span}\{u_n\}$ is dense in H.

Conversely, if H has a countable orthonormal basis $\{u_n\}_1^\infty$, then the finite linear combination of $\{u_n\}$ with coefficients in $\mathbb{Q} + \mathbb{Q}i$ is a countable dense subset of H. So H is separable.

□

Exercises

5.21 Let H be an inner product space. Prove that $S^\perp = \overline{S}^\perp$ for any $S \subset H$.

5.22 Let M be a subspace of a Hilbert space H. Prove that

$$M^{\perp\perp} = \overline{M}.$$

5.23 Let H be a Hilbert space and $T \in L(H, H)$. Prove the following statements.

a. There is a unique $T^* \in L(H, H)$, called the adjoint of T, such that $\langle T(x), y \rangle = \langle x, T^*(y) \rangle$ for all $x, y \in H$, $T^{**} = T$, and $\|T^*\| = \|T\|$. (Hint: Theorem 5.3.10.)

b. $\text{range}(T)^\perp = \text{null}(T^*)$ and $\text{null}(T)^\perp = \overline{\text{range}(T^*)}$.

5.24 Let K be a closed convex set in a Hilbert space H. Prove that

a. any sequence whose norms approach to $\inf_{x\in K} \|x\|$ is a Cauchy sequence, and
b. K has a unique element of minimal norm.

(Hint: The parallelogram law.)

5.25 Let $(X, \mathscr{M}, \mu)$ be a measure space and $\{E_n\}$ be a partition of X. Prove that if $\{f_{nk} : k \in \mathbb{N}\}$ is an orthonormal basis for $L^2(E_n, \mu)$, $n = 1, 2, \ldots$, then $\{f_{nk} : n, k \in \mathbb{N}\}$ is an orthonormal basis for $L^2(\mu)$.

5.26 Let $\{f_m\}$ and $\{g_n\}$ be orthonormal bases for $L^2(\mu)$ and $L^2(\nu)$ over σ-finite measure spaces $(X, \mathscr{M}, \mu)$ and $(Y, \mathscr{N}, \nu)$, respectively. Prove that $\{h_{mn} = f_m(x)g_n(y)\}$ is an orthonormal basis for $L^2(\mu \times \nu)$.

5.27 Prove the following statements.
a. The set of all polynomials is dense in $L^2([0, 1], m)$. (Hint: Theorems 6.1.8 and 4.5.6.)
b. $L^2([0, 1], m)$ is separable. (Hint: Proposition 5.3.16.)
c. $L^2(\mathbb{R}, m)$ is separable. (See Exercise 5.25.)
d. $L^2(\mathbb{R}^n, m)$ is separable. (See Exercise 5.26.)

5.28 Let H be an infinite-dimensional Hilbert space. Prove the following statements.
a. Every orthonormal sequence in H converges weakly to 0.
b. For any $x \in H$ with $\|x\| < 1$, there is a sequence $\{u_n : n \in \mathbb{N}\}$ of unit vectors such that $u_n \to x$ weakly.

L^p Spaces

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6.1 L^p Spaces

Definition 6.1.1

Let $(X, \mathscr{M}, \mu)$ be a measure space and $p > 0$.

a) For any measurable function f, define

$$\|f\|_p = \left(\int |f|^p\right)^{\frac{1}{p}}.$$

b) The L^p space over $(X, \mathscr{M}, \mu)$ is defined as the set of all equivalence classes of measurable functions f such that $\|f\|_p < \infty$, where f and g are equivalent if $f = g$ a.e.

c) If $X = \mathbb{N}$, $\mathscr{M} = \mathscr{P}(\mathbb{N})$, and μ is the counting measure, then $L^p(\mathbb{N}, \mathscr{P}(\mathbb{N}), \mu)$ is denoted by l^p, i.e., l^p is the space of all sequences $\{a = (a_1, a_2, \dots) : a_i \in \mathbb{R}\}$ such that

$$\sum_{i=1}^{\infty} |a|^p < \infty$$

and $\|a\|_p$ is defined as

$$\|a\|_p = \left(\sum_{i=1}^{\infty} |a|^p\right)^{\frac{1}{p}}.$$

Clearly L^p is a vector space because of

$$|f + g|^p \le (|f| + |g|)^p \le (2\max(|f|, |g|))^p \le 2^p(|f|^p + |g|^p).$$

We will show that $\|f\|_p$ is a norm for all $1 \le p < \infty$. The only thing we need to show is the triangle inequality, which is given by the Minkowski inequality. We first prove an inequality.

Lemma 6.1.2 *For all* $0 < \lambda < 1$ *and for all* $t \ge 0$,

$$t^\lambda \le \lambda t + 1 - \lambda.$$

Proof

Consider $f(t) = t^\lambda - \lambda t$. Let $f'(t) = \lambda(t^{\lambda-1} - 1) = 0$, we have that $t = 1$ is the critical point. Since $f''(1) = \lambda(\lambda - 1) < 0$, $t = 1$ is the maximum point, and therefore

$$t^\lambda - \lambda t = f(t) \le f(1) = 1 - \lambda \quad \text{or} \quad t^\lambda \le \lambda t + 1 - \lambda \ \text{ for all } \ t \ge 0.$$

□

Theorem 6.1.3 (Hölder Inequality)
Let $1 < p < \infty$ *and* $\frac{1}{p} + \frac{1}{q} = 1$. *Then for any measurable functions* f, g,

$$\|fg\|_1 \le \|f\|_p \|g\|_q.$$

Proof

Without loss of generality, assume $0 < \|f\|_p < \infty$ and $0 < \|g\|_q < \infty$. Let $t = \frac{|f|^p \|g\|_q^q}{|g|^q \|f\|_p^p}$ and $\lambda = \frac{1}{p}$ in the inequality of Lemma 6.1.2, we have

$$\left(\left| \frac{f}{\|f\|_p} \right|^p \left| \frac{\|g\|_q}{g} \right|^q \right)^{\frac{1}{p}} \le \frac{1}{p} \left| \frac{f}{\|f\|_p} \right|^p \left| \frac{\|g\|_q}{g} \right|^q + \frac{1}{q}.$$

By multiplying both sides by $\left| \frac{g}{\|g\|_q} \right|^q$ and using $q - \frac{q}{p} = 1$, we have

$$\left| \frac{f}{\|f\|_p} \right| \left| \frac{g}{\|g\|_q} \right| \le \frac{1}{p} \left| \frac{f}{\|f\|_p} \right|^p + \frac{1}{q} \left| \frac{g}{\|g\|_q} \right|^q.$$

So

$$\frac{1}{\|f\|_p \|g\|_q} \int |fg| \, d\mu \le \frac{1}{p\|f\|_p^p} \int |f|^p \, d\mu + \frac{1}{q\|g\|_q^q} \int |g|^q \, d\mu = \frac{1}{p} + \frac{1}{q} = 1,$$

i.e., $\|fg\|_1 \le \|f\|_p \|g\|_q$. □

Definition 6.1.4

The positive real numbers $p, q \geq 1$ satisfying

$$\frac{1}{p} + \frac{1}{q} = 1,$$

with the convention that if $p = 1$ then $q = \infty$ and vice versa are called a conjugate pair.

Theorem 6.1.5 (Minkowski Inequality)

For any $1 \leq p < \infty$ and any $f, g \in L^p$,

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Proof

The inequality is clearly true if either $p = 1$ or $\|f + g\| = 0$.

When $p > 1$ and $\|f + g\|_p > 0$, since $|f + g|^p \leq (|f| + |g|)|f + g|^{p-1}$, by the Hölder inequality, we have

$$\begin{aligned}
\int |f + g|^p \, d\mu &\leq \int |f||f + g|^{p-1} \, d\mu + \int |g||f + g|^{p-1} \, d\mu \\
&\leq \|f\|_p \|(f + g)^{p-1}\|_q + \|g\|_p \|(f + g)^{p-1}\|_q \\
&= (\|f\|_p + \|g\|_q) \left(\int |f + g|^{(p-1)q} \, d\mu \right)^{\frac{1}{q}} \\
&= (\|f\|_p + \|g\|_q) \left(\int |f + g|^p \, d\mu \right)^{\frac{1}{q}}
\end{aligned}$$

i.e.

$$\|f + g\|_p = \left(\int |f + g|^p \, d\mu \right)^{1 - \frac{1}{q}} \leq \|f\|_p + \|g\|_p.$$

□

So for all $1 \leq p < \infty$, L^p is a normed space.

Theorem 6.1.6

For all $1 \leq p < \infty$, L^p is a Banach space.

Proof

We need to show that every absolutely convergent series in L^p converges. Let $\{f_k\} \subset L^p$ such that $\sum_{k=1}^{\infty} \|f_k\|_p < \infty$, and define

$$G_n = \sum_{k=1}^{n} |f_k|, \qquad G = \sum_{k=1}^{\infty} |f_k|.$$

Then by the Monotone Convergence Theorem

$$\lim_{n\to\infty} \int |G_n|^p \, d\mu = \int |G|^p \, d\mu.$$

So by the Minkowski inequality, we have

$$\|G\|_p = \lim_{n\to\infty} \|G_n\|_p \le \lim_{n\to\infty} \sum_{k=1}^{n} \|f_k\|_p < \infty.$$

Therefore $G \in L^p$, which means $\sum_{k=1}^{\infty} |f_k(x)| < \infty$ for a.e. x, so $\sum_{k=1}^{\infty} f_k$ converges a.e.

Let

$$F(x) = \begin{cases} \sum_{k=1}^{\infty} f_k(x), & \text{if } G(x) < \infty \\ 0, & \text{otherwise.} \end{cases}$$

Then since $|F| \le G$, we have $F \in L^p$, and

$$\left| F - \sum_{k=1}^{n} f_k \right|^p \le \left(|F| + \sum_{k=1}^{n} |f_k| \right)^p \le (2G)^p.$$

By the Dominated Convergence Theorem

$$\lim_{n\to\infty} \int \left| F - \sum_{k=1}^{n} f_k \right|^p \, d\mu = 0,$$

i.e., $\sum_{k=1}^{\infty} f_k$ converges to F in L^p. □

Proposition 6.1.7 *For any $1 \le p < \infty$, simple functions are dense in L^p.*

Proof

For any $f \in L^p$, there is a sequence of simple functions $\{\phi_n\}$ such that $|\phi_n| \le |f|$ (which means that $\{\phi_n\} \subset L^P$) and $\phi_n \to f$ as $n \to \infty$. Since $|\phi_n - f|^p \le 2^p |f|^p$ for all n and $2^p |f|^p \in L^1$, by Dominant Convergence Theorem

$$\lim_{n\to\infty}\int |\phi_n - f|^p \, d\mu = 0,$$

i.e., $\|\phi_n - f\|_p \to 0$ as $n \to \infty$. □

We have a result similar to Theorem 2.3.8 for L^p functions, $1 \le p < \infty$, over the n-dimensional Lebesgue measure m^n.

Theorem 6.1.8
Continuous function with compact support are dense in $L^p(m^n)$ for all $1 \le p < \infty$.

Proof
We first show that finite linear combinations of characteristic functions of cubes are dense in $1 \le p < \infty$. By Proposition 6.1.7, we only need to show that for any measurable $E \subset \mathbb{R}^n$ with $m^n(E) < \infty$, χ_E can be approximated by finite linear combinations of characteristic functions of cubes. By Proposition 2.7.4, for any $\epsilon > 0$, there exists an $A = \bigcup_{i=1}^n Q_i$, where $\{Q_i\}$ are cubes with disjoint interior, such that $m^n(E \triangle A) < \epsilon$. Since $\chi_A(x) = \sum_{i=1} \chi_{Q_i}(x)$ for every interior point Q_i, and the boundaries of cubes have Lebesgue measure 0, $\chi_A = \sum_{i=1} \chi_{Q_i}$ a.e. So (note that $|\chi_A - \chi_E|^p = \chi_{E\triangle A}$)

$$\left\| \sum_{i=1}^n \chi_{Q_i} - \chi_E \right\|_p^p = \|\chi_A - \chi_E\|_p^p = \int |\chi_A - \chi_E|^p \, dm^n = \int \chi_{E\triangle A} \, dm^n = m^n(E \triangle A) < \epsilon.$$

Therefore we only need to show that for any cube Q, χ_Q can be approximated by continuous functions with compact support. Without loss of generality (because Lebesgue measure is translation invariant), we can assume

$$Q = \{(x_1, \ldots, x_n) : 0 \le x_i \le h, i = 1, 2, \ldots, n\}.$$

For any $k \in \mathbb{N}$, define

$$U_k = \left\{(x_1, \ldots, x_n) : -\frac{1}{k} < x_i < h + \frac{1}{k}, i = 1, 2, \ldots, n\right\}.$$

Then by Urysohn's Lemma Theorem 4.1.10, there exists a continuous function f_k such that $f_k = 1$ on Q and $f_k = 0$ on U_k^c. Clearly $f_k \to \chi_Q$ as $k \to \infty$. So by the Dominated Convergence Theorem (what is the dominate function?)

$$\|f_k - \chi_Q\|_p^p = \int |f_k - \chi_Q|^p \, dm^n \to 0$$

as $k \to \infty$. □

Recall that the translation $\tau_y(f)$ is defined by $\tau_y(f)(x) = f(x - y)$.

Proposition 6.1.9 *Translation is uniformly continuous in $L^p(m^n)$ for any $1 \leq p < \infty$, i.e., for any $f \in L^p(m^n)$, for any z, and for any $\epsilon > 0$, there exists a $\delta > 0$, such that for any $y \in N_\delta(z)$*

$$\|\tau_y(f) - \tau_z(f)\|_p < \epsilon.$$

Proof

By Theorem 6.1.8, there is $g \in C_c(\mathbb{R}^n)$ such that $\|f - g\|_p < \frac{\epsilon}{3}$. Let

$$Q_h = \{x \in \mathbb{R}^n : |x_i| \leq h, \quad i = 1, \ldots, n\}.$$

Since g has compact support, there exists an h such that supp $(g) \subset Q_h$. Then g is uniformly continuous on Q_{2h}, so there is a δ, $0 < \delta < h$, such that

$$|g(y) - g(z)| < \frac{\epsilon}{3m^n(Q_{2h})^{\frac{1}{p}}}$$

for all z, $y \in Q_{2h}$ with $y \in N_\delta(z)$. The same is true for all z, $y \in \mathbb{R}^n$ with $y \in N_\delta(z)$ because if one of y or z is not in Q_{2h}, then both of them are in Q_h^c and $g(z) = g(y) = 0$. So for all $z \in \mathbb{R}^n$ and for all $y \in N_\delta(z)$

$$\begin{aligned}
\|\tau_y(f) - \tau_z(f)\|_p &= \|\tau_{y-z}(f) - f\|_p \\
&\leq \|\tau_{y-z}(f) - \tau_{y-z}(g)\|_p + |\tau_{y-z}(g) - g\|_p + \|g - f\|_p \\
&= 2\|g - f\|_p + \left(\int |g(x - (y - z)) - g(x)|^p \, dx\right)^{\frac{1}{p}} \\
&= 2\|g - f\|_p + \left(\int_{Q_{2h}} |g(x - (y - z)) - g(x)|^p \, dx\right)^{\frac{1}{p}} \\
&< 2\|g - f\|_p + \left(\left(\frac{\epsilon}{3m^n(Q_{2h})^{\frac{1}{p}}}\right)^p m^n(Q_{2h})\right)^{\frac{1}{p}} \\
&< \epsilon.
\end{aligned}$$

□

Next, we define $L^\infty(X, \mathscr{M}, \mu)$.

Definition 6.1.10

Let $(X, \mathscr{M}, \mu)$ be a measure space.

a) For any measurable function f, let

$$M(f) = \{\alpha \geq 0 : \mu(\{x : |f(x)| > \alpha\}) = 0\}, \tag{6.1.1}$$

and define the essential supremum, denoted by $\|f\|_\infty$, as

$$\|f\|_\infty = \operatorname{ess\,sup} f = \begin{cases} \inf M(f), & \text{if } M(f) \neq \emptyset, \\ \infty, & \text{if } M(f) = \emptyset. \end{cases}$$

b) The L^∞ space over $(X, \mathscr{M}, \mu)$ is the set of equivalence classes of measurable functions f such that $\|f\|_\infty < \infty$, where f and g are equivalent if $f = g$ a.e.

c) If $X = \mathbb{N}$, $\mathscr{M} = \mathscr{P}(\mathbb{N})$, and μ is the counting measure. Then the corresponding L^∞ is denoted by l^∞, i.e., l^∞ is the set of all bounded sequences $\{x = (x_1, x_2, \dots)\}$ and

$$\|x\|_\infty = \sup_n |x_n|.$$

From the definition, we have the following properties of essential supremum.

Lemma 6.1.11 *Let $f \in L^\infty$.*

a) If $\alpha \in M(f)$ then $[\alpha, \infty) \subset M(f)$.

b) $\|f\|_\infty \in M(f)$.

c)

$$\mu(\{x : |f(x)| > \alpha\}) = 0$$

if and only if $\alpha \geq \|f\|_\infty$, and

$$\mu(\{x : |f(x)| > \alpha\}) > 0$$

if and only if $\alpha < \|f\|_\infty$.

d) $|f| \leq \|f\|_\infty$ a.e.

Proof

a) For any $\beta \in [\alpha, \infty)$, $\{x : |f(x)| > \beta\} \subset \{x : |f(x)| > \alpha\}$

b) Let $\alpha_1 > \alpha_2 > \cdots$ in $M(f)$ and $\alpha_n \to \|f\|_\infty$ as $n \to \infty$. Then by the continuity from below of μ

$$\mu(\{x : |f(x)| > \|f\|_\infty\}) = \lim_{n\to\infty} \mu(\{x : |f(x)| > \alpha_n\}) = 0.$$

c) By a) and b), if $\alpha \geq \|f\|_\infty$, then $\alpha \in M(f)$, and by the definition of $\|f\|_\infty$, if $\alpha < \|f\|_\infty$, then $\alpha \notin M(f)$.
d) Because of b).

□

Theorem 6.1.12

For any measurable functions f, g,

a) $\|fg\|_1 \leq \|f\|_1 \|g\|_\infty$;
b) $\|\cdot\|_\infty$ is a norm;
c) $f_n \to f$ in L^∞ if and only if $f_n \to f$ uniformly a.e.;
d) L^∞ is a Banach space;
e) Simple functions are dense in L^∞.

Proof

a) By Lemma 6.1.11 d), $|fg| \leq \|g\|_\infty |f|$ a.e. So

$$\|fg\|_1 = \int |fg|\, d\mu \leq \|g\|_\infty \int |f|\, d\mu = \|g\|_\infty \|f\|_1.$$

b) We only need to prove the triangle inequality. Since $|f+g| \leq |f|+|g| \leq \|f\|_\infty + \|g\|_\infty$ a.e., by Lemma 6.1.11 c), $\|f\|_\infty + \|g\|_\infty \geq \|f+g\|_\infty$.
c) Assume $\|f_n - f\|_\infty \to 0$ as $n \to \infty$. Let

$$E_n = \{x : |f_n(x) - f(x)| > \|f_n - f\|_\infty\}, \quad E = \bigcup_{n=1}^{\infty} E_n.$$

Then by Lemma 6.1.11 c), $\mu(E_n) = 0$ and hence $\mu(E) = 0$. For all $x \in E^c$, $|f_n(x) - f((x)| \leq \|f_n - f\|_\infty$ for all n. So $f_n \to f$ uniformly on E^c.

Conversely, if $f_n \to f$ uniformly on E^c for an E with $\mu(E) = 0$, then for any $\epsilon > 0$, there is an N such that $|f_n - f| < \epsilon$ a.e. for all $n \geq N$. By Lemma 6.1.11 c), $\|f_n - f\|_\infty \leq \epsilon$ for all $n \geq N$.
d) Consider a series $\sum_{n=1}^\infty f_n$ in L^∞ such that $\sum_{n=1}^\infty \|f_n\|_\infty < \infty$. We need to show $\sum_{n=1}^\infty f_n$ converges in L^∞. Let

$$E_n = \{x : |f_n(x)| > \|f_n\|_\infty\}, \quad E = \bigcup_{n=1}^{\infty} E_n.$$

Then by Lemma 6.1.11 c), $\mu(E_n) = 0$ and hence $\mu(E) = 0$. So

$$\sum_{n=1}^{\infty} |f_n(x)| \leq \sum_{n=1}^{\infty} \|f_n\|_\infty < \infty$$

for all $x \in E^c$, which means that $\sum_{n=1}^{\infty} f_n(x)$ converges for all $x \in E^c$. Define

$$f(x) = \begin{cases} \sum_{n=1}^{\infty} f_n(x), & x \in E^c \\ 0, & x \in E. \end{cases}$$

Then

$$\left| f(x) - \sum_{n=1}^{m} f_n(x) \right| = \left| \sum_{n=m+1}^{\infty} f_n(x) \right| \le \sum_{n=m+1}^{\infty} |f_n(x)| \le \sum_{n=m+1}^{\infty} \|f_n\|_\infty \quad \text{a.e.}$$

By Lemma 6.1.11 c)

$$\left\| f - \sum_{n=1}^{m} f_n \right\|_\infty \le \sum_{n=m+1}^{\infty} \|f_n\|_\infty.$$

Let $m \to \infty$, we have $\sum_{n=m+1}^{\infty} \|f_n\|_\infty \to 0$, so $\left\| f - \sum_{n=1}^{m} f_n \right\|_\infty \to 0$, i.e., $\sum_{n=1}^{m} f_n \to f$ in L^∞. By Theorem 5.1.5, L^∞ is a Banach space.

e) Let $f \in L^\infty$ and define

$$E = \{x : |f(x)| > \|f\|_\infty\}.$$

Then $\mu(E) = 0$ by Lemma 6.1.11 b). By Theorem 2.2.4, there is a sequence of simple functions $\{\phi_n\}$ which converges to f uniformly on E^c (because f is bounded on E^c). Then by c), $\|\phi_n - f\|_\infty \to 0$.

□

We can also define L^p space for $0 < p < 1$ as the space of all equivalent classes of all measurable functions such that

$$\|f\|_p = \left(\int |f|^p \, d\mu \right)^{\frac{1}{p}} < \infty.$$

However, in these cases, L^p's are not normed spaces as we can see that for $(\mathbb{R}, \mathscr{L}, m)$

$$\left\| \chi_{[0,\frac{1}{2}]} + \chi_{[\frac{1}{2},1]} \right\|_p = 1 > \left(\frac{1}{2}\right)^{\frac{1}{p}} + \left(\frac{1}{2}\right)^{\frac{1}{p}} = \left\| \chi_{[0,\frac{1}{2}]} \right\|_p + \left\| \chi_{[\frac{1}{2},1]} \right\|_p.$$

Nevertheless, we have the following results.

Proposition 6.1.13 *If $0 < p < q < r \le \infty$, then*

$$L^q \subset L^p + L^r,$$

i.e., any $f \in L^q$ can be written as

$$f = g + h, \quad g \in L^p, \ \ h \in L^r.$$

Proof
For any $f \in L^q$, let

$$E = \{x : |f(x)| > 1\}, \quad g = f\chi_E, \ \ h = f\chi_{E^c}.$$

Then $f = g + h$. Since

$$|g|^p = |f|^p \chi_E \le |f|^q \chi_E, \quad |h|^r = |f|^r \chi_{E^c} \le |f|^q \chi_{E^c},$$

we have

$$\int |g|^p \, d\mu \le \int |f|^q \chi_E \, d\mu \le \int |f|^q \, d\mu < \infty$$

and

$$\int |h|^r \, d\mu \le \int |f|^q \chi_{E^c} \, d\mu \le \int |f|^q \, d\mu < \infty,$$

i.e., $g \in L^p$ and $h \in L^r$. □

Proposition 6.1.14 *If $0 < p < q < r \le \infty$, then $\|f\|_q \le \|f\|_p^{\lambda} \|f\|_r^{1-\lambda}$, where*

$$\lambda = \begin{cases} \frac{p(r-q)}{q(r-p)}, & \text{if } r < \infty, \\ \\ \frac{p}{q}, & \text{if } r = \infty. \end{cases}$$

So $L^p \cap L^r \subset L^q$

Proof
If $r = \infty$, then by the Hölder inequality Theorem 6.1.12 a)

$$\begin{aligned} \|f\|_q &= \left(\int \left(|f|^p\right)^{\frac{q}{p}} \right)^{\frac{1}{q}} = \left(\int |f|^p \left(|f|^p\right)^{\frac{q}{p}-1} \right)^{\frac{1}{q}} \\ &\le \left(\|f\|_\infty^{q-p} \|f^p\|_1 \right)^{\frac{1}{q}} = \|f\|_\infty^{1-\frac{p}{q}} \|f\|_p^{\frac{p}{q}} = \|f\|_\infty^{1-\lambda} \|f\|_p^{\lambda}. \end{aligned}$$

If $r < \infty$, then from the definition of λ we derive

$$\frac{1}{\frac{p}{\lambda q}} + \frac{1}{\frac{r}{(1-\lambda)q}} = 1.$$

Then by the Hölder inequality

$$\int |f|^q \, d\mu = \int |f|^{\lambda q} |f|^{(1-\lambda)q} \, d\mu \le \left\| f^{\lambda q} \right\|_{\frac{p}{\lambda q}} \left\| f^{(1-\lambda)q} \right\|_{\frac{r}{(1-\lambda)q}}$$
$$= \left(\int |f|^p \, d\mu \right)^{\frac{\lambda q}{p}} \left(\int |f|^r \, d\mu \right)^{\frac{(1-\lambda)q}{r}}$$

Therefore

$$\|f\|_r \le \|f\|_p^{\lambda} \|f\|_r^{1-\lambda}.$$

□

Proposition 6.1.15 *Let $(\mathbb{N}, \mathscr{P}(\mathbb{N}), \mu)$ be the counting measure space, and write each function x on $\mathbb{N}$ as $x = (x_1, x_2, \dots)$, where $x_n = x(n)$. If $1 < p < q \le \infty$, then $\|x\|_q \le \|x\|_p$. So $l^p \subset l^q$.*

Proof

For counting measure $\mu(E) = 0$ if and only if $E = \emptyset$. Therefore for any $x = (x_1, x_2, \dots)$, $\|x\|_\infty = \sup_i |x_i|$. So for $q = \infty$

$$\|x\|_\infty^p = \sup_i |x_i|^p \le \sum_{i=1}^{\infty} |x_i|^p = \|x\|_p^p,$$

i.e., $\|x\|_\infty \le \|x\|_p$.

For $q < \infty$, let $r = \infty$ in Proposition 6.1.14

$$\|x\|_q \le \|x\|_p^{\lambda} \|x\|_\infty^{1-\lambda} \le \|x\|_p^{\lambda} \|x\|_p^{1-\lambda} = \|x\|_p.$$

□

Proposition 6.1.16 *If $\mu(X) < \infty$ and $0 < p < q \le \infty$, then $\|f\|_p \le \|f\|_q \mu(X)^{\frac{q-p}{pq}}$ with the convention that $\frac{q-p}{pq} = \frac{1}{p}$ if $q = \infty$. So $L_q \subset L_p$.*

Proof

If $q = \infty$, then

$$\|f\|_p^p = \int |f|^p \, d\mu \le \|f\|_\infty^p \int d\mu = \|f\|_\infty^p \mu(X),$$

i.e., $\|f\|_p \le \|f\|_\infty \mu(X)^{\frac{1}{p}}$.

If $q < \infty$, then

$$\frac{1}{\frac{q}{p}} + \frac{1}{\frac{q}{q-p}} = 1.$$

So by Hölder inequality

$$\int |f|^p \, d\mu \le \|f^p\|_{\frac{q}{p}} \|1\|_{\frac{q}{q-p}} = \|f\|_q^p \mu(X)^{\frac{q-p}{q}}.$$

So $\|f\|_p \le \|f\|_q \mu(X)^{\frac{q-p}{pq}}$. □

Exercises

6.1 Let $1 \le p < q < r \le \infty$. Prove that $L^p \cap L^r$ is a Banach space with norm

$$\|f\| = \max(\|f\|_p, \|f\|_r)$$

and the inclusion map $L^p \cap L^r \to L^q$ is continuous. (See Proposition 6.1.14.)

6.2 Let $1 \le p < q < r \le \infty$. Prove that $L^p + L^r$ is a Banach space with norm

$$\|f\| = \inf\{\max(\|g\|_p, \|h\|_r) : f = g + h\}$$

and the inclusion map $L^q \to L^p + L^r$ is continuous. (See Proposition 6.1.13.)

6.3 Let (X, M, μ) be a measure space, and $1 < p, q < \infty$ such that $\frac{1}{p} + \frac{1}{q} \le 1$. Prove that $fg \in L^{\frac{pq}{p+q}}$ for any $f \in L^p$ and $g \in L^q$.

6.4 Let $1 \le p < q \le \infty$. Prove that $L^p \not\subset L^q$ if and only if

$$\inf\{\mu(E) : E \in \mathscr{M}, \mu(E) > 0\} = 0.$$

(Hint: "if" part: find disjoint $\{E_n\}$ such that $\mu(E_n) < \frac{1}{2^n}$ and construct $f \in L^p$ but $f \notin L^q$.)

6.5 Let $1 \le p < q < \infty$. Prove that $L^q \not\subset L^p$ if and only if

$$\sup\{\mu(E) : E \in \mathscr{M}, \mu(E) < \infty\} = \infty.$$

(Hint: "if" part: find disjoint $\{E_n\}$ such that $\mu(E_n) > 1$ and construct $f \in L^q$ but $f \notin L^p$.)

6.6 Let $f \in L^p \cap L^\infty$. Prove that

a. $f \in L^q$ for all $q \ge p$,

b. for any $\epsilon > 0$, the set E_ϵ defined by

$$E_\epsilon = \{x : |f(x)| > \|f\|_\infty - \epsilon\}$$

has the properties that $\mu(E_\epsilon) > 0$ and

$$(\|f\|_\infty - \epsilon)^q \chi_{E_\epsilon} \le |f|^q \le |f|^p \|f\|_\infty^{q-p} \text{ a.e.},$$

c. $\|f\|_\infty = \lim_{q\to\infty} \|f\|_q$.

6.7 Let $1 \le p < \infty$. Prove that
a. if $f_n \to f$ in L^p, then $f_n \to f$ in measure, and
b. if $f_n \to f$ in measure and there exists a $g \in L^p$ such that $|f_n| \le g$ for all n, then $f_n \to f$ in L^p.

6.8 Let $1 \le p < \infty$, $f_n, f \in L^p$, and $f_n \to f$ a.e. prove that $f_n \to f$ in L^p if and only if $\|f_n\|_p \to \|f\|_p$. (Hint: The generalized dominated convergence theorem.)

6.9 Prove that if $\dim(L^p) > 1$, then L^p norm is a norm induced by an inner product if and only if $p = 2$. (Hint: The parallelogram law.)

6.10 Let $1 < p, q < \infty$, $\frac{1}{p} + \frac{1}{q} = 1$, and $k(x,t) \in L^q(\mathbb{R}^2, m^2)$. Prove that for any $f(t) \in L^p(\mathbb{R}, m(t))$, $k(x,t)f(t) \in L^1(\mathbb{R}, m(t))$ for a.e. x, and

$$T(f)(x) = \int k(x,t)f(t)\,dt$$

is a bounded linear operator from $L^p(\mathbb{R}, m)$ to $L^q(\mathbb{R}, m)$ with $\|T\| \le \|k\|_q$.

6.11 Show that $L^p(\mathbb{R}^n, m^n)$ is separable for all $1 \le p < \infty$, and $L^\infty(\mathbb{R}^n, m^n)$ is not separable.

6.12 Let $(X, \mathscr{M}, \mu)$ be a σ-finite measure space and $g \in L^\infty(\mu)$. Prove that the operator defined by $T(f) = fg$ is a bounded linear operator on $L^p(\mu)$ for all $1 \le p \le \infty$, and $\|T\| = \|g\|_\infty$.

6.2 The Dual of L^p

Consider positive real numbers $1 \le p, q \le \infty$ such that

$$\frac{1}{p} + \frac{1}{q} = 1.$$

In this section we will prove the general Riesz Representation theorem, which says that the space of all bounded linear functionals on L^p is L^q (strictly speaking, it is isometrically isomorphic to L^q). The result is also true for $p = 1$ and $q = \infty$.

First of all, by using the Hölder inequality, one can see that every $g \in L^q$ defines a bounded linear functional $\phi_g : L^p \to \mathbb{C}$ by

$$\phi_g(f) = \int fg\,d\mu.$$

ϕ_g is bounded because

$$|\phi_g(f)| \leq \int |fg|\, d\mu \leq \|f\|_p \|g\|_q$$

which means that

$$\|\phi_g\| \leq \|g\|_q. \tag{6.2.2}$$

Actually they are equal.

Proposition 6.2.1 *If* $\frac{1}{p} + \frac{1}{q} = 1$, $1 \leq q < \infty$, *then* $\|\phi_g\| = \|g\|_q$, *and when* μ *is semifinite, the result also holds for* $q = \infty$.

Proof

Because of (6.2.2), we only need to either find one $f \in L^p$ such that

$$\left| \int fg\, d\mu \right| = \|f\|_p \|g\|_q$$

or show that for any $\epsilon > 0$, there is a $f \in L^p$ such that

$$\left| \int fg\, d\mu \right| \geq \|f\|_p (\|g\|_q - \epsilon).$$

For $1 \leq q < \infty$, let

$$f = \frac{\text{sign}\,(g)}{\|g\|_q^{q-1}} |g|^{q-1}.$$

It can be easily checked that $\|f\|_p = 1$, so $f \in L^p$. Then

$$\left| \int fg\, d\mu \right| = \frac{1}{\|g\|_q^{q-1}} \int |g|^q\, d\mu = \|g\|_q = \|f\|_p \|g\|_q.$$

For $q = \infty$ and semifinite μ, and any $\epsilon > 0$, let

$$A = \{x : |g(x)| > \|g\|_\infty - \epsilon\}.$$

Then by Lemma 6.1.11 c), $\mu(A) > 0$. Since μ is semifinite, there is a measurable $B \subset A$ such that $0 < \mu(B) < \infty$. Let

$$f = \frac{\text{sign}\,(g)}{\mu(B)} \chi_B.$$

Then $\|f\|_1 = 1$ and

$$\left|\int fg\, d\mu\right| = \frac{1}{\mu(B)}\int_B |g|\, d\mu \geq (\|g\|_\infty - \epsilon)\frac{1}{\mu(B)}\int_B d\mu$$
$$= \|g\|_\infty - \epsilon = \|f\|_1(\|g\|_\infty - \epsilon).$$

□

Next we prove that if the ϕ_g is a bounded linear functional on L^p-simple functions, then $g \in L^q$.

Theorem 6.2.2
Let $\frac{1}{p} + \frac{1}{q} = 1$, $1 \leq q$, and Σ be the set of all bounded simple functions that vanish outside of a set of finite measure (so $\Sigma \subset L^r$ for all $r > 0$). If for a measurable function g, the set

$$S_g = \{x : g(x) \neq 0\}$$

is σ-finite and the linear functional

$$\phi_g(f) := \int fg\, d\mu$$

is bounded on Σ as a subspace of L^p, then $g \in L^q$ and $\|\phi_g\| = \|g\|_q$.

Proof
Case 1: $q < \infty$.

Since S_q is σ-finite, there are measurable sets $E_1 \subset E_2 \subset \cdots$ such that $\mu(E_n) < \infty$ for all n and $\bigcup_{n=1}^{\infty} E_n = S_g$. Let $\{\psi_n\}$ be a sequence of bounded simple functions such that $|\psi_n| \leq |g|$ and $\psi_n \to g$ pointwise as $n \to \infty$. Define

$$g_n = \psi_n \chi_{E_n}.$$

Then $|g_n| \leq |g|$, $g_n \in \Sigma$, and $g_n \to g$ pointwise as $n \to \infty$. Let

$$f_n = \frac{\text{sign}\,(g_n)}{\|g_n\|_q^{q-1}}|g|^{q-1}.$$

Then f_n is a simple function that vanishes outside of a set of finite measure. So

$$\left|\int f_n g\, d\mu\right| = |\phi_g(f_n)| \leq \|\phi_g\|\|f_n\|_p.$$

Note that for any n,

$$\|f_n\|_p = \frac{1}{\|g_n\|_q^{q-1}} \left(\int |g|^{qp-p} \, d\mu \right)^{1/p} = \frac{1}{\|g_n\|_q^{q-1}} \left(\int |g|^q \, d\mu \right)^{(1/q)(q/p)}$$
$$= \frac{\|g_n\|_q^{q/p}}{\|g_n\|_q^{q-1}} = 1.$$

By the Fatou's lemma

$$\|g\|_q \le \liminf_{n\to\infty} \|g_n\|_q = \liminf_{n\to\infty} \int |f_n g_n| \, d\mu$$
$$\le \liminf_{n\to\infty} \int |f_n g| \, d\mu = \liminf_{n\to\infty} \int f_n g \, d\mu$$
$$\le \liminf_{n\to\infty} \|\phi_g\| \|f_n\|_p = \|\phi_g\|.$$

So $g \in L^q$.

Case 2: $q = \infty$.

For any $\epsilon > 0$, let

$$A = \{x : |g(x)| \ge \|\phi_g\|_\infty + \epsilon\}.$$

Note that $A \subset S_g$. If $\mu(A) > 0$, then there is a measurable $B \subset A$ such that $0 < \mu(B) < \infty$. Let

$$f = \frac{\text{sign}\,(g)}{\mu(B)} \chi_B.$$

Then $\|f\|_1 = 1$ and

$$\|\phi_g\| = \|\phi_g\| \|f\|_1 \ge \left| \int fg \, d\mu \right| = \frac{1}{\mu(B)} \int_B |g| \, d\mu \ge \|\phi_g\| + \epsilon,$$

a contradiction. So $\mu(A) = 0$ for all ϵ. By the definition formula (6.1.1), $\|\phi_g\|_\infty + \epsilon \in M(g)$ for all $\epsilon > 0$. So $\|g\|_\infty = \inf M(g) \le \|\phi_g\|$ and $g \in L^\infty$.

Also for both cases and for any $f \in \Sigma$, by the Hölder inequality,

$$|\phi_g(f)| = \left| \int fg \, d\mu \right| \le \int |fg| \, d\mu \le \|f\|_p \|g\|_q,$$

which means that $\|\phi_g\| \le \|g\|_q$. So $\|\phi_g\| = \|g\|_q$. □

Since we need to use following technique in multiple places in the proof of Riesz Representation Theorem, we list it as a lemma.

Lemma 6.2.3 *Let $\frac{1}{p}+\frac{1}{q}=1$, $1 \le p < \infty$, $g \in L^q$, $\phi \in (L^p)^*$, and $S \subset L^p$ be a subset of L^p such that for every $f \in L^p$, there exists a sequence $\{\psi_n\} \subset S$ with the properties that $|\psi_n| \le |f|$ for all n and $\psi_n \to f$ a.e. as $n \to \infty$. If $\phi(\psi) = \int \psi g \, d\mu$ for all $\psi \in S$, then $\phi(f) = \int fg \, d\mu$ for all $f \in L^p$.*

Proof

For any $f \in L^p$, let $\{\psi_n\} \subset S$ such that $|\psi_n| \le |f|$ for all n and $\psi_n \to f$ a.e. as $n \to \infty$. Since $|\psi_n - f|^p \le |2f|^p$, by the Dominated Convergence Theorem

$$\int |\psi_n - f|^p \, d\mu \to 0 \quad \text{as } n \to \infty.$$

So $\|\psi_n - f\|_p \to 0$, and because ϕ is continuous on L^p, $\phi(\psi_n) \to \phi(f)$, as $n \to \infty$. By the Hölder inequality, $\int |fg| \, d\mu \le \|f\|_p \|g\|_q < \infty$. Then by the Dominated Convergence Theorem again,

$$\int \psi_n g \, d\mu \to \int fg \, d\mu \quad \text{as } n \to \infty.$$

Therefore

$$\phi(f) = \lim_{n\to\infty} \phi(\psi_n) = \lim_{n\to\infty} \int \psi_n g \, d\mu = \int fg \, d\mu.$$

□

Theorem 6.2.4 (Riesz Representation Theorem)
Let $\frac{1}{p}+\frac{1}{q}=1$, $1 < p < \infty$. For all $\phi \in (L^p)^*$, $\phi = \phi_g$ for some $g \in L^q$. If μ is σ-finite, the conclusion is also true for $p = 1$.

Proof

Case 1: μ is finite and $1 \le p < \infty$.

In this case, every finite simple function is in L^p. Let $\phi \in (L^p)^*$. For $E \in \mathcal{M}$ define $\nu(E) = \phi(\chi_E)$. We claim that ν is a signed measure. We only need to prove the countable additivity. For any disjoint $\{E_i\} \subset \mathcal{M}$, let $E = \bigcup_{i=1}^{\infty} E_i$. Then since $\chi_E - \sum_{i=1}^n \chi_{E_i} \to 0$ as $n \to \infty$ and $\left|\chi_E - \sum_{i=1}^n \chi_{E_i}\right|^p \le 1$, by the Dominated Convergence Theorem, $\left\|\chi_E - \sum_{i=1}^n \chi_{E_i}\right\|_p \to 0$ as $n \to \infty$, i.e., $\sum_{i=1}^n \chi_{E_i} \to \chi_E$ in L_p. Since ϕ is a bounded linear functional on L^p,

$$\sum_{i=1}^{\infty} \nu(E_i) = \lim_{n\to\infty} \sum_{i=1}^{n} \phi(\chi_{E_i}) = \lim_{n\to\infty} \phi\left(\sum_{i=1}^{n} \chi_{E_i}\right) = \phi(\chi_E) = \nu(E).$$

If $\mu(E) = 0$, then $\chi_E = 0$ μ-a.e., i.e., $\chi_E = 0$ in L^p. So $\nu(E) = \phi(\chi_E) = 0$, and hence $\nu \ll \mu$. Let $g = \frac{d\nu}{d\mu}$. Then for any measurable simple function $\psi = \sum_{i=1}^k a_i \chi_{E_i}$,

$$\phi(\psi) = \sum_{i=1}^k a_i \phi(\chi_{E_i}) = \sum_{i=1}^k a_i \nu(E_i) = \sum_{i=1}^k a_i \int \chi_{E_i} g \, d\mu = \int \psi g \, d\mu.$$

By Theorem 6.2.2, $g \in L^q$. Then by Lemma 6.2.3, $\phi(f) = \int fg \, d\mu$ for any $f \in L^p$.

Case 2: μ is σ-finite and $1 \le p < \infty$.

Let $E_1 \subset E_2 \subset \cdots$ be measurable sets such that $\mu(E_i) < \infty$ and $X = \bigcup_{i=1}^\infty E_i$. Then by case 1, there are $g_i \in L^q(E_i)$ such that $\phi(f) = \int fg \, d\mu$ for all $f \in L^p(E_i)$ and

$$\|g_n\|_q = \left\| \phi \Big|_{L^p(E_i)} \right\| \le \|\phi\|.$$

Also by the uniqueness of the Radon-Nikodym derivative, $g_i = g_j$ μ-a.e. on E_i if $i < j$. Define $g = g_1$ on E_1 and $g = g_i$ on $E_i \setminus E_{i-1}$ for $i = 2, 3, \ldots$. Then by the Monotone Convergence Theorem

$$\|g\|_q = \lim_{i \to \infty} \|g_n\|_q \le \|\phi\|.$$

So $g \in L^q$. Then for any $f \in L^p$, $\{\psi_i = f\chi_{E_i}\}$ satisfy the condition of Lemma 6.2.3. So $\phi(f) = \int fg \, d\mu$ for any $f \in L^p$.

Case 3: μ is arbitrary and $1 < p < \infty$.

We have $1 < q < \infty$. For any σ-finite subset $E \subset X$, by case 2, there is a $g_E \in L^q(E)$ such that $\phi(f) = \int f g_E \, d\mu$ for all $f \in L^p(E)$. Without loss of generality, we can assume $g_E = 0$ on E^c

Let

$$\alpha = \sup\{\|g_E\|_q : E \subset X \text{ is } \sigma\text{-finite}\}.$$

Then there are $\{E_n\}$ such that

$$\alpha \ge \|g_{E_n}\|_q > \alpha - \frac{1}{n}.$$

Let

$$F = \bigcup_{n=1}^\infty E_n.$$

Then F is σ-finite, and $\|g_F\|_q = \alpha$.

Define

$$g = \begin{cases} g_F, & \text{if } x \in F, \\ 0, & \text{if } x \in F^c. \end{cases}$$

For any $f \in L^p$, the set $A = F \cup \{x : f(x) \neq 0\}$ is σ-finite. So

$$\begin{aligned} \alpha^q \geq \int |g_A|^q \, d\mu &= \int_F |g_A|^q \, d\mu + \int_{A \setminus F} |g_A|^q \, d\mu \\ &= \int_F |g_F|^q \, d\mu + \int_{A \setminus F} |g_A|^q \, d\mu = \alpha^q + \int_{A \setminus F} |g_A|^q \, d\mu, \end{aligned}$$

which means that $g_A = 0$ a.e. on $A \setminus F$. Therefore $g_A = g$ a.e., and since $f\chi_{A^c} \equiv 0$,

$$\phi(f) = \phi(f\chi_A) + \phi(f\chi_{A^c}) = \phi(f\chi_A) = \int f g_A \, d\mu = \int fg \, d\mu.$$

□

Exercises

6.13 Let $(X, \mathscr{M}, \mu)$ be a measure space, and $1 < p, q < \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$. Prove that for any bounded linear operator T on $L^p(\mu)$, there is a bounded linear operator T^* on $L^q(\mu)$ with $\|T^*\| = \|T\|$ such that

$$\int T(f)g \, d\mu = \int f T^*(g) \, d\mu$$

for any $f \in L^p(\mu)$ and $g \in L^q(\mu)$.

6.14 Let $(X, \mathscr{M}, \mu)$ be a semifinite measure space, $1 \leq q < \infty$, $\frac{1}{p} + \frac{1}{q} = 1$, g be measurable such that $\phi g \in L^1$ for every simple function $\phi \in L^p$. Prove that if

$$M = \sup \left\{ \left| \int \phi g \, d\mu \right| : \phi \in L^p \text{ is simple and } \|f\|_p = 1 \right\} < \infty,$$

then the set $\{x : g(x) \neq 0\}$ is σ-finite.

6.15 Suppose $\sup_n \|f_n\|_p < \infty$ and $f_n \to f$ a.e.

a. Prove that if $1 < p < \infty$, then $f_n \to f$ weakly in L^p. (Hint: Exercise 2.23 a., Theorem 2.5.5, Corollary 3.1.10.)

b. Find a counterexample to show that a. is not true for $p = 1$.

6.16 Let $1 < p < \infty$. Prove $f_n \to f$ weakly in l^p if and only if $\sup_n \|f_n\|_p < \infty$ and $f_n \to f$ pointwise.

6.17 a. Show that $\sin nx \to 0$ weakly in $L^2([0, 2\pi], m)$, but not a.e. or in measure. (Hint: Exercise 5.28.)

b. Show that $n\chi_{(0,1/n)} \to 0$ a.e. and in measure, but not weakly in $L^p([0, 1], m)$ for any $1 \le p \le \infty$.

6.3 Some Useful Inequalities

Theorem 6.3.1 (Chebychev's Inequality)
If $f \in L^p$, $< p < \infty$, then for any $\alpha > 0$

$$\mu(\{x : |f(x)| \ge \alpha\}) \le \left(\frac{\|f\|_p}{\alpha}\right)^p.$$

Proof
Let $E_\alpha = \{x : |f(x)| \ge \alpha\}$. Then

$$\|f\|_p^p = \int |f|^p \, d\mu \ge \int_{E_\alpha} |f|^p \, d\mu \ge \alpha^p \mu(E_\alpha),$$

i.e., $\mu(E_\alpha) \le \left(\frac{\|f\|_p}{\alpha}\right)^p$. □

Theorem 6.3.2
Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite measure spaces and $1 \le p \le \infty$, and let $k(x, y)$ be an $\mathcal{M} \otimes \mathcal{N}$-measurable function on $X \times Y$. Suppose that there exists $c > 0$ such that

$$\int |k(x, y)| \, d\mu(x) \le c \text{ for a.e. } y \in Y, \quad \int |k(x, y)| \, d\nu(y) \le c \text{ for a.e. } x \in X.$$

Then for any $f \in L^p(\mu)$ and for almost all of x, $k(x, y) f(y) \in L^1(\nu)$, and

$$T(f)(x) = \int k(x, y) f(y) \, d\nu(y) \in L^p(\mu)$$

is a bounded linear operator on $L^p(\mu)$ and $\|T\| \le c$.

Proof
Case 1: $1 < p < \infty$.

We need to show that for any $f \in L^p(\mu)$, $T(f)$ is an a.e. defined function and $\|T(f)\|_p \le c\|f\|_p$. Let q be conjugate to p. Then by the Hölder inequality

$$\begin{aligned}\int |k(x,y)f(y)|\,d\nu(y) &= \int |k(x,y)|^{\frac{1}{q}}|k(x,y)|^{\frac{1}{p}}|f(y)|\,d\nu(y)\\ &\le \left(\int |k(x,y)|\,d\nu(y)\right)^{\frac{1}{q}}\left(\int |k(x,y)||f(y)|^p\,d\nu(y)\right)^{\frac{1}{p}}\\ &\le c^{\frac{1}{q}}\left(\int |k(x,y)||f(y)|^p\,d\nu(y)\right)^{\frac{1}{p}}\end{aligned}$$

for a.e. x. So by the TonelliŠs theorem,

$$\begin{aligned}&\int\left(\int |k(x,y)f(y)|\,d\nu(y)\right)^p d\mu(x)\\ &\le c^{\frac{p}{q}}\int\int |k(x,y)||f(y)|^p\,d\nu(y)d\mu(x)\\ &= c^{\frac{p}{q}}\int\int |k(x,y)|\,d\mu(x)|f(y)|^p d\nu(y)\\ &= c^{\frac{p}{q}+1}\int |f(y)|^p d\nu(y)\\ &= c^p\|f\|_p^p,\end{aligned}$$

which means that $\int |k(x,y)f(y)|\,d\nu(y) < \infty$ for a.e. x, i.e., $k(x,y)f(y) \in L^1(\nu)$ for a.e. x. Therefore $T(f)(x)$ is a.e. defined function and

$$\begin{aligned}\|T(f)\|_p^p &= \int\left(\left|\int k(x,y)f(y)\,d\nu(y)\right|\right)^p d\mu(x)\\ &\le \int\left(\int |k(x,y)f(y)|\,d\nu(y)\right)^p d\mu(x)\\ &\le c^p\|f\|_p^p,\end{aligned}$$

i.e., $T(f) \in L^p(\mu)$ and $\|T(f)\|_p \le c\|f\|_p$.

Case 2: $p = 1$.

For any $f \in L^1(\mu)$, by the TonelliŠs theorem,

$$\begin{aligned}&\int\int |k(x,y)f(y)|\,d\nu(y)\,d\mu(x)\\ &= \int\int |k(x,y)|\,d\mu(x)|f(y)|\,d\nu(y)\\ &\le c\int |f(y)|\,d\nu(y)\\ &= c\|f\|_1,\end{aligned}$$

which means that $\int |k(x,y)f(y)|\,d\nu(y) < \infty$ for a.e. x, i.e., $k(x,y)f(y) \in L^1(\nu)$ for a.e. x. Therefore $T(f)(x)$ is a.e. defined function, and

$$\begin{aligned}\|T(f)\|_1 &= \int \left| \int k(x,y)f(y)\,d\nu(y) \right| d\mu(x) \\ &\le \int \int |k(x,y)f(y)|\,d\nu(y)\,d\mu(x) \\ &\le c\|f\|_1,\end{aligned}$$

i.e., $T(f) \in L^1(\mu)$ and $\|T(f)\|_1 \le c\|f\|_1$.

Case 3: $p = \infty$.

For any $f \in L^\infty(\mu)$,

$$\begin{aligned}&\int |k(x,y)f(y)|\,d\nu(y) \\ &\le \int |k(x,y)|\,d\nu(y)\|f\|_\infty \\ &\le c\|f\|_\infty \quad \text{for a.e. } x \in X.\end{aligned}$$

So $k(x,y)f(y) \in L^1(\nu)$ for a.e. x, and

$$\begin{aligned}|T(f)| &= \left| \int k(x,y)f(y)\,d\nu(y) \right| \\ &\le \int |k(x,y)f(y)|\,d\nu(y) \\ &\le c\|f\|_\infty \quad \text{for a.e. } x \in X.\end{aligned}$$

Therefore $T(f) \in L^\infty(\mu)$ and $\|T(f)\|_\infty \le c\|f\|_\infty$. □

Next we prove the Minkowski inequality for integrals. To understand why it is called the Minkowski inequality for integrals, let's start with the Minkowski inequality

$$\left\| \sum_{i=1}^n f_i \right\|_p \le \sum_{i=1}^n \|f_i\|_p.$$

We can write $f_i(x) = f(x,i)$. Suppose we can extend it to infinite sum and represent the infinite sum as the integral over the counting measure ν on $\mathbb{N}$, then we have

$$\left\| \int f(x,i)\,d\nu(i) \right\|_p = \left\| \sum_{i=1}^\infty f_i \right\|_p \le \sum_{i=1}^\infty \|f_i\|_p = \int \|f(x,i)\|_p\,d\nu(i),$$

The Minkowski inequality for integrals simply says that it is true for any measure ν.

Theorem 6.3.3 (Minkowski Inequality for Integrals)
Let $(X, \mathscr{M}, \mu)$, $(Y, \mathscr{N}, \nu)$ be σ-finite measure spaces and $p \in [1, \infty]$, and let $\| \cdot \|_{p,\nu(x)}$ be the p-norm of a function of x-variable. Then for any positive $(\mathscr{M} \otimes \mathscr{N})$-measurable function or $\mu(x)$-a.e. $\nu(y)$-integrable function $f(x, y)$,

$$\left\| \int f(x, y)\, d\nu(y) \right\|_{p,\nu(x)} \leq \int \|f(x, y)\|_{p,\nu(x)}\, d\nu(y),$$

that is, for $1 \leq p < \infty$,

$$\left(\int \left| \int f(x, y)\, d\nu(y) \right|^p d\mu(x) \right)^{\frac{1}{p}} \leq \int \left(\int |f(x, y)|^p\, d\mu(x) \right)^{\frac{1}{p}} d\nu(y), \tag{6.3.3}$$

and for $p = \infty$,

$$\operatorname*{ess\,sup}_x \left| \int f(x, y)\, d\nu(y) \right| \leq \int \operatorname*{ess\,sup}_x |f(x, y)|\, d\nu(y).$$

Proof
We only need to prove it for positive $(\mathscr{M} \otimes \mathscr{N})$-measurable functions, because for $\mu(x)$-a.e. $\nu(y)$-integrable function $f(x, y)$

$$\left| \int f(x, y)\, d\nu(y) \right| \leq \int |f(x, y)|\, d\nu(y), \qquad \mu(x) - \text{a.e.}$$

and $|f(x, y)|$ is a positive $(\mathscr{M} \otimes \mathscr{N})$-measurable function.

Let f be a positive $(\mathscr{M} \otimes \mathscr{N})$-measurable function.

We first consider $| \leq p < \infty$. The inequality (6.3.3) is certainly true if the right-hand side is ∞; so without loss of generality, assume right-hand side of (6.3.3) is finite.

$p = 1$ is simply the Tonelli's theorem. For $1 < p < \infty$, let q be conjugate to p. For any $g \in L^q(\mu))$, by the Tonelli's theorem,

$$\int\int f(x, y)\, d\nu(y)|g(x)|\, d\mu(x) = \int\int f(x, y)|g(x)|\, d\mu(x)d\nu(y),$$

and by the Hölder inequality,

$$\int\int f(x, y)|g(x)|\, d\mu(x)d\nu(y) \leq \|g\|_q \int \left(\int |f(x, y)|^p\, d\mu(x)) \right)^{\frac{1}{p}} d\nu(y) < \infty.$$

So the map $\phi : L^q(\mu) \to \mathbb{R}$ defined by

$$\phi(g) = \int\int f(x, y)\, d\nu(y) g(x)\, d\mu(x)$$

is a well-defined (i.e., the integral over $\mu(x)$ is defined), bounded linear functional on $L^q(\mu)$, and

$$\|\phi\| \leq \int \left(\int |f(x, y)|^p\, d\mu(x)) \right)^{\frac{1}{p}} d\nu(y).$$

Then by the Riesz representation theorem

$$\left(\int \left| \int f(x, y)\, d\nu(y) \right|^p d\mu(x) \right)^{\frac{1}{p}} = \|\phi\| \leq \int \left(\int |f(x, y)|^p\, d\mu(x)) \right)^{\frac{1}{p}} d\nu(y).$$

For $p = \infty$, since

$$\int f(x, y)\, d\nu(y) \leq \int \operatorname*{ess\,sup}_x |f(x, y)|\, d\nu(y),$$

and the left-hand side is ≥ 0,

$$\operatorname*{ess\,sup}_x \left| \int f(x, y)\, d\nu(y) \right| \leq \int \operatorname*{ess\,sup}_x |f(x, y)|\, d\nu(y).$$

□

Theorem 6.3.4

Let $k(x, y)$ be Lebesgue measurable function on $(0, \infty) \times (0, \infty)$ such that $k(\lambda x, \lambda y) = \frac{1}{\lambda} k(x, y)$ for all $\lambda > 0$ and

$$\int_0^\infty \frac{|k(z, 1)|}{z^{1/p}}\, dx < \infty$$

for a $p \in [1, \infty]$. Let q be the conjugate exponent to p. Then for $f \in L^p(0, \infty)$ and $g \in L^q(0, \infty)$,

$$T(f)(y) = \int_0^\infty k(x, y) f(x)\, dx, \quad S(g)(x) = \int_0^\infty k(x, y) g(y)\, dy$$

are well-defined bonded linear functional on $L^p(0, \infty)$ and $L^q(0, \infty)$, respectively, and

$$\|T\| \leq \int_0^\infty \frac{|k(z, 1)|}{z^{1/p}}\, dz, \quad \|S\| \leq \int_0^\infty \frac{|k(z, 1)|}{z^{1/p}}\, dz.$$

Proof
Linearities are obvious. Since

$$\int_0^\infty \frac{|k(1,y)|}{y^{1/q}}\,dy = \int_0^\infty \frac{\left|k\left(\frac{1}{y},1\right)\right|}{y^{1+1/q}}\,dy = \int_0^\infty \frac{|k(z,1)|}{z^{1/p}}\,dz,$$

by symmetry, we only need to show that

$$\left(\int_0^\infty \left(\int_0^\infty |k(x,y)f(x)|\,dx\right)^p dy\right)^{\frac{1}{p}} \le \int_0^\infty \frac{|k(z,1)|}{z^{1/p}}\,dz\|f\|_p.$$

Because if the inequality is true, then $\int_0^\infty |k(x,y)f(x)|\,dx < \infty$ for a.e. y, so $T(f)(y)$ is an a.e. defined function, and $\frac{\|T(f)\|_p}{\|f\|_p} \le \int_0^\infty \frac{|k(z,1)|}{z^{1/p}}\,dz$ for all $\|f\|_p \ne 0$.

The case $p = \infty$ is left as a homework. We only show the cases $1 \le p < \infty$.

By changing variables and applying the Minkowski's inequality for integrals, we have

$$\begin{aligned}
&\left(\int_0^\infty \left(\int_0^\infty |k(x,y)f(x)|\,dx\right)^p dy\right)^{\frac{1}{p}} && z = x/y\\
&= \left(\int_0^\infty \left(\int_0^\infty |k(z,1)f(yz)|\,dz\right)^p dy\right)^{\frac{1}{p}}\\
&\le \int_0^\infty \left(\int_0^\infty |k(z,1)f(yz)|^p\,dy\right)^{\frac{1}{p}} dz && u = yz\\
&= \int_0^\infty \left(\int_0^\infty |k(z,1)f(u)|^p\,du\right)^{\frac{1}{p}} \frac{1}{z^{\frac{1}{p}}}\,dz\\
&= \int_0^\infty \frac{|k(z,1)|}{z^{1/p}}\,dz\|f\|_p
\end{aligned}$$

□

Corollary 6.3.5 *For $1 < p \le \infty$ and $1 \le q < \infty$, the operators*

$$T(f)(y) = \frac{1}{y}\int_0^y f(x)\,dx, \quad S(g)(x) = \int_x^\infty \frac{g(y)}{y}\,dy$$

are bounded linear operators on $L^p(0,\infty)$ and $L^q(0,\infty)$, respectively, and

$$\|T\| \le \frac{p}{p-1}, \quad \|S\| \le q.$$

Proof
Just apply Theorem 6.3.4 to $k(x,y) = \frac{1}{y}\chi_E(x,y)$, where $E = \{(x,y) : x < y\}$. Note that

$$\int_0^\infty \frac{|k(x,1)|}{x^{1/p}}\,dx = \int_0^1 \frac{1}{x^{1/p}} = \frac{p}{p-1} = q.$$

□

Exercises

6.18 Let $k(x, y)$ be Lebesgue measurable function on $(0, \infty) \times (0, \infty)$ such that $k(\lambda x, \lambda y) = \frac{1}{\lambda} k(x, y)$ for all $\lambda > 0$ and

$$\int_0^\infty |k(z, 1)|\, dz < \infty.$$

Prove that

$$T(f)(y) = \int_0^\infty k(x, y) f(x)\, dx$$

is a well-defined bonded linear functional on $L^\infty(0, \infty)$, and

$$\|T\| \leq \int_0^\infty |k(z, 1)|\, dz.$$

6.19 (Hilbert's Inequality) Let $1 < p < \infty$. Prove that the operator

$$T(f)(x) = \int_0^\infty \frac{f(y)}{x + y}\, dy$$

is a bounded linear operator on $L^p(0, \infty)$, and

$$\|T\| \leq \int_0^\infty \frac{1}{z^{\frac{1}{p}(z+1)}}\, dz.$$

6.20 Let $1 < p < \infty$, q be conjugate to p, and $k(x)$, $f(x)$, $g(x)$ be positive Lebesgue measurable functions on $(0, \infty)$. Prove that

$$\int_0^\infty \int_0^\infty k(xy) f(x) g(y)\, dxdy \leq \left(\int_0^\infty \frac{k(x)}{x^{1/q}}\, dx \right) \left(\int_0^\infty \frac{f(x)^p}{x^{2-p}}\, dx \right)^{1/p} \left(\int_0^\infty g(x)^q\, dx \right)^{1/q}.$$

(Hint: Apply the Hölder inequality to the y integral, change variable $u = xy$, apply the Minkowski's inequality for integrals, and then change variable $z = u/y$.)

6.21 Let $k(x)$ be a Lebesgue measurable function on $(0, \infty)$ such that $\int_0^\infty \frac{|k(u)|}{u^{1/2}}\, du < \infty$. Prove that

$$T(f)(x) = \int_0^\infty k(xy) f(y)\, dy$$

is a well-defined bounded linear operator on $L^2(0, \infty)$, and $\|T\| \leq \int_0^\infty \frac{|k(u)|}{u^{1/2}}\, du$. (Hint: Show $\left(\int_0^\infty \left(\int_0^\infty |k(xy) f(y)|\, dy \right)^2 dx \right)^{1/2} \leq \int_0^\infty \frac{|k(u)|}{u^{1/2}}\, du \|f\|_2$ by changing variable $u = xy$, applying the Minkowski's inequality for integrals, and changing variable $z = u/x$.)

6.22 (A generalized Hölder inequality) Let $1 \le p_j \le \infty$ and $\sum_{j=1}^n \frac{1}{p_j} = \frac{1}{r} \le 1$. Prove that for any measurable functions $\{f_j\}$,

$$\left\| \prod_{j=1}^n f_j \right\|_r \le \prod_{j=1}^n \|f_j\|_{p_j}.$$

6.4 Convolutions

Definition 6.4.1

Let f and g be measurable functions on $\mathbb{R}^n$. The convolution of f and g, denoted by $f * g$, is the function defined by

$$f * g(x) = \int f(x-y)g(y)\,dy$$

for all x where the integral exists.

Proposition 6.4.2 *Assuming that all integrals exist, then*

a) $f * g = g * f$.

b) $(f * g) * h = f * (g * h)$.

c) For any $s \in \mathbb{R}^n$, $\tau_z(f * g) = (\tau_z f) * g = f * (\tau_z g)$, *where* $\tau_z(f) = f(x - z)$.

d) $supp\,(f * g) \subset \overline{supp\,(f) + supp\,(g)}$.

Proof

a) Let $z = x - y$. Then $f * g(x) = \int f(x-y)g(y)\,dy = \int f(z)g(x-z)\,dz = g * f(x)$.

b) $(f * g) * h(x) = \int\int f(y)g(x-z-y)h(z)\,dydz = \int\int f(y)g(x-z-y)h(z)\,dzdy = f * (g * h)(x)$.

c) $\tau_z(f * g)(x) = \int f(x-z-y)g(y)\,dy = \int \tau_z f(x-y)g(y)\,dy = (\tau_z f) * g(x)$, and $\tau_z(f * g) = \tau_z(g * f) = (\tau_z g) * f = f * (\tau_z g)$.

d) For any $x \in \overline{(\text{supp}\,(f) + \text{supp}\,(g))^c}$, if $x = z + y$, then either $z \notin \text{supp}\,(f)$ or $y \notin \text{supp}\,(g)$. So $f(z)g(y) = f(x-y)g(y) = 0$ and $f * g(x) = \int f(x-y)g(y)\,dy = 0$, which implies that $x \in \{x : f * g(x) \neq 0\}^c$. Therefore $\{x : f * g(x) \neq 0\} \subset \overline{\text{supp}\,(f) + \text{supp}\,(g)}$ and $\text{supp}\,(f * g) = \overline{\{x : f * g(x) \neq 0\}} \subset \overline{\text{supp}\,(f) + \text{supp}\,(g)}$.

□

Theorem 6.4.3 (Young's Inequality)

Let $1 \le p, q, r \le \infty$ *and* $\frac{1}{p} + \frac{1}{q} = \frac{1}{r} + 1$. *Then*

$$\|f * g\|_r \le \|f\|_p \|g\|_q.$$

Proof

We prove the cases $1 < p, q, r < \infty$ here. The other cases are left as homework.

Note that by the given condition

$$\frac{1}{r} + \frac{r-p}{pr} + \frac{r-q}{qr} = 1.$$

By using the generalized Hölder inequality (Exercise 6.21), we have

$$\begin{aligned}
&|f * g(x)| \\
&\le \int |f(x-y)g(y)|\, dy \\
&= \int \left(|f(x-y)|^p |g(y)|^q\right)^{1/r} |f(x-y)|^{(r-p)/r} |g(y)|^{(r-q)/r}\, dy \\
&\le \left(\int |f(x-y)|^p |g(y)|^q\, dy\right)^{1/r} \left(\int |f(x-y)|^p\, dy\right)^{(r-p)/(pr)} \\
&\quad \left(\int |g(y)|^q\, dy\right)^{(r-q)/(pr)} \\
&= \left(\|f\|_p^{r-p} \|g\|_q^{r-q} \int |f(y)|^p |g(x-y)|^q\, dy\right)^{1/r}.
\end{aligned}$$

Therefore

$$\begin{aligned}
\|f * g\|_r &= \left(\int |f * g(x)|\, dx\right)^{1/r} \\
&\le \left(\|f\|_p^{r-p} \|g\|_q^{r-q} \int\int |f(y)|^p |g(x-y)|^q\, dy dx\right)^{1/r} \\
&= \left(\|f\|_p^{r-p} \|g\|_q^{r-q} \int\int |g(x-y)|^q\, dx |f(y)|^p dy\right)^{1/r} \\
&= \left(\|f\|_p^{r-p} \|g\|_q^{r-q} \|f\|_p^p \|g\|_q^q\right)^{1/r} \\
&= \|f\|_p \|g\|_q
\end{aligned}$$

□

Proposition 6.4.4 *If p and q are conjugate exponents, $f \in L^p$, and $g \in L^q$, then $f * g(x)$ exists for every x, $f * g$ is bounded and uniformly continuous, and $\|f * g\|_u \le \|f\|_p \|g\|_q$. If $1 < p, q < \infty$, then $f * g \in C_0(\mathbb{R}^n)$.*

Proof

By Hölder's inequality, for any x, $\int |f(x-y)g(y)|\, dy \le \|f\|_p \|g\|_q < \infty$. So $f * g(x)$ exists for every x. For any x,

$$|f * g(x)| \le \int |f(x-y)g(y)|\, dy \le \|f\|_p \|g\|_q$$

which implies that $\|f * g\|_u \le \|f\|_p \|g\|_q$. Also

$$\begin{aligned} |f * g(z) - f * g(x)| &= \left| \int (f(z-y) - f(x-y))g(y)\, dm^n \right| \\ &\le \int |f(z-y) - f(x-y))g(y)|\, dm^n \\ &\le \|\tau_{x-z}(f) - f\|_p \|g\|_q \to 0 \end{aligned}$$

uniformly as $z \to x$ by Proposition 6.1.9 (if $p = \infty$, then do translation on g).

For $1 < p, q < \infty$, there are $\{f_n\} \subset C_c(\mathbb{R}^n)$ and $\{g_n\} \subset C_c(\mathbb{R}^n)$ such that

$$\|f_n - f\|_p \to 0, \quad \|g_n - g\|_q \to 0, \quad \text{as } n \to \infty.$$

Then

$$\begin{aligned} &|f_n * g_n - f * g| \\ &= \left| \int f_n(y-x)g_n(y) - f(y-x)g(y)\, dy \right| \\ &\le \int |f_n(y-x)g_n(y) - f(y-x)g_n(y)|\, dy + \int |f(y-x)g_n(y) - f(y-x)g(y)|\, dy \\ &\le \|\tau_x(f_n) - \tau_x(f)\|_p \|g_n\|_q + \|\tau_x(f)\|_p \|g_n - g\|_q \\ &\le \|f_n - f\|_p (\|g_n - g\|_q + \|g\|_q) + \|f\|_p \|g_n - g\|_q \to 0 \end{aligned}$$

as $n \to \infty$. Therefore $f * g \in C_0(\mathbb{R}^n)$ by Proposition 4.4.7. □

Definition 6.4.5

For any $\alpha = (\alpha_1, \cdots, \alpha_n) \in \mathbb{N}^n$, define

$$\partial^\alpha = \left(\frac{\partial}{\partial x_1}\right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_n}\right)^{\alpha_n},$$

and

$$|\alpha| = \alpha_1 + \alpha_2 + \cdots \alpha_n.$$

By Theorem 2.3.9b, we have the following result.

Proposition 6.4.6 *If $f \in L^1$, $g \in C^k$, and $\partial^\alpha g$ is bounded for $|\alpha| \le k$, then $f * g \in C^k$ and $\partial^\alpha (f * g) = f * (\partial^\alpha g)$ for all $|\alpha| \le k$.*

Exercises

6.23 Prove that if f and g are Lebesgue measurable over $\mathbb{R}^n$, then $f(x-y)g(y)$ is Lebesgue measurable over $\mathbb{R}^{2n}$. (Hint: $f(x-y)g(y)$ is a composition of $f(u)g(v)$ with the linear transformation $u = x - y$, $v = y$.)

6.24 Prove that if f is integrable over every bounded subset of $\mathbb{R}^n$ and $g \in C^k$ has compact support, then $f * g \in C^k$.

6.25 Let $1 \le p \le \infty$ and q be conjugate to p. Prove that for any $f \in L^p(\mathbb{R})$ and $g \in L^q(\mathbb{R})$, $f * g$ exists, $f * g \in L^\infty(\mathbb{R})$, and

$$\|f * g\|_\infty \le \|f\|_p \|g\|_q.$$

6.26 Prove that if $f \in L^1$ and $g \in L^p$ for $1 \le p \le \infty$, then $f * g(x)$ exists for a.e. x, $f * g \in L^p$ and

$$\|f * g\|_p \le \|f\|_1 \|g\|_p.$$

(Hint: Theorem 6.3.2.)

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